A fixed-target approach to the sharp iid Berry–Esseen inequality*

Bartosz Kołodziejek Kacper Rodziewicz

4 November 2026

Abstract

We prove the sharp iid Berry–Esseen inequality with constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ for every sample size. The proof is computer-assisted: finite exact certificates establish the numerical inequalities on complete parameter regions. The normalized optimal constant is strictly below C_{E} for $n \geq 700$; Esseen's skew-Bernoulli limit makes C_{E} the sharp global constant.

The central idea is to measure distance from the extremal two-point law by a nonnegative support-gap defect. Attainment and the contact condition give a conservation identity relating this defect to an explicit remainder. Successive bounds confine a possible extremizer, and conditioning on the signs of its summands reduces the comparison to binomial counts with centered conditional noise. Polynomial majorants, a physical comparison at four summands and finite count comparisons establish a sharp base through 42. A one-step smoothing estimate places a first strict failure in the domain of either a finite or a uniform local comparison, both of which exclude it. The proof keeps the same marginal throughout these reductions, includes threshold atoms and closed parameter boundaries, and separates the analytic reductions from certificate construction and exact verification.

Contents

1 The problem, results, and proof strategy . . . 3
2 Fixed-target extremizers and contact . . . 7
3 The two-point skeleton and its moment defect . . . 10
4 Conservation and control of the defect . . . 14
5 Quantitative localization . . . 17
6 From small defect to a common support region . . . 19
7 The direct comparison with a binomial law . . . 23
8 What a first failing index must satisfy . . . 33
9 The starting cases . . . 35
10 A common finite count comparison . . . 39
11 The finite induction from five to fourteen . . . 43
12 The finite induction from fifteen to thirty-nine . . . 47
13 The three indices preceding the uniform local range . . . 51
14 A short bridge from a sharp finite base . . . 55
15 Why the intermediate indices reduce to finite local comparisons . . . 56
16 The sharp inequality and strictness above the finite range . . . 62
17 Discussion . . . 63
A A uniform Fourier enclosure with explicit input hypotheses . . . 64
B Elementary Gaussian constants . . . 69
C The identity without contact . . . 71

*This paper was written with substantial assistance from AI systems; see the end of Section 17.

D Details of the coarse localization . . . 71
E How the finite inequalities imply bounds for arbitrary laws . . . 85
F The four-summand physical comparison . . . 86
G A support criterion at a specified index . . . 95
H Support acquisition at the fifth index . . . 97
I Support at the actual indices six through fourteen . . . 99
J The common moment information in the Fourier bounds . . . 101
K Fourier refinements at a small fixed index . . . 107
L A five-summand corner without a support assumption . . . 110
M The finite-frequency sine bound used at indices 26–30 . . . 111
N Moment constraints in finite count comparisons . . . 113
O Finite acquisition for one fixed marginal . . . 118
P The fixed-index positive allowance at 39 . . . 125
Q One support estimate, uniform in the threshold . . . 127
R Entry as a resource enclosure . . . 132
S The minimal contact–Fourier foundation . . . 139
T Coefficient atlas for the enclosure . . . 147
U One confinement lemma with two concentration choices . . . 159
V One local comparison theorem . . . 165
W Noise and inward displacement share one budget . . . 174
X The sign of the cubic correction . . . 178
Y Keeping phase and resource information in the local comparison . . . 180
Z The signed Gaussian correction in nearest-count comparison . . . 183
AAA larger domain for paired Fourier cancellation . . . 185
ABCertificates and independent reproduction . . . 187

1 The problem, results, and proof strategy

1.1 The problem and the candidate constant

Let $\mathcal{P}$ be the laws on $\mathbb{R}$ with mean zero, variance one and finite third absolute moment $\rho(P) = \mathbb{E}|X|^3$. For independent copies $X_1, \dots, X_n$ of P, put $S_n = \sum_i X_i$ and

$$\Delta_n(P) = \sup_{H \in \mathbb{R}} |\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})|, \qquad C_n = \sup_{P \in \mathcal{P}} \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}.$$

Here Φ and ϕ are the standard normal distribution function and density. The sharp-constant conjecture is

$$\sup_{n \ge 1} C_n = C_{\mathrm{E}}, \qquad C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}} = 0.409732\ldots. \tag{1}$$

Throughout, H is a threshold for the unnormalized sum S_n, and a terminating decimal in an inequality denotes the corresponding exact rational number. Affine standardization gives the equivalent result for any mean and positive variance, with third absolute central moment divided by the cube of the standard deviation.

Berry and Esseen established the classical order $n^{-1/2}$ [3–5]. Esseen's lower bound [6] explains the proposed constant. If $p_* = (4 - \sqrt{10})/2$, $q_* = 1 - p_*$ and $X = (B - p_*)/\sqrt{p_* q_*}$ for $B \sim \mathrm{Bernoulli}(p_*)$, the fixed-law lattice limit gives

$$\lim_{n \to \infty} \frac{\sqrt{n}\Delta_n(P)}{\rho(P)} = \frac{3 + q_* - p_*}{6\sqrt{2\pi}(p_*^2 + q_*^2)} = C_{\mathrm{E}}. \tag{2}$$

Indeed $p_*^2 + q_*^2 = 10 - 3\sqrt{10}$ and $(3 + q_* - p_*)/(p_*^2 + q_*^2) = 3 + \sqrt{10}$. Every constant below C_{E} therefore fails for a sufficiently large sum of this one law. The upper conjecture is associated with Zolotarev [20, 21]; Schulz proves the strict finite-index bound for every nondegenerate Bernoulli law [12].

1.2 Results and their hypotheses

Theorem 16.1 proves $C_n \le C_{\mathrm{E}}$ for every $n \ge 1$, with strict inequality for $n \ge 700$. The Bernoulli limit above proves that the global constant cannot be lowered. The proof uses three parts.

Part	Conclusion and role
Finite induction through 42	The first four indices use polynomial majorants and a physical comparison. For $5 \le n \le 42$, actual-index support and Fourier bounds acquire the hypotheses of finite count comparisons. Each step uses sharp earlier indices and retains one common marginal throughout.
First-failure comparison, Theorem 14.3	At a least index with $C_n > C_{\mathrm{E}}$, a one-step smoothing estimate gives $n(C_n - C_{\mathrm{E}}) < 0.3$, sharp deleted predecessors and zero predecessor loss. A finite comparison excludes $43 \le n \le 48$; a uniform comparison excludes every $n \ge 49$. This proves the weak bound for all n.
Compact strict comparison, Theorem 16.2	For $n \ge 700$, a fixed-target maximizer with nonnegative excess would have $nD \le 32.20665378$ and $\lvert H\rvert \le 1.83605680$. These bounds place it in an explicit conditional-support region where the binomial comparison is strict, including the zero-noise case.

The small-index and localization calculations are computer-assisted. Their proofs combine continuous analytic bounds, exact inequalities on closed parameter regions, and checks that

successive bounds concern the same law. Appendix AB identifies the inputs, complete executions and acceptance procedures. The published structural and finite-binomial results used in the compact comparison are stated at (74) and (29), with their hypotheses and references.

There is no assumption of a sharp bound through 699. The finite induction stops at 42; the first-failure argument supplies all later indices at once. The separate compact comparison proves strictness from 700 onward. Nor is eventual two-point optimality asserted: the proof bounds every possible extremal witness without replacing an unrestricted law by a two-point law.

1.3 Why the defect controls a maximizer

At a fixed n, maximize the difference between the two sides of the candidate inequality:

$$a(P,H)=\sqrt{n}\{\mathbb{P}_P(S_n\le H)-\Phi(H/\sqrt{n})\}-C_{\mathrm{E}}\rho(P). \tag{3}$$

Proposition 2.1 proves attainment. Its first-order condition bounds $y\mapsto\mathbb{P}(S_{n-1}\le H-y)$ by an explicit piecewise cubic function that agrees with it on the closed support. This contact condition allows us to use the geometry of that support.

Let $\tau(t)=\mathbb{E}[X\mathbf{1}_{\{X\ge t\}}]$ be the zero-bias density and $J=\sum_g\tau_g|g|^2$, where g runs over bounded support gaps and τ_g is the constant value of τ there. A lattice law of span h has $J\ge h$; if every successive gap has length h, then $J=h$. Thus J measures gap size with zero-bias weights. The inequality

$$\gamma+3J\le(3+\sqrt{10})\rho,\qquad \gamma=\mathbb{E}X^3,$$

proved in Lemma 3.1, extends the moment–span bound from lattice laws to arbitrary standardized laws. Its slack

$$D=(3+\sqrt{10})\rho-\gamma-3J$$

is nonnegative and vanishes exactly at the positive-skew Bernoulli law in (2). The lattice consequence and its reflection are proved in Corollary 3.3.

Integrating the contact relation against τ in two ways gives the conservation identity of Theorem 4.1. Since $C_{\mathrm{E}}=(3+\sqrt{10})\phi(0)/6$, its two leading moment terms combine into a multiple of D. Solving for the excess gives

$$a=\frac{\mathcal{R}_c-nC_{\mathrm{E}}\rho(1-e)-n\phi(0)eD/6-n^{3/2}L}{n+H^2},\qquad e=\exp(-H^2/(2n)).$$

Here $L\ge0$ is a weighted contribution from the law of $H-S_{n-1}$ inside the support gaps, and $\mathcal{R}_c$ is an explicitly defined remainder. The three subtracted terms are nonnegative. A nonnegative excess therefore requires $n\phi(0)eD/6\le\mathcal{R}_c$. Once localization bounds both $\mathcal{R}_c$ and $|H|$, this bounds nD. The uniform bounds come from Theorem 5.1, not from a fixed-law asymptotic expansion.

The predecessor hypothesis supplies an upper bound for $\mathbb{P}(S_{n-1}\le H-t)$ in the contact condition, which excludes remote support points; see (76). This support argument is used both in coarse localization and in the final two-ray estimate. With $nD\le32.20665378$, the geometry at $n\ge700$ gives $3/8<p<93/200$, inward support displacement below $1/250$ and $\lambda<7/20$. Count conditioning then compares the law with a binomial sum. One threshold count controls noise up to $1/100$; five neighboring counts control the remaining interval. Both comparisons keep all probabilities entering from other counts.

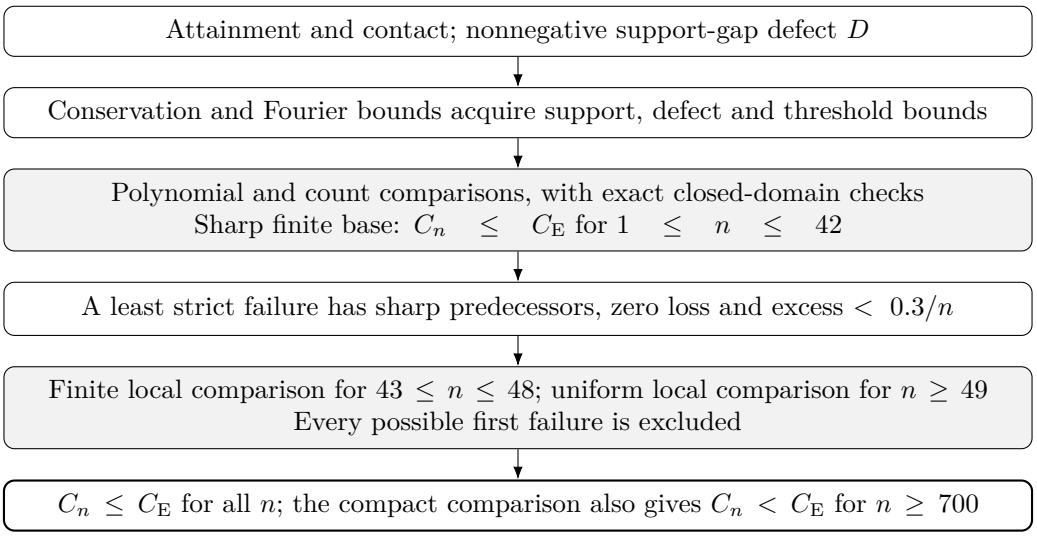


Figure 1: The proof's main dependencies. The shaded comparisons combine analytic inequalities with complete exact finite checks. The uniform comparison is proved for every index in its range; it is not an extrapolation from the finite calculation.

1.4 What the certificates prove

A certificate is finite exact data for a specified mathematical inequality, together with rules for checking those data. Here the claim is always about an entire region of possible laws or parameters. It is not a claim about only the distributions or points tried during a numerical search.

The proof uses three forms of certificate. For the first few indices, rational polynomial coefficients majorize the closed sum event on a whole support product; independence expresses the expectation through moments of one common marginal. For localization and Fourier bounds, rational boxes cover the complete admissible parameter region. A box is either excluded by a necessary inequality or assigned bounds valid everywhere on it. For the count comparisons, a finite tree subdivides the acquired probability, phase and noise region until each terminal cell has a strict score bound or a proved contradiction. The checker verifies the initial domains, every subdivision and all terminal inequalities, including shared boundaries. Thus every possible witness is accounted for.

Finding these data and verifying them are separate tasks. A search can choose polynomial coefficients, subdivisions, prices or refinement passes. It may use floating-point calculations to suggest those choices. Acceptance uses the resulting exact rational data and checks every claimed inequality; a successful search or a saved success message has no independent authority. Some of the present candidates are supplied objects whose complete acceptance is repeated. Other tables are constructed and then recomputed. The reproduction record distinguishes these cases. Where the constructor and acceptor share numerical routines, this is described as exact replay, not as an independent implementation of the same calculation.

Acceptance uses integer and rational arithmetic. Roots, exponentials, trigonometric functions and Gaussian probabilities are enclosed by rational bounds with explicit analytic remainders. The main checking programs use Python integers; the three-summand Bernstein checker uses C++ with Boost's arbitrary-precision integers. No floating-point approximation is accepted as an exact sign test. The trusted implementation includes the checking programs, their arithmetic libraries and their execution environment; this is a computer-assisted proof, not a proof-assistant formalization.

A passed certificate still needs its mathematical hypotheses. For example, a count

certificate can establish a bound for every law in a prescribed support and moment region, while a separate localization argument proves that the same hypothetical maximizing law lies in that region. The resource-to-count checks verify this implication without changing the index, the marginal, or the discrepancy orientation. Continuous arguments such as attainment, contact, Fourier inversion and the least-failure induction are proved in the text; they do not follow from an arithmetic success flag.

The appendices give the inequalities implemented by the checkers and the rules for their domain coverage. Appendix AB maps the full argument to its numerical families and explains the source package, commands and execution records. An independent verification needs these derivations, the exact inputs, the complete acceptors and the checks matching each output to the next theorem's hypotheses. Hashes identify the files used; they do not prove the inequalities.

Subdivision and rigorous enclosure have established precedents in computer-assisted proofs [16]. Numerical optimization also appears in Berry–Esseen estimates [13, Section 2]. The custom JSON files organize this proof's exact data; their format is not a standard theorem-proving language. The mathematical justification comes from the inequalities, complete coverage and correct implementation of the checking rules.

1.5 Related work

Fourier and structural moment inequalities progressively improved the universal upper bound [10, 13, 15, 17, 18]. We use Shevtsova's structural inequality explicitly in localization. Recent manuscripts report smaller upper constants [9, 19], and He and Cheng report eventual sharpness beyond a uniform, very large threshold [7]. Work on independent arrays and extremizers addresses the related non-iid problem [1, 2]. These recent reported results are not used in our proof. An earlier AI-assisted working draft is on Hexagon [11].

1.6 Organization and notation

Section 2 gives attainment, contact and reflection. The next two sections develop the defect and the conservation identity. These are the shared analytic tools. The remaining argument has two routes. The compact route in Sections 5–7 gives strict exclusion at $n \geq 700$ from a sharp predecessor. The extension supplies a finite induction and a different local comparison that applies at a first strict failure below that threshold as well.

For the logic of the extension, read the one-step estimate in Section 8 and the first-failure argument in Section 14 before the numerical tables. Their conclusion requires a sharp base through 42 and two local exclusions. Section 9 treats the first four indices; Sections 10–13 describe the finite induction through 42. Section 15 explains the two local exclusions. The continuous estimates and full numerical interfaces are in the appendices. The assembly criteria state the numerical premises explicitly, so each can be matched to its reproduction record.

Two optimization problems are used. The finite induction maximizes the ratio, with attained score $C = C_n$. The compact tail argument maximizes the excess above the fixed coefficient $c = C_{\mathrm{E}}$. Both give contact, but their excesses and normalization differ; Corollary 2.2 records the precise relation. Neither route changes the marginal law while acquiring its successive bounds.

The following table distinguishes the three scales used in the proof. Gap-specific parameters are introduced after the existence of a central gap has been established.

Symbol	Meaning and scale
H, $H/\sqrt{n}$	Threshold for the original sum; its normal-standardized value.
$B_0 = \mathbf{1}_{\{X>0\}}$, p, q	Sign class and its probabilities for the original marginal.
m, $\ell = m/(pq)$	$m = \mathbb{E}X_+ = \mathbb{E}X_-$; distance between conditional means.
W, $\theta = \mathbb{E}W^2$	$W = X - \mathbb{E}(X \mid B_0)$; conditional noise in original units.
$U = W/\ell$, $s = \mathbb{E}U^2$, $\lambda = ns$	Noise in count units; marginal and total noise variances.
$t = np + H\sqrt{pq+s} = k + u$	Count threshold, integer anchor, and phase.
a_0, a_1, $d = a_0 + a_1$	Inward support displacements in count units.
J, D	Weighted support-gap sum and defect of the original law.

The exact normalization is proved in (6). It identifies the count-coordinate law with the original X and leaves H as its raw sum threshold. The signed moments are $\gamma = \mathbb{E}X^3$ and $\beta = \mathbb{E}[X|X|]$.

2 Fixed-target extremizers and contact

We maximize the difference between the two sides of the desired inequality. Its sign detects failure of the target bound, while the coefficient of the third absolute moment is fixed in advance. Throughout, sums have independent identically distributed summands with mean zero, variance one and finite third absolute moment ρ. Fix $n \geq 2$, $c > 0$, and a real raw sum threshold H. Set

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \leq H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho,$$
$$\phi_z = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

In particular a is not divided by ρ.

2.1 Attainment, contamination and the entire closed support

Proposition 2.1. *For each fixed n and $c > 0$, the supremum of a over these laws and thresholds is attained, and the closed-CDF discrepancy at every attaining pair is positive. At every maximizing pair,*

$$I(y) = n^{3/2}\{h(y) - F\} + ny\phi_z + \tfrac{1}{2}H\phi_z(y^2-1) + \tfrac{3}{2}c\rho(y^2-1) - c\{|y|^3 - \rho - 3y\beta\} \leq 0, \qquad h(y) = \mathbb{P}(S_{n-1} \leq H - y), \tag{4}$$

with equality on the whole closed support. That support is compact. Neither statement assumes $a \geq 0$ or $H \geq 0$.

Equivalently, $\Delta = Q - h \geq 0$ everywhere and $\Delta = 0$ on the closed support, where

$$Q(y) = F + \frac{H\phi_z + c\rho}{2n^{3/2}} - \frac{n\phi_z + 3c\beta}{n^{3/2}}y - \frac{H\phi_z + 3c\rho}{2n^{3/2}}y^2 + \frac{c}{n^{3/2}}|y|^3. \tag{5}$$

The contact condition is a first-order optimality condition: replacing one summand by a fixed value y gives the conditional probability $h(y)$. After recentering and rescaling the perturbed law, the directional change of the excess is nonpositive. Its average against P is zero, which forces equality on the closed support. The function Q is piecewise cubic because it contains $|y|^3$.

Proof. Write a^* for the supremum. Since $\rho \geq 1$ and the discrepancy is at most one, $a^* \leq \sqrt{n} - c$. Symmetric Bernoulli evaluated at $H = n$ gives $a^* \geq \sqrt{n}(1 - \Phi(\sqrt{n})) - c > -c$. For a maximizing sequence, $a_j \to a^*$ and $c\rho_j \leq \sqrt{n} - a_j$ give a common third-moment ceiling. The

discrepancies satisfy $\sqrt{n}(F_j - \Phi(H_j/\sqrt{n})) = a_j + c\rho_j \geq a_j + c > \eta$ eventually for some $\eta > 0$. Chebyshev bounds the negative threshold tail by n/H_j^2; the positive discrepancy is at most the normal tail, also at most n/H_j^2. Thus H_j is bounded.

Tightness gives a subsequence $P_j \Rightarrow P$, $H_j \to H$ and $\rho_j \to b$. The common third-moment ceiling M gives tail bounds $\int_{|x|>K} x^2 dP_j \leq M/K$ and $\int_{|x|>K} |x| dP_j \leq M/K^2$. Bounded continuous truncations therefore preserve mean and variance. Lower semicontinuity gives $1 \leq \rho(P) \leq b$. For fixed n, the sum laws converge weakly. Applying the closed-set inequality first at $H + t$ and then letting $t \downarrow 0$ gives $\limsup F_j(H_j) \leq F_P(H)$, including an atom at H. Consequently

$$a(P, H) \geq a^* + c\{b - \rho(P)\} \geq a^*.$$

Maximality forces equality and $\rho(P) = b$. Thus third moments converge along this maximizing sequence. The discrepancy remains positive since $a^* + c\rho \geq a^* + c > 0$.

The contact inequality is a first-variation condition. To obtain it, contaminate the raw marginal by $(1 - \epsilon)P + \epsilon\delta_y$. Its mean is ϵy, variance $v_\epsilon = 1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$, and centered third absolute moment has derivative $|y|^3 - \rho - 3y\beta$ at zero. The last assertion follows by dominated differentiation of $|x - \epsilon y|^3$, with bound $3|y|(|x| + |y|)^2$. Evaluate the standardized law at the standardized image of the fixed raw event $S_n \leq H$. The raw event probability is a polynomial in ϵ, with derivative $n(h(y) - F)$ even at an atom collision. The normal argument is $(H - n\epsilon y)/\sqrt{nv_\epsilon}$, whose derivative is $-\sqrt{n}y - H(y^2 - 1)/(2\sqrt{n})$. The one-sided derivative of the excess is exactly (4); maximality makes it nonpositive.

Its integral against P is zero. All terms are integrable. The closed-event map h is upper semicontinuous; at an atom it takes the larger one-sided value. Thus I is upper semicontinuous. Strict negativity at a closed-support point would hold with a uniform negative margin on a neighborhood of positive P-mass, contradicting its zero integral. Finally $I(y) \leq -c|y|^3 + C_2 y^2 + C_1|y| + C_0$ for constants depending on this pair. It is strictly negative on both sufficiently distant rays, proving compact support. □

2.2 Reflection, equality, and the first unresolved index

The sign of the attained maximum detects failure of the strict bound $C_n < c$, including equality. If C_n denotes the supremum of the normalized absolute discrepancy, reflection gives, including an atom at H,

$$\mathbb{P}(-S_n \leq -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \geq \Phi(z) - F.$$

Thus each negative closed-CDF error is bounded by a positive closed-CDF error of the reflected law; equality at an atom is not assumed. Consequently C_n is also the supremum of positive closed scores r. Since $a = \rho(r - c)$, $C_n < c$ gives $a^* \leq C_n - c < 0$, and $C_n > c$ gives $a^* > 0$. If $C_n = c$, choose $r_j \uparrow c$ with $r_j \geq c/2$. Then $\rho_j \leq 2\sqrt{n}/c$, so $\rho_j(r_j - c) \to 0$ and $a^* = 0$. Hence $C_n \geq c$ supplies an attained pair with $a \geq 0$.

Corollary 2.2 (Ratio extremizers from fixed-target attainment)**.** *For every $n \geq 2$, the constant C_n is attained as a positive closed-CDF score by a compactly supported standardized law and a raw threshold H. This pair maximizes the fixed-target excess at $c = C_n$, with maximum zero. In particular, it satisfies* (4) *with $c = C_n$.*

Proof. The inequality $\rho \geq 1$ gives $C_n \leq \sqrt{n}$. Symmetric Bernoulli at $H = n$ gives $C_n \geq \sqrt{n}\{1 - \Phi(\sqrt{n})\} > 0$. Set $c = C_n$ in the preceding argument. Every excess is nonpositive, and scores $r_j \uparrow C_n$ with $r_j \geq C_n/2$ satisfy $\rho_j \leq 2\sqrt{n}/C_n$, so their excesses tend to zero. Proposition 2.1 attains this zero maximum. At an attaining pair, $\rho(r - C_n) = 0$ and $\rho > 0$, hence $r = C_n$. The same proposition gives compact support and contact. □

This corollary provides the extremizers used by the finite-index arguments. Their target coefficient is the unknown extremal ratio C_n, whereas the large-index argument fixes the coefficient at C_{E}. The contact expressions agree after the normalization $I_{\mathrm{fixed}} = \rho I_{\mathrm{ratio}}$ at $c = C_n$. An extremizer of the ratio need not maximize the excess at C_{E}: its excess there is $\rho(C_n - C_{\mathrm{E}})$. Each estimate below specifies which of these two optimization problems it uses.

Corollary 2.3 (First failure of a strict bound)**.** *Assume a genuine finite base $C_j \le c$ for $1 \le j < N$. If a strict bound $C_n < c$ fails at any index $n \ge N$, there is a least such index, an attained fixed-target pair with $a \ge 0$, and predecessor bounds $C_j \le c$ for every $j < n$. The bounds are strict for $N \le j < n$.*

Proof. The nonempty set of failing indices at least N has a least member. The preceding equivalence supplies its maximizing pair. Indices below N use the finite base; the other earlier indices use minimality. □

Thus an argument excluding every such pair for $n \ge N$ would give strict bounds above the finite base, including exclusion of equality. The subsequent entry argument uses only $C_{n-1} \le c$; it does not use an upper bound for the actual score $c + a/\rho$. The finite base and the eventual exclusion remain separate obligations.

2.3 Two elementary moment domains

These two inequalities restrict the signed moments in terms of ρ. They hold for every standardized law with finite third moment, including laws of unbounded support.

Theorem 2.4 (Signed and absolute third moments)**.** *If $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, and $\gamma = \mathbb{E}X^3$, then*

$$\rho \ge \frac{\gamma^2 + 2}{\sqrt{\gamma^2 + 4}}.$$

This is the sharp moment relation of Shevtsova [14]; in the cited arXiv version its equivalent form is Theorem 2. The following factorization gives a self-contained proof.

Proof. Set $s = \sqrt{\gamma^2 + 4}$, $a = (s - \gamma)/2$, $b = (s + \gamma)/2$ and $c_\gamma = \gamma(\gamma^2 + 6)/s^3$. Then $a, b > 0$, $ab = 1$ and $a + b = s$. Define

$$w(y) = |y|^3 - c_\gamma y^3 - \frac{12y^2}{s^3} + \frac{6\gamma y}{s^3} + \frac{4}{s^3}.$$

For $y, t \ge 0$, direct multiplication gives

$$w(y) = \frac{2a^2}{s^3}(y - b)^2\{2 + (a + 3b)y\}, \qquad w(-t) = \frac{2b^2}{s^3}(t - a)^2\{2 + (3a + b)t\}.$$

Thus $w \ge 0$ on the whole real line. Taking expectations and using the first three signed moments gives

$$0 \le \mathbb{E}w(X) = \rho - c_\gamma\gamma - 8/s^3 = \rho - (\gamma^2 + 2)/s.$$

All terms are integrable under the stated third-moment hypothesis. □

Lemma 2.5 (The signed second moment)**.** *With the same normalization and $\beta = \mathbb{E}[X|X|]$,*

$$|\beta| < 1, \qquad \rho \ge \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}.$$

Proof. Both signs have positive probability. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. If $p_\pm = \mathbb{P}(\pm X > 0)$, Cauchy's inequality gives $m^2 \le p_\pm a_\pm$. Since $p_+ + p_- \le 1$, this implies

$$m^2(1/a_+ + 1/a_-) \le 1, \qquad 2m \le \sqrt{1-\beta^2}.$$

Another application of Cauchy's inequality, to $X_\pm^{1/2}$ and $X_\pm^{3/2}$, gives $m\mathbb{E}X_\pm^3 \ge a_\pm^2$. Summing yields $m\rho \ge (1+\beta^2)/2$, which proves the claim. A possible zero atom enters only through $p_+ + p_- \le 1$ and causes no exception. □

3 The two-point skeleton and its moment defect

Every centered law can be separated into its positive and nonpositive parts. Replacing each part by its conditional mean gives a two-point law; the remaining conditional fluctuations measure its departure from that law. We first make this decomposition exact, then identify a moment defect that vanishes at Esseen's extremal distribution.

3.1 The sign split and its units

Fix $P \in \mathcal{P}$ and put

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad B_0 = \mathbf{1}_{\{X>0\}}.$$

Both p, q are positive; a possible atom at zero is assigned to the nonpositive class. Define $V = \mathbb{E}[X \mid B_0]$ and $W = X - V$, so that

$$V = -m/q \quad (B_0 = 0), \qquad V = m/p \quad (B_0 = 1), \qquad \theta = \mathbb{E}W^2 = 1 - m^2/v < 1.$$

Write $\theta_- = \mathbb{E}[W^2; B_0 = 0]$ and $\theta_+ = \mathbb{E}[W^2; B_0 = 1]$. The distance between the two conditional means is

$$\ell = m/v = \sqrt{(1-\theta)/v} > 0.$$

Set $U = W/\ell$, $Y = B_0 + U$, and $s = \mathbb{E}U^2$. Then

$$X = \ell(Y - p) = \frac{B_0 + U - p}{\sqrt{pq + s}}, \qquad \mathbb{E}[U \mid B_0] = 0, \tag{6a}$$

$$1 = pq\ell^2 + \theta, \qquad s = \theta/\ell^2, \qquad v + s = \ell^{-2}. \tag{6b}$$

Let U_0, U_1 have the two conditional laws, and put

$$w_i = \mathbb{E}U_i^2, \quad \mu_i = \mathbb{E}U_i^3, \quad s = qw_0 + pw_1 = \frac{v\theta}{1-\theta}, \quad \lambda = ns.$$

The moments μ_i are finite because $\rho < \infty$. Higher moments will be used only after bounded support is known: for extremizers this follows from Proposition 2.1, and for the local comparison it is an explicit hypothesis.

If $s = 0$, both nonnegative conditional variances vanish, so $U_0 = U_1 = 0$ almost surely. This is exactly the standardized Bernoulli law. If $s > 0$, the parameter $\alpha = pw_1/s$ lies in $[0, 1]$ and gives $w_0 = s(1-\alpha)/q$, $w_1 = s\alpha/p$.

The signs of X are still those defining the two classes, so expansion of the actual conditional moments gives

$$\begin{aligned}
\rho/\ell^3 &= v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1, \\
\gamma/\ell^3 &= v(q-p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1, \\
\beta/\ell^2 &= v(q-p) + pw_1 - qw_0.
\end{aligned} \tag{7}$$

For example, on the nonpositive class $|X|^3 = \ell^3(p-U_0)^3$; on the positive class it is $\ell^3(q+U_1)^3$. Conditional centering removes the linear terms and gives the first identity; the other two follow in the same way. The three identities share the same pair (μ_0, μ_1).

3.2 A nonnegative moment defect

The constant $3+\sqrt{10}$ has a simple geometric explanation. The signed third moment and the first absolute moment satisfy a circle inequality; the lengths of the support gaps are controlled by the same first absolute moment. Neither fact requires bounded support.

Define

$$\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X\geq t\}}], \qquad \sigma = \sqrt{10}.$$

For $t<0$, centering gives $\tau(t) = -\mathbb{E}[X\mathbf{1}_{\{X<t\}}] \geq 0$; for $t \geq 0$ nonnegativity is immediate. Signwise Tonelli integration gives

$$\int_{\mathbb{R}} \tau(t)\,dt = 1, \qquad 2\int_{\mathbb{R}} |t|\tau(t)\,dt = \rho. \tag{8}$$

On each bounded complementary interval (a,b) of the closed support, τ is constant in the interior, with value $\tau_{a,b}$. Put

$$J = \sum_{(a,b)} \tau_{a,b}(b-a)^2, \qquad D = (3+\sigma)\rho - \gamma - 3J. \tag{9}$$

There are at most countably many gaps; the next proof shows that the nonnegative sum is finite. Endpoint atoms do not affect its integrals.

Lemma 3.1 (A moment circle and the gap inequality)**.** *For every $P \in \mathcal{P}$,*

$$\gamma^2 + (2m)^2 \leq \rho^2, \qquad J \leq \rho + 2m, \qquad \gamma + tJ \leq \bigl(t+\sqrt{1+t^2}\bigr)\rho \quad (t>0). \tag{10}$$

The last constant is sharp. Equality holds exactly at the standardized Bernoulli law with positive mass $p_t = (1+t-\sqrt{1+t^2})/2$. In particular, $D \geq 0$. If zero belongs to the support, then $D \geq (\sigma-1)\rho$; hence $D < (\sigma-1)\rho$ implies a central support gap $(-A_-, A_+)$ with $A_-, A_+ > 0$.

Proof. Write $a_\pm = \mathbb{E}X_\pm^2$ and $b_\pm = \mathbb{E}X_\pm^3$, so $a_+ + a_- = 1$, $b_+ + b_- = \rho$ and $b_+ - b_- = \gamma$. Cauchy–Schwarz on each sign class gives $a_\pm^2 \leq m b_\pm$. It also gives $m^2 \leq p a_+$ and $m^2 \leq q a_-$, where an atom at zero is included in q. Thus $m^2(1/a_+ + 1/a_-) \leq 1$, or $m^2 \leq a_+ a_-$. Consequently $b_+ b_- \geq (a_+a_-)^2/m^2 \geq m^2$, proving the circle inequality since $\rho^2 - \gamma^2 = 4b_+b_-$.

Every gap on one side of zero satisfies $(b-a)^2 \leq 2\int_a^b |x|\,dx$. If a central gap $(-A_-, A_+)$ exists, its weight is m and its excess over this integral is $2mA_-A_+$. Moreover $0 \leq \mathbb{E}[(X+A_-)(X-A_+)] = 1 - A_-A_+$. Summing the gap bounds by Tonelli proves $J \leq \rho + 2m$. If zero is in the support there is no central gap, so $J \leq \rho$. Finally Cauchy–Schwarz in $\mathbb{R}^2$ gives $\gamma + 2tm \leq \sqrt{1+t^2}\rho$, proving the family and the assertions about D.

Equality in the circle forces equality in the two signwise Cauchy bounds: the positive and negative parts are each concentrated at one nonzero value. Equality in $m^2 \leq a_+a_-$ excludes mass at zero. For such a law, with positive mass p and $q = 1-p$, one has $J = \rho + 2m$. Equality in the last Cauchy bound requires $\gamma = 2m/t$, equivalently $t(1-2p) = 2p(1-p)$. Its unique solution in $(0,1)$ is the stated p_t. □

Corollary 3.2 (The zero-defect law)**.** *For $P \in \mathcal{P}$, $D = 0$ if and only if*

$$X = (B_0 - p_*)/\sqrt{p_*q_*}, \qquad p_* = (4-\sqrt{10})/2, \quad q_* = 1-p_*, \quad B_0 \sim \text{Bernoulli}(p_*).$$

Proof. Apply the equality case of Lemma 3.1 with $t = 3$. □

Corollary 3.3 (The lattice consequence)**.** *Let $P \in \mathcal{P}$ be a lattice law with maximal span $h > 0$. Then*

$$J \geq h, \qquad |\gamma| + 3h \leq (3 + \sqrt{10})\rho.$$

Equality in the second inequality holds exactly for the law of Corollary 3.2 and its reflection. If successive support points are all separated by h, then $J = h$.

Proof. The support is a closed subset of a translate of $h\mathbb{Z}$, so its points have Lebesgue measure zero and every bounded gap has length $k_g h$ for an integer $k_g \geq 1$. The density τ vanishes outside the convex hull of the support. Hence, by nonnegative integration,

$$\sum_g \tau_g k_g h = \int_{\mathbb{R}} \tau(t)\, dt = 1, \qquad J = \sum_g \tau_g k_g^2 h^2 \geq h.$$

These identities also hold for unbounded support: the gaps are countable and their contributions increase to the full integral. If all $k_g = 1$, the inequality is an equality. Under reflection the zero-bias density becomes $\tau(-t)$ almost everywhere. Thus reflection preserves ρ, J and h, and changes the sign of γ. Apply Lemma 3.1 to the orientation with nonnegative third moment. Equality forces $D = 0$, and the equality law has $J = h$, so Corollary 3.2 proves both directions. □

For a lattice law, J is therefore a weighted average of support-gap lengths, with weights $\tau_g|g|$ summing to one. The same functional is defined for arbitrary support, which is why it can replace the lattice span in the defect inequality.

In particular the defect separates two losses:

$$D = \underbrace{\sigma\rho - \gamma - 6m}_{\text{moment loss}} + 3\underbrace{(\rho + 2m - J)}_{\text{gap loss}},$$

each nonnegative. The point $(\gamma/\rho, 2m/\rho)$ lies in the upper unit disk. The line $x + 3y = \sqrt{10}$ supports that disk at $(1/\sqrt{10}, 3/\sqrt{10})$, explaining the coefficient in the moment loss. For a standardized Bernoulli law with positive mass p, direct calculation gives

$$J = \frac{1}{\sqrt{pq}}, \qquad \rho_B(p) = \frac{1 - 2pq}{\sqrt{pq}},$$
$$\gamma_B(p) = \frac{q - p}{\sqrt{pq}}, \qquad D_B(p) = \frac{(6 + 2\sigma)(p - p_*)^2}{\sqrt{pq}}. \tag{11}$$

Indeed its zero-bias density is $\sqrt{pq}$ on the interval between its two support points. The probability displacement therefore has a quadratic cost. The equality proof above is what rules out other laws with zero defect.

For quantitative estimates we need more information than their signs. Suppose a central gap $(-A_-, A_+)$ has been established, and write

$$T_+ = \mathbb{E}X_+^3 - mA_+^2, \quad T_- = \mathbb{E}X_-^3 - mA_-^2, \qquad R_+ = \mathbb{E}X_+^2 - mA_+, \quad R_- = \mathbb{E}X_-^2 - mA_-.$$

These four quantities are nonnegative. For the outer gaps put $J_o = J - m(A_- + A_+)^2$. A positive outer gap satisfies $(b - a)^2 \leq 2\int_a^b (t - A_+)\, dt$, so Tonelli gives

$$J_o^+ \leq 2\int_{A_+}^{\infty} (t - A_+)\tau(t)\, dt = \mathbb{E}[X_+(X_+ - A_+)^2] = T_+ - 2A_+R_+ \leq T_+.$$

Similarly $J_o^- \le T_- - 2A_-R_- \le T_-$. Expanding D, using $\sqrt{\sigma+1}\sqrt{\sigma-1} = 3$, gives

$$D = m\{\sqrt{\sigma+1}A_- - \sqrt{\sigma-1}A_+\}^2 + (\sigma-1)T_+ + (\sigma+1)T_- + 3(T_+ + T_- - J_o). \tag{12}$$

Every term is nonnegative. Cauchy–Schwarz also gives $1/(2\rho) \le m \le 1/2$: indeed $1 \le \rho\mathbb{E}|X| = 2m\rho$ and $\mathbb{E}|X| \le 1$. Since $\sigma - 1 > 2$, the decomposition yields $T_+ + T_- \le D/2$. If $D < 2$, then

$$m(A_-^2 + A_+^2) = \rho - T_+ - T_- \ge 1 - D/2, \qquad A_- + A_+ \ge \sqrt{2-D}.$$

3.3 One joint budget for endpoint motion and noise

Let $\bar{A}_- = m/q = \ell p$, $\bar{A}_+ = m/p = \ell q$ be the conditional mean magnitudes. For a central support gap $(-A_-, A_+)$ define

$$\Delta_- = \bar{A}_- - A_-, \quad \Delta_+ = \bar{A}_+ - A_+, \qquad a_0 = \Delta_-/\ell, \quad a_1 = \Delta_+/\ell, \quad d = a_0 + a_1.$$

These quantities are nonnegative, $d < 1$, and $U_0 \le a_0$, $U_1 \ge -a_1$. The physical displacement $\Delta_- + \Delta_+$ is ℓd; the displacement d is in count units. Put $w = 1 - \theta = pq\ell^2$ and

$$\mathcal{Q}(p) = (\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2.$$

Lemma 3.4 (Joint defect budget)**.** *For every $P \in \mathcal{P}$ whose closed support has a central gap $(-A_-, A_+)$ with $A_-, A_+ > 0$,*

$$\begin{aligned} D \ge{}& \ell w\mathcal{Q}(p) + 6\ell w\{d(1-d) + a_0a_1\} \\ &+ \mathbb{E}[\{(\sigma-1)(X + 2\bar{A}_+) + 6A_+\}W^2; X > 0] \\ &+ \mathbb{E}[\{(\sigma+1)(-X + 2\bar{A}_-) + 6A_-\}W^2; X < 0]. \end{aligned} \tag{13}$$

All terms on the right are nonnegative. In order, they charge $p \ne p_*$, inward motion of the gap endpoints away from their conditional means, and the spread within each sign class. The last two expectations retain the actual locations of the conditional fluctuations.

Proof. Write $M_{j,\pm} = \mathbb{E}X_\pm^j$. The outer-gap bounds proved above give

$$J_o^+ \le M_{3,+} - 2A_+M_{2,+} + A_+^2 m, \qquad J_o^- \le M_{3,-} - 2A_-M_{2,-} + A_-^2 m.$$

Adding the central term $m(A_- + A_+)^2$ and substituting in D yields

$$\begin{aligned} D \ge{}& (\sigma-1)M_{3,+} + (\sigma+1)M_{3,-} + 6A_+M_{2,+} + 6A_-M_{2,-} \\ &- 6m(A_-^2 + A_-A_+ + A_+^2). \end{aligned}$$

Conditional centering gives the exact identities

$$M_{2,+} = m\bar{A}_+ + \mathbb{E}[W^2; X > 0], \qquad M_{3,+} = m\bar{A}_+^2 + \mathbb{E}[(X + 2\bar{A}_+)W^2; X > 0],$$

and the analogous identities with $-X, \bar{A}_-$ on the negative class. After substituting $A_- = \ell(p - a_0)$ and $A_+ = \ell(q - a_1)$, the deterministic terms are

$$\ell w\{\mathcal{Q}(p) + 6[d(1-d) + a_0a_1]\};$$

the remaining terms are exactly the two expectations in (13). The gap sums are nonnegative, so monotone convergence justifies summing countably many gaps. Each expectation is integrable under the finite third-moment assumption; endpoint atoms do not affect the Lebesgue integrals. □

Corollary 3.5 (Simultaneous gap inequality)**.** *For this central gap,*

$$D \geq \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p-p_*)^2 + 6m\{(A_- + A_+)(\Delta_- + \Delta_+) + \Delta_-\Delta_+\} + 3(\sigma+1)A_+\theta_+ + 3(\sigma+3)A_-\theta_-. \tag{14}$$

Proof. On the positive support $X \geq A_+$ and $\bar{A}_+ \geq A_+$, so the positive kernel in (13) is at least $3(\sigma+1)A_+$. On the negative support $-X \geq A_-$ and $\bar{A}_- \geq A_-$, the corresponding kernel is at least $3(\sigma+3)A_-$. Finally $m = pq\ell$ and $A_- + A_+ = \ell(1-d)$ identify the two deterministic terms with those in (14). □

4 Conservation and control of the defect

The equality case in Section 3 identifies the extremal Bernoulli law. To connect that geometry to the excess, we integrate the nonnegative contact difference $\Delta = Q - h$ against the zero-bias density. This integral has two useful descriptions. Since Δ vanishes on the support, integration over the gaps produces J and a nonnegative leakage term. The identity $\int \tau g' = \mathbb{E}[Xg(X)]$ gives a second description in terms of moments and the sum distribution. Equating them yields conservation. The coefficient $C_{\mathrm{E}} = (3+\sqrt{10})\phi(0)/6$ then combines the third-moment penalty with the gap term into the nonnegative defect D.

Retain the definitions of a, Q, h and $\Delta = Q - h$ from Section 2. For now let (P, H) be an attained pair at any $c > 0$, so the law has compact support K and $\Delta = 0$ on K.

4.1 Zero-bias integration and the conservation identity

Put $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \geq t}]$. Then $\tau \geq 0$, $\int \tau = 1$. Over the bounded complementary intervals (u, v) of the closed support K, define

$$J = \sum \tau_{u,v}(v-u)^2, \qquad J_1 = \sum \tau_{u,v} \int_u^v (t-u)t\,dt,$$
$$J_2 = \sum \tau_{u,v} \int_u^v (t-u)t|t|\,dt, \qquad L = \sum \tau_{u,v} \int_{[u,v)} (t-u)\,d\nu(t).$$

Here $\tau_{u,v}$ is the constant interior value; J and τ agree with the definitions in Section 3. The leakage L is a weighted contribution of the law of $H - S_{n-1}$ inside support gaps, and is nonnegative because its measure and integrand are nonnegative. This sign needs no estimate for C_{n-1}. The integral uses $[u, v)$: the right endpoint belongs to $h(v)$, whereas an atom at the left endpoint has zero weight. If $K \subset [-R, R]$, then $J \leq 2R$, $|J_1| \leq RJ/2$, $|J_2| \leq R^2 J/2$, and $0 \leq L \leq 2R$, so all gap sums converge.

The remainder has four parts: a constant moment and gap term, a linear threshold term, a quadratic threshold term, and the Fourier error E_n. Specifically, let $\phi_z = \phi(H/\sqrt{n})$, $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and put

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$

$$\Omega = J_1 + (M_4 - 3)/6, \quad \varepsilon = \mathbb{E}q_z(S_n/\sqrt{n}) - \mathbb{E}q_z(Z_0) - 2\gamma\phi_z/(3\sqrt{n}), \quad q_z(x) = (x-z)|x-z|,$$

where $z = H/\sqrt{n}$ and Z_0 is standard normal. Define $E_n = -n^{3/2}\varepsilon/2$ and

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \phi_z\Omega) - H^2(c\rho + \gamma\phi_z/2) + E_n. \tag{15}$$

Theorem 4.1 (Conservation at contact)**.** *At an attained pair with $c = C_{\mathrm{E}}$, put $e = \exp(-H^2/(2n))$. Then*

$$\mathcal{R}_c = (n + H^2)a + nc\rho(1 - e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{16}$$

More generally, at an attained pair for any $c > 0$,

$$(n + H^2)a + nc\rho - \frac{n\phi_z}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c. \tag{17}$$

The terms in (16) have separate roles:

Term	Role
$(n + H^2)a$	The excess, weighted by $n + H^2$.
$nc\rho(1 - e)$	The loss from moving the normal threshold away from zero.
$n\phi(0)eD/6$	The moment and gap defect from Section 3.
$n^{3/2}L$	A weighted contribution inside the support gaps.
$\mathcal{R}_c$	The moment, threshold and Fourier terms to be bounded.

The four terms on the right of (16) are nonnegative when $a \geq 0$. Solving for the excess makes the role of the remainder explicit:

$$a = \frac{\mathcal{R}_c - nC_{\mathrm{E}}\rho(1 - e) - n\phi(0)eD/6 - n^{3/2}L}{n + H^2}. \tag{18}$$

In particular, fix $T \geq 0$, $H_0 \geq 0$ and $N \geq 2$. For any such pair with $a \geq 0$, $n \geq N$, $\mathcal{R}_c \leq T$ and $|H| \leq H_0$, one has

$$D \leq \frac{6T \exp(H_0^2/(2N))}{n\phi(0)}.$$

Theorem 5.1 supplies such numerical bounds under its stated hypotheses. Thus the $1/n$ control of the defect comes from uniform quantitative estimates, rather than from a fixed-law limit.

Proof. 1. Integration over the gaps. Let $\mathcal{A} = \int \tau(Q - h)$. Since $h(t) = \nu([t, \infty))$, one has $h(t) = h(v) + \nu([t, v))$ inside a gap. Thus bounded Fubini gives

$$\int_u^v \Delta(t)dt = (v - u)\Delta(v) + \int_u^v (t - u)(-Q'(t))dt - \int_{[u,v)} (t - u)d\nu(t).$$

An atom at v belongs to $h(v)$ and is not in the last integral; an atom at u is included but multiplied by zero. Because $\Delta = 0$ on the entire closed support, the endpoint terms and the integral over K vanish. Summing gives

$$\mathcal{A} = \frac{\phi_z J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\phi_z + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L. \tag{19}$$

2. Zero-bias integration. Signwise Fubini proves $\int \tau g' = \mathbb{E}[Xg(X)]$ for absolutely continuous g with bounded derivative on the relevant compact range. In particular

$$\int t\tau = \gamma/2, \quad \int t^2\tau = M_4/3, \quad \int |t|^3\tau = M_5/4.$$

Apply that same identity conditionally to $g_s(x) = \min(s+x, H)$ and then use exchangeability. The choice of derivative at its single corner is immaterial under the density τ. It follows directly that

$$\int \tau(t)h(t)dt = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T,$$

where $T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z\le z}]$, $Z = S_n/\sqrt{n}$. The last expression vanishes at an endpoint atom. Integrating (5) against τ therefore yields

$$\mathcal{A} = F - \frac{\gamma\phi_z}{2\sqrt{n}} + \frac{H\phi_z(3-M_4)/6 + cK_q}{n^{3/2}} - T. \tag{20}$$

3. The first-moment identity. Set $M = \mathbb{E}[Z\mathbf{1}_{Z\le z}]$. Since $\mathbb{E}[Xh(X)] = M/\sqrt{n}$, integration of $XQ(X)$ against P gives

$$M + \phi_z = -\frac{\gamma H\phi_z}{2n} + \frac{cK_1}{n}. \tag{21}$$

4. Gaussian comparison and assembly. The pointwise identity $q_z(x) = (x-z)^2 - 2(x-z)^2\mathbf{1}_{x\le z}$, with the same identity for a normal variable, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\phi_z/(3\sqrt{n}) + z(M+\phi_z) - z^2(F - \Phi(z)).$$

Inserting (21) and $\sqrt{n}(F - \Phi(z)) = c\rho + a$ into (20) gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\phi_z/6 + P_0/n + \sqrt{n}\varepsilon/2,\\ P_0 &= H^2(c\rho + a + \gamma\phi_z/2) - H\{cK_1 + \phi_z(M_4-3)/6\} + cK_q.\end{aligned}$$

Equate this with (19) multiplied by $\sqrt{n}$, then multiply by n and collect terms. This is (17). Substituting $c = C_{\mathrm{E}} = (3+\sqrt{10})\phi(0)/6$, $\phi_z = \phi(0)e$, and $\gamma + 3J = (3+\sqrt{10})\rho - D$ proves (16). □

The compensated Fourier representation of this same E_n is proved in Appendix A. Its comparator has transform $e^{-nt^2/2}(1 - in\gamma t^3/6)$. Appendix C retains the stronger identity for arbitrary compactly supported laws, including the residuals that disappear here by contact.

4.2 Why the remainder bound must come first

By Lemma 3.1,

$$D = 3(\rho + 2m - J) + (\sqrt{10}\rho - \gamma - 6m) \ge 0. \tag{22}$$

Specialize from now on to $c = C_{\mathrm{E}}$ and $a \ge 0$. The sharper estimate below keeps the threshold loss. Its coefficient is not known positive until a bound on $\mathcal{R}_c$ has first been established.

Corollary 4.2 (From a remainder bound to a defect bound)**.** *At such a contact pair with $a \ge 0$, assume $n \ge N \ge 2$, $\rho \ge l > 0$, $0 < c_- \le c$, and a proved remainder bound $\mathcal{R}_c \le T < Nc_-l$. A negative T excludes the pair. Otherwise*

$$\begin{gathered}\gamma + 3J > 0, \qquad e \ge 1 - T/(Nc_-l) > 0,\\ \mathcal{U} := (n+H^2)a + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}_c,\\ \mathcal{U} \le \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2.\end{gathered}$$

No condition $D < 2$ or own-score upper bound is required.

Proof. Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma + 3J)/6$. The contact identity gives

$$eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \geq c\rho - T/n > 0.$$

Thus $M_c > 0$ and, since $D \geq 0$, $M_c \leq c\rho$. The density bound follows. Regrouping (16) gives $\mathcal{R}_c = \mathcal{U} + n(1-e)M_c$, proving the unweighted bound. Finally, put $w = H^2/(2n)$, so $e = e^{-w}$. Then

$$n(1-e)M_c = n(e^w - 1)eM_c \geq \frac{H^2}{2}(c\rho - T/n) \geq \frac{H^2}{2}(c_l - T/N).$$

This is a bound on $\mathcal{U}$, not an improved bound on the raw remainder. □

The corollary is used in two steps. First, an upper bound for $\mathcal{R}_c$ proves $M_c > 0$ and a positive lower bound for e. Second, the now nonnegative term $n(1-e)M_c$ can be subtracted to improve the bound for $\mathcal{U}$ by a multiple of H^2. This is the improvement that helps bound the threshold. It is a bound on $\mathcal{U}$, so it cannot be substituted for the initial upper bound on $\mathcal{R}_c$. Every inequality holds separately at each n, H; common bounds on a closed moment cell therefore also apply when the maximizing law varies with n.

5 Quantitative localization

The zero-defect law is now known. We need a quantitative statement strong enough to control a possible extremizer. Conservation reduces this to an upper bound for its remainder. Contact bounds the support, which permits higher-moment and Fourier estimates. Iterating these estimates gives the following theorem; its two principal outputs for the next section are $nD \leq 32.20665378$ and $|H| \leq 1.83605680$.

Theorem 5.1 (Fixed-target coarse localization)**.** *Let $n \geq 42$ be an actual index and let an attained fixed-target pair at C_{E} have $a \geq 0$. Assume $C_{n-1} \leq C_{\mathrm{E}}$ and the external structural inequality* (74)*. Then the complete probability cover specified in Appendix D applies to that pair, and simultaneously*

$$\begin{gathered}\mathcal{R}_c \leq 2.1995174, \qquad nD \leq 32.20665378, \qquad D \leq 0.76682509,\\ \rho \leq 71527/48000, \qquad |\beta| \leq 0.50549963, \qquad |H| \leq 1.83605680,\\ \operatorname{supp} P \subset (-157/50, 157/50).\end{gathered}$$

The analytical proof and the computer-assisted Propositions D.3 and D.5 are in Appendix D. The Fourier bounds in Appendix A apply uniformly to every actual $n \geq 42$, even when the law and threshold vary with n. Evaluating their scalar majorants at the floor $N = 42$ is therefore sufficient; it does not set the actual sample size equal to 42.

5.1 How the predecessor bound controls support

Lemma 5.2 (Contact bound from the predecessor inequality)**.** *Let (P, H) maximize the excess at $c = C_{\mathrm{E}}$, with $n \geq 42$, $a \geq 0$ and $C_{n-1} \leq c$. Put $F_v^{\mathrm{G}}(x) = \Phi(x/\sqrt{v})$ and $\eta_n = n(\sqrt{n/(n-1)} - 1) - 1/2$. Define*

$$\frac{H_n(H,t)}{n^{3/2}} = F_{n-1}^{\mathrm{G}}(H-t) - F_n^{\mathrm{G}}(H) + t(F_n^{\mathrm{G}})'(H) - \frac{t^2-1}{2}(F_n^{\mathrm{G}})''(H).$$

Every support point satisfies

$$0 \leq H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho\eta_n\} - na.$$

Proof. At a support point the contact expression (4) equals zero. Substitute

$$\mathbb{P}(S_{n-1} \le H - t) \le F^{\mathrm{G}}_{n-1}(H - t) + c\rho/\sqrt{n-1}$$

and $F = F^{\mathrm{G}}_n(H) + (c\rho + a)/\sqrt{n}$ into that equality. Collecting the Gaussian terms gives the displayed H_n; the remaining terms give the asserted inequality. The estimate uses the closed distribution function at the same threshold $H - t$. □

A support point is impossible when the right side is negative. The Gaussian expansion and radial derivative estimates in the proof of Theorem D.1 make this exclusion effective on entire exterior rays. This is how the predecessor assumption enters both coarse localization and Proposition 6.2.

5.2 How the bounds improve

The steps have distinct purposes:

1. The structural inequality bounds ρ. Contact and the sharp predecessor then bound the entire support, uniformly in H.

2. Bounded support supplies higher-moment bounds and an event for $X - X'$ under two independent copies of the original law. This event controls the nonzero-frequency peaks in the Fourier integral.

3. Two coarse passes bound the remainder and then the defect, proving that zero is outside the support. Thus the original law has an actual central gap.

4. Four signed-moment passes retain the gap geometry and the signs of the moment terms. They improve the same-law bounds and give the outputs of the theorem.

Choose rational constants $0 < c_- < C_{\mathrm{E}} < c_+$ and $0 < \phi_l < \phi(0)$; their explicit values and proofs are given in Appendices D and B. For example, suppose the coefficient bounds on a cell with $\rho \ge l$ give, for $v = c_-, c_+$,

$$\mathcal{R}_c \le \max_v \{B_v + F_v|H| - Q_v H^2\}, \qquad Q_v > 0.$$

Completing the square bounds this by $\max_v(B_v + F_v^2/(4Q_v))$. An outward rational upper bound T with $T < Nc_-l$ first establishes the positivity needed in Corollary 4.2. Only then do the larger curvatures $Q_v + (c_-l - T/N)/2$ give a bound U for $\mathcal{U}$, and hence $nD \le 6U/\phi_l$. This explains both the update and its order: U bounds a nonnegative expression, not the original remainder.

5.3 Why the finite verification covers all witnesses

Proposition 5.3 (Soundness of the finite localization)**.** *Suppose the enclosure and exclusion rules proved in Appendices A and D are evaluated in the stated directions, every required inequality passes, and every closed-cover incidence is checked. Then the resulting bounds hold simultaneously for every attained nonnegative-excess pair in the initial domain at every actual $n \ge 42$ with the sharp predecessor.*

Proof. The initial 24 closed ρ-cells cover the moment range supplied by Theorem D.1. Fix one possible witness in one of these cells. Every subdivision covers its parent. Intersecting necessary moment inequalities keeps this witness; the original β-interval is restored separately for each γ-slice. A child is discarded only by an incompatible necessary inequality. In

particular, a positive numerator with zero denominator is treated using the original moment matrix, and a zero-over-zero correction is dropped.

For each surviving child, a support bound is proved before higher moments are used. The actual-event and Fourier estimates then bound the remainder, and the ordered update bounds the nonnegative expression in conservation. Taking maxima of upper bounds and minima of lower bounds can enlarge the possible set but cannot remove the fixed witness. A parent disappears only if all descendants have been excluded. Induction over the two coarse and four signed passes preserves this invariant.

The final closed probability panels cover the surviving probability intervals. At a shared endpoint the cell contributes to every incident panel. Pooling the simultaneous bounds therefore retains the conclusions for the same law. Finally, the analytical all-index comparisons in the Fourier appendix justify each floor evaluation at every actual $n \geq 42$. The numbers of processed boxes are accounting checks, not substitutes for this coverage argument. □

6 From small defect to a common support region

The localization theorem bounds nD and $|H|$. Lemma 3.4 bounds D below by nonnegative costs, so a small defect makes each cost small: the sign probability must be close to p_*, the support gap must nearly reach the conditional means, and the spread within each sign class must be small. Proposition 6.1 makes these conclusions quantitative using the defect alone. Proposition 6.2 then uses contact and the sharp predecessor bound to control the entire conditional support. Together with the threshold bound, these conclusions give the domain needed by Theorem 7.1. Throughout this section set

$$\sigma = \sqrt{10}, \qquad p_* = (4 - \sigma)/2, \qquad B_* = 32.20665378.$$

The law, support gaps and sign split are those of Section 3; in particular D is the actual gap defect, not a moment surrogate. Write the central gap, once acquired, as $(-A_-, A_+)$, using the notation of Section 3. The first use of $n \geq 700$ is the inequality $B_*/700 < .047$ below; the count comparison also uses this sample-size bound in its error estimates.

Proposition 6.1 (Direct geometry)**.** *Let X be compactly supported, centered and of variance one. If $n \geq 700$ and $nD \leq B_*$, its closed support has a gap $(-A_-, A_+)$ across zero, with $A_-, A_+ > 0$. In the actual decomposition $X = \ell(B_0 + U - p)$, put*

$$a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1,$$
$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2, \qquad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2.$$

Then $a_0, a_1 \geq 0$, $\mathbb{E}(U \mid B_0) = 0$, $U_0 \leq a_0$, $U_1 \geq -a_1$, and

$$\frac{3}{8} < p < \frac{93}{200}, \qquad \theta < .0036, \qquad d < .004, \qquad \ell > 1.996, \tag{23}$$
$$\rho < \frac{9}{8}, \qquad .066148 < \beta < .2527, \qquad \lambda < \frac{7}{20}. \tag{24}$$

These conclusions require no contact or predecessor hypothesis.

The estimates are obtained in the following order. Each preliminary bound supplies a hypothesis for a sharper one later in the proof.

Input	Purpose of the output
$D < .047$ and the zero-in-support bound	Establish a central gap and nonnegative inward displacements.
The joint defect budget	Obtain $.36 < p < .48$, $\theta < .006$ and $\ell > 1.99$ before taking derivatives of the refined bounds.
The inward-motion term	Give $d < .004$.
Monotonicity in d and θ	Give the final probability interval, $\theta < .0036$ and $\ell > 1.996$.
The conditional cubic costs	Control the total count variance λ, rather than only the variance of X.
Contact and the predecessor bound	Exclude the two outer support rays in Proposition 6.2.

Proof. 1. A central gap and preliminary bounds. We have $D < D_0 := .047$, because $700D_0 > B_*$. If zero were in the closed support, Lemma 3.1 would give $D \geq (\sigma - 1)\rho \geq \sigma - 1 > D_0$. Thus the central gap exists; centering supplies both signs. Conditional means imply $a_j \geq 0$ and $0 \leq d < 1$. Put $v = pq$ and $w = 1 - \theta = v\ell^2$, and use the probability polynomial $\mathcal{Q}$ from Section 3. We first retain all the costs of the actual conditional noises. Let

$$\int f \, d\mu_{\pm} = \mathbb{E}\big[(X - \mathbb{E}(X \mid B_0))^2 \mathbf{1}_{\{\pm X > 0\}} f(|X|)\big].$$

Their masses sum to θ. In this notation the joint budget (13) reads

$$D \geq \ell w \mathcal{Q} + E_0 + \int k_+ \, d\mu_+ + \int k_- \, d\mu_-, \tag{25}$$

where

$$E_0 = 6\ell w\{d(1-d) + a_0 a_1\},$$
$$k_+(x) = (\sigma - 1)(x + 2\ell q) + 6A_+, \qquad k_-(x) = (\sigma + 1)(x + 2\ell p) + 6A_-.$$

The measures $\mu_{\pm}$ abbreviate the displayed variance-weighted expectations under the same marginal P.

Hereafter use $3.162277660 < \sigma < 3.162277661$. The weaker gap estimate $J \leq \rho + 2mA_-A_+$, together with $\rho \geq 1$, $\gamma \leq \rho$ and $m \leq 1/2$, gives

$$A_-A_+ \geq z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \geq A_-A_+ \geq z.$$

The second inequality follows from $A_- \leq \ell p$, $A_+ \leq \ell q$. All costs in (25) are nonnegative, so

$$D \geq \frac{w^{3/2}\mathcal{Q}}{\sqrt{v}} \geq 2z^{3/2}\mathcal{Q}.$$

Let $\mathcal{Q}_-$ replace σ in its polynomial expression by the lower rational bound. The two exact tests

$$4z^3\mathcal{Q}_-(.36)^2 > D_0^2, \qquad 4z^3\mathcal{Q}_-(.48)^2 > D_0^2$$

exclude $p \leq .36$ and $p \geq .48$, respectively: the squared distance to $p_* \in (.36, .48)$ is monotone outside that interval.

2. Preliminary noise and inward-displacement bounds. On $.36 < p < .48$, $\ell \geq 2\sqrt{z} > 1.67$, and the noise coefficients satisfy

$$k_+(x)/\ell \geq q\{(\sigma + 5)z/w + 2(\sigma - 1)\} > 5.2,$$
$$k_-(x)/\ell \geq p\{(\sigma + 7)z/w + 2(\sigma + 1)\} > 5.2.$$

For the first use $x \geq A_+$ and $A_+ \geq z/(\ell p)$; for the second use their negative counterparts. Since $w \leq 1$, the constants follow by taking $q = .52$ and $p = .36$. Therefore $D \geq 1.67(5.2)\theta$, whence $\theta < .006$, $w > .994$ and $\ell > 1.99$. Also $A_- + A_+ \geq 2\sqrt{z}$ gives

$$d \leq 1 - 2\sqrt{zv/w} < .2, \qquad v \geq .36(.64).$$

Evaluating the kernels at their lower support endpoints in (25) now yields

$$D \geq \ell\{w\mathcal{Q} + 6wd(1-d) + \theta \min(C_+ - b_+ d, C_- - b_- d)\}, \tag{26}$$

where

$$C_+ = 3(\sigma+1)q, \quad C_- = 3(\sigma+3)p, \quad b_+ = \sigma + 5, \quad b_- = \sigma + 7.$$

Each branch increases in d on the current domain, since $6(.994)(.6) - .006(3.162277661 + 7) > 0$; both noise prices are positive. The inward cost alone excludes $d \geq .004$, because

$$1.99(.994)6(.004)(.996) > .047.$$

3. The final probability and variance bounds. After replacing d by zero, the derivative in θ of either branch $\sqrt{(1-\theta)/v}\{(1-\theta)\mathcal{Q} + \theta C\}$ has the sign of $(2-3\theta)C - 3(1-\theta)\mathcal{Q}$. It is positive: $C > 6.49$, $\mathcal{Q} < .05$ and $\theta < .006$. Hence $D \geq \mathcal{Q}/\sqrt{v}$. Since $\sqrt{v} \leq 1/2$, it follows that $D \geq 2\mathcal{Q}(p)$. Moreover

$$\frac{3}{8} < p_* < \frac{21}{50} = \frac{1}{2}\left(\frac{3}{8} + \frac{93}{200}\right).$$

Every p outside $(3/8, 93/200)$ is therefore at least as far from p_* as $3/8$, so $\mathcal{Q}(p) \geq \mathcal{Q}(3/8)$. The single rational comparison

$$\mathcal{Q}_-(3/8) = \frac{37936011}{1600000000} > \frac{47}{2000} = \frac{D_0}{2}$$

contradicts $D < D_0$ on both outside rays, including the endpoints. Thus $3/8 < p < 93/200$. Since

$$C \geq \min\{3(\sigma+1)(107/200),\ 3(\sigma+3)(3/8)\},$$

the bound $D \geq \ell\theta C$, with $\ell > 1.99$, improves $\theta < .0036$ and consequently $\ell > 1.996$. Here the d-dependent terms may be dropped together, since $6w(1-d) - \theta b_\pm > 0$ on the current domain. This keeps the displacement cost together with its signed noise correction.

4. The remaining moment bounds. The exact conditional cubic expansion gives

$$\rho = w^{3/2}\rho_B(p) + \int (x + 2\ell q)\, d\mu_+ + \int (x + 2\ell p)\, d\mu_-, \qquad \rho_B(p) = \frac{1-2v}{\sqrt{v}}.$$

Because the two integrands are priced by at least $\sigma - 1$ in (25),

$$\rho \leq \rho_B(p) + D/(\sigma - 1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \geq 15/64$ implies $\rho_B < 1.1$ by squaring positive quantities. The signed-second-moment identity gives

$$1 - 2p - 2q\theta \leq \beta \leq 1 - 2p + 2p\theta;$$

its endpoints imply $.066148 < \beta < .2527$.

5. From conditional spread to total count variance. A bound on θ alone would not control $\lambda = n\theta/\ell^2$ uniformly in n. We therefore retain more of the cubic cost. A one-sided moment inequality supplies a variance-squared term divided by the inward displacement; pairing

it with the displacement cost by AM–GM will give a larger linear coefficient of λ. If Z is centered, $Z \le a$, $a > 0$, and $t = \mathbb{E}Z^2$, then

$$0 \le \mathbb{E}[(a - Z)(Z + t/a)^2] = at - \mathbb{E}Z^3 - t^2/a.$$

Apply this to U_0 and $-U_1$. A zero endpoint forces the corresponding noise to vanish, so no division by zero is needed. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$, the same gap-budget expansion, with these two cubic bounds inserted, gives

$$\begin{aligned} nD/\ell^3 \ge\ & nv\mathcal{Q} + 6nv\{d(1-d) + a_0a_1\} \\ & + \{3(\sigma+3)p - (\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q - (\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

Terms with $a_j = 0$ are absent. Set $\bar{d} = .004$ and use $d(1-d) + a_0a_1 \ge (1-\bar{d})(a_0 + a_1)$. The reciprocal term is essential: pairing its variance-squared cost with the inward-displacement cost gives a linear cost in λ. Specifically, AM–GM gives

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \ge 2\lambda_0\sqrt{6(1-\bar{d})(\sigma+1)p}.$$

Thus

$$nD \ge \ell^3\{nv\mathcal{Q} + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{27}$$

where

$$\begin{aligned} Y_0 &= 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p}, \\ Y_1 &= 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}. \end{aligned}$$

Each increases in its respective argument p or q. At $p = 3/8$ and $q = 107/200$, lower bounds 3 and 2.6 for the roots give $Y_0 > 12.8$, $Y_1 > 11.8$. Since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \ge nD > 1.996^3(11.8)\lambda \quad (\lambda > 0), \qquad 1.996^3(11.8)(.35) > B_*.$$

This proves $\lambda < .35$; at zero noise it is immediate. □

Proposition 6.2 (Two contact rays)**.** *Under the hypotheses of Proposition 6.1, suppose in addition that $C_{n-1} \le C_{\mathrm{E}}$ and that (P, H) maximizes the excess at C_{E}, with nonnegative maximum. Then*

$$-.6 < U_0 \le a_0, \qquad -a_1 \le U_1 < .6.$$

The support conclusion is uniform in the raw threshold H.

Proof. A support point too far from its conditional mean would make the contact upper expression (76) negative. The all-threshold Gaussian heat estimate and its radial derivative bound from the proof of Theorem D.1 give, for $t > \sqrt{3}$,

$$H_n(H, \pm t) \le \overline{H}(t) := S^3\left\{\frac{2}{5}\left(\frac{t^3 - 3t}{6} + \frac{|t^4 - 6t^2 + 3|}{72}\right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720}\right\},$$

$$|\partial_t H_n(H, \pm t)| < g_{\mathrm{rad}}(t) := \frac{2}{5}(t^2/2 + 1/2 + \eta), \qquad S = 81/80, \quad \eta = 8/875.$$

The first bound follows directly from (77): $\|\phi'''\|_\infty < 2\phi(0)$, $n - 1 \ge 41$, and $\sqrt{n-1} > 6$. No derivative of the absolute-value majorant is taken.

At a support point of magnitude t, contact requires a nonnegative upper expression $H_n(H, \pm t) + C_{\mathrm{E}} Q_b(t)$, where

$$Q_b(t) = -t^3 + \tfrac{3}{2} r t^2 + 3bt + \eta r, \quad r = 9/8, \quad b = \begin{cases} -.066148 & \text{negative ray,} \\ .2527 & \text{positive ray.} \end{cases}$$

These replacements follow with their signs from (24); the discarded excess term is nonpositive. Take $c_0 = 819/2000 < C_{\mathrm{E}}$ and the two fixed boundaries

$$t_- = 1.996(3/8 + .6) = 1.9461, \qquad t_+ = 1.996(107/200 + .6) = 2.26546.$$

For a fixed raw threshold H, abbreviate the appropriate radial function $H_n(H, \pm t)$ by $H_n(t)$, with derivative taken in the positive magnitude t. Set

$$\Xi(t) = g_{\mathrm{rad}}(t) + c_0 Q_b'(t).$$

Exact rational substitution gives the following conservative bounds:

Boundary quantity	t_-, negative ray	t_+, positive ray
$\overline{H}(t) + c_0 Q_b(t)$	$< -2/5$	$< -13/100$
$\Xi(t)$	< -1	$< -8/5$
$\Xi'(t)$	$< -13/5$	$< -16/5$

Both boundaries satisfy $t^2 > 3$. The constant second derivative is

$$\Xi'' = \frac{2}{5} - 6c_0 = -\frac{2057}{1000} < 0.$$

Consequently Ξ' and then Ξ remain negative on each outward ray. Because $g_{\mathrm{rad}} > 0$, the identity $Q_b' = (\Xi - g_{\mathrm{rad}})/c_0$ gives $Q_b' < 0$ there. At the boundary $\overline{H} \geq 0$, so the first table entry gives $Q_b < 0$; this sign also persists outward. Since $C_{\mathrm{E}} > c_0$, we obtain

$$H_n(t_\pm) + C_{\mathrm{E}} Q_b(t_\pm) < 0, \qquad H_n'(t) + C_{\mathrm{E}} Q_b'(t) \leq g_{\mathrm{rad}}(t) + c_0 Q_b'(t) = \Xi(t) < 0.$$

The contact upper expression therefore cannot be nonnegative anywhere on either ray. This excludes every support point there, including boundary atoms. We differentiate H_n itself; the absolute-value majorant $\overline{H}$ is used only at the two boundary points.

On the negative class, $X > -t_-$ and $\ell > 1.996$ give $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. On the positive class, $X < t_+$ gives $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inward endpoints and conditional centering are the actual ones from Proposition 6.1. □

By (6), the law compared in the next section is exactly the original law X, with the same threshold H. The sign probability and the signed moments refer throughout to this chosen law.

7 The direct comparison with a binomial law

In count coordinates, conditioning on the number of positive summands expresses the sum as an integer count plus centered noise. We compare its CDF with the corresponding binomial CDF. Noise can move mass out of a count near the threshold and bring mass in from counts above it. The proof shows that the outgoing loss, together with the normal and third-moment corrections, exceeds all incoming contributions. The quantity $\lambda = n\mathbb{E}U^2$ measures the total noise variance in count units.

Theorem 7.1 (The common-strip comparison)**.** *Let $n \geq 700$ be an integer, $B \sim \mathrm{Bernoulli}(p)$, and $\mathbb{E}(U \mid B) = 0$, where $p \in [3/8, 93/200]$. Write U_j for the conditional noise given $B = j$, and suppose*

$$-3/5 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq 3/5, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 1/250.$$

Put $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (B + U - p)/\sqrt{v + s}$ and $\rho = \mathbb{E}|X|^3$. For independent copies and every raw threshold $|H| \leq 46/25$,

$$\sqrt{n}\left\{\mathbb{P}\left(\sum_{i=1}^{n} X_i \leq H\right) - \Phi(H/\sqrt{n})\right\} - C_{\mathrm{E}}\rho < -\lambda/100 \qquad (0 < \lambda \leq 7/20). \tag{28}$$

At $\lambda = 0$ the same excess is strictly negative.

To prove the theorem, condition on $K = \sum_{i=1}^{n} B_i$, which has law $\mathrm{Bin}(n, p)$. Given $K = j$, the count-unit noise W_j is the sum of j independent copies of U_1 and $n - j$ independent copies of U_0. The threshold in these units is

$$t = np + H\sqrt{v + s}, \qquad \left\{\sum X_i \leq H\right\} = \{K + W_K \leq t\}.$$

We compare this probability with the binomial CDF and then account for the changes in the normal term and the third absolute moment. A count below t can leave the event, while a count above t can enter. The aim is to show that the loss and the favorable corrections exceed the total gain from all counts above the threshold.

For $0 < \lambda \leq 1/100$, the argument keeps both the loss and every error proportional to λ. It uses one nearby count and a fourth-moment bound for all entering counts. For $1/100 \leq \lambda \leq 7/20$, a finite summation identity separates the crossing loss from errors in interpolating the binomial CDF. A periodic sine function then gives a loss from five counts that is uniform in the phase $t - \lfloor t \rfloor$. Its interpolation error has an upper bound of order $1/n$ that does not vanish with λ, which explains the separate small-noise argument. The two ranges share their endpoint. At $\lambda = 0$ the law is exactly Bernoulli. These comparisons use the explicit hypotheses of this theorem; the preceding sections obtain those hypotheses for a maximizer.

7.1 Three elementary ingredients

Write $b_j = \binom{n}{j} p^j q^{n-j}$, $F_B(j) = \sum_{i \leq j} b_i$, and $\rho_B = (1 - 2v)/\sqrt{v}$. We use Schulz's strict finite-binomial theorem [12, Theorem 1] in the form

$$\sqrt{n}\{F_B(j) - \Phi((j - np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B \quad (0 < p < 1, n < \infty). \tag{29}$$

Its strictness concerns every finite binomial law; it is not inferred from the limiting lower approximation for the universal constant.

The factorial estimates used throughout are finite Stirling bounds:

$$\log(j!) = (j + \tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For completeness, with $x = 1/(2j + 1)$, $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{l \geq 1} x^{2l}/(2l + 1)$ lies strictly between 0 and $1/(12j(j + 1))$. Wallis's product fixes the limiting constant, so $r_j \to 0$; summing these differences proves the bounds. One useful consequence is

$$\max_j \mathbb{P}\{\mathrm{Bin}(M, p) = j\} < \frac{41}{100\sqrt{Mv}} \quad (M \geq 43, p \in [1/3, 2/3]). \tag{30}$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \le 2/3$. Since $v \ge 2/9$ and $|1-2p| \le 1/3$, $z(1-z) \ge v(1 - 1/M - 2/M^2) > 39v/40$. Stirling, the nonnegative relative entropy, and $e^{1/516} < 516/515$ bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv}$, which is smaller than the right side of (30) by a rational square comparison.

Finally the cubic correction has the favorable sign on the full strip. Let $s_j = \mathbb{E}U_j^2$, so $s = qs_0 + ps_1$, and set $r = 3/8$, $d = 1/250$, $v_* = 15/64$. The conditional supports have fixed signs relative to p, since $d < r$. Expansion of the raw third absolute moment gives

$$\mathbb{E}|B + U - p|^3 = v(1-2v) + 3v(s_0 + s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds imply $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$. Together with $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, this yields

$$\begin{aligned}(v+s)^{3/2}(\rho - \rho_B) &\ge s\left\{3r - d - \frac{3}{2}(1-2v) - \frac{3(1-2v)s}{8v}\right\} \\ &\ge s\left\{\frac{2593}{8000} - \frac{17}{20}s\right\} > 0, \qquad 0 < s \le \frac{1}{2000}. \end{aligned} \tag{31}$$

Here $s \le (7/20)/700 = 1/2000$, and the displayed coefficient is $3r - d - \frac{3}{2}(1-2v_*)$. Thus every nonnegative coefficient of $\rho_B - \rho$ can be discarded in an upper comparison.

7.2 Small noise: one count and the whole incoming tail

Assume $0 < \lambda \le \lambda_s = 1/100$. Choose an integer k with

$$t = np + H\sqrt{v+s} = k + u, \qquad -3/10 \le u \le 7/10, \qquad \delta = k - np.$$

The phase interval has length one, so such an integer always exists. It need not be the floor anchor used in the larger-noise proof. Let W_j have the law of the sum of j independent U_1 variables and $n-j$ independent U_0 variables. For $C \ge 0$, subtract the signed binomial excess at k from the actual excess:

$$\begin{aligned}\Delta_C = \sqrt{n}\left\{\mathbb{P}\left(\sum X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B.\end{aligned}$$

Exact conditioning gives

$$\begin{aligned}\mathbb{P}\left(\sum X_i \le H\right) - F_B(k) = &-\sum_{i=0}^{k} b_{k-i}\mathbb{P}(W_{k-i} > u + i) \\ &+ \sum_{i=1}^{n-k} b_{k+i}\mathbb{P}(W_{k+i} \le u - i).\end{aligned}$$

The event is closed: count $k-i$ is lost when $W_{k-i} > u + i$, whereas count $k+i$ enters when $W_{k+i} \le u - i$. Thus an atom at the threshold remains in the event in both sums. Define

$$g_k(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{b_k}, \qquad O = \mathbb{P}(W_k > u) + g_k(u),$$

$$T_{\mathrm{in}} = \sum_{i=1}^{n-k}\mathbb{P}(W_{k+i} \le u - i), \qquad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Keeping the own-count loss and discarding the other nonpositive outgoing terms gives

$$\Delta_C \le -m_k(O - r_{\max}T_{\mathrm{in}}) + \mathcal{R}, \qquad m_k = \sqrt{n}\, b_k, \tag{32}$$

whenever $b_{k+i}/b_k \le r_{\max}$ for all legal incoming counts.

Uniform normalization.

Set $D_* = 5/3$, $b_* = 1/20$, $T_* = 101/10000$ and $c_* = 2/525$. Because $v \geq v_*$ and $v + s < 1/4$, $|t - np| < 23/25$. Consequently $|\delta| < D_*$ globally and $|\delta| < 1$ when $|u| \leq b_*$. Write $\mathcal{K}(z, p) = z \log(z/p) + (1-z)\log((1-z)/q)$ for binomial relative entropy; this is distinct from the physical gap defect D. The segment between p and k/n lies inside $[.37, .47]$; in particular $1 \leq k \leq n-1$. On this segment

$$z(1-z) > 23/100, \quad |\partial_z^3 \mathcal{K}(z,p)| < 5, \quad |(\log(z(1-z)))'| < 6/5, \quad (12z(1-z))^{-1} < 3/8.$$

Stirling and entropy Taylor expansion therefore give

$$m_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 3D_0/5 + 3/8 + D_0^3/840 \tag{33}$$

for $|\delta| \leq D_0 \leq D_*$. The three terms respectively bound the logarithmic prefactor, factorial remainder and entropy remainder. Also

$$\log g_k'(z) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+z)^2}{2n(v+s)} - e,$$

so, on an integration interval $|z| \leq b_0$,

$$n|\log g_k'(z)| \leq \frac{11}{5}\lambda + \frac{2D_0 b_0 + b_0^2}{2v_*} + Z(D_0).$$

The coefficient $11/5$ exceeds $1/(2v_*) + D_*^2/(1400v_*^2)$. Using $e^{-x} \geq 1 - x$, $e^x \leq (1-x)^{-1}$ and $\phi(0) > 199/500$ now gives

	$\lvert u\rvert \leq 1/20$	$-3/10 \leq u \leq 7/10$
m_k lower bound	$3/4$	$3/4$
g_k' lower bound	$99/100$	$49/50$
g_k' upper bound	$101/100$	$51/50$
$r_{\max}$ upper bound	$51/50$	$6/5$

(34)

The derivative estimates hold on the whole segment from 0 to u. For the global ratio, (30) gives $\sqrt{n}\max_j b_j < 17/20$, hence $r_{\max} < (17/20)/(3/4) < 6/5$. In the inner interval the first ratio is

$$\frac{b_{k+1}}{b_k} = 1 + \frac{p - \delta - 1}{nv + q(\delta+1)} < 1 + \frac{93/200}{700 v_*} < 51/50;$$

every later ratio is at most one, since $p - \delta < 2$. By (31) the moment correction is nonpositive. Differentiating the normal CDF and using $\phi(0)/(2v_*^{3/2}) < 2$ therefore gives

$$\mathcal{R} < 2\lambda/n \quad (|u| \leq b_*), \qquad \mathcal{R} < 4\lambda/n \quad \text{on the full phase interval.} \tag{35}$$

Indeed the positive displacement is bounded by 1 or $D_* < 2$.

No truncation of the incoming tail.

Since $|s_1 - s_0| \leq (8/3)s$, the actual own-count variance $t_0 = \mathbb{E}W_k^2$ satisfies $.99\lambda < t_0 < 1.01\lambda \leq T_*$. For each incoming count,

$$t_i = \mathbb{E}W_{k+i}^2 \leq \lambda\{1 + c_*(D_0 + i)\}, \qquad \mathbb{E}W_{k+i}^4 \leq A^2 t_i + 3t_i^2, \quad A = 3/5,$$

where $D_0 = 1$ in the inner interval and $D_0 = D_*$ otherwise. The moment bound follows from independence and $|U_j| \le A$. For $a = 1 + c_* D_0$ put

$$K(a) = A^2 \left(\frac{13a}{12} + \frac{5c_*}{4} \right) + 3\lambda_s \left(\frac{13a^2}{12} + \frac{5ac_*}{2} + \frac{5c_*^2}{3} \right).$$

Fourth-moment Markov at distance $i-u$, and extension of the nonnegative sum to all positive integers, give

$$\begin{aligned} T_{\mathrm{in}} &< \frac{K(1+c_*)}{(1-b_*)^4}\lambda < .53\lambda \quad && (|u| \le b_*), \\ T_{\mathrm{in}} &< \frac{.43\lambda}{(1-u)^4} && (0 \le u \le .7), \\ T_{\mathrm{in}} &< .43\lambda && (u \le 0). \end{aligned} \tag{36}$$

Here $K(1+c_*D_*) < .43$, and $\sum i^{-4} < 13/12$, $\sum i^{-3} < 5/4$, $\sum i^{-2} < 5/3$. These series bounds follow by retaining respectively five, three and five terms and estimating the remaining sum by an integral. For $u \ge 0$ use $i - u \ge (1-u)i$; for $u \le 0$ use $i - u \ge i$.

The loss at the threshold count.

The variance of the count-unit noise already predicts the scale of this loss. Cauchy–Schwarz and Hölder applied to the centered sum W_k give

$$\mathbb{P}(W_k > 0) \ge \frac{(\mathbb{E}|W_k|)^2}{4t_0} \ge \frac{t_0^2}{4\mathbb{E}W_k^4}.$$

Indeed $\mathbb{E}(W_k)_+ = \mathbb{E}|W_k|/2$ and $t_0 \le (\mathbb{E}|W_k|)^{2/3}(\mathbb{E}W_k^4)^{1/3}$. The fourth-moment bound $\mathbb{E}W_k^4 \le A^2 t_0 + 3t_0^2$ makes this at least $t_0/[4(A^2+3t_0)] > (16/25)t_0$ on $t_0 \le T_*$. This zero-threshold estimate explains why a loss proportional to the variance is possible. We use a quartic minorant to obtain a bound that also pays for moving the threshold and for the normal increment.

Lemma 7.2 (Loss at a nearby count threshold)**.** *Under the small-noise hypotheses and for* $|u| \le b_*$,

$$O > \frac{3}{5} t_0 > .59\lambda. \tag{37}$$

Proof. We first use a moment minorant when $|u| \le 3t_0/2$, then a one-sided variance bound for the remaining phases. Set $\alpha = 1/2$ and $m_h = 5\alpha^4 = 5/16$. The factored quartic

$$h_\alpha(x) = \frac{-16x^4 + 12x^2 + 4x}{5} = \frac{4}{5} x(1-x)(2x+1)^2 \le \mathbf{1}_{(0,\infty)}(x) \tag{38}$$

is a global minorant for the strict event. Its factorization makes it nonpositive for $x \le 0$ and $x \ge 1$. On $(0,1)$ the maximum occurs at $x = (1+\sqrt{3})/4$ and equals $(9+6\sqrt{3})/20 < 1$; the last comparison follows from $6\sqrt{3} < 11$. Thus it may be integrated against the conditional sum W_k. Each summand is bounded by $3/5$, whereas the sum can exceed that bound; the minorant is valid on the entire real line. The expectation of the quartic uses moments through order four, supplied by independence and the bounded summands. Let $r = zt_0$ and set

$$\begin{aligned} F_{\varepsilon,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{m_h} + \left(\frac{2\varepsilon}{5\alpha} + b - \frac{4At}{m_h} \right) z \\ + \frac{3\alpha^2 t - 6t^2}{m_h} z^2 - \frac{t^3 z^4}{m_h}. \end{aligned}$$

Expanding $h_\alpha(W_k + r)$ for $u = -r$, or $h_\alpha(W_k - r)$ for $u = r$, and using $|\mathbb{E}W_k^3| \le At_0$, $\mathbb{E}W_k^4 \le A^2 t_0 + 3t_0^2$, yields

$$O/t_0 \ge F_{1,-101/100}(z, t_0) \quad (u = -r), \qquad O/t_0 \ge F_{-1,99/100}(z, t_0) \quad (u = r).$$

Both functions exceed $3/5$ on $0 \le z \le 3/2$, $0 \le t \le T_*$. Indeed $\partial_t^2 F = -(12z^2 + 6tz^4)/m_h \le 0$, so it suffices to bound the two edges $t = 0, T_*$. Replacing z^4 by $(9/4)z^2$ in its negative term gives a lower quadratic whose coefficient of z^2 is

$$q_{\mathrm{quad}}(t) = \frac{3\alpha^2 t - 6t^2 - (9/4)t^3}{m_h}, \qquad q_{\mathrm{quad}}(0) = 0, \quad q_{\mathrm{quad}}(T_*) = \frac{27842487291}{1250000000000} > 0.$$

For the negative phase, its derivative at the right endpoint is $-21/100$ when $t = 0$, and is less than $-1/5$ when $t = T_*$. The quadratic therefore decreases throughout the interval. For the positive phase, its derivative at zero is $19/100$ or $7027/62500$, so it increases. The resulting endpoint minima satisfy

	$t = 0$	$t = T_*$
$F_{1,-101/100}$ lower bound at $z = 3/2$	$933/1000$	$> 769/1000$
$F_{-1,99/100}$ lower bound at $z = 0$	$156/125$	$3597/3125$

All four exceed $3/5$. Concavity in t then covers the entire rectangle.

For negative phases $3t_0/2 \le r \le b_*$, Cantelli gives

$$O \ge \frac{r^2}{t_0 + r^2} - \frac{101}{100} r > \frac{3}{5} t_0.$$

Its derivative at $r = 3t_0/2$ is positive and first increases, then decreases. The minimum is therefore at an endpoint. At the first use $t_0 \le T_*$ after division by t_0; at the other use $b_*^2/(T_* + b_*^2) - (101/100)b_* > (3/5)T_*$. For positive phases beyond this switch, $O \ge (99/100)r \ge (297/200)t_0 > (3/5)t_0$. The switch lies below b_* because $3T_*/2 < b_*$. The value $u = 0$ is included, with $h_\alpha(0) = 0$. □

The outer phases and the final margin.

For $u = -r$, $b_* \le r \le 3/10$, Cantelli gives

$$O \ge \frac{r^2}{T_* + r^2} - \frac{51}{50} r > 7/50 \ge 14\lambda.$$

The derivative is positive at b_* and has at most one downward zero; both endpoint values exceed $7/50$. For $b_* \le u \le 7/10$,

$$O - r_{\max} T_{\mathrm{in}} \ge \frac{49}{50} u - \frac{(6/5)(43/100)\lambda_s}{(1-u)^4} > 1/25 \ge 4\lambda.$$

This function is concave and both endpoints exceed $1/25$. In the inner interval, (36) and (37) give $O - r_{\max} T_{\mathrm{in}} > (.59 - (51/50).53)\lambda$. Substitute these three margins and (35) into (32), using $m_k > 3/4$. The smallest final margin is the inner one, and

$$\frac{3}{4}\left(\frac{59}{100} - \frac{51}{50}\frac{53}{100}\right) - \frac{2}{700} = \frac{4787}{140000} > \frac{1}{100}.$$

Hence $\Delta_C < -\lambda/100$. Taking $C = C_{\mathrm{E}}$ and applying (29) proves (28) for $0 < \lambda \le 1/100$.

7.3 Larger noise: five counts and two endpoint checks

Now assume $1/100 \le \lambda \le 7/20$ and choose $k = \lfloor t \rfloor$, $u = t - k \in [0, 1)$, $w = t - np$. The following constants will be used only in this subsection:

$$n_0 = 700, \quad p_- = 3/8, \quad p_+ = 93/200,$$
$$r = 3/8, \quad q_{\min} = 107/200,$$
$$v_- = 15/64, \quad v_+ = 9951/40000, \quad A = 3/5, \quad d = 1/250,$$
$$\eta = 2/5, \quad c_{\mathrm{mass}} = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5,$$
$$\kappa = 3, \quad \xi = 93/100, \quad c_{\mathrm{var}} = 6, \quad \zeta = 23/100, \quad m_* = 3/4, \quad V_* = 113/1000,$$
$$\bar{c} = 79/200, \quad j_0 = 1/5, \quad j_1 = 5/2.$$

In particular $|w| < \xi < 1$, since $(46/25)^2(v_+ + (7/20)/n_0) < \xi^2$. Also $n_0(r - d) > \xi$, so $na_0 \le t < n(1 - a_1)$.

An exact weighted crossing identity.

On an auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \qquad \mathcal{L}_j(t) = \mathbb{E}(t - T_j)_+, \qquad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh is increasing because $T_j - T_{j-1} \ge 1 - a_0 - a_1 > 0$. Each T_j has the conditional convolution given j successes. Set

$$\chi(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad \chi_i = \mathbb{E}\chi(t - i + 1 - V_i).$$

Lemma 7.3 (Count interpolation and crossing loss)**.** *With the preceding definitions,* $\mathcal{B}_j = \mathcal{L}_j(t) - (t - j)_+ \ge 0$ *and*

$$\mathbb{P}\left(\sum X_i \le H\right) = (1 - u)F_B(k) + uF_B(k + 1) + \sum_{j=1}^{n-1}(b_{j+1} - b_j)\mathcal{B}_j - \sum_{i=1}^{n} b_i\chi_i. \tag{39}$$

Proof. Jensen's inequality for the convex positive-part function and $\mathbb{E}T_j = j$ give $\mathcal{B}_j \ge 0$. Independence of the omitted coordinate gives $\chi_i = \mathcal{L}_{i-1}(t) - \mathcal{L}_i(t) - \mathbb{P}(T_i \le t)$. Here $\mathcal{L}_0 = t$, $\mathcal{L}_n = 0$ and $\mathbb{P}(T_0 \le t) = 1$: the support bounds give $T_0 \le na_0 \le t < T_n$. Multiplication by b_i and finite summation by parts now gives the identity, since replacing $\mathcal{L}_j$ by $(t - j)_+$ in $b_0 + b_1 t + \sum(b_{j+1} - b_j)\mathcal{L}_j$ telescopes to the stated interpolation. All CDF events are closed, including atoms. □

The functions $\mathcal{L}_j(t)$ are expectations of the convex positive-part function; their Jensen gaps $\mathcal{B}_j$ measure how noise changes those expectations. Thus the three terms in (39) are the interpolated binomial probability, an error from variation in the binomial masses, and the crossing loss. The next argument proves $\chi_i \ge 0$ using the separated conditional supports and conditional centering; this does not follow merely from $\mathcal{B}_j \ge 0$.

Five kernels with four centered replacements.

The sine bump below serves three purposes. Its translates recover a single periodic function, its Lipschitz derivative makes centered replacement errors proportional to variance, and the identity for $\sin^2$ turns independence into a product of characteristic functions. Five translates will supply the main loss; the omitted translates are bounded using a fourth moment. Let $\psi(x) = \sin^2(\pi x)$ on $[0, 1]$ and zero outside. It is continuously differentiable with derivative Lipschitz constant $2\pi^2$.

Lemma 7.4 (A nonnegative crossing kernel)**.** *For every real x,*

$$\chi(x) \geq \eta\psi(x) \geq 0. \tag{40}$$

Proof. For $x < 1 - a_1$, $\chi(x) = \mathbb{E}(x - U_0)_+ \geq 0$, and Jensen gives $\chi(x) \geq x$ when $x \geq 0$. On $[0, 1]$, $\psi(x) \leq (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $\psi \leq 1$ thereafter. For $x \geq 1 - a_1 > a_0$, put $D_1 = 1 + U_1$ and $L_1 = \mathbb{E}(D_1 - x)_+$. Put–call parity gives $\chi(x) = \mathbb{P}(D_1 > x) - L_1 \geq 0$, since $D_1 \leq 1 + A$ and $x > A$. When $1 - a_1 \leq x \leq 1$, $L_1 \geq 1 - x$ and

$$\chi(x) \geq \frac{x - A}{1 + A - x} L_1 \geq \frac{1 - A - d}{A + d}(1 - x) \geq \eta\pi^2(1 - x)^2 \geq \eta\psi(x).$$

The penultimate inequality uses $\eta\pi_+^2 d \leq (1 - A - d)/(A + d)$. At $x = 0, 1$ both comparisons are weak, as required for atoms. □

The useful feature of the sine bump is its phase-uniform average: for any real random variable V,

$$\mathbb{E}\sin^2(\pi(u - V)) = \frac{1 - \Re\{e^{2\pi i u}\mathbb{E}e^{-2\pi i V}\}}{2} \geq \frac{1 - |\mathbb{E}e^{2\pi i V}|}{2}.$$

The characteristic function factorizes for a sum of independent noises. To use this observation, we compare five neighboring count sums with one fixed sum, paying for four centered replacements.

Lemma 7.5 (Five-count crossing loss)**.** *Under the larger-noise hypotheses,*

$$\sum_{i=k-1}^{k+3} \chi_i \geq \eta \left\{ \frac{1 - e^{-\kappa(1-6/n)\lambda}}{2} - \frac{A^2(1 + 3/n)\lambda + 3(1 + 3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr} \right\}. \tag{41}$$

Proof. Fix $V = V_{k+1}$ and let $V^{(j)}$ have the law of V_{k+1+j}. For $j > 0$ it replaces j copies of U_0 by U_1. With $\psi_j(x) = \psi(u - j - x)$, regroup the two positive changes as

$$\begin{aligned} &\mathbb{E}\psi_1(V^{(1)}) + \mathbb{E}\psi_2(V^{(2)}) - \mathbb{E}\psi_1(V) - \mathbb{E}\psi_2(V) \\ &\quad = \mathbb{E}(\psi_1 + \psi_2)(V^{(1)}) - \mathbb{E}(\psi_1 + \psi_2)(V) + \mathbb{E}\psi_2(V^{(2)}) - \mathbb{E}\psi_2(V^{(1)}). \end{aligned}$$

There is an identical grouping for the negative changes. Adjacent bumps have disjoint open supports and zero derivatives at the joins, so the derivative Lipschitz constant remains $2\pi^2$. For a centered Z, $|\mathbb{E}F(a + Z) - F(a)| \leq \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Conditioning on all unchanged summands, and applying this twice to each replacement, bounds the total cost by $4\pi^2(s_0 + s_1)$. Hence

$$\sum_{i=k-1}^{k+3} \chi_i \geq \eta \left\{ \sum_{j=-2}^{2} \mathbb{E}\psi(u - j - V) - 4\pi^2(s_0 + s_1) \right\}.$$

These five indices are interior, because $|i - np| < \xi + 3 < 4$. The summands within each sum are independent; neighboring count sums share the unchanged summands in the replacement argument.

The infinite translates of ψ sum to $\sin^2(\pi(u - V))$. An omitted nonzero translate requires $|V| > 2$, including endpoint collisions because the bump vanishes there. Thus the omitted part is at most $\mathbb{E}V^4/16$. With $\delta = k - np \in (-1 - \xi, \xi)$, the variance of the fixed sum is

$$\sigma_V^2 = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad (1 - 6/n)\lambda \leq \sigma_V^2 \leq (1 + 3/n)\lambda.$$

The lower estimate uses $c_{\mathrm{var}} r \geq 1 + \xi$; the upper uses $r \geq 1/3$. Also $s_0 + s_1 \leq \lambda/(nr)$ and $\mathbb{E}V^4 \leq A^2\sigma_V^2 + 3\sigma_V^4$. For either conditional noise Z, an independent copy satisfies $|Z - Z'| \leq$

$A + d < 2/3$. Concavity of sine and $\sin(2\pi/3) = \sqrt{3}/2 > 17/20$ give $\sin(\pi x) \geq (51/40)x$ on $[0, A + d]$; hence $1 - \cos(2\pi x) \geq 3x^2$ on $|x| \leq A + d$. Therefore

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \geq 2\kappa \operatorname{Var} Z,$$

and $|\mathbb{E}e^{2\pi i Z}| \leq e^{-\kappa \operatorname{Var} Z}$. Multiplying over the independent summands of V shows $\mathbb{E}\sin^2(\pi(u - V)) \geq (1 - e^{-\kappa\sigma_V^2})/2$. Combining the replacement cost, the omitted translates, and the characteristic-function bound proves the displayed inequality. □

The weights and the complete curvature tail.

For $k - 1 \leq i \leq k + 3$, the segment between i/n and p has variance product greater than ζ; indeed $\zeta < \min_{z=p_- - 4/n_0,\, p_+ + 4/n_0} z(1 - z)$. Relative entropy is bounded by $(i - np)^2/(2n\zeta) < 8/(n\zeta)$, and the factorial remainder is greater than $-1/(12n\zeta)$. Stirling therefore gives

$$\sqrt{n}\, b_i > 2\phi_- \left(1 - \frac{8 + 1/12}{n_0\zeta}\right) > m_*.$$

Every other χ_i is nonnegative. The right-hand side of (41) increases with n. Replace n by n_0 first, then use $1 - e^{-x} \geq x - x^2/2 + x^3/6 - x^4/24$ for $x \geq 0$. Writing $b = 1 + 3/n_0$ and $\kappa_0 = \kappa(1 - 6/n_0)$, we obtain

$$\begin{gathered} \sqrt{n} \sum_i b_i \chi_i \geq \mathcal{W}(\lambda), \\ \mathcal{W}(\lambda) := \eta m_* \left\{ \frac{\kappa_0\lambda - \kappa_0^2\lambda^2/2 + \kappa_0^3\lambda^3/6 - \kappa_0^4\lambda^4/24}{2} - \frac{A^2 b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{n_0 r} \right\}. \end{gathered} \tag{42}$$

Lemma 7.6 (Variation of the binomial weights)**.** *Put $c = (np - q - t)_+ \leq \bar{c}$. If T and M_4 bound respectively the second and fourth moments of W_j whenever $b_{j+1} > b_j$, then*

$$\sqrt{n} \sum_j (b_{j+1} - b_j)\mathcal{B}_j \leq \frac{c_{\text{mass}}}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M_4}{256} \left(\frac{5}{3} + \frac{5c}{4} \right) \right\}. \tag{43}$$

Proof. For $a = t - j$ and $W_j = T_j - j$, centering gives

$$\mathcal{B}_j = (\mathbb{E}|a - W_j| - |a|)/2 \geq 0, \qquad \mathcal{B}_j \leq \sqrt{T}/2, \qquad a_+\mathcal{B}_j \leq T/4,$$

whenever $T \geq \mathbb{E}W_j^2$. The factorization $27x^4 - 256x + 256 = (3x - 4)^2(3x^2 + 8x + 16)$ gives $\mathcal{B}_j \leq 27M_4/(256|a|^3)$ for $a \neq 0$, if $M_4 \geq \mathbb{E}W_j^4$. The exact adjacent-mass formula is

$$b_{j+1} - b_j = \frac{np - q - j}{(n + 1)v}\, \mathbb{P}\{\text{Bin}(n + 1, p) = j + 1\}.$$

Discard its nonpositive terms and bound the positive numerator by $(a + c)_+$. Since $c \leq \bar{c} < 1$, at most one mesh point lies in each of $(-c, 0)$ and $[0, 1)$; together they cost $c\sqrt{T} + T/4$. The rest is bounded by the complete series $\sum i^{-2} < 5/3$, $\sum i^{-3} < 5/4$. Using (30) and $\sqrt{n}/(n + 1)^{3/2} \leq 1/n$ proves the asserted bound. □

A positive adjacent mass requires $j < np$, where its actual variance is at most $\lambda/q \leq T = \lambda/q_{\min}$ and $\mathbb{E}W_j^4 \leq M_4 = A^2T + 3T^2$. Moreover $L_{\text{var}}(\lambda) = j_1\lambda + j_0 \geq \sqrt{T}$, since $4j_0j_1 \geq 1/q_{\min}$.

Both signs of the normal correction.

Apply (29) at $k, k+1$ and interpolate. For $G(x) = \Phi((x-np)/\sqrt{nv})$, $G''(x) \le \phi(0)(np - x)_+/(nv)^{3/2}$. The interpolation remainder is

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}} u(1-u)(u-w)_+.$$

If $H \le 0$, changing normal variance from v to $v+s$ is favorable. Using $u(1-u) \le 1/4$ and $u^2(1-u) \le 4/27$, the total normal cost is at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$, and $c = (-q-w)_+ \le \bar{c}$. If $H \ge 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance change costs at most $\phi(0)ws/(2v^{3/2})$ by $1 - (1+x)^{-1/2} \le x/2$. The cubic correction is favorable by (31).

Define the nonnegative quadratics

$$S_c = \frac{T}{4} + cL_{\text{var}}(\lambda) + \frac{27M_4}{256}\left(\frac{5}{3} + \frac{5c}{4}\right), \qquad E_- = \frac{c_{\text{mass}} S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V_*},$$

$$E_+ = \frac{c_{\text{mass}} S_0 + \phi_+(2/27 + \xi\lambda/2)}{V_*}.$$

Because $V_*^2 < v_-^3$, combining (39)–(43) with these normal bounds makes the full excess strictly less than $-\mathcal{W}(\lambda) + E_\pm(\lambda)/n$, with the sign chosen according to H. A single envelope suffices for both signs, because

$$\begin{aligned} V_*(E_- - E_+) &= c_{\text{mass}}\bar{c}L_{\text{var}}(\lambda) + c_{\text{mass}}\frac{27M_4}{256}\frac{5\bar{c}}{4} + \phi_+\xi\left(\frac{1}{8} - \frac{\lambda}{2}\right) \\ &= \frac{7889}{100000} + \frac{1751}{8000}\lambda + c_{\text{mass}}\frac{27M_4}{256}\frac{5\bar{c}}{4} > 0 \qquad (\lambda \ge 0). \end{aligned}$$

Set $E = E_-$. The full excess is strictly less than $-\mathcal{W}(\lambda) + E(\lambda)/n$ for either sign of H.

Concavity reduces the interval to two endpoints.

Set $\Pi(\lambda) = \mathcal{W}(\lambda) - E(\lambda)/n_0 - \lambda/100$. The nonnegative polynomial E is a convex quadratic, whereas direct differentiation gives

$$\mathcal{W}''(\lambda) = -\eta m_* \left\{ \frac{\kappa_0^2}{4}\left[(\kappa_0\lambda - 1)^2 + 1\right] + \frac{3b^2}{8} \right\} < 0.$$

Thus Π is strictly concave. Substitution of the rational constants above gives the exact rational lower bounds

$$\Pi(1/100) > .002314, \qquad \Pi(7/20) > .061304.$$

Both are positive, proving $\Pi > 0$ on the whole closed interval. Since $n \ge n_0$ and $E \ge 0$, this proves (28) for $1/100 \le \lambda \le 7/20$. The endpoint calculations are rational after using $199/500 < \phi(0) < 2/5$ and $25/8 < \pi < 22/7$. Concavity supplies the bounds between the endpoints, while (36) and (43) retain all entering counts.

The two positive-noise proofs have identical support and probability hypotheses, include their shared endpoint, and cover every phase. At $\lambda = 0$, conditional centering and $\mathbb{E}U^2 = 0$ give $U = 0$ almost surely, so (29) applies directly. This completes the proof of Theorem 7.1.

8 What a first failing index must satisfy

The finite arguments use the ratio extremizer of Corollary 2.2. A simple recurrence supplies the same small excess at any first failure of the sharp bound. This lets us use local exclusions directly, without first proving a separate sequence of positive upper barriers.

Lemma 8.1 (Gaussian derivative bounds)**.** *The standard normal density satisfies*

$$\|\phi''\|_\infty = \phi(0) < \frac{2}{5}, \qquad \|\phi'''\|_\infty \le \frac{2}{\pi}, \qquad \frac{1}{4\pi} < \frac{2}{25}, \qquad C_{\mathrm{E}} < \frac{41}{100}.$$

Proof. Since $\phi''(x) = (x^2-1)\phi(x)$, its absolute maximum is at zero: the other candidates $x = \pm\sqrt{3}$ have value $2e^{-3/2}\phi(0) < \phi(0)$. The absolutely convergent Gaussian Fourier integral gives

$$|\phi'''(x)| \le \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\,dt = \frac{2}{\pi}.$$

The bounds for $\phi(0)$ and $1/(4\pi)$ follow from $\pi > 25/8$. For the last comparison one may use $\sqrt{10} < 31623/10000$ and $\pi > 314159/100000$, square the positive sides, and compare rational numbers. □

Theorem 8.2 (One-step smoothing)**.** *For every integer $n \ge 2$,*

$$C_n \le \sqrt{\frac{n}{n-1}}\,C_{n-1} + \frac{\sqrt{n}}{15(n-1)^{3/2}} + \frac{2\sqrt{n}}{25(n-1)^2}. \tag{44}$$

Proof. Fix one standardized marginal P, write $v = n-1$, and put $G_v(x) = \Phi(x/\sqrt{v})$. Conditioning on the last summand yields

$$|\mathbb{P}(S_n \le H) - \mathbb{E}G_v(H-X)| \le C_{n-1}\rho/\sqrt{v}.$$

Taylor expansion of the smooth Gaussian CDF, followed by expectation under this same marginal, gives

$$|\mathbb{E}G_v(H-X) - G_v(H) - \tfrac{1}{2}G_v''(H)| \le \frac{\phi(0)\rho}{6v^{3/2}} < \frac{\rho}{15v^{3/2}}.$$

Here $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ cancel the first two Taylor terms. For the change of variance, $\partial_v G_v = G_v''/2$ and $\partial_v^2 G_v = G_v^{(4)}/4$. Thus

$$G_{v+1}(H) - G_v(H) - \tfrac{1}{2}G_v''(H) = \frac{1}{4}\int_0^1 (1-t)G_{v+t}^{(4)}(H)\,dt.$$

Since $G_w^{(4)}(H) = w^{-2}\phi'''(H/\sqrt{w})$, this remainder has absolute value at most $1/(4\pi v^2) < 2/(25v^2)$. Combine the three estimates, multiply by $\sqrt{n}/\rho$, and use $\rho \ge 1$. The bound is uniform in P and H and treats both signs of the discrepancy, so it passes to the two suprema defining C_n. □

Corollary 8.3 (Entry from a sharp predecessor)**.** *If $n \ge 40$ and $C_{n-1} \le C_{\mathrm{E}} \le C_n$, then*

$$0 \le n(C_n - C_{\mathrm{E}}) < \frac{1433}{4875} < \frac{3}{10}, \qquad C_n < \frac{167}{400} < 0.4226, \qquad n(C_{n-1} - C_n)_+ = 0.$$

For $n \ge 42$ the excess bound improves to $3003/10250$ and the score bound to $73/175$. For every standardized marginal P, the same-law predecessor loss $n(\sqrt{n-1}\Delta_{n-1}(P)/\rho(P) - C_n)_+$ is zero. At a first failing index $n \ge 43$ above a sharp base, both deleted constants C_{n-1} and C_{n-2} are below 0.4253.

Proof. Put $s = \sqrt{n/(n-1)}$. The recurrence and $n(s-1) = \{n/(n-1)\}/(s+1)$ give

$$n(C_n - C_{\mathrm{E}}) \le \frac{n}{n-1}\left\{\frac{C_{\mathrm{E}}}{s+1} + \frac{s}{15} + \frac{2s}{25\sqrt{n-1}}\right\}.$$

For $n \ge 40$, $n/(n-1) \le 40/39$, $\sqrt{n-1} > 6$, and $1 < s < 51/50$, since $39 \cdot 51^2 - 40 \cdot 50^2 = 1439 > 0$. The right side is therefore less than

$$\frac{40}{39}\left(\frac{41}{200} + \frac{17}{250} + \frac{17}{1250}\right) = \frac{1433}{4875} < \frac{3}{10}.$$

This gives $C_n < 0.41 + 0.3/40 = 167/400$. For $n \ge 42$, $n/(n-1) \le 42/41$ and $1 < s < 81/80$; the last comparison follows from $41 \cdot 81^2 - 42 \cdot 80^2 = 201 > 0$. Using $C_{\mathrm{E}} < 41/100$ and $s + 1 > 2$, the right side is less than

$$\frac{42}{41}\left(\frac{41}{200} + \frac{27}{400} + \frac{27}{2000}\right) = \frac{3003}{10250} < \frac{3}{10}.$$

Consequently $C_n < 0.41 + 0.3/42 = 73/175$. Every law's predecessor ratio is bounded by $C_{n-1} \le C_{\mathrm{E}} \le C_n$, so both loss assertions follow. At a first failure the two deleted indices are sharp and $C_{\mathrm{E}} < 0.41 < 0.4253$. □

8.1 The loose ceilings needed after index twenty-four

Lemma 8.4. *If $n \ge 25$ and $C_{n-1} \le C_{\mathrm{E}} \le C_n$, then*

$$n(C_n - C_{\mathrm{E}}) < \frac{4177}{13824} < \frac{303}{1000}, \qquad C_n < \frac{145873}{345600} < \frac{423}{1000}.$$

For each marginal the same-law predecessor allowance is zero.

Proof. Set $s = \sqrt{n/(n-1)}$. Theorem 8.2 gives

$$n(C_n - C_{\mathrm{E}}) \le \frac{n}{n-1}\left(\frac{C_{\mathrm{E}}}{s+1} + \frac{s}{15} + \frac{2s}{25\sqrt{n-1}}\right).$$

For $n \ge 25$ one has $n/(n-1) \le 25/24$, $1 < s < 49/48$ and $\sqrt{n-1} > 24/5$. The latter strict comparisons follow from $24 \cdot 49^2 - 25 \cdot 48^2 = 24 > 0$ and $24 > 576/25$. Together with $C_{\mathrm{E}} < 41/100$, these imply

$$n(C_n - C_{\mathrm{E}}) < \frac{25}{24}\left(\frac{41}{200} + \frac{49}{720} + \frac{49}{2880}\right) = \frac{4177}{13824}.$$

Dividing by $n \ge 25$ proves the score bound. The exact positive comparison margins are $303/1000 - 4177/13824 = 1459/1728000$ and $423/1000 - 145873/345600 = 1579/1728000$. Finally each marginal's predecessor score is at most $C_{n-1} \le C_{\mathrm{E}} \le C_n$. □

In the original floor-25 initial and resource calculations, the own-score ceiling is $434477/10^6$ and the retained predecessor allowance is $5/16$. In the floor-31 calculations they are respectively $432313/10^6$ and $31/100$. Both own-score ceilings exceed $423/1000$, by respectively $11477/10^6$ and $9313/10^6$. Both allowances are nonnegative. Thus Lemma 8.4 supplies strictly stronger entry information at a first failure. The native targets and numerical inputs remain exactly as originally specified. The stronger information licenses their weaker hypotheses; it does not change any certificate inequality.

Using a floor-N resource calculation at an actual index $n > N$ additionally requires its weighted invariant and its actual-index support hypotheses. Lemma 8.4 supplies only the scalar entry conditions; it does not transfer support or the bound $D \le d$ from one index to another. The transfer used here is the proved $nD \le Nd$ from the common finite resource calculus, together with separately verified actual-index support.

9 The starting cases

9.1 The first three sample sizes

The small cases provide quantitative starting bounds for the later induction. The second and third cases use the same idea: a hypothetical violating extremizer has bounded support, and a polynomial majorizes its closed sum event on the whole support product. Independence then evaluates the polynomial using moments of the same marginal. No restriction on the number of atoms is imposed.

Theorem 9.1. *For every standardized law with finite third absolute moment,*

$$C_1 \le \frac{2}{5}, \qquad C_2 \le \frac{2\sqrt{2}}{7}, \qquad C_3 \le \frac{59\sqrt{3}}{250}.$$

All three upper bounds are strictly smaller than C_{E}.

One summand.

The certificate consists of 150 consecutive rational cells $[a, b]$ covering $[-2, 1]$, with two rational coefficients u, v in each cell. It proves

$$L_a - v + uy + vy^2 + \frac{2}{5}|y|^3 - \mathbf{1}_{\{y \le b\}} \ge 0 \quad (y \in \mathbb{R}), \qquad L_a \le \Phi(a). \tag{45}$$

Taking expectations cancels the linear and quadratic moment terms. For $t \in [a, b]$ this gives

$$F_P(t) - \Phi(t) \le F_P(b) - L_a \le \frac{2}{5}\rho.$$

For $t \le -2$, Cantelli gives $F_P(t) \le 1/(1 + t^2) \le 1/5$. For $t \ge 1$, the discrepancy is at most $1 - \Phi(1) < 2/5$. Since $\rho \ge 1$, these bounds cover both exterior rays. Reflection gives the negative branch, including atoms.

For completeness, the verification of (45) splits the real line at $-10, 0, b, 10$. On each finite interval the expression is a cubic, checked in the Bernstein basis with subdivision; on either infinite ray its translated coefficients are nonnegative. The value at b uses the included, closed-event polynomial. The normal lower bound uses Machin's identity, integer square-root brackets and an alternating Gaussian series. All 150 cells and 1,128 terminal Bernstein intervals pass exact rational verification.

A common support argument for two and three summands.

Let $K_2 = 2\sqrt{2}/7$, $K_3 = 59\sqrt{3}/250$ and suppose $C_n > K_n$, where $n = 2$ or 3. Corollary 2.2 supplies a positive closed-CDF maximizing pair. Write

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \le H), \quad D_0 = F - \Phi(z), \quad \lambda = D_0/\rho, \quad \beta = \mathbb{E}[X|X|].$$

Here D_0 denotes the discrepancy, not the support-gap defect used in the main argument. The contact condition is equivalently

$$n\mathbb{P}(S_{n-1} \le H - y) \le Q_n(y),$$
$$Q_n(y) = \lambda|y|^3 - \frac{z\phi(z) + 3D_0}{2}y^2 - (\sqrt{n}\phi(z) + 3\lambda\beta)y + nF + \frac{z\phi(z) + D_0}{2}, \tag{46}$$

with equality on the entire closed support. The published affine estimate

$$\frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)} \le \frac{1661}{5000}\left(1 + \frac{429}{1000\rho(P)}\right) \tag{47}$$

from Shevtsova [15] rules out $\rho \geq 2$ at $n = 2$ and $\rho \geq 15/8$ at $n = 3$. Together with $|\beta| \leq 1$, $\phi(z) < 2/5$ and $|z\phi(z)| < 1/4$, it gives the following lower bounds $Q_n(y) \geq P_n(|y|)$:

n	ρ ceiling	$P_n(r)$
2	2	$\frac{2}{7}r^3 - \frac{67}{56}r^2 - \frac{23}{14}r - \frac{1}{8}$
3	15/8	$\frac{8}{35}r^3 - \frac{4}{5}r^2 - \frac{8}{5}r - \frac{1}{8}$

Indeed, for $n = 2$, $\lambda > 2/7$, $D_0 < 5/7$ and $\lambda < 5/14$ suffice: $\sqrt{2} < 10/7$ gives $\sqrt{2}\phi(z) + 3\lambda < 4/7 + 15/14 = 23/14$. For $n = 3$, the corresponding bounds are $\lambda > 8/35$, $D_0 < 9/20$ and $\lambda < 2/7$. In the latter case (47) gives $C_3\rho < 77/100$, and $\sqrt{3} > 12/7$ then gives $D_0 = C_3\rho/\sqrt{3} < 9/20$. The shifted differences are

$$P_2(6+s) - 2 = \frac{2}{7}s^3 + \frac{221}{56}s^2 + \frac{104}{7}s + \frac{373}{56},$$
$$P_3(6+s) - 3 = \frac{8}{35}s^3 + \frac{116}{35}s^2 + \frac{472}{35}s + \frac{2197}{280}.$$

Every coefficient is positive. Equality in (46) therefore excludes every support point with $|y| \geq 6$, because its left side is at most n.

The two-summand majorants.

Put $\gamma = \mathbb{E}X^3$ and

$$v = (1, y, y^2, y^3, |y|^3), \qquad \mu = \mathbb{E}v(X) = (1, 0, 1, \gamma, \rho).$$

For a rational symmetric coefficient array indexed by $i \leq j$, let

$$M_c(x, y) = \sum_i c_{ii} v_i(x) v_i(y) + \sum_{i<j} c_{ij}\{v_i(x)v_j(y) + v_j(x)v_i(y)\},$$

and $q_c(\gamma, \rho) = \sum_i c_{ii}\mu_i^2 + 2\sum_{i<j} c_{ij}\mu_i\mu_j$. Independence gives $\mathbb{E}M_c(X_1, X_2) = q_c(\gamma, \rho)$. Theorem 2.4 puts the actual moments in

$$\mathcal{D}_2 = \{(\gamma, \rho) \in [-2, 2] \times [1, 2] : \rho^2(\gamma^2 + 4) - (\gamma^2 + 2)^2 \geq 0\}.$$

The 70 certificate rows use $a_j = -5/2 + j/20$, $b_j = a_j + 1/20$, $0 \leq j < 70$. Their exact inequalities are

$$M_{c^{(j)}}(x, y) \geq \mathbf{1}_{\{x+y \leq b_j\}} \qquad ((x, y) \in [-6, 6]^2),$$
$$q_{c^{(j)}}(\gamma, \rho) \leq \Phi(a_j/\sqrt{2}) + \frac{2}{7}\rho \qquad ((\gamma, \rho) \in \mathcal{D}_2).$$

Each sign quadrant is cut by the event line and triangulated into closed triangles. Degree-six Bernstein coefficients, with exact subdivision, establish the first inequality. A second checker independently homogenizes the original polynomial on every triangle and checks integer coefficients. Both checkers accept all 70 rows: 35,878 triangle nodes and 134 moment boxes. The second inequality is checked on moment rectangles; a rectangle is discarded only if $\rho_+^2(g_* + 4) - (g_* + 2)^2 < 0$, where g_* is the minimum of γ^2 on that rectangle. This is valid because the expression increases with ρ and decreases with γ^2 when $\rho \leq 2$.

For a threshold in $[a_j, b_j]$, expectation of the first inequality and the second give discrepancy at most $2\rho/7$. For $H \leq -5/2$, Cantelli gives $2/(2 + H^2) \leq 8/33 < 2/7$; for $H \geq 1$, $1 - \Phi(H/\sqrt{2}) < 2/7$. These bounds contradict the chosen violating maximizer. This proves the second assertion of Theorem 9.1.

The three-summand majorants.

Use the features

$$(v_0,\dots,v_6)(y)=(1,y,y^2-1,y^3,|y|^3,y^4,y^5).$$

Let $\mathcal{I}$ be the 59 nondecreasing triples from $\{0,\dots,6\}$ whose largest entry is less than 5 or which contain an entry 1 or 2. For $\alpha\in\mathcal{I}$, average the associated product over its distinct coordinate permutations, obtaining V_α. Set $M_c=\sum_\alpha c_\alpha V_\alpha$. The bounded-support argument above makes every feature integrable. Every term involving a fourth or fifth power contains a zero-mean factor X_i or X_i^2-1 in a different coordinate. Independence therefore gives the exact identity

$$\mathbb{E}M_c(X_1,X_2,X_3)=q_c(\gamma,\rho)=\sum_{\substack{\alpha\in\mathcal{I}\\ \alpha\text{ uses only }0,3,4}} c_\alpha\gamma^{\#3(\alpha)}\rho^{\#4(\alpha)}.$$

There are ten surviving monomials and 49 zero-expectation controls. The controls improve the pointwise majorant without adding moment parameters.

The 111 rational rows form a consecutive closed cover of $[-7/2,3/2]$. They satisfy

$$\begin{aligned} M_{c^{(j)}}(x_1,x_2,x_3)&\ge \mathbf{1}_{\{x_1+x_2+x_3\le b_j\}} && ((x_1,x_2,x_3)\in[-7,7]^3),\\ q_{c^{(j)}}(\gamma,\rho)&\le \Phi(a_j/\sqrt{3})+\frac{59}{250}\rho && ((\gamma,\rho)\in\mathcal{D}_3),\end{aligned}$$

where $\mathcal{D}_3$ replaces 2 by 15/8 in the rectangular bounds defining $\mathcal{D}_2$. Symmetry reduces the sign patterns to four ordered-coordinate orthants. A box is discarded only when it has no ordered point. On every other box, degree-five tensor Bernstein bounds establish the pointwise inequalities. On a box crossing the event plane, the included side is checked by $M-1\ge s(x_1+x_2+x_3-b_j)$ with $s\le 0$, and the other side by the analogous inequality with $s\ge 0$. Existence of a slope is checked by intersecting exact rational intervals; zero plane coefficients are checked directly. This retains the closed event boundary.

Fresh exact verification accepts all 111 cells, with 501,156 cube nodes, 219,860 terminal leaves and 349 moment boxes. An additional 95,904 exact interpolation identities compare the initial polynomial expansion with direct evaluation of the original features. Those identities check the expansion; the cube verification proves positivity. As in the two-summand case, expectation bounds the discrepancy on each threshold interval. The exterior bounds are $3/(3+H^2)\le 12/61<59/250$ for $H\le -7/2$ and $1-\Phi(H/\sqrt{3})<59/250$ for $H\ge 3/2$. They contradict the violating maximizer and prove the third assertion. The first assertion was proved above. Rational square-root and Machin bounds give $K_2<0.405<0.409<C_{\mathrm{E}}$ and $K_3<0.409<C_{\mathrm{E}}$.

9.2 The fourth case: localization before numerical comparison

The third bound is needed quantitatively. Merely knowing $C_3\le C_{\mathrm{E}}$ would not give the following support reduction. Write

$$K_3=\frac{59\sqrt{3}}{250},\quad c_0=\frac{256}{625},\quad T=\frac{4097}{10000};\qquad K_3<c_0<T<C_{\mathrm{E}}.$$

For any standardized law set $\beta=\mathbb{E}[X|X|]$, $u=\mathbb{E}X_+=\mathbb{E}X_->0$, $v_\pm=(1\pm\beta)/2$, and

$$a=\frac{v_-}{u},\quad b=\frac{v_+}{u},\quad w=\frac{4u^2}{1-\beta^2},\quad Q=\frac{u}{a}\delta_{-a}+(1-w)\delta_0+\frac{u}{b}\delta_b. \tag{48}$$

Cauchy's inequality gives $0<w\le 1$ and makes Q a standardized probability law. Put $r=\rho(Q)=(1+\beta^2)/\sqrt{w(1-\beta^2)}$.

Lemma 9.2. *The nonnegative quantity*

$$W(X) = X_+(X_+ - b)^2 + X_-(X_- - a)^2$$

has expectation $\eta = \rho(P) - r$. Thus $r \le \rho(P)$, with equality exactly when $P = Q$. Moreover, writing $c(X) = b$ for $X > 0$ and $c(X) = a$ for $X < 0$,

$$\begin{aligned} \mathbb{E}X^3 &= b - a + \mathbb{E}[\operatorname{sgn}(X)W(X)], \\ \mathbb{E}X^4 &= a^2 - ab + b^2 + \mathbb{E}[(|X| + 2c(X))W(X)] \end{aligned} \tag{49}$$

whenever the fourth moment is finite.

Proof. On either half-line put $y = |X|$ and $c = a$ or b. The identities $y^3 = y(y-c)^2 + 2cy^2 - c^2y$ and $y^4 = (y+2c)y(y-c)^2 + 3c^2y^2 - 2c^3y$ integrate using $\mathbb{E}[y; \text{half-line}] = u$ and $\mathbb{E}[y^2; \text{half-line}] = uc$. If $\mathbb{E}W = 0$, the support is contained in $\{-a, 0, b\}$ and the first two moments fix the masses. Conversely $P = Q$ has $W = 0$. □

The three-point law is a reference; the original law remains arbitrary.

Proposition 9.3. *If $C_4 > c_0$, a positive closed-CDF maximizing pair satisfies*

$$\operatorname{supp} P \subset (-101/25, 101/25), \quad \rho(P) < 46/25, \quad 625/2116 < w \le 1, \quad |\beta| < 321/500.$$

In particular it belongs to the closed numerical class with $|X| \le 17/4$ and $|\beta| \le 13/20$.

Proof. Use Corollary 2.2. The affine inequality (74) first gives $\rho < 979013/532500 < 46/25$, without a support assumption. Since $r \le \rho$ and $w \le 1$, the inequality

$$\frac{(1 + (321/500)^2)^2}{1 - (321/500)^2} > (46/25)^2$$

gives $|\beta| < 321/500$. Put $C = C_4$, $z = H/2$, $s = \sqrt{4/3}$. The contact expression, using the quantitative bound on the three-summand predecessor, obeys

$$\begin{aligned} I_{\text{new}}(y) \le{}& \mathcal{H}_4(z, y) + 4\rho(K_3 - C) + 4\rho(s-1)K_3 \\ &+ C\{-|y|^3 + \tfrac{3}{2}\rho y^2 + 3\beta y - \rho/2\}, \mathcal{H}_4(z, y) = \quad 8\left\{\Phi\left(\frac{z - y/2}{\sqrt{3/4}}\right) - \Phi(z)\right\} \\ &+ 4y\phi(z) + z\phi(z)(y^2 - 1). \end{aligned}$$

Taylor expansion in the translation and the variance gives, uniformly in z, y,

$$|\mathcal{H}_4(z, y)| < \frac{\phi(0)}{6}|y|^3 + \frac{|y|}{5} + \frac{1}{15}.$$

Here $\sup|\phi''| = \sup|(x\phi(x))'| = \phi(0) < 2/5$ and $\sup|x(3 - x^2)\phi(x)| < 3/5$ suffice. For the last bound, on $t = x^2 \le 3$ compare e^t with its cubic Taylor polynomial and use

$$\tfrac{9}{4}(1 + t + t^2/2 + t^3/6) - t(3 - t)^2 = \tfrac{3}{4}\{7(t - 9/14)^2 + 3/28\} + \tfrac{5}{8}t^2(3 - t) > 0.$$

For $t \ge 3$, use $|x|^3\phi(x) < 1/2$. The second derivative of $\Phi((z - y/2)/\sqrt{1 - \lambda})$ has absolute value below $4/15$ on $0 \le \lambda \le 1/4$, giving the remainder $1/15$ after multiplication by eight.

Discard $4\rho(K_3 - C) < 0$ and use $4(s-1) < 5/8$. For $R = |y| \ge 4$, the remaining coefficient of C is at most $B(R) = -R^3 + 69R^2/25 + 963R/500 + 23/100$. All coefficients of $B(4 + t)$ are negative, so substituting $C > c_0$ is now valid. Consequently

$$I_{\text{new}}(y) < -\frac{643}{1875}R^3 + \frac{17664}{15625}R^2 + \frac{77257}{78125}R + \frac{7541}{46875}.$$

The coefficients after translation by $R = 101/25 + t$ are all negative. This excludes $|y| \ge 101/25$ from the closed support by contact. Finally $r^2 \ge 1/w$ gives $w > 625/2116$. □

10 A common finite count comparison

The finite argument follows the number of positive summands. Conditional on that number, the sum is a binomial lattice point plus a centered residual. The residual can move mass across a threshold in both directions. Keeping its outgoing and incoming contributions together is essential near a binomial jump. Everything in this section concerns one common marginal; no finite-atom assumption is made.

10.1 Exact conditioning and the common moments

For a standardized law put

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+, \quad h = v/m.$$

Centering and positive variance imply $0 < p < 1$ and $m > 0$. With $B = \mathbf{1}_{\{X>0\}}$, define $U = hX - B + p$. Then $\mathbb{E}(U \mid B) = 0$, and

$$hX = B - p + U, \qquad h^2 = v + s, \qquad s = qw_0 + pw_1, \quad w_i = \mathbb{E}(U^2 \mid B = i).$$

The atom at zero belongs to $B = 0$. If $s > 0$, let $\alpha = pw_1/s$ and $\lambda = ns$; thus $w_0 = \lambda(1-\alpha)/(nq)$ and $w_1 = \lambda\alpha/(np)$. When $s = 0$, both conditional residuals vanish and these formulas hold with any $\alpha \in [0, 1]$.

Write W_j for a sum of $n - j$ independent copies of $U \mid B = 0$ and j independent copies of $U \mid B = 1$, all independent. For $b_j = \binom{n}{j} p^j q^{n-j}$ and $z = np + hH = k + u$,

$$\mathbb{P}(S_n \le H) = \sum_{j=0}^{n} b_j \mathbb{P}(W_j \le k + u - j), \tag{50}$$

$$\mathbb{P}(S_n \le H) \le F_{\mathrm{Bin}(n,p)}(k) - b_k \mathbb{P}(W_k > u) + \sum_{j>k} b_j \mathbb{P}(W_j \le u - (j-k)). \tag{51}$$

The second line discards nonnegative outgoing probabilities at $j < k$. The outgoing event is strict and the incoming event is closed. In particular, an atom exactly on the threshold is retained on the CDF side. Also

$$V_j := \mathbb{E}W_j^2 = (n-j)w_0 + jw_1 = \lambda\left(1 + \frac{(j - np)(\alpha - p)}{npq}\right).$$

Suppose support acquisition gives $U_0 \in [-O_0, a_0]$ and $U_1 \in [-a_1, O_1]$, with nonnegative endpoints. For a centered $T \in [-L, R]$ of variance w,

$$w \le LR, \quad w^2/L - Lw \le \mathbb{E}T^3 \le Rw - w^2/R, \quad \mathbb{E}T^4 \le (R-L)\mathbb{E}T^3 + LRw. \tag{52}$$

The first inequality integrates $(T + L)(R - T) \ge 0$. The two cubic bounds integrate $(T + L)(T - w/L)^2 \ge 0$ and $(R - T)(T + w/R)^2 \ge 0$. Multiplying the first polynomial by T^2 gives the fourth-moment bound. If $L = 0$ or $R = 0$, centering forces $T = 0$; the divisions are replaced by this deterministic case.

Put $\mu_i = \mathbb{E}U_i^3$. Expanding on the two actual sign classes gives

$$\rho h^3 = v(1 - 2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1, \tag{53}$$

$$\gamma h^3 = v(1 - 2p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1, \tag{54}$$

$$\beta h^2 = v(1 - 2p) + s(2\alpha - 1). \tag{55}$$

Here $\gamma = \mathbb{E}X^3$ and $\beta = \mathbb{E}[X|X|]$. The same (μ_0, μ_1) must be used in these identities and in every probability bound below. Optimizing each term at a different pair would lose the constraint that all terms arise from one law.

10.2 Outgoing and incoming probability bounds

A useful outgoing minorant is

$$G_a(x) = x(x+a)^2(2a-x), \qquad a > 0.$$

It is nonpositive outside $[0, 2a]$, and its maximum is Ka^4, where $K = 9/4 + 3\sqrt{3}/2$. Consequently

$$\mathbb{P}(W_k > u) \geq \frac{\mathbb{E}G_a(W_k - u)}{Ka^4}.$$

This is a global polynomial inequality, including $W_k = u$. Its expectation uses

$$\mathbb{E}W_k^3 = (n-k)\mu_0 + k\mu_1,$$

$$\mathbb{E}W_k^4 = (n-k)\mathbb{E}U_0^4 + k\mathbb{E}U_1^4 + 3\{V_k^2 - (n-k)w_0^2 - kw_1^2\}.$$

Thus (52) makes the outgoing bound affine in the two common cubic moments. Negative lower bounds remain valid; they are not mistaken for negative probabilities.

For an incoming count $j > k$ and phase $u \leq 1/2$, put $d = j - k - u > 0$. Cantelli gives

$$\mathbb{P}(W_j \leq -d) \leq \frac{V_j}{V_j + d^2}.$$

For completeness, for a centered Y of variance V and $a > 0$, $\mathbb{P}(Y \geq d) \leq \mathbb{E}(Y+a)^2/(d+a)^2 = (V+a^2)/(d+a)^2$. Choosing $a = V/d$ proves the formula; $V = 0$ is deterministic. Exponential bounds can be sharper. If $\mathbb{E}Y = 0$, $Y \leq B$, $B > 0$, and $\mathbb{E}Y^2 \leq w$, then for $t \geq 0$,

$$\mathbb{E}e^{tY} \leq M(w; B, t) := \frac{B^2 e^{-tw/B} + we^{tB}}{B^2 + w}. \tag{56}$$

Indeed, the quadratic interpolating e^{tx} at $-w/B$ with multiplicity two and at B majorizes it on $x \leq B$: the Hermite remainder has sign $(x + w/B)^2(x - B) \leq 0$. Its quadratic coefficient is nonnegative. The centered two-point law on $\{-w/B, B\}$ with variance w evaluates its expectation exactly. This proves the formula also with an upper variance bound. At zero variance use $Y = 0$. Differentiation shows that M is increasing and concave in w: with $x = t(B^2 + w)/B$,

$$M''(w) = \frac{B^2 e^{-tw/B}}{(B^2+w)^3}(x^2 + 2x + 2 - 2e^x) \leq 0.$$

Also

$$M'(w) = \frac{B^2}{(B^2+w)^2}\{e^{tB} - e^{-tw/B}(1+x)\} \geq 0,$$

since $e^x \geq 1 + x$. Its tangents therefore give affine upper bounds throughout the variance half-line. Applying (56) to $-U_0, -U_1$ and multiplying the corresponding factors bounds $\mathbb{E}e^{-tW_j}$. Chernoff then gives $e^{-td}\mathbb{E}e^{-tW_j}$ as an incoming upper bound.

One can also retain the common cubic moments. On $[A, B]$, $A < 0 < B$, the function

$$r_t(x) = \frac{e^{tx} - 1 - tx}{x^2} = t^2 \int_0^1 (1-r)e^{trx}\, dr$$

is increasing and convex. Its chord gives $e^{tx} \leq 1 + tx + c_2x^2 + c_3x^3$, with $c_2, c_3 \geq 0$. Applied to $-U_i$, choose $C_2 \geq c_2$ and $0 \leq C_{3,-} \leq c_3 \leq C_{3,+}$, and put

$$Q_i(\mu_i) = \max\{0, 1 + C_2w_{i,+} - C_{3,-}\mu_i, 1 + C_2w_{i,+} - C_{3,+}\mu_i\}.$$

This is a nonnegative convex upper bound for its MGF. The maximum with zero also makes it nonnegative at infeasible vertices of a relaxed moment polygon. For fixed positive pivots P_0, P_1, weighted AM–GM yields

$$Q_0^{n-j} Q_1^j \le P_0^{n-j} P_1^j \left(\frac{n-j}{n} \frac{Q_0}{P_0} + \frac{j}{n} \frac{Q_1}{P_1} \right)^n .$$

The right side is convex in (μ_0, μ_1). A fixed exponential rate and its phase chord produce a convex common-moment bound on an entire closed phase interval. Choose a strategy on that whole interval and moment polygon before comparing different strategies; a pointwise minimum of convex functions need not be convex.

10.3 Normalization and a finite closed cover

Let $\mathcal{M}$ be a compact polygon containing the actual common cubic moments, obtained from proved moment and defect restrictions. Suppose F is convex on $\mathcal{M}$ and L is affine and strictly positive there. Then

$$\frac{F(x)}{L(x)} \le \max_{v \text{ vertex of } \mathcal{M}} \frac{F(v)}{L(v)}. \tag{57}$$

To see this, write $x = \sum_v \theta_v v$. Convexity bounds the numerator by $\sum_v \theta_v F(v)$; affinity identifies the denominator with $\sum_v \theta_v L(v)$. The ratio is therefore bounded by the largest vertex ratio. Segments and single points are included. If the lower bound for ρ is $\max\{r_0, L\}$, split $\mathcal{M}$ along $L = r_0$ and retain both closed pieces.

Proposition 10.1 (Whole-cell count certificate)**.** *Fix a closed box of acquired parameters, a count k, and a closed phase interval. Let B_k bound the binomial CDF above, b_k^- the own mass below, and b_j^+ the incoming masses above. Let L_k be a proved outgoing lower bound, $A_j \ge 0$ proved incoming upper bounds, and N a lower bound for $\Phi((k+u-np)/(\sqrt{n}h))$. Suppose the combination*

$$B_k - b_k^- L_k + \sum_{j>k} b_j^+ A_j - N$$

is bounded on the whole phase interval by convex Bernstein control functions $F_i(\mu_0, \mu_1)$. Let $r > 0$ be the affine lower bound for ρ on each closed normalization piece and let $n_+ \ge \sqrt{n}$. Then the positive normalized CDF error is at most

$$n_+ \max_{i,\, v \text{ vertex}} \frac{\max\{0, F_i(v)\}}{r(v)}.$$

Proof. Insert the probability bounds into (51). For the outgoing term, $b_k \mathbb{P}(W_k > u) \ge b_k^- \mathbb{P}(W_k > u) \ge b_k^- L_k$, even if $L_k < 0$. The nonnegativity of the incoming bounds justifies the upper mass substitutions. The Bernstein basis is nonnegative and sums to one. Take the positive part and apply (57); the positive part of a convex function is convex. Finally enlarge $\sqrt{n}$ to n_+. □

The acquired p, h, H bounds must first put every actual threshold into one of the original roots $k = 0, \ldots, n$, $u \in [-1/2, 1/2]$. If they do not, prove the remaining phase tails separately. A tied nearest count belongs to both closed roots. Apply Proposition E.3 only after every leaf is either a proved necessary-condition contradiction or satisfies the whole-cell score bound. The finite tree then covers infinitely many parameters. Its leaves never assert that every feasible moment tuple is an iid law; they need only contain all actual laws.

10.4 The predecessor input and index transfers

At a ratio maximizer from Corollary 2.2, $C = C_n > C_{\mathrm{E}}$ and $C_{n-1} \le C_{\mathrm{E}}$ imply

$$n\left(\frac{\sqrt{n-1}\Delta_{n-1}(P)}{\rho(P)} - C\right)_+ = 0.$$

This supplies the zero predecessor allowance for the same law. It is not a monotonicity assertion about the sequence C_n.

When a resource calculation is transferred from a floor N to an actual index $m \ge N$, the invariant is

$$\mathcal{R}_{\mathrm{actual}} \le r, \qquad mD_{\mathrm{actual}} \le Nd.$$

For example, if the same-law defect inequality is $D \ge b + gd_{\mathrm{in}} + (N/m)h_N\lambda_m$ with nonnegative b, g, h_N, multiplying by m and dividing by N gives $(m/N)(b + gd_{\mathrm{in}}) + h_N\lambda_m \le d$. Since $m/N \ge 1$ and $b + gd_{\mathrm{in}} \ge 0$, this implies $b + gd_{\mathrm{in}} + h_N\lambda_m \le d$. The noise is $\lambda_m = ms$, and cannot be replaced by Ns. Every downstream bound must retain its original index interval. In particular, $I/m \le \max\{I/40, I/42\}$ for a signed I is a valid floor transfer only on $40 \le m \le 42$.

10.5 How a physical defect bound becomes a count domain

This step explains the origin of the count coordinates and their joint constraint. A small defect must be acquired for the same extremal law before the following argument is used. Put $\sigma = \sqrt{10}$ and use the physical defect $D = (3+\sigma)\rho - \gamma - 3J$ from the conserved identity, where J is the gap sum.

Suppose $\operatorname{supp} P$ is bounded and has a central gap $(-A, B)$ with $A, B > 0$. Let $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$, $V = \mathbb{E}(X \mid \mathbf{1}_{\{X>0\}})$, and $W = X - V$. Write

$$\theta_+ = \mathbb{E}[W^2; X > 0], \quad \theta_- = \mathbb{E}[W^2; X < 0], \quad \theta = \theta_+ + \theta_-, \quad \overline{B} = m/p, \quad \overline{A} = m/q.$$

Then $m^2 = (1-\theta)pq$. Set $\Delta B = \overline{B} - B$ and $\Delta A = \overline{A} - A$; both are nonnegative. The gap decomposition gives the joint inequality

$$\begin{aligned} D &\ge b(p,\theta) + 6m\{(A+B)(\Delta A + \Delta B) + \Delta A \Delta B\} \\ &\quad + 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-, \qquad (58) \\ b(p,\theta) &= (6+2\sigma)(1-\theta)^{3/2}\frac{(p-p_*)^2}{\sqrt{pq}}, \qquad p_* = (4-\sigma)/2. \end{aligned}$$

Here the binary charge, the inward displacement, and the conditional variance all spend one common defect budget.

For clarity, the derivation needs only polynomial inequalities. Put $a_\pm = \mathbb{E}X_\pm^2$, $b_\pm = \mathbb{E}X_\pm^3$, and $T_+ = b_+ - mB^2$, $T_- = b_- - mA^2$. Gap integration on each side of the central gap gives

$$J - m(A+B)^2 \le T_+ + T_- - 2B(a_+ - mB) - 2A(a_- - mA).$$

Substitute this bound in the definition of D. For a nonnegative variable Y of conditional mean c one has exactly

$$\mathbb{E}Y^3 = c^3 + \mathbb{E}[(Y-c)^2(Y+2c)]$$

under the conditional probability measure. Since $X \ge B$ on the positive class and $-X \ge A$ on the negative class, this yields $b_+ \ge m\overline{B}^2 + 3B\theta_+$ and $b_- \ge m\overline{A}^2 + 3A\theta_-$. Also $a_+ = m\overline{B} + \theta_+$ and $a_- = m\overline{A} + \theta_-$. Substitution and collection of terms give (58); the constant part is

$$m\{(\sigma-1)\overline{B}^2 + (\sigma+1)\overline{A}^2 - 6\overline{AB}\} = b(p,\theta),$$

because $(\sigma+1)p^2+(\sigma-1)q^2-6pq=(6+2\sigma)(p-p_*)^2$.

Suppose a closed probability panel and a prior resource calculation provide $p\in[l,u]\subset(0,1)$, $\theta\le\Theta<1$, $A\ge A_l>0$, $B\ge B_l>0$, and $D\le d$. Let $v_u=\max_{p\in[l,u]}p(1-p)$, $k_+=3(\sigma+1)B_l$, $k_-=3(\sigma+3)A_l$, and let $b_l\ge 0$ be any proved lower bound for $b(p,\theta)$ on this closed panel. These quantities may be replaced by rational bounds in their indicated directions. With

$$\ell=\frac{m}{pq}=\sqrt{\frac{1-\theta}{pq}},\qquad X=\ell(\mathbf{1}_{\{X>0\}}-p+U),$$

put $d_Y=(\Delta A+\Delta B)/\ell$ and $\lambda=n\mathbb{E}U^2=npq\theta/(1-\theta)$. Then

$$b_l+gd_Y+h\lambda\le D\le d,\qquad g=6(1-\Theta)(A_l+B_l),\quad h=\frac{\min(k_+,k_-)(1-\Theta)}{nv_u}. \tag{59}$$

Indeed $m\ell=1-\theta$, so the displacement term in (58) is at least gd_Y. Its variance term is at least $\min(k_+,k_-)\theta\ge h\lambda$. All coefficients are positive. Thus $d<b_l$ excludes the panel; if $d=b_l$, the residual is deterministic; if $d>b_l$, the admissible triangle is

$$\frac{d_Y}{d_0}+\frac{\lambda}{\Lambda}\le 1,\qquad d_0=\frac{d-b_l}{g},\qquad \Lambda=\frac{d-b_l}{h}.$$

The separate intercept bounds $d_Y\le d_0$ and $\lambda\le\Lambda$ are weaker consequences of this one inequality. They must not be used to invent an additional independent defect budget.

The construction is finite: each new moment, gap, or support bound must follow from previously acquired simultaneous inequalities for this same law. Intersecting those inequalities preserves the law. No assertion about convergence of the refinement procedure is needed.

11 The finite induction from five to fourteen

The fifth index starts a different part of the proof. The polynomial majorants for two and three summands and the physical comparison at four are followed by the conditional count comparison of Section 10. The comparison itself is the same at every index; what changes is the numerical region in which the law has first been placed. This section describes how those regions are acquired and how they cover all third moments.

11.1 Closing the fifth index

The fifth-index calculation has three complementary parts: a count comparison at small third moments, an elementary corner estimate where the signed moment is large and positive, and a Fourier estimate for the remaining third moments. The corner is a genuine part of the cover; it cannot be discarded as a small exceptional region.

Proposition 11.1 (Fifth-index assembly criterion)**.** *Assume $C_4\le C_{\mathrm{E}}$, the continuous inequalities proved in this paper, and complete verification of the finite inputs and their implications described below. Then $C_5\le C_{\mathrm{E}}$.*

Proof. Suppose $C_5>C_{\mathrm{E}}$ and choose a positive closed-event ratio maximizer. Its same-law predecessor score is at most C_4, so its contact allowance is zero. The support calculations in Appendix H therefore apply to this very marginal. They precede the supported Fourier estimates. The unrestricted auxiliary Fourier estimate first supplies $C_5<429/1000$; this is the input to the two lowest-moment initializers, not their conclusion. The higher-moment

initializers use separately acquired supported own-score bounds. Their scalar input ceilings are

region	own score	HIGH
$1 \le \rho \le 11/10$	429/1000	337/1000
$11/10 \le \rho \le 23/20$	429/1000	369/1000
$23/20 \le \rho \le 237/200$	52/125	393/1000
$237/200 \le \rho \le 6/5,\ \beta \le 1/5$	207/500	193/500
$237/200 \le \rho \le 6/5,\ \beta \ge 1/5$	103/250	193/500

Here HIGH is the compensated high-frequency contribution, as in Lemma J.1. Each such input is used only on a support and moment domain contained in the domain of its Fourier proof. In particular a better score bound requiring a smaller support radius is not transferred to a larger-radius profile. The last two rows are restricted to the count rectangles specified below; the separate corner does not require these inputs.

Here is the exact signed-moment partition behind the small-moment count calculation. Set

$$b_2 = \frac{84729828287999}{250000000000000}, \qquad b_3 = \frac{35139006743001}{10^{14}}.$$

The sharp signed-moment inequality bounds $|\beta|$ by b_2 on $\rho \le 237/200$ and by b_3 on $\rho \le 6/5$. The rational endpoints are checked by $(1 + b_i^2)^2 \ge \rho_i^2(1 - b_i^2)$ at the respective upper third moments, using the monotonicity of this expression in $b_i \ge 0$.

The two lowest moment intervals each form one full signed region. On $[23/20, 237/200]$, split ρ at 467/400 and β at 1/5, giving four closed rectangles. On $[237/200, 6/5]$, split ρ at 477/400. In the lower half use the three signed intervals $[-b_3, 1/5]$, $[1/5, 3/10]$, and $[3/10, b_3]$. In the upper half use $[-b_3, 1/5]$, $[1/5, 3/10]$, and $[3/10, 17/50]$. These are the other six count regions. The remaining rectangle is

$$[477/400, 6/5] \times [17/50, b_3] \quad \text{in } (\rho, \beta). \tag{60}$$

It is covered by the whole-line elementary estimate in Appendix L, including its two unbounded threshold rays. This estimate has no support or predecessor premise. Its upper bound is below $4097/10000 < C_{\mathrm{E}}$.

In each of the twelve count regions, an unseeded finite initializer first acquires the raw remainder and then the physical defect, gap and threshold constraints. The whole final moment cover is exported. For each probability panel $[j/100, (j+1)/100]$, the subsequent resource calculation concerns that same cover and that same law. Its signed cubic, conditional variance and inward-displacement inequalities give a joint count domain as in Section 10.5. Every inequality in the numerical count contract must follow in its stated direction from this acquired domain; matching a filename or an isolated scalar maximum is insufficient.

The twelve regions have $1,200$ probability-panel instances. Of these, 905 are excluded by their resource inequalities. The other 295 are covered by six-root count forests with $1,770$ roots and $75,178$ terminal leaves. Every accepted leaf has score strictly below the fixed count target, and that target is strictly below C_{E}. The interval boundaries in ρ, β and p are included. Consequently these twelve regions and (60) cover all $1 \le \rho \le 6/5$, without assuming that the sign of β can be changed while keeping the event fixed.

For completeness, the supported Fourier complement has six closed bands, each supplied

with a previously acquired support radius:

ρ	R
$[6/5, 5/4]$	$13/5$
$[5/4, 7/5]$	$117/40$
$[7/5, 3/2]$	$25/8$
$[3/2, 8/5]$	$133/40$
$[8/5, 17/10]$	$7/2$
$[17/10, 7361/4000]$	$15/4$

The first band uses the additional radius-13/5 support calculation. The second restricts the wider support interval $[6/5, 7/5]$. All six implications retain both support signs and every real threshold. The Fourier proof subdivides these bands into 41 closed strips and bounds the score below $4097/10000$ on each. Its recorded 42 mesh attempts include the unsuccessful coarser attempt; only the successful complete enclosures are used as inequalities. For $\rho \geq 7361/4000$, the decreasing published affine bound is at most $819/2000 < 4097/10000$.

Thus the count regions, the elementary corner, the six Fourier bands and the infinite tail exclude the chosen maximizing witness. Closed-event attainment and reflection account for either sign of the original Kolmogorov discrepancy. This proves the implication. □

11.2 Induction from six through fourteen

Proposition 11.2 (Finite induction, with explicit numerical premises)**.** *Assume $C_5 \leq C_{\mathrm{E}}$. For each integer $n = 6, \ldots, 14$, assume that the support, Fourier, initializer, resource, and count checks described below pass on their complete domains, and that every output is joined to the input of the next check for the same index and marginal law. Then $C_n \leq C_{\mathrm{E}}$ for $6 \leq n \leq 14$.*

The complete checks and their input joins are recorded in Appendix AB. We state the assembly implication with explicit numerical premises so that each computation can be matched to the precise hypothesis it supplies.

Proof. Suppose there is a first failure in this range and let n be its index. Corollary 2.2 supplies an attained positive closed-event ratio maximizer (P, H) with $C = C_n > C_{\mathrm{E}}$. Write $\rho = \mathbb{E}|X|^3$ and $A = \sqrt{n-1}\Delta_{n-1}(P)/\rho$. The induction hypothesis gives

$$A \leq C_{n-1} \leq C_{\mathrm{E}} < C, \qquad d_P = n(A - C) < 0.$$

This is the only predecessor estimate needed here. In particular it implies the nonnegative allowances retained in the original numerical formulas: $579/5000$ at $n = 6$, and $99/1000$ or $63/1000$ in the two low-moment bands at $n = 10$. At other indices the allowance is zero. Keeping a larger allowance weakens an upper bound; it does not introduce an assumption about a different marginal or a later sample size.

The low-moment bands.

The required data are as follows. An own-score ceiling means an upper bound on $\sqrt{n}\Delta_n(P)/\rho$. A HIGH ceiling bounds the compensated Fourier integral defined in Lemma J.1; it is not by itself a score bound.

indices	ρ interval	support radius	own-score ceiling	HIGH ceiling
$6, \ldots, 9$	$[1, 11/10]$	$23/10$	$423/1000$	$69/200$
$6, \ldots, 9$	$[11/10, 23/20]$	$12/5$	$209/500$	$3/8$
$6, \ldots, 9$	$[23/20, 237/200]$	$5/2$	$83/200$	$83/200$
$10, \ldots, 14$	$[1, 11/10]$	$23/10$	$1047/2500$	$69/200$
$10, \ldots, 14$	$[11/10, 23/20]$	$12/5$	$519/1250$	$93/250$

The support column is acquired first, by Proposition I.1. The two Fourier columns are then evaluated with the actual integer n and that supported law, using Appendix J. A Fourier estimate may cover a larger moment interval or support radius than the row requires; the join checks inclusion of the whole smaller region. In particular, the third row may use the wider interval $[23/20, 6/5]$ with radius $13/5$ for its Fourier inputs. No estimate at index nine is transferred numerically to index ten.

The order after these three inputs is important. First an unseeded initializer proves a raw remainder bound, and only then feeds the resulting defect and threshold bounds back into subsequent passes. Appendix O proves this finite enclosure induction. The resource calculation consumes the complete initializer and acquires a central gap, signed moments, and the common conditional variance budget. Its export supplies the hypotheses of the count comparison. Thus support, the own-score and HIGH bounds, the initializer, the resource, and the count contraction all concern one law. An initializer or count certificate alone does not supply the earlier hypotheses. The additional outer-BN initializer at $n = 6$ uses exactly the same support, own-score, HIGH and predecessor premises as the ordinary BN initializer. Its output is joined separately to the subsequent $n = 6$ resource calculation.

For $6 \le n \le 9$ the three bands give twelve families, each covering the one hundred closed probability panels $[j/100, (j+1)/100]$, $0 \le j \le 99$. Of these 1,200 panels, 910 are excluded by the resource inequalities, six by the separate whole-panel sign-mass inequality of Lemma N.1, and 284 by complete count forests. For $10 \le n \le 14$ the two bands give 1,000 panels: 798 resource exclusions and 202 retained parent panels. The latter have 217 count forests, because five parent panels are covered by four closed subintervals each. These are coverage censuses, not estimates obtained by sampling. Each retained parent rectangle is checked against the entire corresponding count input, including its signed moment and inward-displacement bounds. Shared endpoints belong to both adjacent panels and cannot be discarded.

The complementary moment intervals.

For $6 \le n \le 9$ a supported Fourier comparison covers $[237/200, 6/5]$ at radius $13/5$. Five further supported comparisons cover the following closed intervals:

ρ	$[6/5, 7/5]$	$[7/5, 3/2]$	$[3/2, 8/5]$	$[8/5, 17/10]$	$[17/10, 7361/4000]$
R	3	16/5	17/5	18/5	19/5.

Proposition I.2 acquires each radius by an actual-index Gaussian calculation, using Lemma G.1. The ensuing full Fourier calculation excludes a score above $4097/10000 < C_{\mathrm{E}}$ throughout the stated band. It does not assume the low-moment resource bound outside its domain.

For $10 \le n \le 14$ the supported corner is $[23/20, 71/50]$, with radius three acquired first. An unrestricted Fourier comparison then covers $[71/50, 15/8]$. Both comparisons have target $4097/10000$. Beyond the displayed intervals the published affine Berry–Esseen bound already used in the entry argument gives

$$\frac{\sqrt{n}\Delta_n(P)}{\rho} \le \frac{3031}{10000}\left(1 + \frac{323}{500\rho}\right) \le \frac{819}{2000} < \frac{4097}{10000} < C_{\mathrm{E}} \quad (\rho \ge 7361/4000).$$

For the second group this tail overlaps the unrestricted Fourier interval, since $7361/4000 < 15/8$. Hence every $\rho \ge 1$ belongs to a low-moment band, a complementary band, or the affine tail. All boundaries are included. Reflection and closed-event attainment cover the original negative-discrepancy branch as well. Every possible first failure is therefore excluded, completing the finite induction. □

What must be checked at an interface.

The numerical record must bind complete source and result objects, not just a success word or a maximum value. In the original fifth- and sixth-index chains some immutable

candidate inputs contain earlier receipt bytes. Their mathematical contents must agree with the entire newly verified parent result, apart from explicitly identified timing metadata. Later chains bind directly to the fresh initializer and cap receipts. In either representation the join must preserve the actual index, the full moment interval, both support signs, every probability panel, and the direction of every rounded inequality. These checks are the finite proof of applicability of the count comparison; their meaning is independent of the operating system used to run the exact arithmetic.

12 The finite induction from fifteen to thirty-nine

The same argument is repeated at each index, with different rational bounds. Suppose that n is the first index in this range for which $C_n > C_{\mathrm{E}}$. By Corollary 2.2, choose an unrestricted positive closed ratio maximizer P at this index. Its score is C_n, and the same marginal satisfies $C_{n-1}(P) \le C_{\mathrm{E}} < C_n$. Consequently its contact predecessor allowance is zero. The positive allowances retained in some calculations below are weaker valid inputs. The law is never replaced when passing from a support calculation to a moment calculation or to a conditional count probability.

There are two ways to exclude this law. For sufficiently large ρ, a Fourier estimate directly bounds its score below C_{E}. In the remaining moment intervals, support and Fourier estimates first give necessary constraints. A finite moment calculation then gives a central support gap and, on each probability panel, a joint bound on inward displacement and conditional variance. The count calculation excludes every distribution obeying those constraints. Its domain is larger than the set of distributions, which is harmless in this direction.

12.1 Moment intervals and the order of their inputs

Write $P_j = [j/100, (j+1)/100]$. All moment intervals and probability panels in the tables are closed; common endpoints are checked in both adjacent pieces. The value $4097/10000$ is strictly below C_{E}. The following are the exact low-moment input contracts. An own-score ceiling bounds C_n, whereas a HIGH ceiling bounds only the named high-frequency term in the compensated Fourier formula.

actual n	ρ	initial radius	own-score ceiling	HIGH ceiling
15–19	$[1, 11/10]$	23/10	417/1000	191/500
15–18	$[11/10, 23/20]$	12/5	827/2000	409/1000
19	$[11/10, 23/20]$	23/10	827/2000	409/1000
20	$[1, 11/10]$	23/10	831/2000	cellwise
20	$[11/10, 9/8]$	23/10	103/250	cellwise
21–24	$[1, 11/10]$	23/10	419/1000	cellwise
21–24	$[11/10, 9/8]$	12/5	419/1000	cellwise
25–30	$[1, 11/10]$, $[11/10, 6/5]$	actual support table	$434477/10^6$	cellwise
31–34	$[1, 11/10]$, $[11/10, 6/5]$	47/20, 13/5	$432313/10^6$	cellwise

For an entry marked “cellwise”, the compensated Fourier formula is evaluated directly on each moment cell, without a separate scalar HIGH ceiling. The floor-25 and floor-31 entries require the actual-index support calculations in addition to the listed scalar ceilings. They do not assert that support at index 25 or 31 transfers to another index. The direct Gaussian criterion supplies each such calculation.

For $15 \le n \le 19$, support of radius 3 is first established on $1 \le \rho \le 33/25$. Supported Fourier bounds then establish the first three own-score ceilings in the table; their hypotheses therefore do not depend on the later, smaller support radii. The HIGH computation uses this already acquired radius 3. At index 15 the retained contact allowances in the two low bands

are 219/2000 and 57/1000; at indices 16–19 they are zero. The exact Gaussian support tests then establish the smaller radii used by the moment calculation.

At index 20, the direct radius-3 result on $\rho \le 6/5$ supplies the support premise for the two supported Fourier own-score estimates 831/2000 and 103/250. The two subsequent moment calculations retain contact allowances 29/250 and 23/500, respectively. They start without a numerical moment seed, perform eight refinements, and export a ninth complete cover. There is no need to assume the older coarse index-20 calculation in order to start this iteration. The original coarse count contracts on $P_{37}, \dots, P_{40}$ do, however, require a separate simultaneous-containment check against the newly acquired constraints; their historical headers are not a proof of that implication.

For $21 \le n \le 24$, the unrestricted auxiliary Fourier calculation covers each actual index and gives $C_n < 419/1000$. Its acceptance checks this strict per-index ceiling, not merely the looser discovery target 21/50. Each band then uses eight refinements and a ninth export, and all 100 probability panels are retained. The direct radius-3 theorem supplies the supported complement below 6/5.

The high-moment alternatives are as follows:

actual n	moment interval	extra hypothesis	score ceiling
15–19	$[23/20, 33/25]$	$\lvert X\rvert \le 3$	4097/10000
15–19	$[33/25, 19/10]$	none	4097/10000
20–24	$[9/8, 6/5]$	$\lvert X\rvert \le 3$	4097/10000
20–24	$[6/5, 19/10]$	none	4097/10000
25–39	$[6/5, 19/10]$	none	4097/10000

The published decreasing Berry–Esseen majorant supplies $\rho \ge 19/10$. Every unrestricted row uses the full admissible auxiliary-moment interval, not a selected value of that moment. Thus the last row removes all need for the older index-25 band $[6/5, 13/10]$ and the subsequent high-moment resource chains. The unused computations may be retained as supplementary evidence but are not premises of this shorter proof.

12.2 Finite acquisition and the common probability law

The Fourier and contact bounds contain a few signed moments. Split their complete admissible rectangle into closed cells. On a cell, use outward bounds for the Fourier terms and for the signed Gaussian maximum in the contact inequality. If the resulting upper bound is negative, no maximizing witness lies in the cell. Otherwise retain all necessary constraints established there. Taking minima of valid upper bounds and maxima of valid lower bounds improves the same witness's constraints. It does not require a contraction or convergence of the iteration. A finite number of passes, followed by one complete export pass, is sufficient.

The exported global defect upper bound is smaller than the positive central-gap threshold. Hence zero is excluded from the support, and sign conditioning is legitimate. In a probability panel P_j, the resource calculation retains bounds for both central endpoints, both outer endpoints, both signs of the phase, the signed moments, and one common physical defect. Section N and the resource-to-count calculation turn these into a simultaneous count domain. In particular, if

$$D \le d, \qquad D \ge b + g d_{\mathrm{in}} + h\lambda,$$

then $g d_{\mathrm{in}} + h\lambda \le d - b$. The triangle's original variance intercept is retained even when a separate scalar estimate further clips λ. Its two intercepts must not be independently improved and then treated as a new joint triangle.

The floor-25 calculation is used at actual indices 26–30, and the floor-31 calculation at actual indices 32–34. Their invariant is $nD \le Nd$, where N is the floor. This is stronger than simply reusing $D \le d$. The count contract uses $D \le Nd/n$ and the corresponding noise

coefficient Nh/n, with the actual index in every binomial probability and Gaussian argument. The actual-index support checks are separate inputs. These facts explain why the reuse is valid and why it is limited to the stated finite ranges.

The floor-31 low-moment calculation has two special local refinements in the second band, on P_{35} and P_{36}. The full global acquisition supplies every other panel. At indices 32–34 the same weighted physical inequalities are used, with individually checked directed count domains.

12.3 The extra own-score bound at index 39

Indices 35–39 use actual-index INITIAL calculations, rather than the floor-31 profile. Their input pairs (support radius, own-score ceiling) are

n	35	36	37	38	39
R	39/10	19/5	19/5	19/5	19/5
c	53539/125000	1699/4000	84867/200000	211449/500000	41449/100000.

Lemma 8.4, retaining its sharper expression $C_n < 41/100 + 4177/(13824n)$, implies the first four score ceilings. The fifth requires an additional argument: the same-law local comparison at actual index 39 with

$$K = 741/4000, \qquad q = 33/100, \qquad C_{39} < 2113/5000,$$

excludes $39(C_{39} - C_{\mathrm{E}}) \geq 741/4000$. Its scalar entry conditions follow from sharpness at 38 and Lemma 8.4. Since $C_{\mathrm{E}} < 40974/10^5$ and $741/(4000 \cdot 39) = 19/4000$, this gives $C_{39} < 41449/10^5$, the required native ceiling. Thus this positive-excess calculation is an internal prerequisite of the selected sharp-39 calculation. Omitting it merely because sharp-39 will later be available would be circular.

The actual-index traversals then cover every probability panel. At 35 they combine four local refinements with separate lower and upper probability ranges; at 36 only one interval needs a local refinement. The remaining indices use their global INITIAL exports. Appendix O.5 gives the complete union and explains how the acquired bounds imply the 120 surviving count contracts, including their signed phase inequalities.

12.4 Exact count coverage and assembly

The following table specifies the surviving probability panels. Every omitted panel must be explicitly excluded by its acquired resource or scalar inequality. A failed resource evaluation is not an exclusion. Multiple forests in one panel form a closed subdivision in probability or another displayed coordinate; their union is checked separately.

n	first low band	second low band	selected forests
15	37–56	33–44	32
16	37–56	33–43	31
17	37–55	33–43	30
18	37–54	34–43	28
19	37–53	34–42	26
20	37–58	35–45	49
21	37–57	35–44	31
22	37–56	35–43	29
23	37–55	35–43	28
24	37–54	35–42	26
25	37–53	32–42	52
26	37–53	Fourier	17
27	37–52	Fourier	16
28	37–52	Fourier	16
29	37–52	Fourier	16
30	37–51	Fourier	15
31–34	36–49	30–40	25 per index
35	23–48		26
36	24–48		25
37	24–47		24
38	25–47		23
39	26–47		22

At $26 \le n \le 30$, the second-band resource survivors are $P_{32}, \dots, P_{42}$; the selected supported Fourier calculation checks 23 closed moment strips, at each of the five actual indices, using the acquired support and moment constraints on these panels.

Five panels excluded before the count calculation.

The first-band count list shortens as n increases because the upper bound $D \le 25d/n$ improves while the physical binary-asymmetry cost does not decrease. The floor-25 calculation leaves the 17 panels $P_{37}, \dots, P_{53}$. Five of the 85 resulting index–panel pairs at $26 \le n \le 30$ are excluded as follows.

The complete acquisition gives these convenient outward bounds for the same actual law:

j	d in $D \le 25d/n$	Θ in $\theta \le \Theta$
52	3/10	1/500
53	5/16	1/2500.

The physical variance bound is transferred by the argument in Appendix O.6; it is not a bound for an unrelated optimizer at index 25. Set $s = 316227766/10^8 < \sqrt{10}$ and $l = j/100$. Since $p \ge l > (4-s)/2 \ge p_*$ and $\sqrt{p(1-p)} \le 1/2$, (58) gives

$$D \ge (12+4s)(1-\Theta)^2 \left(l - \frac{4-s}{2}\right)^2 > \begin{cases} 251/1000, & j = 52, \\ 38/125, & j = 53. \end{cases}$$

We used $(1-\theta)^{3/2} \ge (1-\Theta)^2$; the two final strict comparisons are rational arithmetic. But on P_{53} for $n = 27, 28, 29, 30$ the upper bound gives $D \le 125/432 < 38/125$, and on P_{52} for $n = 30$ it gives $D \le 1/4 < 251/1000$. These contradictions exclude all five closed panels, including their endpoints. The remaining 80 selected cases have full actual-index count certificates. Thus no missing certificate is treated as an exclusion.

At index 20, P_{36} in the first band and P_{34} in the second are excluded by a strict Bernoulli-moment lower bound. These two panels must not be silently omitted from the probability union.

For each selected forest, the source-to-domain join has four distinct checks. First, the fresh resource bounds contain the same witness on the entire original moment and probability domain. Second, all scalar and phase constraints in the count contract are consequences of those bounds, with upper bounds enlarged, lower bounds diminished, and each signed phase quadratic dominated on its full phase interval. Third, the native count calculation uses the actual n and verifies every closed leaf and every root count. Finally, the accepted leaves and split panels form the complete closed domain. The common-count identity then bounds the original convolution probability; it is not an inference from the feasibility of a relaxation.

The direction of the phase comparison deserves emphasis. The count kernel uses the maximum of its listed quadratic bounds, since an exported list can represent alternative surviving moment cells. If the acquired quadratics are $q_i(H)$ and the count quadratics are $\widetilde{q}_j(H)$, the join verifies, for each i, some j with $q_i(H) \le \widetilde{q}_j(H)$ throughout the relevant signed interval. It follows that

$$\max_i q_i(H) \le \max_j \widetilde{q}_j(H).$$

The full-interval comparison checks the two endpoints and any interior minimum of their quadratic difference. Replacing either maximum by a minimum would require a different argument.

Proposition 12.1 (Finite assembly criterion)**.** *Assume $C_{14} \le C_{\mathrm{E}}$, the continuous support, Fourier, finite-moment, resource, and count inequalities stated above, and the complete exact arithmetic and source-to-domain joins for the displayed families. Then $C_n \le C_{\mathrm{E}}$ for every $15 \le n \le 39$.*

Proof. If the conclusion failed, choose the least such n and its positive closed ratio maximizer. The moment tables cover its ρ. In a Fourier region its score is already strictly below C_{E}, a contradiction. In every remaining region the support and own-score premises precede the moment acquisition; at $n = 39$ include the positive-excess prerequisite just described. The full exported probability cover either excludes its panel or places the same marginal in an accepted count domain. The count identity and the strict leaf bound again give a score below C_{E}. This contradiction proves the conclusion, and supplies the sharp predecessor for the next actual index. □

13 The three indices preceding the uniform local range

The local argument at $n = 40, 41, 42$ uses the same contact, Fourier and conditional-count identities as Section 15. There are two extra issues. A bound computed with index 40 must remain valid at the actual index, and a signed correction divided by n cannot always be enlarged by replacing n by 40.

Suppose that $40 \le n \le 42$ is a first strict failure above a sharp base through 39. The recurrence of Theorem 8.2 gives

$$0 < n(C_n - C_{\mathrm{E}}) < \frac{40}{39}\left(\frac{41}{200} + \frac{17}{250} + \frac{17}{1250}\right) = \frac{1433}{4875} < \frac{3}{10}, \qquad C_n < \frac{167}{400} < .4226.$$

Here $\sqrt{n/(n-1)} < 51/50$ and $\sqrt{n-1} > 6$ were used. Both deleted constants are at most $C_{\mathrm{E}} < .4253$ and the predecessor loss is zero. In particular all bounds remain valid if the local estimates use the weaker excess allowance $.54/n$ and deletion allowance .19.

13.1 Why a computation at floor 40 can be reused

The invariant on one fixed actual law is

$$\mathcal{R}_{\mathrm{actual}} \le r, \qquad nD_{\mathrm{actual}} \le 40d. \tag{61}$$

The second inequality implies $D_{\mathrm{actual}} \le d$, which may be used in index-free moment and endpoint bounds. Its converse is not used. Every contact update bounds $\kappa n D$ by a phase budget B, where $\kappa = 997/15000$, and therefore exports $d = B/(40\kappa)$. A negative budget excludes the law. Taking minima with established bounds or maxima over a complete cover preserves this invariant.

For example, suppose the joint resource gives

$$D \ge b + 6(1-\theta)(A+B)d_{\mathrm{in}} + K\theta, \qquad b, K \ge 0,$$

and $\theta \le \Theta < 1$, $pq \le v_+$. Since $\lambda = npq\theta/(1-\theta)$, multiplication by n yields

$$40(d-b) \ge 40(1-\Theta)6(A+B)d_{\mathrm{in}} + \frac{K(1-\Theta)}{v_+}\lambda.$$

This proves the saved joint intercepts for the actual noise λ. On a noise band $[a, b']$, the further simultaneous restrictions are

$$d_{\mathrm{in}} \le d_0(1 - a/\Lambda), \qquad \theta \le \min\left\{\Theta, \frac{b'}{40v_- + b'}\right\}.$$

No different law is selected when one of these bounds improves another. If a saved harmonic geometry bound has the form

$$\theta \le \left(1 + \frac{nv_-}{\Lambda} + \frac{v_-}{B_0 d_0}\right)^{-1}, \qquad B_0, d_0, \Lambda > 0,$$

its value at 40 is an upper bound. This substitution bounds θ; the actual count noise remains $\lambda = npq\theta/(1-\theta)$.

The nonnegative Fourier error coefficients decrease with the actual index. The low-frequency factors are products and sums of $(n/(n-1))^{3/2}$, $(n-1)^{-1}$, $n^{-1/2}$ and inverse powers of $(n-1)/(2n)$ and $(45-M)(1-2/n)/96$. The latter bases are positive and increasing on their retained domains. For the high-frequency terms, $\sqrt{n}e^{-cn}$ decreases when $2 \cdot 40c > 1$; the other factors are n^{-1}, $e^{-knb^2}/\sqrt{n}$ and the positive decreasing Mills bound $(x+2)/(x^2+4x+2)$. The optional product $\sqrt{x}e^{-x}(x+2)/(x^2+4x+2)$ decreases for $x \ge 1/4$: the numerator of its logarithmic derivative, after multiplication by a positive denominator, is $4 - 10x - 22x^2 - 13x^3 - 2x^4 < 0$ there. Each use retains its executable positivity and domain gates.

The density lower bound $\phi(0)(1 - 5r/80)$ remains valid later. Signed coefficients are minimized over the whole density interval; in particular the paired curvature is

$$C\rho + \frac{1}{2}\min\{\phi_l g_0, (2/5)g_0\}, \qquad g_0 = (79/25)\rho - 3 - d.$$

This is increasing with ϕ_l if $g_0 \ge 0$, and constant at the upper-density endpoint otherwise. This verifies the sign-dependent direction of the floor substitution. The positive conserved phase price and both signed linear endpoints are retained. The general, unpaired curvature has lower-density derivative $(g_0 + (d - 4/25)_+)/2 \ge 0$: for $d \le 4/25$ one has $g_0 \ge 4/25 - d$, and otherwise the numerator is $(79/25)(\rho - 1)$. The conserved addition has coefficient $(.4095\rho - d/15)(1 - 5r/80)/2 > 0$ on each admitted cell. The own-score numerator

$$\frac{r}{40} - \phi_{0,l}(1 - 5r/80)\frac{D_l}{6}$$

also bounds its actual-index counterpart when $r, D_l \ge 0$. For a strict failure, the additional sharp-score charge satisfies

$$n(C - C_{\mathrm{E}})\rho \ge 40\max\{0, C - .40974\}\rho_l.$$

Indeed $C_{\mathrm{E}} < .40974$, and each replacement decreases a nonnegative factor on the part where $C > .40974$. The score cover includes that breakpoint. The charge is affine on either side; hence the coherent (C, ρ) corner argument still applies before maximizing over the threshold. These statements concern the zero-barrier sharp target.

The first support proof is evaluated separately on the full band $40 \le n \le 42$, with zero loss and radius $96/25$. Its favourable fifth-derivative term uses the upper endpoint 42; the value at index 40 alone would not suffice. Every later left radius contraction also carries this two-endpoint proof. In the central contact expansion, all index-dependent error numerators are nonnegative after taking positive parts, so their values at 40 bound their later values. The already acquired support closes the exterior of every such expansion. In particular its complete cell bound is

$$L_P + \frac{.54Q_+}{n} + .4\left(\sum_{j=1}^{5}\frac{M_j}{n^j} + \frac{T}{n^6}\right) + \rho\left(\sqrt{\frac{n}{n-1}}\delta_n + \frac{.384(.4226)}{n}\right),$$

where $Q_+, M_j, T \ge 0$, L_P is independent of n and may be negative, and $\delta_n \le .19$. Every positive term is bounded by its value at 40. This discards favourable Gaussian terms and is therefore distinct from the two-endpoint initial support polynomial.

For concentration, $\sqrt{\max\{(40/39)^3, 40/38\}}$ bounds both deletion factors. The binomial mode interval at $n-2$ is contained in the wider interval at 38. The positive inverse-index errors increase when evaluated at 40. A retained negative penalty is kept outside this substitution: an expression $(zr + \mathrm{Dist})/n - zP$ uses $zr + \mathrm{Dist} \ge 0$ and keeps $-zP$ unchanged. For $d \ge 0$, $\tau > 0$, its bad-pair factor satisfies

$$\frac{d}{n\tau + d} \le \frac{d}{40\tau + d} \le \frac{40\tau}{(40\tau + s)^2}d + \frac{s^2}{(40\tau + s)^2} \qquad (s \ge 0),$$

the tangent bound for a concave function. Thus the same actual moment/noise triangle and every positive-width and signed derivative gate license the confinement and its feedback at all three indices.

Here the numerical constants in the confinement proof of Section U are parameters. More precisely, replace its floor 47 by $N \ge 40$, its initial strip .55 by a number $w \in (0, 1)$, and its conditional cubic coefficient by $c_\rho > 0$. Retain a probability panel $[l, h] \subset (0, 1/2)$ and assume

$$\operatorname{supp} Y \subset (-w, 1 + w), \quad d_0 < w, \quad c_\rho^2[l(1-l)]^3 > [3(1-l) + w]^2.$$

The strip and the inward bounds $U_0 \le d_0$, $U_1 \ge -d_0$ together give $|U_i| < w$. The displayed conditional cubic identity gives $\rho \le \rho_B + \ell^3(3q + w)s \le \rho_B + c_\rho s$. Use $J_0 = N - 2$ for the binomial mode interval and the deletion factor $\sqrt{\max\{(N/(N-1))^3, N/(N-2)\}}$. In the interval geometry use $g = 1 - w - 2d_0 > 0$ and $r \le w + d_0$; in the upper contradiction use $L_g - wU_+ > \mathcal{E}$. The other width, disjointness, positive-slope and budget gates are unchanged. These substitutions follow directly through the proof: w enters only the conditional cubic bound and the endpoints of the known strip, while the index enters only the decreasing factors already bounded above. They therefore prove the same confinement implication whenever all the resulting gates pass.

The central sequence uses $N = 40$, $w = .55$ initially and $c_\rho = 24$. The left rows start with their separately acquired strip $w \in \{.55, .575, .6\}$ and use $c_\rho = 30$. After a strict contraction, the proved amplitude becomes the next strip width. Thus every later use has an already established strip, and the left sequence does not assume the narrower uniform-tail strip.

13.2 The full acquired domain and its local cover

The initializer covers the whole moment range from $\rho = 1$ to the Shevtsova upper cap, not merely a low-moment subinterval. Its fourteen refinements and final export each cover all signed second and third moments and the common first moment; its probability table has 100 closed cells. The left acquisition covers its 38 cells below .38, refines all twelve surviving cells into sixty children and excludes the first eleven children. The remaining 49 rows cover $[.282, .38]$. The central acquisition covers the other 62 coarse cells, excludes every cell above .48, and refines $[.38, .48]$ into fifty children; the final two children are excluded. Its 48 remaining rows cover $[.38, .476]$. Thus the probability and moment complements are part of this acquisition itself.

Each local row uses the same-law joint resource and both signs of the raw threshold. Zero noise is covered by the Bernoulli theorem. At positive noise, nearest-count and whole-count witnesses come from the common exact identity (66). The central phase cover is

λ	nearest-count phase	whole-count phase
$(0, .05]$	all	$--$
$[.05, .08]$	all	$--$
$[.08, .10]$	$\lvert u\rvert \geq .19$	$\lvert u\rvert \leq .19$
$[.10, .12]$	$\lvert u\rvert \geq .27$	$\lvert u\rvert \leq .27$
$[.12, \Lambda]$	$--$	all.

The left rows require a different cover:

λ	nearest-count phase	whole-count phase
$(0, .05]$	all	$--$
$[.05, .0925]$	all	$--$
$[.0925, .10]$	$\lvert u\rvert \geq .16$	$\lvert u\rvert \leq .16$
$[.10, .12]$	$\lvert u\rvert \geq .25$	$\lvert u\rvert \leq .25$
$[.12, .135]$	$u \leq -.27$	$-.27 \leq u \leq .5$
$[.135, \Lambda]$	$--$	all.

Every band is intersected with the row's acquired Λ and all seams are closed. The actual computations further split noise into intervals of width $1/400$ and phase into intervals of width $1/1000$ where needed. The full-count families have 70 central signed rows and 82 left signed rows; the other rows end below the respective starting noise levels and are already covered by the nearest-count tests.

At these three finite indices the proof uses a positive lower bound ϵ_p for $\min_{n=40,41,42}(C_{\mathrm{E}} - C^{\mathrm{bin}}_{n,p})$, uniform on its probability cell. The complete binomial check covers $p \in [.28, .48]$, both discrepancy signs and every count at each actual index. A signed inverse correction I is bounded by

$$I/n \leq \max\{I/40, I/42\}.$$

For normalized mass $b = \sqrt{n}\pi_k \in [b_0, 1]$, a bracket J of either sign is bounded above by $\max\{b_0 J, J\}$. These replacements explain why this finite comparison is not an unrestricted floor-40 theorem. They also give the finite-slack version of the whole-count test. Suppose S bounds $\rho_B(C - C_B)/\lambda$ on a closed noise band $[a, b]$ with $a > 0$, and $\rho_B \geq \rho_{B,l} > 0$. It is enough to check

$$S < \rho_{B,l}\epsilon_p/b.$$

If $S < 0$, then $C < C_B$; otherwise multiplication by $\lambda \leq b$ gives $C < C_B + \epsilon_p \leq C_{\mathrm{E}}$. The saved full-count formulas have a bracket affine in λ plus a positive multiple of $(\alpha\lambda + \beta)^{-1}$, and an inverse correction that is constant plus a nonnegative multiple of $1/\lambda$. The two

sign-sensitive maxima above preserve convexity. Thus their values at the two noise endpoints bound the entire band. The finite check may use positive binomial slack even where strict comparison with C_B itself is unavailable. For the nearest-count comparisons, the entropy segment $p \pm \delta/40$ contains the actual segment $p \pm \delta/n$. All log-factorial, variance and ratio errors have positive numerators. The signed Gaussian derivative factors have the forms $1 - A_n/n$ and $(1 - \eta_n/n)^{-1}$, with $0 \le A_n \le A_{40}$ and $0 \le \eta_n \le \eta_{40} < 40$. Thus the lower bound for the first and upper bound for the second are valid at the floor. Their phase and threshold domains and the gate $40pq > \delta^2$ are retained. A signed cubic correction is minimized over its whole positive multiplier interval before it is used in a lower repair bound; its sign is not discarded by changing the index.

The finite binomial slack itself has a direct one-dimensional proof. On a probability cell $[a, b] \subset (0, 1/2)$ put $F_j(p) = \Pr(\mathrm{Bin}(n, p) \le j)$ and $z_j(p) = (j - np)/\sqrt{np(1-p)}$. Both functions decrease with p, while $\sqrt{np(1-p)}/(1 - 2p(1-p))$ increases. Therefore both jump sides of the normalized discrepancy are bounded by

$$\frac{\sqrt{nb(1-b)}}{1 - 2b(1-b)} \max\{F_j(a) - \Phi(z_j(b)),\, \Phi(z_j(a)) - F_{j-1}(b),\, 0\}.$$

Here $F_{-1} = 0$. Taking every $j = 0, \dots, n$ covers all thresholds, because between jumps the distribution function is constant and Φ increases. Each of the 100 probability panels is divided into 128 closed subcells. The binomial probabilities at the endpoints are exact rational numbers. On $|z| \le 5$, an integrated degree-158 Taylor polynomial for $e^{-z^2/2}$ with its explicit next-term integral remainder $|z|^{161}/(2^{80}80!161)$ encloses Φ after multiplication by $\phi(0)$; outside this interval the proved Mills bound is $3 \cdot 10^{-7}$. Directed integer square roots and a rational Machin enclosure for π complete the enclosure. Subtracting these upper scores from a rational lower bound for C_{E} gives the saved positive ϵ_p with both atom conventions included.

Binding the actual inputs.

The comparison's amplitude profiles are the exact rational-string acquisition object, renamed `profiles.json`; their complete bytes must agree with `profile_candidate.json`. The same source binding is checked for the probability targets and for every actual-index binomial table. Thus the numerical comparisons consume the data already proved for the same marginal.

Numerical completion.

The finite conclusion requires the ten fresh acquisition actions, the separate indexed support check, all nine companion checks, all 152 full-count rows with their separate arithmetic replays and two complete binders. The numerical status of these branches is recorded separately; source identities and historical success fields do not supply any of these inequalities.

14 A short bridge from a sharp finite base

The smoothing estimate removes the need for the historical sequence of positive upper barriers. At the first strict failure of the sharp bound, the predecessor is already sharp. This supplies a much smaller excess than either local exclusion requires, and makes the predecessor loss zero.

Proposition 14.1 (Finite local exclusion)**.** *For an integer $42 \le n \le 48$, the conjunction*

$$0 < n(C_n - C_{\mathrm{E}}) \le \frac{54}{100}, \qquad n(C_{n-1} - C_n)_+ \le \frac{19}{100},$$

$$C_n < \frac{4226}{10000}, \qquad C_{n-1}, C_{n-2} < \frac{4253}{10000}$$

is impossible.

Proposition 14.2 (Uniform local exclusion)**.** *For an integer $n \geq 47$, the conjunction*

$$0 < n(C_n - C_{\mathrm{E}}) \leq \frac{7}{10}, \qquad n(C_{n-1} - C_n)_+ \leq \frac{19}{100},$$

$$C_n < \frac{4247}{10000}, \qquad C_{n-1}, C_{n-2} < \frac{4253}{10000}$$

is impossible.

Theorem 14.3 (First-failure composition)**.** *Assume the two local exclusions above and $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 42$. Then $C_n \leq C_{\mathrm{E}}$ for every $n \geq 1$. In particular, these inputs fill the range $43 \leq n \leq 699$ needed to join the finite base to the compact manuscript's tail theorem.*

Proof. Suppose otherwise and choose the least integer n with $C_n > C_{\mathrm{E}}$. The finite base gives $n \geq 43$, and both deleted predecessors satisfy $C_{n-1}, C_{n-2} \leq C_{\mathrm{E}}$. Corollary 8.3 gives

$$0 < n(C_n - C_{\mathrm{E}}) < \frac{3003}{10250} < \frac{3}{10}, \qquad C_n < \frac{73}{175} < \frac{4226}{10000}, \qquad n(C_{n-1} - C_n)_+ = 0.$$

Also $C_{\mathrm{E}} < 41/100 < 4253/10000$. If $43 \leq n \leq 48$, every hypothesis of Proposition 14.1 holds. If $n \geq 49$, every hypothesis of Proposition 14.2 holds. Both alternatives are impossible. □

The finite local exclusion concerns actual indices up to 48; it cannot be extrapolated to a larger index. The second exclusion is uniform in n. This distinction is what makes the dependency structure acyclic: the finite argument uses no previously established sharp tail, and the uniform argument uses no conclusion from the present induction. The sharp base may be weak. Strict local comparisons are used only to exclude a strict failure above that base.

15 Why the intermediate indices reduce to finite local comparisons

The argument has two distinct parts. First, contact at an extremizer forces its marginal into a finite list of neighborhoods of two-point laws. Second, a direct comparison of the actual iid sum with its own binomial count excludes every neighborhood. The first part uses extremality; the second part uses only the explicit support and moment bounds already acquired for that same marginal. This order is essential.

The two index ranges use the same analytic argument. The finite range $42 \leq n \leq 48$ permits a stronger Gaussian support estimate and has 92 surviving probability intervals. The uniform range $n \geq 47$ uses a different support estimate and has 83 surviving intervals. An interval table supplies simultaneous bounds on one law. Its coordinates are not independently selected laws, and it is not an assertion that every point of a rectangular relaxation is realizable. The complete hypotheses for the two applications are as follows, with $d_n = n(C_{n-1} - C_n)_+$:

index range	$n(C_n - C_{\mathrm{E}})$	d_n	C_n	C_{n-1}, C_{n-2}
$42 \leq n \leq 48$	$(0, .54]$	$\leq .19$	$< .4226$	$< .4253$
$n \geq 47$	$(0, .7]$	$\leq .19$	$< .4247$	$< .4253.$

Every support and acquisition claim in this section is applied under the corresponding full row of hypotheses.

15.1 Contact first gives bounded support

Let $C = C_n > C_{\mathrm{E}}$ and choose the standardized attained positive ratio maximizer from Corollary 2.2. Write $\rho = \mathbb{E}|X|^3$, $\beta = \mathbb{E}[X|X|]$, and let H be its raw sum threshold. Set $d_n = n(C_{n-1} - C_n)_+$ and $s_n = \sqrt{n/(n-1)}$. The contact inequality and the predecessor bound give, at every support point t,

$$0 \le \mathcal{H}_n(H,t) + C\left\{-|t|^3 + \frac{3}{2}\rho t^2 + 3\beta t + \rho e_n\right\} + \rho s_n d_n, \qquad e_n = n(s_n - 1) - \frac{1}{2}, \tag{62}$$

where

$$\mathcal{H}_n(H,t) = n^{3/2}\left\{\Phi\left(\frac{H-t}{\sqrt{n-1}}\right) - \Phi\left(\frac{H}{\sqrt{n}}\right)\right\} + nt\phi\left(\frac{H}{\sqrt{n}}\right) + \frac{H}{2}(t^2-1)\phi\left(\frac{H}{\sqrt{n}}\right).$$

This estimate requires only a finite third moment. In particular, it can exclude a tiny exterior atom before any fourth or higher moment is used.

Here is the finite calculation underlying the support check. Put $v = n - 1$, $u = |t|$, $z = H/\sqrt{v}$, and

$$A = \frac{u^3 - 3u}{6}, \quad B = \frac{u^4 - 6u^2 + 3}{24\sqrt{v}}, \quad D_5 = \frac{u^5 - 15u}{120v}.$$

Taylor's theorem in the argument and the variance of the Gaussian CDF gives

$$s_n^{-3}\mathcal{H}_n(H, \pm u) = \pm A(1-z^2)\phi(z) + B\phi'''(z) \mp D_5\phi^{(4)}(z) + R_5,$$

$$|R_5| \le \frac{51u^6}{14400v^{3/2}} + \frac{51(3u^2-2)}{960v^{3/2}} + \frac{u}{8v^2}.$$

Here $u \ge 3.84$, the range needed for the exterior-support test, so in particular $3u^2 - 2 > 0$. The displayed remainder follows from the more general expansion for $u > \sqrt{3}$ and is uniform over all real thresholds. Gaussian Fourier inversion proves the derivative bounds used here. For $|z| \le 1$ use $\phi^{(4)}(z) \ge \phi(0)(3 - 15z^2/2)$ and complete a square; on $|z| \ge 1$ use the uniform derivative bounds. The resulting polynomial inequalities in u are checked on adjacent closed intervals from 3.84 to 4.5. The exterior rays use the cubic bound $|\mathcal{H}_n(H,t)| \le u^3/15 + u/5 + 1/2$ and its derivative signs. Under the corresponding local hypotheses stated above, the accepted polynomial signs exclude all support with $|t| \ge 3.84$.

For a finite index band $N \le n \le L$, the favorable term $-3D_5$ is evaluated at L, and the adverse terms at N. For an infinite band its lower bound is zero. Thus finite support estimates must keep their upper endpoint. The finite campaign uses $(N, L) = (42, 48)$. The uniform campaign covers

$$[45, 49],\ [50, 59],\ [60, 89],\ [90, 134],\ [135, 199],\ [200, \infty),$$

using only indices at least 47. These bands have no uncovered integer. The full support polynomials and their derivation are retained in the support supplement; no asymptotic remainder is used.

15.2 A single nonnegative resource controls distance from two points

Now all moments exist. For the closed support gaps $g = (a, b)$ put $\tau_g = \mathbb{E}[X\mathbf{1}_{X \ge t}]$, constant for $a < t < b$, and define

$$J = \sum_g \tau_g (b-a)^2, \qquad D = (3 + \sqrt{10})\rho - \mathbb{E}X^3 - 3J.$$

Let ν be the law of $H - S_{n-1}$ and

$$L = \sum_{g=(a,b)} \tau_g \int_{[a,b)} (t-a)\,\nu(dt) \geq 0.$$

The half-open convention fixes the contribution of every atom. Contact and compensated inversion of the smooth function $(x-z)|x-z|$ give the exact identity

$$n(C - C_{\mathrm{E}})\rho + nC_{\mathrm{E}}\rho(1-e^{-w}) + \frac{n\phi(0)e^{-w}}{6}D + n^{3/2}L = \mathcal{R}, \qquad w = H^2/(2n). \tag{63}$$

For completeness, the right side is the following explicit expression. Set $\gamma = \mathbb{E}X^3$, $M_j = \mathbb{E}|X|^j$ and $\varphi = \phi(H/\sqrt{n})$, and define

$$J_1 = \sum_{g=(a,b)} \tau_g \int_a^b (t-a)t\,dt, \qquad J_2 = \sum_{g=(a,b)} \tau_g \int_a^b (t-a)t|t|\,dt,$$

$$K_1 = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma, \qquad K_q = M_5/4 - \rho M_4/2 - \tfrac{3}{2}\beta\gamma + \rho/2, \qquad U = J_1 + (M_4 - 3)/6.$$

Then

$$\begin{aligned}\mathcal{R} = {} & C\{3\beta J/2 + 3\rho J_1 - 3J_2 - K_q\} + H(CK_1 + \varphi U) \\ & - H^2(C\rho + \gamma\varphi/2) + E_n.\end{aligned} \tag{64}$$

If $f(t) = \mathbb{E}e^{itX}$ and $V_n(t) = f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6)$, the exact error is

$$E_n = \frac{2\sqrt{n}}{\pi} \int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt.$$

The integral is absolutely convergent. A complete Fourier estimate gives a coherent bound $|E_n| \leq K_F + s_F|H|$; a constant and a slope from unrelated estimates may not be combined. The low-frequency part retains the signed quartic/fifth-order cancellation. The high-frequency part includes the whole half-line and uses an event under the product of the same marginal law. The detailed formulas appear in the Fourier supplement. There is no inversion of a discontinuous CDF and no hidden lattice midpoint convention.

The key geometric inequality is elementary. If the central support gap is $(-A_0, B_0)$, let $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$, $V = \mathbb{E}[X \mid \mathbf{1}_{X>0}]$ and $\theta = \mathbb{E}(X - V)^2$. Then $m^2 = (1-\theta)pq$. Put $\Delta_A = m/q - A_0$ and $\Delta_B = m/p - B_0$ and $p_* = (4 - \sqrt{10})/2$. Direct conditional expansion gives

$$\begin{aligned} D \geq {} & \frac{(6 + 2\sqrt{10})(1-\theta)^{3/2}}{\sqrt{pq}}(p - p_*)^2 \\ & + 6m\{(A_0 + B_0)(\Delta_A + \Delta_B) + \Delta_A\Delta_B\} \\ & + 3(\sqrt{10} + 1)B_0\theta_+ + 3(\sqrt{10} + 3)A_0\theta_-, \end{aligned} \tag{65}$$

where $\theta_\pm = \mathbb{E}[(X - V)^2; \pm X > 0]$. Every term is nonnegative. If zero belongs to the support, instead $D \geq (\sqrt{10} - 1)\rho$. Consequently a sufficiently small bound on D proves that the central gap exists, then simultaneously bounds the probability parameter, conditional noise and inward support displacement.

A first raw bound $\mathcal{R} < r < .4n$ licenses removal of the Gaussian weight in (63). Indeed, before making that change, its rearrangement proves $\phi(H/\sqrt{n})(\mathbb{E}X^3 + 3J)/6 > 0$. The resulting resource includes

$$n(C - C_{\mathrm{E}})\rho + \frac{997}{15000}nD + n^{3/2}L \leq \min\left\{T, \max_j(A_j + F_{j,\pm}|H| - Q_jH^2)\right\}.$$

The four corner quadratics retain the common score/moment alternatives. The favorable Gaussian phase term is already charged in Q_j and must not be charged again.

Each acquisition step intersects previously established enclosures with one of these inequalities. A split preserves the union of the closed cells; an empty necessary interval excludes the cell. A support improvement is proved before its radius enters a later moment estimate. Induction over this finite schedule establishes all exported bounds. No convergence of an iteration or existence of a fixed point is needed. In particular, the weighted inequality $nD \le Nd_D$ is retained; the unweighted inequality $D \le d_D$ alone would not bound $n\theta$.

15.3 The comparison uses the actual binomial count

On each surviving row write

$$X = \ell(B + U - p), \qquad \mathbb{P}(B = 1) = p, \qquad \mathbb{E}(U \mid B) = 0,$$
$$v = p(1-p), \qquad s = \mathbb{E}U^2, \qquad \lambda = ns, \qquad \ell = (v+s)^{-1/2}.$$

The conditional noises obey

$$|U_j| \le A, \qquad U_0 \le a_0, \qquad U_1 \ge -a_1, \qquad d = a_0 + a_1 < \min(p, 1-p).$$

The row supplies these quantities and both signed threshold ranges. Choose the nearest binomial count so that $np + H/\ell = k + u$, $|u| \le 1/2$, and put $\delta = k - np$. The acquired ranges ensure $1 \le k \le n-1$. Let $\pi_j = \binom{n}{j} p^j (1-p)^{n-j}$ and let W_j be the sum of j independent copies of U_1 and $n-j$ independent copies of U_0. Then the exact difference from the Bernoulli CDF is

$$Q = -\sum_{i=0}^{k} \pi_{k-i} \mathbb{P}(W_{k-i} > u + i) + \sum_{i=1}^{n-k} \pi_{k+i} \mathbb{P}(W_{k+i} \le u - i). \tag{66}$$

Notice the strict outgoing and closed incoming events. This is simply conditioning on the actual iid labels, so it holds at atom collisions.

Put $\rho_B = (1-2v)/\sqrt{v}$ and

$$C_B^+ = \frac{\sqrt{n}}{\rho_B} \{\mathbb{P}(\mathrm{Bin}(n,p) \le k) - \Phi(\delta/\sqrt{nv})\},$$

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

Subtracting the two normalized scores yields

$$\rho_B(C - C_B^+) = \sqrt{n}\pi_k \{Q/\pi_k - f(u)\} + \mathcal{R}_c, \tag{67}$$

$$\mathcal{R}_c = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

The count k and δ are held fixed when differentiating f. This convention matters in the signed Gaussian refinement.

Lemma 15.1 (Two sufficient local tests)**.** *Suppose $\sqrt{n}\pi_k \ge m > 0$ and $\lambda > 0$. The following are sufficient to prove $\rho_B(C - C_B^+) < -\varepsilon\lambda$.*

1. *Bounds O, T, R_0, κ satisfy*

$$\mathbb{P}(W_k > u) + f(u) \ge O, \quad \sum_{i \ge 1} \mathbb{P}(W_{k+i} \le u - i) \le T, \quad \pi_{k+i}/\pi_k \le R_0, \quad \mathcal{R}_c \le \kappa\lambda/n.$$

If $D_c = O - R_0 T > 0$ and $mD_c - \kappa\lambda/n > \varepsilon\lambda$, the conclusion holds.

2. Valid functions $B(\lambda) < 0$, $I(\lambda) \geq 0$ obey

$$\rho_B(C - C_B^+) \leq \sqrt{n}\pi_k \lambda B(\lambda) + \lambda I(\lambda)/n.$$

For $n \geq N$, it is sufficient that $mB(\lambda) + I(\lambda)/N < -\varepsilon$.

Proof. For the first test discard only outgoing terms with $i > 0$ in (66); the discarded terms are nonpositive. Use (67), then the positive lower mass bound. For the second test use $B < 0$ before replacing $\sqrt{n}\pi_k$ by m, and $I \geq 0$ before replacing n by N. □

The nearest-count estimates use a quartic minorant of the strict event, Cantelli's inequality and a Poisson/Bennett upper-tail comparison. For vanishing noise the quartic is rescaled by the actual variance; its expression extends to zero noise. Thus the first interval is $(0, .05]$, with no missing positive lower cutoff.

The whole-count estimate couples all conditional noises independently and defines $K = \max\{j : j + W_j \leq k + u\}$. Since every increment of $j + W_j$ is at least $1 - d > 1/2$, this is a genuine random cutoff and the actual sum probability is $\mathbb{E}F_B(K)$. Binomial curvature, marked crossings and the nonzero fractional-part remainder give the second test. The bound on marked crossings uses one common deleted background law; it does not maximize two conditional probabilities using unrelated marginals. A positive harmonic gain is obtained from

$$1 - \cos(2\pi x) \leq \frac{4553}{1000} x \quad (0 \leq x \leq 1)$$

in the finite refinement, and its weaker constant $23/5$ in the uniform campaign. All resulting error bounds are proportional to λ. This is why the comparison remains valid for arbitrarily small defects.

The joint inequality (65) is also retained in the local comparison. In the finite refinement it gives a resource of the form $d/d_* + \lambda/\Lambda_* \leq 1$. On a noise interval $\lambda \geq L_0$, one may therefore use $d \leq d_*(1 - L_0/\Lambda_*)$. The two separate ceilings would lose the information needed near the lower end of the probability range.

15.4 The two complete acquisitions and their phase covers

The finite campaign starts with 100 closed probability cells. Refining the surviving cells gives the following partition, including all seams:

$[0, .23],\ [.63, 1]$	initial exclusion
$[.23, .27]$	4 coarse exclusions
$[.27, .29]$	10 fine exclusions
$[.29, .38]$	45 left rows
$[.38, .474]$	47 central rows
$[.474, .48]$	3 central exclusions
$[.48, .63]$	15 upper exclusions.

There are 184 closed cells altogether. Endpoint probabilities zero and one cannot occur for a centered variance-one law. The left support radius is first contracted to 3.025 and then row by row. Contact proves the strip $(-.575, 1.575)$ in Bernoulli coordinates before concentration supplies the conditional-noise amplitude. The ensuing feedback and four noise-dependent profiles use already acquired data.

On the left, the local witnesses have the exact domain cover

λ	nearest-count test	whole-count test
$(0, .05]$	all u	$--$
$[.05, .08]$	all u	$--$
$[.08, .10]$	$\lvert u\rvert \geq .19$	$\lvert u\rvert \leq .19$
$[.10, .12]$	$\lvert u\rvert \geq .27$	$\lvert u\rvert \leq .27$
$[.12, \Lambda]$	$--$	all u.

The central rows have complementary phase radii $.09, .14, .19, .29$ on the successive bands $[.05, .06]$, $[.06, .08]$, $[.08, .10]$, $[.10, .12]$; beyond .12 the whole-count test covers all phases. Every interval is intersected with the row's actual Λ, and both signs of H are present. The amplitude profiles on $[.05, .08]$, $[.08, .10]$, $[.10, .12]$, $[.12, \Lambda]$ are $.40, .35, .30, .275$ on the left and $.375, .325, .275, .25$ centrally, each intersected with the previously proved global amplitude.

The uniform campaign starts with radius 3.84 and 13 signed resource passes. Central and complementary acquisition are followed by support contraction and concentration; the central antichain step consumes the original central data, then intersects the first improved table. This prevents a circular use of its own output. A signed resource step excludes one left probability interval. The final cover has 38 left rows on $[.304, .38]$ and 45 central rows on $[.38, .47]$. All complementary probabilities have already been excluded. The initialization stores a floor $N = 47$ and an old upper-index field 98. The latter is used only to check that the stored band is ordered and in the lower-score expression $.4095 + K/98$, with $K = 0$ here. It therefore imposes no upper bound on the actual index. The analytic licenses still matter: the all-threshold support cover ends in an infinite band, the bounded-threshold series controls the entire interval $0 \leq 1/n \leq 1/47$, the Fourier bounds have explicit monotonicity gates, and the local tests retain the signs needed before replacing n by 47. Together these facts, rather than the name of a table, establish the uniform index scope.

Its nearest-count comparison covers $(0, .05]$ completely. The central MID phase radii on the next four bands are $.09, .13, .18, .29$; the left radii are $.09, .14, .19, .29$. On the final left band the positive phases $[.29, .5]$ use a further whole-count test. The central, phase-resolved left and extra positive-left whole-count objects cover the complementary phases and every phase above .12. All seams are closed. These uniform constants are not substituted into the finite campaign, or conversely.

The specified target for both campaigns is the strict local inequality with $\varepsilon = .001$ on every remaining positive-noise cell. Each calculation is a finite collection of rational inequalities, with radicals and transcendental terms replaced by proved outward rational enclosures. The checkers test all required signs before division or multiplication, all rectangle corners, and complete probability/noise/phase coverage.

At $\lambda = 0$, conditional centering implies $U = 0$ almost surely. The marginal is Bernoulli and the all-probability finite-n Bernoulli theorem gives $C \leq C_{\mathrm{E}}$. For $\lambda > 0$ the same theorem gives $C_B^+ \leq C_{\mathrm{E}}$, while Lemma 15.1 gives $C < C_B^+$. Thus each acquired neighborhood contradicts $C > C_{\mathrm{E}}$. This completes the mathematical reduction of the two local exclusions to their specified finite arithmetic predicates and the Bernoulli theorem. The executable correspondence must retain those precise predicates and scopes.

16 The sharp inequality and strictness above the finite range

Theorem 16.1 (The sharp iid Berry–Esseen inequality)**.** *Let $X_1, X_2, \dots$ be independent copies of a random variable with $\mathbb{E}X_1 = 0$, $\mathbb{E}X_1^2 = 1$ and $\rho = \mathbb{E}|X_1|^3 < \infty$. For every integer $n \geq 1$,*

$$\sup_{H\in\mathbb{R}} \left| \mathbb{P}\left(\sum_{i=1}^{n} X_i \leq H \right) - \Phi(H/\sqrt{n}) \right| \leq \frac{C_{\mathrm{E}}\rho}{\sqrt{n}}, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

Equivalently, $C_n \leq C_{\mathrm{E}}$ for every $n \geq 1$. Moreover, $C_n < C_{\mathrm{E}}$ for every $n \geq 700$, and $\sup_{n\geq 1} C_n = C_{\mathrm{E}}$.

Proof of the upper bound. The small-index comparisons and the finite induction establish $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 42$; their complete numerical checks and same-law input joins are recorded in Appendix AB. Suppose that the upper bound fails, and choose the least index n with $C_n > C_{\mathrm{E}}$. Then $n \geq 43$, both deleted predecessors are sharp, and Corollary 8.3 gives

$$0 < n(C_n - C_{\mathrm{E}}) < \frac{3003}{10250} < \frac{3}{10}, \qquad C_n < \frac{73}{175} < \frac{4226}{10000}, \qquad n(C_{n-1} - C_n)_+ = 0.$$

Since $C_{\mathrm{E}} < 41/100 < 4253/10000$, both deleted-predecessor ceilings also hold. These bounds are stronger than the hypotheses of the finite local exclusion when $43 \leq n \leq 48$, and of the uniform local exclusion when $n \geq 49$. Propositions 14.1 and 14.2 exclude both alternatives. Thus the upper bound holds at every index. This is the first-failure composition of Theorem 14.3. □

The weak upper bound already proves the conjectured inequality. The compact fixed-target comparison supplies the stated strictness. We give the interface explicitly because it uses a different optimization problem from the finite induction.

Theorem 16.2 (Exclusion of a nonnegative-excess maximizer)**.** *Let $n \geq 700$ and suppose $C_{n-1} \leq C_{\mathrm{E}}$. No pair (P, H) attaining the fixed-target maximum at C_{E} has $a(P, H) \geq 0$.*

Proof. The structural inequality (74) and Theorem 5.1 give $nD \leq 32.20665378$ and $|H| \leq 1.83605680$ for the same compactly supported law. Propositions 6.1 and 6.2 then give a central gap and the bounds below. In (6a), identify the local theorem's B with $B_0 = \mathbf{1}_{\{X>0\}}$. The variable $(B + U - p)/\sqrt{pq + s}$ is exactly the original X, and H remains its raw sum threshold.

Bound just obtained	Hypothesis of Theorem 7.1
$3/8 < p < 93/200$	$p \in [3/8, 93/200]$
$a_0, a_1 \geq 0$, $a_0 + a_1 < 1/250$	Nonnegative inward endpoints, sum at most $1/250$
$-3/5 < U_0 \leq a_0$, $-a_1 \leq U_1 < 3/5$	The stated closed conditional-support bounds
$0 \leq \lambda < 7/20$	Positive-noise interval or separate zero-noise case
$\lvert H\rvert \leq 1.83605680 < 46/25$	The full threshold interval
Independent copies of X	The iid sum in the local comparison

Positive noise gives $a(P, H) < -\lambda/100$. At zero noise the law is Bernoulli, so Schulz's strict finite-binomial bound gives $a(P, H) < 0$. Both contradict $a(P, H) \geq 0$. □

Proof of strictness and sharpness in Theorem 16.1. The proved upper bound supplies $C_{n-1} \leq C_{\mathrm{E}}$ for every $n \geq 700$. If $C_n \geq C_{\mathrm{E}}$, fixed-target attainment and reflection give a maximizer with $a \geq 0$, contrary to the preceding theorem. Hence $C_n < C_{\mathrm{E}}$ for $n \geq 700$. Finally, Esseen's fixed skew-Bernoulli limit (2) gives $\sup_n C_n \geq C_{\mathrm{E}}$. Together with the upper bound, this proves equality for the global supremum. □

The two local comparisons have different purposes. The first-failure comparison proves the weak bound at every index; the compact comparison excludes equality from 700 onward. Neither argument assumes that an unrestricted extremizer is a two-point law. The certificates verify inequalities on whole parameter regions for the original marginal, after the analytic acquisition of those regions.

17 Discussion

The common mathematical object is the support-gap defect. Contact and zero-bias integration turn it into a quantity controlled by the normal approximation remainder. This makes the geometry of a possible extremizer accessible: the support separates into two clusters, and their conditional fluctuations can be compared with binomial counts. Every acquired bound concerns the same law. This is essential because an optimizer of a relaxed moment problem need not be an iid marginal.

The smallest indices need separate arguments. Their sum distributions are too far from the limiting two-point geometry for the compact tail comparison to apply directly. Polynomial majorants, a four-summand physical comparison and finite count arguments provide a sharp base through 42. Beyond that base, the one-step estimate supplies the small excess and zero predecessor loss at a first failure. The finite and uniform local comparisons then exclude every remaining index.

The number 700 is a sufficient floor for the additional strict comparison. The defect bound gives $32.20665378/700 < 0.047$, and the binomial-mass and error estimates hold in the resulting law region. At small noise the errors remain proportional to λ; on $1/100 \le \lambda \le 7/20$ an additive error of order $1/n$ is absorbed by the comparison margin. At zero noise the strict finite-binomial theorem applies. A smaller strictness threshold would require checking both the geometric acquisition and the comparison. It is not needed for the sharp global inequality proved here.

The certificate format is specific to these arguments. Exact arithmetic removes rounding uncertainty, but the continuous reductions, coverage and implementation still require review. The reproduction record distinguishes a supplied candidate whose acceptance was rerun from a table constructed and recomputed during this verification. It also distinguishes shared-code replay from independently derived checks. No proof-assistant formalization or eventual two-point optimality is asserted.

AI assistance.

AI systems were used extensively for mathematical exploration, exact checking programs, source and bibliography comparisons, and drafting and revision. Separate AI review roles examined component arguments; their agreement is not a substitute for mathematical proof or external human review. Failed arguments remain diagnostics, not premises. The authors are responsible for the content and presentation.

Sources and reproduction.

The earlier all-index manuscript and certificate release [8] supply the finite proof components reconstructed here. The present extension gives their continuous reductions, a shorter first-failure composition and fresh numerical acceptance with complete input joins. The original sources are pinned at commit `3d053204f30eab39da39bfb7b85e380d83ec0646` in the earlier certificate repository. The compact starting manuscript is preserved at its immutable source snapshot. The extended sources are in `manuscript/extended-v17/`. Appendix AB identifies the proof inputs, checking commands and recorded executions. A document build checks presentation; it does not verify a theorem.

A A uniform Fourier enclosure with explicit input hypotheses

We prove the compensated Fourier formula and the complete low- and high-frequency estimates used in Section 5. Every assertion concerns an actual marginal with the displayed moment and event bounds. Appendix D establishes these inputs for the maximizers to which the bound is applied, under its stated entry conditions.

For each integer $n \geq N \geq 42$, let $X = X_n$ be a real random variable with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\mathbb{E}|X|^3 < \infty$. Neither the law nor the real raw threshold $H = H_n$ must be fixed as n varies. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n.$$

The two complete integrals are

$$I_0(H) = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n(t))|}{t^3}\, dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

A.1 The inversion edge, including atoms and moving thresholds

The sum law and its signed Gaussian approximation have the same mass and first moment. Subtracting these two terms before integration makes the Fourier expression absolutely integrable. The following elementary identity also explains why no continuity assumption at the threshold is needed.

Lemma A.1. *If a finite signed measure μ has mass and first moment zero and $\int x^2\, d|\mu|(x) < \infty$, then, for every $H \in \mathbb{R}$,*

$$\int (x - H)|x - H|\, d\mu(x) = -\frac{4}{\pi}\int_0^\infty \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

The integral on the right is absolutely convergent.

Proof. For real y, substitution and integration by parts give

$$\int_0^\infty \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|.$$

The boundary terms vanish: near zero the numerator is $O(t^3)$, and at infinity it is $O(t)$. The integrand has the sign of y, since $v - \sin v \geq 0$ for $v \geq 0$. To calculate the constant, nonnegative Tonelli and elementary Laplace integration give

$$\int_0^\infty \frac{1 - \cos t}{t^2}\, dt = \int_0^\infty s\left(\frac{1}{s} - \frac{s}{s^2+1}\right) ds = \frac{\pi}{2}.$$

The absolute double integral of the compensated expression against $d|\mu|(x)$ equals

$$\frac{\pi}{4}\int (x - H)^2 d|\mu|(x) < \infty.$$

Absolute Fubini is therefore valid. The linear compensation disappears because $\int (x-H)d\mu = 0$, proving both assertions. No continuity of a CDF at H is involved. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of the signed density

$$\phi(x)\{1 + \gamma(x^3 - 3x)/(6\sqrt{n})\}.$$

This signed density has mass one, first moment zero and finite absolute second moment. Its transform after scaling is e_n. Gaussian integration by parts, using $q_z(x) = (x-z)|x-z|$ and $q_z''' = 4\delta_z$, yields

$$\varepsilon_n := \mathbb{E}q_{H/\sqrt{n}}(S_n/\sqrt{n}) - \mathbb{E}q_{H/\sqrt{n}}(Z) - \frac{2\gamma\phi(H/\sqrt{n})}{3\sqrt{n}} = -\frac{4}{\pi n}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt.$$

Consequently the error $E_n = -n^{3/2}\varepsilon_n/2$ satisfies

$$|E_n| \le \frac{2\sqrt{n}}{\pi}\{I_0(H) + I_\infty\}. \tag{68}$$

The identity and bound hold for each n and H separately.

A.2 Low frequencies: Taylor cancellation and moment bounds

Near zero, the Gaussian approximation cancels the first terms of the characteristic function. The fourth- and fifth-order coefficients must still be bounded, and the sixth moment controls the remaining Taylor error. These higher moments are explicit hypotheses of the estimate; in its application they become available only after support has been bounded.

Assume common bounds

$$\mu_4 := \mathbb{E}X^4 \le M \le 111/10, \quad |\gamma| \le g \le 37/20, \quad \mu_6 \le M_6,$$

and put

$$a = (3-\mu_4)/24, \quad z = (10\gamma - \mathbb{E}X^5)/120, \quad |a| \le A, \quad |z| \le Z.$$

Here A, Z, M_6 are nonnegative common ceilings, not quantities inferred from bounded third moments. Define

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$

$$a_0 = \frac{N-1}{2N}, \quad b_0 = \frac{45-M}{96}(1-2/N) > 0, \quad \lambda_N = \sqrt{\pi/2}\left(\frac{N}{N-1}\right)^{3/2}.$$

Proposition A.2. *For all the laws and thresholds above,*

$$I_0(H) \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where, for any $c_0 \ge \lambda_N$ and $c_1 \ge \lambda_N/2$,

$$\begin{aligned} L_0 &= c_0 Z + c_1 A g + 251 g^3/17280, \\ L_H &= c_0 A + 251 g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3). \end{aligned}$$

Proof. Let $h(t) = \exp(-t^2/2 - i\gamma t^3/6)$. Taylor's integral remainder for e^{itX}, followed by expectation, gives its expansion through degree five with error at most $\mu_6 t^6/720$. For $w = -t^2/2 - i\gamma t^3/6$ and $0 \le t \le 1/2$,

$$|w| \le t^2(1/2 + g^2/144) < 1/6, \quad |e^w - 1 - w - w^2/2| \le |w|^3 e^{|w|}/6 \le |w|^3/5.$$

The first bound follows from $\sqrt{1/4+y} \le 1/2 + y$; for the last use $e^{1/6} \le (1-1/6)^{-1} = 6/5$. Comparing the two polynomials gives

$$f - h = -at^4 - izt^5 + r_6(t), \qquad |r_6(t)| \le Et^6, \qquad |f-h| \le Bt^4. \tag{69}$$

The degree-six term in $w^2/2$ costs $g^2t^6/72$; it has not been discarded.

Four integrations of $1-\cos u \ge 0$ show $\cos u \le 1-u^2/2+u^4/24$ for every real u. Independence gives $\mathbb{E}(X-X')^4 = 2\mu_4 + 6$, and hence

$$|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45-M)t^2/48.$$

Thus $|f|, |h| \le e^{-(45-M)t^2/96}$ throughout the low interval. Every point on their complex joining segment obeys the same bound. Taylor's integral formula for the polynomial $v \mapsto v^n$ proves

$$f^n - h^n = n(f-h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8 e^{-(n-2)(45-M)t^2/96}.$$

The leading remainder in (69) and this power remainder cost at most L_2/n after division by t^3 and integration: use $n-1 \ge 2a_0 n$, $(n-2)(45-M)/96 \ge b_0 n$, $n(n-1) \le n^2$, and $\int_0^\infty t^3 e^{-At^2}dt = 1/(2A^2)$, $\int_0^\infty t^5 e^{-At^2}dt = A^{-3}$.

For the remaining leading terms let $b = n-1$ and $v = b\gamma t^3/6$. Their imaginary part after rotation by Ht is bounded by

$$ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}.$$

The Gaussian moments of orders two and four, after dividing by t^3, give $n\sqrt{\pi/2}b^{-3/2}(Z + A|H| + Ag/2)$. This is at most $\lambda_N(Z + A|H| + Ag/2)/\sqrt{n}$. Finally put $u = n\gamma t^3/6$. Direct sine and cosine remainders give

$$|\Im[e^{-itH}(h^n - e_n)]| \le e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2).$$

The Gaussian moments of orders six and four respectively give $\sqrt{\pi/2}g^3/(432\sqrt{n})$ times 5, and $\sqrt{\pi/2}g^2|H|/(24\sqrt{n})$. Using $\sqrt{\pi/2} < 251/200$ gives precisely the two displayed rational coefficients. Every extended integrand was nonnegative. The only index comparisons were the displayed rational inequalities; in particular the law and H can vary arbitrarily with n. □

In the selected enclosure one has actual support $|X| \le R$ with $1 \le R \le 6$ and $\mathbb{E}|X|^5 \le E_5$. Then $M_6 = RE_5$ and $A = \max(2, M-3)/24$ are valid. Its sharper Z is also licensed: for $Q = X^2 - \gamma X - 1$ direct multiplication gives

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \quad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)].$$

Reflect this auxiliary calculation if necessary so that $t = |\gamma| \ge 0$. For any $r \ge \max(R, 2g)$ the two signs of $\mathbb{E}X^5 - 10\gamma$ are at most

$$P_r(t) = r(M-1) + (10-2M)t - rt^2 + t^3, \quad Q_r(t) = r(M-1) + (2M-10)t - rt^2 - t^3.$$

The coefficients of μ_4 before replacing it by M are $r - 2t, r + 2t \ge 0$. If $M \le 5$, $P_r \ge Q_r$ and $P_r \le r(M-1) + (10-2M)t - (r-g)t^2$, whose maximum on $[0, g]$ is at the clipped vertex used in the code. If $M > 5$, $P_r \le r(M-1)$ and $Q_r \le r(M-1) + (2M-10)t - rt^2$, again with the stated clipped vertex. For $r = 6$, $P_6'' = -12 + 6t < 0$ on $[0, g]$. If $M \le 5$ and $P_6'(g) \ge 0$, its maximum is exactly $P_6(g)$; otherwise the preceding quadratic gives $6(M-1) + (10-2M)^2/[4(6-g)]$. For $M > 5$ the unrestrained vertex bound is $6(M-1) + (2M-10)^2/24$. Taking the minimum of these proved ceilings and $E_5 + 10g$, then dividing by 120, gives a valid coefficient ceiling Z. The coefficient ceiling therefore uses the known support radius R.

A.3 High frequencies: six contributions and uniform index bounds

A nearly lattice law can retain characteristic-function peaks away from zero. The high-frequency argument therefore uses an event involving the actual difference $X - X'$ and partitions the oscillations around their zeros. The six contributions below cover intermediate frequencies, nonzero-cell cores, nonzero-cell wings, the small-angle part of the zero cell, the rest of the zero cell, and the signed Gaussian comparator. Each optional improvement has its own stated positivity and monotonicity conditions.

For independent copies of the same marginal suppose an actual event $\mathcal{E}$ satisfies

$$\mathbb{P}(\mathcal{E}) \geq \alpha > 0, \quad |X - X'| \geq d > 0 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \leq M_E.$$

Fix $b = 9/10$, $0 < r \leq 1/4$, and

$$h = 333/106 - (22/7)^3 r^2/6 > 0, \quad \kappa = \alpha(23/48)^2 d^2, \quad L(x) = \frac{x+2}{x^2+4x+2},$$

$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + \frac{1}{4(20-r)^4}.$$

Let $c(u) = u/2 - (M+3)u^2/24$ and set

$$c_* = \min\{c(1/4), c(81/100)\}, \quad c_q = 1/8 - (M+3)/384, \quad s = [1 - (M+3)53/600]/2.$$

Proposition A.3. *Suppose $c_* > 0$. For every $n \geq N$, $\sqrt{n} I_\infty$ is at most the sum of the following six terms:*

$$\begin{aligned}
T_1(n) &= 2\sqrt{n} e^{-nc_*}, \\
T_2(n) &= \frac{M_E}{4\pi^{3/2} h \sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\}, \\
T_3(n) &= \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \\
T_4(n) &= \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}}, \\
T_5(n) &= \frac{\sqrt{n}}{2b^2} e^{-n\alpha(23/48)^2}, \\
T_6(n) &= \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.
\end{aligned}$$

If $s > 0$, T_1 may be replaced by its minimum with $2\sqrt{n} e^{-nc_q} L(ns/4)$. The fourth term may be replaced by its minimum with $\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)/(2b^2)$. The common-floor sum at N is valid if

$$2Nc' > 1 \quad \text{for } c' \in \{c_*, \alpha h^2 r^2, \alpha(23/48)^2, 1/8\}.$$

For the optional first term also require $2Nc_q > 1$; for the optional fourth term require $N\kappa b^2 \geq 1/4$. If an optional gate fails, the simpler term remains available.

Proof. The triangle inequality separates $|f^n|$ and the entire comparator. On $[1/2, b]$, symmetrization gives $|f|^n \leq e^{-nc(t^2)}$. The function c is concave, so $c(t^2) \geq c_*$ and the integral of t^{-3} is less than two. This gives T_1. Also

$$c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \geq 0.$$

When $s > 0$ this chord may be integrated to infinity.

For completeness, the tail calculation used here is, for $x = nab^2 > 0$,

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt \le \frac{\sqrt{n}e^{-x}}{2b^2}L(x). \tag{70}$$

To see where the rational function L comes from, let U have density e^{-u} on $u \ge 0$. Substitution followed by positive Fubini gives

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt = \frac{\sqrt{n}e^{-x}}{2b^2}\mathbb{E}\frac{U}{x+U}.$$

For example, after putting $v = na(t^2 - b^2)$ the remaining factor is $\int_0^\infty xe^{-v}(x+v)^{-2}dv$; integration by parts makes it $\mathbb{E}[U/(x+U)]$. For $x > 0$ put

$$J_x = \mathbb{E}\frac{1}{x+U}, \qquad v_0 = \begin{pmatrix}1\\1\end{pmatrix}, \qquad G = \mathbb{E}\left[(x+U)\begin{pmatrix}1 & U\\ U & U^2\end{pmatrix}\right] = \begin{pmatrix}x+1 & x+2\\ x+2 & 2x+6\end{pmatrix}.$$

The entries use $\mathbb{E}U = 1$, $\mathbb{E}U^2 = 2$, and $\mathbb{E}U^3 = 6$. For $q(U) = c_1 + c_2U$, weighted Cauchy–Schwarz in its squared-residual form gives

$$0 \le \mathbb{E}\frac{[1-(x+U)q(U)]^2}{x+U} = J_x - 2v_0^Tc + c^TGc.$$

The matrix G is positive definite: its leading entry is positive and $\det G = x^2 + 4x + 2 > 0$. Choosing $c = G^{-1}v_0$ yields

$$J_x \ge v_0^TG^{-1}v_0 = \frac{x+3}{x^2+4x+2}, \qquad \mathbb{E}\frac{U}{x+U} = 1 - xJ_x \le \frac{x+2}{x^2+4x+2} = L(x).$$

This proves (70) for every $a, b, n > 0$ with the same function L. Applying it at endpoint $1/2$ and rate s proves the optional first term.

For $t \ge b$, symmetrization, the actual-event probability bound, and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-n\alpha\sin^2(tD/2)} \mid \mathcal{E}], \quad D = |X - X'|.$$

Positive Tonelli allows a separate partition for every $D > 0$. With $\ell = 2\pi/D$, use the zero cell $[0, \ell/2]$ and nonzero cells $\ell[j - 1/2, j + 1/2]$, $j \ge 1$. In a nonzero cell write $t = \ell(j + v)$. On its core $|v| \le r$, $|\sin \pi v| \ge h|v|$; on its wings $r \le |v| \le 1/2$ it is at least hr. These follow from $\sin x \ge x - x^3/6$ and monotonicity of sin on $[0, \pi/2]$, with directed bounds on π. For $|v| \le r$ Taylor's integral remainder gives

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-r)^{-5}v^2.$$

Extend the nonnegative Gaussian majorant over the whole real line. Its zeroth and second moments, the Jacobian $t^{-3}dt = D^2(4\pi^2)^{-1}(j+v)^{-3}dv$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$ give T_2. The two series tails are bounded by their decreasing integrals after 20. The total wing weight is at most $\int_{\ell/2}^\infty t^{-3}dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their nonnegative bounds.

In the zero cell put $u = tD/2 \le \pi/2$. Where $u \le 1/2$, $\sin u \ge 23u/24$, and $D \ge d$ yields rate κ. Extending the set to $[b, \infty)$ gives T_4 from $t^{-3} \le t/b^4$; (70) gives its optional improvement. Where $u \ge 1/2$, monotonicity gives $\sin u \ge 23/48$ and the total weight is at most $1/(2b^2)$, proving T_5. Endpoints have zero Lebesgue measure even if the conditional distance distribution has atoms.

Finally $|e_n(t)| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. The first part on the entire $[1/2, \infty)$ uses (70); the second uses $\int_{1/2}^\infty e^{-nt^2/2}dt \le 2e^{-n/8}/n$. Pearson's identity $\mu_4 - 1 - \gamma^2 = \mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator tail is dropped.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases for $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for every $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. Thus the optional first term decreases under its own gate, and the comparator decreases for $n \geq 5$. The sign of the derivative of $\sqrt{x}e^{-x}L(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4,$$

which is negative at $1/4$ and strictly decreasing on $(0, \infty)$. This proves the optional zero-cell gate. Every comparison is of a common scalar majorant, not of characteristic functions for different laws. □

A.4 The precise closed Fourier conclusion

Under the preceding moment, actual-event and scalar gates, let K_H be any complete HIGH sum evaluated at N, including its permitted optional improvements. Then

$$|E_n| \leq \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \leq \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{71}$$

The second inequality uses $\pi > 25/8$ and nonnegative coefficients. This is the constant Fourier budget and raw-threshold slope used in Section 5. Complete HIGH bounds for several valid windows may be minimized. The LOW coefficients must remain one complete valid triple.

The rational enclosures $\pi > 333/106$, $\sqrt{\pi} > 4431/2500$, $\pi < 22/7$ and $\sqrt{\pi/2} < 251/200$ are proved together in Appendix B. The arithmetic routines and their rounding directions are described in Appendix AB. The floor comparisons concern the analytical majorants under their stated positivity and monotonicity conditions; the rounded routines need not themselves be monotone. The resulting bounds hold for each law and threshold satisfying the hypotheses, even when they vary with n.

B Elementary Gaussian constants

Lemma B.1 (Elementary Gaussian constants)**.** *For the standard normal density,*

$$\|\phi''\|_\infty = \phi(0) < 2/5, \qquad \|\phi'''\|_\infty \leq 2/\pi, \qquad 1/(4\pi) < 2/25.$$

Moreover $819/2000 < C_{\mathrm{E}} < 20487/50000 < 41/100$ *and* $\phi(0) > 0.39886 > 0.3988 = 997/2500$.

We give the exact comparisons used in Lemma B.1.

Proof. Since $\phi''(x) = (x^2 - 1)\phi(x)$, the norm on $x^2 \leq 1$ is at most $\phi(0)$. On $x^2 \geq 1$, the maximum of $(x^2 - 1)e^{-x^2/2}$ is $2e^{-3/2} < 1$, attained when $x^2 = 3$. Thus the norm is exactly $\phi(0)$.

For completeness, the Gaussian transform can be obtained without an inversion theorem for the distribution being studied. Differentiating $J(t) = \int_{\mathbb{R}} e^{-x^2/2}e^{itx}\,dx$ under the integral and integrating by parts give $J'(t) = -tJ(t)$. The Gaussian integral, evaluated by squaring it and using polar coordinates, gives $J(0) = \sqrt{2\pi}$. Hence $J(t) = \sqrt{2\pi}e^{-t^2/2}$; applying the same formula with the variables interchanged gives

$$\phi(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-t^2/2}e^{-itx}\,dt.$$

Three differentiations are justified by the integrability of $|t|^3e^{-t^2/2}$. Consequently

$$\|\phi'''\|_\infty \leq \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\,dt = \frac{2}{\pi}.$$

The last integral equals four after the substitution $w = t^2/2$ on the positive half-line.

Here are elementary rational comparisons for all numerical constants used in this section. Put $a = \arctan(1/5)$ and $b = \arctan(1/239)$. Tangent addition gives $\tan(2a) = 5/12$, $\tan(4a) = 120/119$ and $\tan(4a - b) = 1$. Also

$$0 < 4(1/5 - 1/(3 \cdot 5^3)) - 1/239 < 4a - b < 4/5 < 1 < \pi/2.$$

The arctangent inequalities follow by integrating $1 - t^2 < 1/(1 + t^2) < 1$, while $\pi/2 = 2\arctan(1) > 1$ follows by integrating $1/(1 + t^2) > 1/2$ on $[0, 1)$. It follows that $\pi = 16a - 4b$ and

$$\pi > 16(1/5 - 1/(3 \cdot 5^3)) - 4/239 = 281476/89625 > 25/8.$$

This proves $\phi(0) < 2/5$ and $1/(4\pi) < 2/25$.

To bound C_{E} more closely, define

$$s_k(x) = \sum_{j=0}^{k-1} \frac{(-1)^j x^{2j+1}}{2j+1}, \quad \pi_- = 16s_6(1/5) - 4s_1(1/239), \quad \pi_+ = 16s_5(1/5) - 4s_2(1/239).$$

The remainder in the finite geometric expansion of $1/(1 + t^2)$ has its stated alternating sign, so integration gives $\pi_- < \pi < \pi_+$. Let $r_- = 3162277660/10^9$ and $r_+ = 3162277661/10^9$. Squaring shows $r_-^2 < 10 < r_+^2$. The following comparisons are exact rational ones:

$$\left(\frac{819}{2000}\right)^2 < \frac{19 + 6r_-}{72\pi_+} < C_{\mathrm{E}}^2 = \frac{19 + 6\sqrt{10}}{72\pi} < \frac{19 + 6r_+}{72\pi_-} < \left(\frac{20487}{50000}\right)^2.$$

For example, each outer comparison follows by multiplying by its positive rational denominator; the displayed finite sums specify those denominators completely. All quantities are positive, so taking square roots proves the claimed bounds. In particular, $0.4095 < C_{\mathrm{E}} < 0.40974 < 0.41$. The same exact endpoints also satisfy

$$\pi_- > \frac{333}{106} > \left(\frac{4431}{2500}\right)^2, \qquad \pi_+ < \frac{22}{7} < 2\left(\frac{251}{200}\right)^2.$$

Thus $\sqrt{\pi} > 4431/2500$ and $\sqrt{\pi/2} < 251/200$, as used in the Fourier bounds. Since $(0.39886)^2 < 7/44$, we also obtain $\phi(0) > 0.39886 > 0.3988$. □

Lemma B.2 (The narrow entry enclosure)**.** *The target constant satisfies*

$$c_- := \frac{4097321837023}{10^{13}} < C_{\mathrm{E}} < \frac{4097321837025}{10^{13}} =: c_+.$$

Proof. Use the same finite sums s_k and Machin identity proved above, now with

$$\pi_L = 16s_{10}(1/5) - 4s_5(1/239), \qquad \pi_U = 16s_9(1/5) - 4s_4(1/239).$$

An even number of terms bounds each arctangent from below and an odd number bounds it from above. The negative coefficient of the second arctangent reverses its choice of endpoint, so $0 < \pi_L < \pi < \pi_U$. Put

$$r_L = \frac{3162277660168379}{10^{15}}, \qquad r_U = \frac{3162277660168380}{10^{15}}.$$

Integer squaring gives $r_L^2 < 10 < r_U^2$. Direct rational comparisons give

$$c_-^2 < \frac{19 + 6r_L}{72\pi_U} < C_{\mathrm{E}}^2 < \frac{19 + 6r_U}{72\pi_L} < c_+^2.$$

The finite sums specify every rational in this display; each outer inequality can be checked by cross-multiplying positive integers. All terms are positive, so taking square roots gives the assertion. This directed enclosure, rather than the coarser bounds in the preceding lemma, supplies the precision used by the entry calculation. □

C The identity without contact

The main proof uses the contact identity. This stronger extension applies to every compactly supported standardized law and also provides an identity for checking numerical implementations on nonextremal laws. Retain the definitions of $a, Q, h, \Delta, J, J_1, J_2, L$, and $\mathcal{R}_c$ from the main text, for arbitrary $H \in \mathbb{R}$ and $c > 0$. Introduce the contact residuals

$$\xi = \mathbb{E}[X\Delta(X)], \qquad R_{\mathrm{supp}} = \int_K \tau(t)\Delta(t)\,dt + \sum_{(u,v)} \tau_{u,v}(v-u)\Delta(v). \tag{72}$$

The first integral is with respect to Lebesgue measure on the closed support K, not to P. Since Δ is bounded on the convex hull and $\sum \tau_{u,v}(v-u) \le 1$, the series is absolutely convergent.

Theorem C.1 (Conservation with contact residuals)**.** *For every such law, threshold, and positive coefficient,*

$$(n+H^2)a + nc\rho - \frac{n\phi_z}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(R_{\mathrm{supp}} - H\xi). \tag{73}$$

At a fixed-target maximizing pair, both residuals vanish.

Proof. The gapwise identity in the proof of Theorem 4.1 holds without contact. Keeping its endpoint terms and the integral over K adds exactly R_{supp} to the right side of (19). The zero-bias evaluation (20) is unchanged. In the first-moment calculation, $\mathbb{E}[Xh(X)] = \mathbb{E}[XQ(X)] - \xi$, so (21) becomes

$$M + \phi_z = -\frac{\gamma H \phi_z}{2n} + \frac{cK_1}{n} - \sqrt{n}\xi.$$

The same pointwise identity for q_z therefore gives

$$\sqrt{n}\mathcal{A} = c\rho + a - \gamma\phi_z/6 + P_0/n + \sqrt{n}\varepsilon/2 + H\sqrt{n}\xi,$$

with exactly the P_0 defined in the main proof. Equating the two expressions and multiplying by n gives (73). At contact, $\Delta = 0$ on all of K, so every term in both residuals vanishes, including the Lebesgue integral. □

Why the support integral is needed.

Let X be uniform on $[-\sqrt{3}, \sqrt{3}]$ and take $H = 0$. There are no support gaps, $F = T = 1/2$, $\beta = \gamma = 0$, and $K_q = 3\sqrt{3}/40$. Thus

$$R_{\mathrm{supp}} = \int_K \tau\Delta = cK_q/n^{3/2} > 0.$$

Dropping the support integral would give a false identity even though all gap sums are empty. This example tests the extension without contact; it does not contradict the contact theorem. The leakage integral retains $[u, v)$ throughout: a right-endpoint atom belongs to $h(v)$, and a left-endpoint atom has weight zero.

D Details of the coarse localization

This appendix proves the analytical estimates and the finite computational propositions used by Theorem 5.1. The calculation has two coarse quadratic passes followed by four signed-moment passes. Each improvement is used only after its hypotheses have been established. The cover starts with 24 closed intervals for ρ. A signed pass splits each surviving interval

into 8 intervals for β, 8 for γ and 2 for m. Four such passes process 11264 nominal boxes; the number is smaller than $4 \cdot 24 \cdot 128$ because excluded moment cells need no later pass. Closed shared endpoints belong to both neighboring intervals. Appendix AB describes what the certificate stores and how this cover is reconstructed. Let P be a centered, unit-variance law, S_j a sum of j independent copies, and set

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad c = C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The threshold H always has the units of the raw sum. A fixed-target maximizer maximizes

$$a(P,H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

We use Proposition 2.1 and its closed-support contact formula (4). In particular its closed-support contact equality, not ratio optimality, is the variational premise here. We consider a maximizer with $a \ge 0$. The actual normalized score is $c + a/\rho$; no upper bound on it is assumed.

D.1 The first support bound

Theorem D.1 (Initial moment and support entry)**.** *Assume $n \ge 42$, $C_{n-1} \le c$, and the universal structural inequality*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3031}{10000}\left(\rho + \frac{323}{500}\right) \tag{74}$$

for standardized laws with finite third moment. Then a fixed-target maximizer with $a \ge 0$ satisfies

$$1 \le \rho < \frac{7361}{4000} < \frac{1841}{1000}, \qquad |\beta| < \frac{321}{500}, \qquad \operatorname{supp} P \subset (-96/25, 96/25). \tag{75}$$

The theorem is uniform in all real H and in both signs of a support point. It requires neither C_{n-2} nor an excess bound of order $1/n$.

The structural inequality (74) is the external iid theorem of Shevtsova [15]. Only this affine bound is needed for the displayed moment radius; it is not a consequence of the finite arithmetic in this section.

Proof. Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, elementary division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The other affine bound $0.3322(\rho + 0.429)$ alone would give the larger number $712569/386500$; it must not be used to claim $7361/4000$. Lemma 2.5 gives

$$\rho \ge \frac{1+\beta^2}{\sqrt{1-\beta^2}}.$$

This function increases with $|\beta|$. For $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0,$$

which proves the second assertion without assuming bounded support.

Write $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $\eta_n = n(s-1) - 1/2$. Closed-support contact, the upper predecessor bound $F_{n-1,P}(H-t) \le F_v(H-t) + c\rho/\sqrt{v}$, and $\sqrt{n}(F_{n,P}(H) - \Phi(H/\sqrt{n})) = c\rho + a$ give

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho\eta_n\} - na \quad (t \in \operatorname{supp} P), \tag{76}$$

where exactly

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2-1}{2}F''_{v+1}(H).$$

The extra $-na$ can be discarded in this upper bound. This is the precise reason the predecessor assumption is $C_{n-1} \le c$, not a comparison with the actual score. No continuity assumption on the distribution function at $H-t$ is inserted.

Rationalization gives

$$n\eta_n = \frac{s^2(2s+1)}{2(s+1)^2}, \qquad s < 81/80 =: S, \qquad n\eta_n < \frac{793881}{2073680} < \frac{48}{125}.$$

The first expression increases for $s \ge 1$, as its logarithmic derivative is $2/s + 2/(2s+1) - 2/(s+1) > 0$. Thus $\eta_n < 8/875$.

For completeness we rederive the Gaussian estimate used at the radius. The heat identity $\partial_v F_v = F''_v/2$ permits spatial expansion of $F_v(H-t)$ to order four, variance expansion of F_{v+1} to order two, and variance expansion of F'_{v+1}, F''_{v+1} to order one. The cubic coefficient is $(-t^3+3t)/6$, and the quartic coefficient is $(t^4-6t^2+3)/24$. Therefore, for $t = \pm u$, $z_v = H/\sqrt{v}$,

$$H_n(H, \pm u)/s^3 = \pm A(1-z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{77}$$

where $A = (u^3-3u)/6$, $H_4 = u^4 - 6u^2 + 3$ and $B = H_4/(24\sqrt{v})$. The four error bounds after division by s^3 are respectively

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives norms at most $3\phi(0) < 6/5$ and $8/\pi < 51/20$. Since $v \ge 41$ and $\sqrt{v} > 6$, for $u \ge 1$ this gives

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

These errors are uniform in H; they use no higher moment of P.

If $A, B > 0$ and $B \le A/4$, the leading part of (77) is at most $\phi(0)(A + 9B^2/(4A))$ for both signs. Indeed, for the positive sign and $|z_v| \le 1$, it is at most $\phi(0)(A - Az_v^2 + 3B|z_v|)$, and complete the square. Outside that interval the first term is nonpositive. For the negative sign the first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$. In both cases use $\|\phi'''\|_\infty < 2\phi(0)$: on $[0, \sqrt{3}]$ the polynomial $3x - x^3$ is at most two, and beyond $\sqrt{3}$ use $x^3 e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$.

For frozen r, b above define

$$Q(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r,$$
$$P(u) = \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256\cdot 41}\right) + S^3E(u)A + \frac{819}{2000}Q(u)A.$$

At $R = 96/25 < 6$ we have $A > 0$, $0 < H_4 < 36A$, hence $B < A/4$. The sign $Q(R) < 0$ is checked before replacing c by its lower endpoint. The upper bound in (76), multiplied by A, is then strictly below $P(R)$, and exact arithmetic gives

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

To cover each entire exterior ray, differentiation of the exact Gaussian expression gives

$$|\partial_t H_n(H,t)| < \tfrac{2}{5}(t^2/2 + 1/2 + \eta_n).$$

For verification put $y = (H-t)/\sqrt{n}$, $z = H/\sqrt{n}$; the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The second-order Taylor error between $\phi(y)$ and $\phi(z)$ costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Define $L(u) = \frac{2}{5}(u^2/2 + 1/2 + 8/875) + (819/2000)Q'(u)$. At R, the six numbers Q, Q', Q'', L, L', L'' are strictly negative. Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, all their required signs persist for $u \geq R$. Thus both radial upper bounds decrease from a strictly negative boundary value. The contact inequality excludes every support point with $|t| \geq R$. □

D.2 Higher moments and an actual Fourier event

Higher moments are used only after Theorem D.1. For $x = |X|$ the nonnegative measure $d\mu_0 = (x-1)^2(x+1/2)\,dP$ has mass $\rho - 1$. Direct expansion gives

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2}\,d\mu_0,$$
$$M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x+1/2}\,d\mu_0,$$

where $M_j = \mathbb{E}|X|^j$. Both ratios increase for $x \geq 0$: they equal $x + 3/2 + 1/[4(x+1/2)]$ and $x^2 + 3x/2 + 9/4 + 3/[8(x+1/2)]$, whose derivatives are nonnegative. Writing their values at $R = 96/25$ as $k_4(R), k_5(R)$, we obtain

$$M_4 \leq \min(R\rho, 1 + k_4(R)(\rho-1)), \quad M_5 \leq \min(RM_4, 1 + k_5(R)(\rho-1)), \quad M_6 \leq RM_5. \tag{78}$$

Also $\gamma^2 \leq M_4 - 1$, by expanding $\mathbb{E}(X^2 - 1 - \gamma X)^2 \geq 0$.

There is no need to assume a central support gap before the first Fourier bound. For independent copies let $E = \{(X - X')^2 \geq 9/25\}$ and let M be any valid fourth-moment ceiling. Then

$$\mathbb{P}(E) \geq \alpha := \frac{(2 - 9/25)^2}{2M + 6} > 0, \quad |X - X'| \geq 3/5 \text{ on } E, \quad \mathbb{E}[(X - X')^2 \mid E] \leq 2/\alpha. \tag{79}$$

Indeed $\mathbb{E}[(X-X')^2; E] \geq 2 - 9/25$ and $\mathbb{E}(X - X')^4 = 2M_4 + 6 \leq 2M + 6$, so Cauchy proves the mass bound. The conditional second-moment bound follows from its unconditional value two. This is an event under one actual iid marginal, not a replacement characteristic function.

The same support proof will be used again on a smaller closed moment cell. Freeze upper bounds r, b for $\rho, |\beta|$ in its polynomial Q, and test a proposed radius before changing any moment bound. For $R \in [5/2, 96/25]$, the gates $A > 0$, $0 < H_4 < 36A$, $Q(R) < 0$, $P(R) < 0$, and the six negative ray signs suffice. Their strictness excludes the boundary atoms as well as the open exterior rays. A radius that fails a gate is not used. The selected finite search is $R = 2.50, 2.51, \ldots, 3.84$; it only chooses among proved implications.

D.3 The Fourier enclosure used by every pass

Let $f(t) = \mathbb{E}e^{itX}$ and $e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6)$. For the remainder in (15), the compensated inversion proved in Appendix A gives

$$|E_n| \leq \frac{2\sqrt{n}}{\pi}\left\{\int_0^{1/2} \frac{|\Im[e^{-itH}(f(t)^n - e_n(t))]|}{t^3}dt + \int_{1/2}^{\infty} \frac{|f(t)^n - e_n(t)|}{t^3}dt\right\}.$$

The comparator is signed, has the correct mass and first moment, and its entire high-frequency tail is included. This formula applies at atoms and for arbitrary real H.

For clarity, the exact input and output of the selected Fourier step are as follows. Acquire $|X| \le R$ first, and choose common ceilings $M \le 11.1$, $g \le 1.85$, M_6 for $M_4, |\gamma|, \mathbb{E}X^6$, together with

$$|(3 - M_4)/24| \le A_4, \qquad |(10\gamma - \mathbb{E}X^5)/120| \le Z_5.$$

Put

$$\begin{aligned} E_6 &= M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \\ B_6 &= A_4 + Z_5/2 + E_6/4, \\ a_0 = (N-1)/(2N), \quad b_0 &= (45 - M)(1 - 2/N)/96, \\ c_0 \ge \sqrt{\pi/2}(N/(N-1))^{3/2}, \qquad c_1 &\ge c_0/2. \end{aligned}$$

A complete admissible low-frequency triple is

$$\begin{aligned} L_0 &= c_0 Z_5 + c_1 A_4 g + 251 g^3/17280, \\ L_H &= c_0 A_4 + 251 g^2/4800, \\ L_2 &= E_6/(2a_0^2) + B_6^2/(2b_0^3). \end{aligned}$$

One may take $A_4 = \max(2, M-3)/24$ and $M_6 = RM_5$. The sharper fifth-order coefficient used by the finite cover is proved in Appendix A; it follows from the exact identity for $X^2 - \gamma X - 1$, not from an assumed fifth moment.

The high-frequency input is an actual event for two iid copies:

$$\mathbb{P}(\mathcal{E}) \ge \alpha > 0, \qquad |X - X'| \ge d_E > 0 \text{ on } \mathcal{E}, \qquad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \le M_E.$$

Proposition A.3 supplies a complete six-term bound K_H at $N = 42$, subject to its explicit positive damping and all-index gates. The first event is (79); an opposite-sign event is used only after an actual central gap is acquired below. Different valid windows may be minimized only after all six terms, including the comparator, are present. No characteristic function of an artificial binary law is substituted for f.

The exact scalar output is

$$|E_n| \le K_F + s_F|H|, \qquad K_F = \frac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \frac{16}{25}L_H. \tag{80}$$

All square roots and exponential bounds in the finite evaluation are directed outward. The proof in the appendix applies to every actual $n \ge N$ and allows the law and threshold to vary with n. Thus no fixed-law limit is being used as a uniform bound.

D.4 Coarse coefficients before a central gap is known

Write the non-Fourier constant charge in (15) as

$$A_c^{\mathrm{law}} = \frac{3}{2}\beta(J + \gamma) + 3(\rho J_1 - J_2) + \frac{\rho}{2}M_4 - \frac{1}{4}M_5 - \frac{\rho}{2}.$$

Here A_c^{law} does not depend on c; the subscript records its role as the coefficient of c. We will bound it from above, retaining all signs in the later passes.

The following gap inequalities need no gap around zero:

$$-M_4/6 \le J_1 \le M_4/2, \qquad -3J_2 - M_5/4 \le 0, \qquad J \le \rho + 1. \tag{81}$$

For a gap (u, v) of length h, the first pair follows from

$$\int_u^v (t-u)t\,dt + \frac{1}{2}\int_u^v t^2 dt = hv^2/2, \quad \frac{3}{2}\int_u^v t^2 dt - \int_u^v (t-u)t\,dt = h(2u+v)^2/6.$$

Sum with the nonnegative gap weights and use $\int t^2\tau = M_4/3$. On a negative gap, reflection and

$$\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3 \geq 0 \quad (0 \leq u \leq v)$$

prove the second assertion; the positive portion only improves it. The last bound follows from $J \leq \rho + 2m$ and $2m \leq 1$.

For $1 \leq r \leq \rho$ and $R \leq 6$, choose $\Gamma(R, r) \geq 0$ with

$$\Gamma(R,r) \geq \max_{r \leq v \leq R} \left\{ -\frac{v^2}{4} + \frac{(R+r)v}{3} - \frac{Rr}{2} + \frac{r^3(2R-r)}{12v^2} \right\}_+ . \tag{82}$$

Then

$$3(\rho J_1 - J_2) \leq \frac{3}{2}\Gamma(R,r)(\rho - \gamma) + \frac{3}{8}\rho^2 J. \tag{83}$$

On the negative portion of a gap discard $|t| < \rho$, where its charge is nonpositive, and put $x = -t$. Since $1 \leq r \leq \rho$ and $|X| \leq R$, the remaining disjoint contributions are bounded by

$$\begin{aligned} 3\int_r^R \tau(-x)(R-x)x(x-r)\,dx &= 3\mathbb{E}\left[X_- \mathbf{1}_{\{X_- > r\}} \int_r^{X_-} (R-x)x(x-r)\,dx \right] \\ &\leq 3\Gamma(R,r)\mathbb{E}X_-^3. \end{aligned}$$

Nonnegativity permits the countable gap sum and these integrals to be interchanged by Tonelli. For $v \in [r, R]$, the inner integral divided by v^2 is exactly the expression in (82). Since $\mathbb{E}X_-^3 = (\rho - \gamma)/2$, this gives the first term of (83). On the positive portion, $t(\rho - t) \leq \rho^2/4$ and $\sum \tau_{u,v} \int_u^v (t-u)dt = J/2$ give the other term. This argument includes gaps crossing zero and atoms at either endpoint.

The finite bound for Γ partitions $[r, R]$ into 64 closed intervals. On each, maximize the concave quadratic at its clipped vertex $2(R+r)/3$ and its nonnegative reciprocal term at the left endpoint. The alternative ceiling $5/2$ follows at $r = 1, R = 6$ from $3v^4 - 28v^3 + 66v^2 - 11 > 0$ on $[1, 6]$: the endpoint values are $30, 205$ and the two interior critical values are $(85 \mp 35\sqrt{5})/2 > 0$. The positive integral representation is increasing in R and decreasing in r, so that endpoint choice encloses the present domain.

The elementary square

$$V = \mathbb{E}[|X|^3(|X| - \rho)^2] = M_5 - 2\rho M_4 + \rho^3 \geq 0$$

gives $\rho M_4/2 - M_5/4 \leq \rho^3/4$. Thus on $\rho \in [l, u]$, $|\beta| \leq b$, $M_4 \leq M$, and with $0 \leq \rho - \gamma \leq \Delta$, two complete constant bounds are

$$\begin{aligned} A_1(\Delta) &= \tfrac{3}{2}b(2u+1) + \tfrac{3}{2}\Gamma(R,l)\Delta + \tfrac{3}{8}u^2(u+1) + u^3/4 - l/2, \\ A_2 &= \tfrac{3}{2}b(2u+1) + 2uM - l/2. \end{aligned} \tag{84}$$

Use $A = \min(A_1, A_2)$. The first uses (83); the second uses (81). They are alternative complete bounds, not partial costs to be mixed.

For the linear term K_1, the following three bounds are valid:

$$\begin{aligned} K_{\text{rad}} &= 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \\ K_{\text{univ}} &= 6u - 3u^2/2 + 3b, \\ K_R &= Ru/2 + \max_{r \in \{l,u\}} |R/2 - 3r/2|g + 3b. \end{aligned}$$

For the first use the positive measure $d\mu_0 = (|X|-1)^2(|X|+1/2)dP$, of mass $\rho - 1$, and the exact identity

$$K_1 + 3\rho\beta = \int \operatorname{sgn}(X)\frac{x(x+2-3\rho/2)}{x+1/2}\,d\mu_0, \qquad x = |X|.$$

The ratio is convex in $x \geq 0$ and decreases in ρ. Its negative part is at most $31/40$ on $\rho \leq 1.85$; require its displayed endpoint ceiling to exceed that number. The second bound uses $|\mathbb{E}[X|X|^3] - 3\gamma| \leq 3\rho$ for $R \leq 6$, and $\rho \leq 2$. For the third use the signwise fourth-moment bounds $0 \leq \mathbb{E}X_\pm^4 \leq R(\rho \pm \gamma)/2$. Finally (81) gives

$$|\phi_z\Omega| \leq v_H := \frac{2}{5}\max\{1/2, 2M/3 - 1/2\}, \qquad \Omega = J_1 + (M_4 - 3)/6.$$

Lemma D.2 (A stronger signed fourth-moment charge)**.** *Let $|X| \leq R$, $\rho \in [l,u]$, $|\beta| \leq b < 1$. Put $h_* = 1 + b$ and $a(r) = (3r - R)/2$. A valid bound for $|K_1|$ is the maximum on $[l,u]$ of*

$$B_R(r,b) = Rr - \tfrac{3}{2}r^2 + 3b + \tfrac{1}{2}(1+b)a(r)^2.$$

If the three gates

$$h_* \geq a(u), \qquad 3 + 2a(l)h_* - \tfrac{3}{2}h_*^2 \geq 0, \qquad R - 3u + \tfrac{3}{2}h_*^2 \geq 0$$

hold, it may be intersected with the smaller bound

$$Ru - \tfrac{3}{2}u^2 - 3 + 3h_* + a(u)h_*^2 - h_*^3/2.$$

Proof. For $\epsilon = \pm 1$ set $h = 1 - \epsilon\beta > 0$ and $\delta = \rho - \epsilon\gamma$. Signwise Cauchy and $m \leq 1/2$ give $\delta \geq h^2$. The fourth moments on the two signs give

$$\epsilon K_1 \leq R\rho - \tfrac{3}{2}\rho^2 - 3 + 3h + a(\rho)\delta - \delta^2/(2h).$$

Completing the square in unrestricted δ and using $h \leq h_*$ proves the first bound. With $\delta \geq h^2$, the last three terms maximize to

$$\psi_a(h) = \begin{cases} h(3 + a^2/2), & 0 < h \leq a, \\ 3h + ah^2 - h^3/2, & h \geq a. \end{cases}$$

The first branch is increasing. The second has derivative $3 + 2ah - 3h^2/2$, a concave quadratic whose endpoint values on $[0, h_*]$ are nonnegative under the second gate. The branches agree at $h = a$; the first gate places h_* in the second branch. The remaining derivative in ρ is $R - 3\rho + 3h_*^2/2$, nonnegative by the third gate. This proves the claim, including all endpoints. The quadratic maximum in the first bound is evaluated at its endpoints and, when present, its concave vertex. □

Take the minimum of the applicable bounds for $|K_1|$ and call it K. No assertion about the signs of $\beta\gamma$ or $K_1\gamma$ is used.

D.5 Two quadratic passes and the first actual gap

Use the 24 closed cells with endpoints

$$r_i = 1 + 3361i/96000, \quad i = 0, \ldots, 24, \qquad N = 42,$$

and directed constants

$$c_- = 0.4097321837023 < C_{\mathrm{E}} < c_+ = 0.4097321837025,$$
$$\phi_l = 997/2500 < \phi(0), \quad \sigma_l = 3.162277660 < \sqrt{10}.$$

The Gaussian constants follow from the rational Machin estimates in Lemma B.1 and Appendix A; the narrow c enclosure has the explicit directed recipe in Lemma B.2. The interval encloses the fixed coefficient c, not the actual normalized score.

On a cell $[l, u]$, form the moments and the universal actual event above, and compute the complete Fourier pair K_F, s_F. In pass 1 use $\Delta = 2u$, $A = \min(A_1(2u), A_2)$, and, for $t \in [c_-, c_+]$,

$$f(t) = tK + v_H + s_F, \qquad q(t) = tl - g/5.$$

Require $q > 0$. The bound $\mathcal{R}_c \le K_F + cA + f(c)|H| - q(c)H^2$ is affine in c at each fixed $|H|$, so its maximum is at an endpoint before completing squares. Let an overline denote upward rounding to 10^{-10}. Compute

$$\begin{aligned}
T_1 &= \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/(4q(t))\}},\\
\eta_1 &= (c_- l - T_1/N)/2,\\
U_1 &= \min\left\{T_1, \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/[4(q(t)+\eta_1)]\}}\right\},\\
d_1 &= \overline{6U_1/(N\phi_l)}.
\end{aligned} \tag{85}$$

The strict gate $0 \le T_1 < Nc_- l$ is checked before η_1 or the resource bound is used. Corollary 4.2 then proves $\mathcal{R}_c \le T_1$, $\mathcal{U} \le U_1$, and $nD \le Nd_1$. The scalar bounds hold at every actual $n \ge N$.

For the second pass the circle in Lemma 3.1 gives

$$D \ge \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}, \qquad \sigma = \sqrt{10}.$$

Put $a_0 = \sigma_l - d_1/l$. A valid lower bound for γ/ρ is

$$Y(a_0) = \begin{cases} -1, & a_0 \le -1,\\ (a_0 - 3\sqrt{10 - a_0^2})/10, & -1 < a_0 \le \sqrt{10}. \end{cases} \tag{86}$$

Indeed $y + 3\sqrt{1-y^2}$ increases from -1 to $\sqrt{10}$ on $[-1, 1/\sqrt{10}]$. Solving its level equation on that branch gives the smaller quadratic root in (86); it has $a_0 - y \ge 0$, so squaring introduced no false branch. Every y below it violates the required lower level. A downward rational bound y_l is obtained by rounding the square root upward. Thus

$$\begin{gathered}
g_l = \max\{-g, \sigma_l l - 3 - d_1, \min(ly_l, uy_l)\} \le \gamma,\\
\Delta = \min\{2u, (1-\sigma_l)l + 3 + d_1, u - g_l\}, \qquad \phi_- = \phi_l(1 - T_1/(Nc_- l)) > 0,\\
A = \min\{A_1(2u), A_1(\Delta), A_2\},\\
q(t) = \max\{tl - g/5, tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\}.
\end{gathered}$$

The first two bounds also use $2m \le 1$. Each value of $q(t)$ is affine in t, since its two alternatives have the same slope l. Repeat the raw bound first, intersect it with T_1, then price phase using this new raw bound, intersect the resource with U_1, and finally retain $d_2 \le d_1$. A resource bound is never substituted for a raw bound in the density or phase license.

Proposition D.3 (The two coarse passes, computer-assisted)**.** *The preceding coefficient domains, Fourier gates, endpoint enclosures and strict raw gates hold on every one of the 24 closed cells. Outward evaluation gives the following ceilings, and in every second-pass cell* $d_2 < (\sigma_l - 1)l$*:*

pass	$\max T$	$\max U$	$\max d$	*central gap proved*
1	13.2818901598	10.9330610254	3.9164138937	23
2	11.5164967977	9.9790315907	3.5746638454	24.

The last column counts the cells satisfying $d_j < (\sigma_l - 1)l$*, which excludes zero from the support by the argument below. The minimum second-pass gap margin is at least* $526024602543/1600000000000 > 0$*.*

Proof. Evaluate the displayed formulas with rational arithmetic on the 24 endpoint pairs $[r_i, r_{i+1}]$, using the directed Fourier enclosures in Appendix A. The selected routine is `one_pass` in `mathematics/fixed-target-v6/2026-10-27/signed-two-pass-check.py`. It checks the coefficient and Fourier domains, $q > 0$, and $0 \le T < Nc_- l$ before each resource update. Taking the maxima of its 24 outputs gives each row of the table. The minimum of $(\sigma_l - 1)r_i - d_2$ is the positive rational margin displayed above.

The exact replay and a separate reconstruction of all 48 quadratic updates are recorded in Appendix AB. The reconstruction also checks that the implementation's retained older minima never improve the fresh ceilings in this cover. Thus no earlier resource-refinement schedule is needed for these bounds. The preceding analytical estimates justify every coefficient; Proposition 5.3 supplies the closed-cover implication. This establishes the assertions for every witness in the initial domain, rather than only for the cell endpoints. □

If zero belonged to the closed support, every gap would be on one side of zero and $J \le \rho$. Hence $D \ge (\sigma - 1)\rho$, contradicting the strict second-pass test. Centering and positive variance therefore give an actual central gap $(-A, B)$, with positive endpoints. This acquires an event, rather than supposing an artificial two-point kernel.

D.6 Signed moments and physical gap costs

Write $\mu_4 = M_4$, $\eta_5 = M_5$, $\sigma = \sqrt{10}$. Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Both signs have positive probability and positive first moment, since a centered variance-one law cannot be supported on one half-line. Thus $|\beta| < 1$ and the signwise denominators below are positive whenever used; degenerate correction denominators are handled separately. Cauchy gives $m^2 \le a_+ a_-$ and $b_\pm m \ge a_\pm^2$. Consequently

$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}, \qquad m \ge \max\left\{\frac{1+\beta^2}{2\rho}, \frac{a_+^2}{b_+}, \frac{a_-^2}{b_-}\right\}.$$

These statements also hold with a zero atom. Projection of $X|X|$ onto the span of X and $|X|$ gives

$$\mu_4 \ge \rho^2 + \frac{(\gamma - \rho\beta)^2}{1-\beta^2}.$$

For completeness, this follows by minimizing the nonnegative expectation $\mathbb{E}(X|X| - uX - v|X|)^2$ over u, v; its Gram matrix has diagonal entries one and off-diagonal entry β, and is positive definite.

Two support-localizing moment matrices give the additional upper bounds as follows. If $D_\pm = Ra_\pm - b_\pm > 0$, then

$$2m + \sum_\pm \frac{(Rm - a_\pm)^2}{D_\pm} \le R, \qquad \mu_4 \le R\rho - \sum_\pm \frac{D_\pm^2}{Rm - a_\pm}.$$

Indeed the first follows by summing the determinant inequalities for

$$\mathbb{E}[(R-|X|)(1,|X|)^{\mathsf{T}}(1,|X|);\pm X>0].$$

The second follows from the corresponding matrices with an additional factor $|X|$. One more factor $|X|$ yields, sign by sign,

$$\mathbb{E}[X_{\pm}^5]\le R\mathbb{E}[X_{\pm}^4]-\frac{(Rb_{\pm}-\mathbb{E}[X_{\pm}^4])^2}{D_{\pm}}.$$

Zero denominators are treated by dropping their nonnegative correction; they never imply exclusion without a separate contradictory numerator. The common m is retained throughout.

A second useful upper bound for m follows by integrating $(x-a)^2(R-x)\ge 0$ for $x=|X|\in[0,R]$. For every fixed $a>0$,

$$2m\le\frac{R-\rho+2a+Ra^2}{a(2R+a)}.$$

The cover uses only the eleven rational choices $a=2/10,\dots,12/10$.

Write $\delta=\gamma-\rho\beta$. Cauchy and a square completion on each sign give the signed constant bound

$$\frac{\rho}{2}\mu_4-\frac{1}{4}\eta_5\le\frac{\rho^3}{4}-\frac{(\rho+\gamma)(\delta_+)^2}{8(1+\beta)^2}-\frac{(\rho-\gamma)((-\delta)_+)^2}{8(1-\beta)^2}.$$

Indeed on either sign, write $a=\mathbb{E}X_s^2$, $b=\mathbb{E}X_s^3$, $c=\mathbb{E}X_s^4$, $f=\mathbb{E}X_s^5$. Cauchy gives $c\ge b^2/a$, $f\ge c^2/b$, and $\rho c/2-f/4\le\rho^2 b/4-(c-\rho b)^2/(4b)$. Distance to the half-line $c\ge b^2/a$ proves the stated penalty on summing both signs. Independently, $\mu_4^2\le\rho\eta_5$ and the projection bound above give

$$\frac{\rho}{2}\mu_4-\frac{1}{4}\eta_5\le\frac{\rho^3}{4}-\frac{(\gamma-\rho\beta)^4}{4\rho(1-\beta^2)^2}.$$

Each is a complete deduction from the same square. There is a third independent deduction which is often stronger. For $Y=|X|$ and $\delta=\gamma-\rho\beta$, put

$$V=\mathbb{E}[Y(Y^2-\rho Y)^2]=\eta_5-2\rho\mu_4+\rho^3.$$

Since $\delta=\mathbb{E}[Y\operatorname{sgn}(X)(Y^2-\rho Y)]$, weighted Cauchy gives $\delta^2\le 2mV$. Hence

$$\frac{\rho}{2}\mu_4-\frac{1}{4}\eta_5\le\frac{\rho^3}{4}-\frac{\delta^2}{8m}. \tag{87}$$

The maximum of these three separately valid losses can be subtracted. Their sum cannot. A zero atom contributes zero to the weighted measure.

An actual central gap without assuming $D<2$

The physical decomposition (12) shows that $D<(\sigma-1)\rho$ implies an actual central gap. Use its same nonnegative terms, writing $T=T_++T_-$ and $Q=(\sqrt{\sigma+1}A-\sqrt{\sigma-1}B)^2$. Hence $T\le D/(\sigma-1)$. The additional inequality $J\le\rho+2mAB$ gives $6mAB\ge(\sigma-1)\rho-D$. Consequently, if $\rho\ge l$, $D\le d<(\sigma_l-1)l$ and $m\le m_u$,

$$(A+B)^2\ge\frac{l-d/(\sigma_l-1)+\{(\sigma_l-1)l-d\}/3}{m_u}>0. \tag{88}$$

This is the distance lower bound used in the complete high-frequency event. It uses the actual moment m and the actual support endpoints.

Moreover $A \le a_-/m$, $B \le a_+/m$ and $J \le \rho + 2m$. The last bound gives the compulsory defect

$$D \ge \sigma\rho - \gamma - 6m.$$

The inequalities $T - J_o \ge 2BR_+ + 2AR_-$ give the useful additional endpoint constraint

$$D \ge (\sigma - 1)b_+ + 6a_+B - (\sigma + 5)mB^2.$$

If $D < (\sigma - 1)b_+$, this downward quadratic forces B to be at least its positive root. The reflected constraint has a_-, b_-, A and the coefficients $\sigma + 1, \sigma + 7$. In a closed moment box, positive coefficients are rounded down, negative coefficients up, and the lower square-root endpoint is used in the resulting root bound.

The actual opposite-sign event has probability $2pq$. There is no atom at zero. Cauchy gives the necessary probability interval $2m_-^2/(1 + \beta_+) \le p \le 1 - 2m_-^2/(1 - \beta_-)$ on a moment cell. In particular, $2pq \ge (1 + \beta^2)^2/(2\rho^2)$, using $m \ge (1 + \beta^2)/(2\rho)$ and $pq \ge m^2$. The minimum of $p(1 - p)$ on a closed interval is attained at an endpoint.

For independent copies one has exactly

$$\mathbb{E}[(X - X')^2 \mid XX' < 0] = \frac{1 + \beta(1 - 2p) + 4m^2}{2pq} = \frac{1 + 4m^2 + \beta}{2p} + \frac{1 + 4m^2 - \beta}{2q}.$$

For an upper bound, use the upper m endpoint and both endpoints of p and β. The expression is convex in p and affine in β. This improves the former $2/\mathbb{P}(XX' < 0)$ bound inside the complete high-frequency estimate. Both copies still have the same actual marginal.

Preserving central and outer integral costs

Let

$$P_1(A, B) = B^3/3 + AB^2/2 - A^3/6, \quad P_2(A, B) = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

Then $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A + B)(\rho - B)$,

$$G(A, B; \rho) \le G(A, \rho; \rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

The last expression increases with A because its derivative is $(\rho - A)^2(\rho + 2A)/6$, and is convex in ρ because its second ρ derivative is $\rho^2 + \rho A$. Its closed-rectangle maximum is therefore bounded using the upper A endpoint and both ρ endpoints.

Positive outer gaps satisfy $(t - a)(\rho - t) \le (\rho - B)_+^2/4$. On negative outer gaps the positive part is bounded by $(R - \rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \le \frac{3}{8}\{(\rho - B)_+^2 T_+ + (R - \rho)^2 T_-\}.$$

The available weighted outer budget is $D - mQ$, not the entire D. The cover intersects this budget with the individual bounds $T_+ \le b_{+,u} - m_l B_l^2$ and $T_- \le b_{-,u} - m_l A_l^2$.

For the signed phase term $\Omega = J_1 + (\mu_4 - 3)/6$, retain

$$-RT_-/2 \le J_{1,o} \le RT_+/2.$$

The central minimum of P_1 occurs at B_l and an A endpoint; its maximum occurs at B_u and A clipped to the interval at $A = B_u$. A nonnegative polynomial bound is multiplied by m_u for an upper bound and m_l for a lower bound; a negative one uses the opposite endpoint. For example, the binary endpoints $A = 3/2$, $B = 2/3$ make this maximum negative; the sign switch is necessary even when $D < 2$. The fourth-moment sign intervals also sharpen $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$. The complete K_1, Ω, ϕ_z rectangles are used in the signed expression $|cK_1 + \phi_z\Omega|$, without replacing it by two separate absolute values.

A compulsory first moment before each signed evaluation.

The physical inequality $D \geq \sigma\rho - \gamma - 6m$ is especially useful inside a cell. If its prior ceiling is $D \leq d$, and $\rho \geq l$, $\gamma \leq \gamma_+$, then

$$m \geq m_* := \max\{0, (\sigma_l l - \gamma_+ - d)/6\}. \tag{89}$$

If $m_* > 1/2$, the cell is empty. Otherwise the common sign-moment inequality $4m^2 + \beta^2 \leq 1$ forces $|\beta| \leq \sqrt{1 - 4m_*^2}$. This restriction must be intersected with the beta interval for this gamma cell, not carried from a different gamma cell. Recompute all dependent sign-moment ranges after the intersection. These are necessary conditions on one law; no sufficiency or moment-realizability assertion is made.

For completeness, the event distance and conditional moment refinements fit the Fourier input exactly. On the acquired actual gap, use $\mathcal{E} = \{XX' < 0\}$, a downward square root of (88), and the larger of the two valid probability lower bounds

$$\alpha = \max\left\{\frac{(1 + \min|\beta|^2)^2}{2u^2}, \quad 2\min_{p=p_-,p_+} p(1-p)\right\}.$$

The conditional second-moment ceiling is the smaller of $2/\alpha$ and the corner maximum of the exact expression displayed above. Its numerators are positive, it increases with m, is affine in beta, and is convex in p, so these corner choices are valid. Each event option is inserted into all six HIGH terms. Taking the minimum with the complete universal-event bound preserves the same actual Fourier integral. None of these statements follows merely from declaring a moment rectangle feasible.

D.7 The signed enclosure and its ordered update

On one closed moment cell retain bounds for (ρ, β, γ, m), an already proved support radius R, an incoming raw ceiling T, and the weighted defect invariant $nD \leq Nd$. Thus $D \leq d$ for every actual $n \geq N$. Intersect all necessary moment conditions just proved, with directed endpoint operations. A positive numerator with a zero denominator proves incompatibility only through its original positive-semidefinite inequality; a zero-over-zero correction is dropped. No division is made before this check. Bisection of the monotone first-moment necessary inequality keeps its upper endpoint, so its rounding cannot remove a law.

Here is an explicit description of the coefficient enclosures. Let M be the resulting upper fourth moment and let $[J_-, J_+]$ contain J; one can use

$$J_- = \max\{0, ((3 + \sigma_l)l - \gamma_+ - d)/3\}, \quad J_+ = \min\{u + 2m_+, ((3 + \sigma_u)u - \gamma_-)/3\},$$

with a rational $\sigma_u > \sqrt{10}$. For the beta term take the full corner maximum of $3\beta(J + \gamma)/2$. For the gap charge take the minimum of the complete coarse charge (83) and the central-plus-outer charge just proved. Write this bound as G_* and the beta-term maximum as B_*. Let $P_* \geq 0$ be any lower bound for the maximum of the three independent fifth-moment penalties. Then a valid complete constant bound is

$$A = \min\{B_* + G_* + u^3/4 - l/2 - P_*, \quad B_* + 2uM - l/2\}. \tag{90}$$

The fourth semimoment inequalities and the signed product rectangle for $\rho\gamma$ give an interval $[k_-, k_+]$ containing $K_1 = \mathbb{E}X_+^4 - \mathbb{E}X_-^4 - 3\beta - 3\rho\gamma/2$. The central polynomial mP_1 and the signed outer bounds give an interval $[\omega_-, \omega_+]$ containing $\Omega = J_1 + (M_4 - 3)/6$. Intersect them with every previous valid interval. When multiplying by a moment endpoint, the sign of the coefficient is retained. These constructions explain every coefficient in the next quadratic, without assuming that unrelated corner maxima are attained by one common law.

Require the incoming raw gate $0 \le T < Nc_l$. By Corollary 4.2,

$$\phi_z = \phi(H/\sqrt{n}) \in [\phi_-, 2/5], \qquad \phi_- = \phi_l(1 - T/(Nc_l)) > 0.$$

For a formal coefficient $v \in [c_-, c_+]$ set

$$\begin{aligned} B_v &= K_F + vA, \\ F_v &= \max_{k\in\{k_-,k_+\},\ \omega\in\{\omega_-,\omega_+\},\ \psi\in\{\phi_-,2/5\}} |vk + \psi\omega| + s_F, \\ Q_v &= vl + \tfrac{1}{2}\min\{\phi_-\gamma_-, (2/5)\gamma_-\}. \end{aligned} \tag{91}$$

The actual raw remainder is at most $B_c + F_c|H| - Q_cH^2$. The maximum over the signed rectangle is valid by successive affine maximization, and the absolute value preserves convexity. For fixed $|H|$, this expression is convex in c: the constant and curvature parts are affine and F_c is convex. Thus the two endpoints c-,c+ suffice, even if A is negative. A negative lower gamma must be priced with the upper density endpoint; it cannot be reflected inside the contact law.

Lemma D.4 (One ordered signed step)**.** *Suppose all preceding cell bounds hold and $Q_v > 0$ at both endpoints. Let upward rounding be applied to every proposed numerical ceiling. Set*

$$\begin{aligned} T' &= \min\{T, \max_{v=c_-,c_+}(B_v + F_v^2/(4Q_v))\}, \\ \eta &= (c_l - T'/N)/2, \\ U' &= \max_{v=c_-,c_+}(B_v + F_v^2/[4(Q_v + \eta)]). \end{aligned}$$

A negative T' or U' excludes the cell. Otherwise $\mathcal{R}_c \le T'$, $\mathcal{U} \le U'$, and the new weighted defect bound is

$$nD \le Nd', \qquad d' = \min\{d, 6U'/(N\phi_l)\}.$$

A valid raw-threshold ceiling is

$$|H| \le \max_{v=c_-,c_+} \frac{F_v + \sqrt{F_v^2 + 4(Q_v + \eta)\max(0, B_v)}}{2(Q_v + \eta)}. \tag{92}$$

Proof. Complete the two raw squares first. The incoming strict gate persists under $T' \le T$, and gives $\eta > 0$. Only now apply Corollary 4.2 and complete the two resource squares with curvature $Q_v + \eta$. A negative complete ceiling is incompatible with the nonnegative raw or resource expression. The resource contains $n\phi(0)D/6$, so the displayed weighted bound follows without setting the actual sample size equal to N. If the maximum of both resource quadratics at a phase were negative, no witness could have that phase. Replacing B by its positive part enlarges each positive root and gives (92). Previous valid resource or phase bounds may also be intersected, but an unlicensed resource bound cannot be fed back as a raw ceiling. □

D.8 Closed coverage and the selected acquisition theorem

For each of the original 24 rho cells, acquire a support radius with the independent heat/ray test before recomputing higher moments. Each signed pass then partitions a full necessary beta enclosure into eight closed intervals, each resulting necessary gamma enclosure into eight closed intervals, and each necessary common-m enclosure into two closed intervals. The original beta interval is restored for every gamma slice before applying (89). Discarding an outer interval counts all its descendants, and requires one of the proved necessary conditions to be inconsistent. In every processed rho cell the active and excluded counts sum to 128.

Each surviving actual law satisfies the cell update of Lemma D.4. Pool upper bounds by maxima and lower bounds by minima before the next pass. This may enlarge the possible set, but cannot lose an actual witness. A rho cell disappears from later passes only when all of its descendants have been excluded. Degenerate closed intervals may duplicate boundary work; they do not leave holes.

For export, intersect each surviving necessary probability interval with all 100 closed panels $[j/100, (j+1)/100]$, $j = 0, \dots, 99$. A cell meeting a shared endpoint contributes to both panels. From $\theta = 1 - m^2/[p(1-p)]$ obtain a panel upper bound using its lower m endpoint and the maximum of $p(1-p)$ on the intersection, including $1/4$ if it crosses $1/2$. Pool this simultaneously with the raw, weighted defect, rho, beta, and phase bounds of the same cells.

Proposition D.5 (The four signed passes, computer-assisted)**.** *Starting with Proposition D.3, all support, moment, actual-event, Fourier, denominator, signed-curvature and ordered raw-license gates hold on the complete four-pass cover just described. Its exact rational evaluation contains 11264 nominal signed boxes in total. The complete 100-panel export satisfies, on every nonempty panel,*

$$\begin{aligned} &\mathcal{R}_c \le 2.1995174, && nD \le 32.20665378, && D \le 0.76682509, \\ &\rho \le 71527/48000, && |\beta| \le 0.50549963, && |H| \le 1.83605680. \end{aligned}$$

Fourteen rho cells survive, and thirty probability panels are nonempty. The largest support radius among the surviving final rows is $157/50$*; thus the export also supplies*

$$\operatorname{supp} P \subset (-157/50, 157/50). \tag{93}$$

Proof. The acceptor specified in Appendix AB reconstructs the four signed passes and their complete panel incidences from the fixed proof object. Its rational bounds implement the moment, gap, event and Fourier inequalities above, followed by Lemma D.4. At each stage the recomputed upper bounds are no larger than the stored ceilings. Every exclusion uses a necessary inequality, with the zero-denominator cases treated before division. The exact replay verifies these comparisons on the complete cover; Proposition 5.3 proves their universal meaning.

For the displayed constants, read the maxima of the accepted final rows and panels, rather than merely the acceptor's generic target limits. The row defect maximum is 0.76682509, so the weighted bound is $42 \cdot 0.76682509 = 32.20665378$. The largest surviving row index is 13, giving the exact endpoint $r_{14} = 71527/48000$, and the largest stored radius is $157/50$. The exported panel value for ρ is rounded upward to 1.49014584; the sharper rational bound stated here comes from the original row endpoint. The raw, $|\beta|$ and $|H|$ maxima are respectively 2.1995174, 0.50549963 and 1.83605680. The separate exact output check in Appendix AB verifies these values against the fixed object. The strict support gates exclude boundary atoms, giving the open support interval as stated. □

Proof of Theorem 5.1. Theorem D.1 supplies the actual support before higher moments are used. Appendix A and the actual-event constructions prove the complete Fourier bounds on their specified domains. The coarse coefficient inequalities and the two ordered quadratic steps establish the first actual gap. Every signed cell thereafter uses the moment and physical-gap bounds of this same law, followed by Lemma D.4. Induction over the four passes preserves the weighted invariants. The complete closed cover and outward pooling prove that every witness reaches an incident output panel with the stated bounds. □

The argument contains no upper bound for the witness's actual score, no scalar score recurrence, and no second deleted-score assumption. It uses only the quadratic improvement

in the threshold term. Sections 6 and 7 replace downstream panel arguments by one support region and two direct comparisons at $n \geq 700$. The internal entry cover remains intact. The finite induction through 42 and the first-failure composition supply the remaining indices independently of this compact entry calculation.

E How the finite inequalities imply bounds for arbitrary laws

The finite calculations bound functions on continuous parameter domains. The first task in each application is to prove that the parameters of the same extremal law belong to that domain. The calculations then verify pointwise majorants or exclude all possible parameter values. They do not replace the marginal by a finitely supported law.

Lemma E.1 (Tensor expectation for a common marginal)**.** *Let K contain the support of a standardized law P, and let $f_0 = 1, f_1, \dots, f_d$ be integrable under P. Put $m_j = \mathbb{E} f_j(X)$. For a finite tensor sum*

$$T(x_1, \dots, x_n) = \sum_{\alpha} t_\alpha \prod_{i=1}^{n} f_{\alpha_i}(x_i), \qquad M_T = \sum_{\alpha} t_\alpha \prod_{i=1}^{n} m_{\alpha_i},$$

one has $\mathbb{E}T(X_1, \dots, X_n) = M_T$. If

$$T \geq \mathbf{1}_{\{x_1 + \cdots + x_n \leq H\}} \quad \text{on } K^n, \qquad M_T - \Phi(H/\sqrt{n}) \leq b\rho/\sqrt{n},$$

then the positive closed-CDF score of P at H is at most b.

Proof. Independence makes the absolute expectation of each product equal to the product of its finite absolute expectations. Fubini's theorem and linearity therefore evaluate the finite sum. Integrate the pointwise inequality and multiply by $\sqrt{n}/\rho$, which is positive. □

A product with a centered factor has expectation zero if each factor uses a different summand. Adding such products can improve a majorant without changing M_T. The same marginal supplies every m_j; one cannot choose different moment vectors for different factors. Polynomial features of degree above three are used only after a support bound has supplied their integrability.

Lemma E.2 (Exact positivity on a box)**.** *Let $p(t) = \sum_\alpha a_\alpha t^\alpha$ have coordinate degrees at most d_i. On $[0,1]^r$ its tensor Bernstein coefficients are*

$$b_\nu = \sum_{\alpha \leq \nu} a_\alpha \prod_{i=1}^{r} \frac{\binom{\nu_i}{\alpha_i}}{\binom{d_i}{\alpha_i}}.$$

If all $b_\nu \geq 0$, then $p \geq 0$ throughout the closed box.

Proof. The identity

$$t^k = \sum_{j=k}^{d} \frac{\binom{j}{k}}{\binom{d}{k}} \binom{d}{j} t^j (1-t)^{d-j}$$

follows from the binomial theorem after extracting t^k. Apply it in each coordinate. The resulting basis functions are nonnegative and sum to one, so p is a convex combination of the b_ν at each point of the box. □

Rational affine changes map general rational boxes to the unit box, and exact subdivision can strengthen the coefficient test. The polynomial identity, coordinate degrees and every retained child box are part of the check. For a closed event the majorant must satisfy $T-1 \geq 0$ on the event plane as well as its lower side. Checking only $T \geq 0$ on that plane would lose threshold atoms.

Proposition E.3 (Closed-cover soundness)**.** *Suppose every law and threshold under consideration has a proved map into a domain covered by finitely many closed root cells. At each root there is a finite subdivision tree, and every split retains closed children whose union contains the parent. Suppose that on every leaf either a proved necessary condition excludes all actual mapped points, or a valid whole-leaf inequality bounds their positive scores by b. Then all actual scores in the root domain are at most b. If instead every leaf excludes an assumed extremal score $C \geq b$, no such extremizer belongs to the domain.*

Proof. Every point of a parent belongs to a child, including boundary points. Following children down a finite tree reaches a leaf. Its conclusion therefore applies to the point. The root cover proves the assertion on the whole domain. □

Each application must specify its support and moment bounds, its root domain, and the inequality checked at a leaf. An empty-cell test must exclude actual laws, not merely inconvenient numerical parameters. Transcendental quantities require proved outward enclosures; divisions require a positive denominator on the entire cell. Missing roots, unresolved leaves and omitted endpoint cases prevent acceptance.

F The four-summand physical comparison

F.1 A majorant that includes the boundary atom

The Fourier bound below concerns a closed event. Its value at equality must therefore be one, even when the law has an atom at the threshold. The following construction proves both this endpoint convention and the normalization used in the four-summand certificate.

Lemma F.1 (Closed-half-line majorant)**.** *Put*

$$K(t) = \frac{1-t}{2} + \frac{i}{2}\left((1-t)\cot(\pi t) + \frac{1}{\pi}\right), \qquad 0 < t < 1.$$

For every integrable real random variable Y, every real h and every $A > 0$,

$$\mathbb{P}(Y \leq h) \leq \frac{1}{2} + 2 \lim_{\epsilon \downarrow 0} \Re \int_{\epsilon}^{1} K(t) e^{-iAth} \mathbb{E} e^{iAtY}\, dt. \tag{94}$$

Only the real-part improper limit is intended. The unmodified complex integral need not converge.

Proof. Define

$$W(x) = \left(\frac{\sin \pi x}{\pi x}\right)^2, \quad W(0) = 1, \qquad H_B(x) = \sum_{m \geq 1} \{W(x-m) - W(x+m)\} + 2xW(x).$$

The series converges absolutely and locally uniformly, since the shifted terms are bounded by a constant times m^{-2} on each bounded x interval. The function H_B is odd, $H_B(0) = 0$ and $H_B(j) = \operatorname{sgn}(j)$ at every nonzero integer.

We first show $|H_B(x) - \operatorname{sgn}(x)| \le W(x)$. The partial-fraction identity

$$\sum_{m\in\mathbb{Z}} (x-m)^{-2} = \pi^2 \csc^2(\pi x) \qquad (x \notin \mathbb{Z})$$

follows by integrating $\pi\cot(\pi z)/(z-x)^2$ around the square with vertices $\pm(N+1/2) \pm i(N+1/2)$. The cotangent is uniformly bounded on its edges, so the integral is $O(N^{-1})$; the residues are $(m-x)^{-2}$ at its integer poles and $-\pi^2\csc^2(\pi x)$ at x. For positive noninteger x, subtracting this identity from the series for H_B gives

$$H_B(x) - 1 = \left(\frac{\sin \pi x}{\pi}\right)^2 \left\{ \frac{2}{x} - \frac{1}{x^2} - 2\sum_{m\ge 1} (x+m)^{-2} \right\}.$$

Monotonicity of $s \mapsto (x+s)^{-2}$ gives

$$\sum_{m\ge 1} (x+m)^{-2} \le \frac{1}{x} \le \frac{1}{x^2} + \sum_{m\ge 1} (x+m)^{-2}.$$

The braces consequently lie between $-x^{-2}$ and x^{-2}. Integer values, oddness and $W(-x) = W(x)$ prove the claimed bound on the whole real line. Hence

$$M(x) := \frac{1 - H_B(x) + W(x)}{2} \ge \mathbf{1}_{\{x\le 0\}}, \qquad M(0) = 1, \qquad 0 \le M(x) - \mathbf{1}_{\{x\le 0\}} \le W(x).$$

In particular M is bounded, and the boundary point is included.

It remains to identify its Fourier kernel. Elementary integration gives

$$W(x) = 2\int_0^1 (1-t)\cos(2\pi t x)\, dt, \qquad 2xW(x) = \frac{2}{\pi}\int_0^1 \sin(2\pi t x)\, dt.$$

Insert the first identity into the Abel-weighted shifted series with weight r^m, $0 < r < 1$. Absolute convergence allows termwise integration, and the difference of the two cosines yields

$$\sum_{m\ge 1} r^m \{W(x-m) - W(x+m)\} = 4\int_0^1 (1-t)\sin(2\pi t x) \frac{r\sin(2\pi t)}{1 - 2r\cos(2\pi t) + r^2}\, dt.$$

The fraction tends to $\frac{1}{2}\cot(\pi t)$ and has absolute value at most $\frac{1}{2}|\cot(\pi t)|$: its denominator equals $(1-r)^2 + 4r\sin^2(\pi t)$. The resulting majorant is integrable, since the sine vanishes at zero and $(1-t)\cot(\pi t)$ is bounded at one. Dominated convergence therefore gives

$$H_B(x) = 2\int_0^1 \left((1-t)\cot(\pi t) + \frac{1}{\pi}\right)\sin(2\pi t x)\, dt.$$

Combining these formulas proves

$$M(x) = \frac{1}{2} + 2\lim_{\epsilon\downarrow 0} \Re \int_\epsilon^1 K(t) e^{2\pi i t x}\, dt = 1 + 2\Re\int_0^1 K(t)(e^{2\pi i t x} - 1)\, dt.$$

The second integral is absolutely convergent. For an integrable U, $\mathbb{E}|e^{2\pi i t U} - 1| \le 2\pi t \mathbb{E}|U|$, while $K(t) = O(t^{-1})$ at zero and is bounded near one. Absolute Fubini thus justifies expectation of this compensated identity. Apply it to $U = A(Y-h)/(2\pi)$ and use $M(U) \ge \mathbf{1}_{\{Y\le h\}}$ to obtain (94). □

For $Z = (X_1 + \cdots + X_4)/2$, take $A = 6$. Independence gives $\mathbb{E}e^{6itZ} = \varphi_P(3t)^4$, so (94) has exactly the factor six, raw frequency $3t$ and positive imaginary kernel used below. Its assumptions are only integrability and independence; atoms and infinite supports cause no change to the closed-event bound.

F.2 Exact elementary enclosures

The four-summand certificate uses rational intervals, not floating-point values as premises. Addition and multiplication take the extrema of the appropriate endpoint expressions; division first verifies that its denominator interval excludes zero. At integer scale $S = 10^{40}$, lower and upper endpoints are rounded using floor and ceiling. If $v = p/q \geq 0$, a square-root bracket is checked by the integer inequalities $qL^2 \leq pS^2 \leq qU^2$. Thus the true value stays in the recorded interval under every permitted arithmetic operation.

The transcendental enclosures have explicit finite remainders. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and the alternating series for arctan give rational endpoints for π; the first omitted term bounds each error. The identity follows by the tangent addition formula, with the angle in $(0, \pi/2)$. Positive square-root and reciprocal brackets then give $\phi(0) = 1/\sqrt{2\pi}$. For real z,

$$\left| e^{iz} - \sum_{k=0}^{60} \frac{(iz)^k}{k!} \right| \leq \frac{|z|^{61}}{61!},$$

by Taylor's integral remainder and $|e^{is}| = 1$. For $y \geq 0$, integration of the Taylor remainder for e^{-v}, $v \geq 0$, gives

$$\left| \int_0^y e^{-s^2/2}\, ds - \sum_{k=0}^{60} \frac{(-1)^k y^{2k+1}}{2^k k!(2k+1)} \right| \leq \frac{y^{123}}{2^{61} 61!\, 123}.$$

Together with $\Phi(-y) = 1 - \Phi(y)$ this encloses every required normal CDF value. The proof uses $|d^{61}e^{-v}/dv^{61}| \leq 1$ on $v \geq 0$, so it does not presume decreasing terms at the start of the series.

Finally, for $0 < u < 1$ and an integer $m \geq 1$,

$$\cot(\pi u) = \frac{1}{\pi u} - \frac{2u}{\pi} \sum_{j \geq 1} \frac{1}{j^2 - u^2}, \qquad \frac{1}{m+1} \leq \sum_{j > m} \frac{1}{j^2 - u^2} \leq \frac{2m+1}{2m(m+1)}.$$

The lower tail bound follows by comparison with $\int_{m+1}^{\infty} s^{-2}\, ds$; the upper bound follows from the telescoping sum of $(j^2 - 1)^{-1}$. The partial-fraction identity is obtained by integrating the absolutely convergent derivative identity proved in Lemma F.1, with the constant fixed at $u = 1/2$. These formulas enclose every midpoint kernel value. All strict comparisons used below are comparisons of the resulting rational endpoints.

F.2.1 One defect measure for all frequencies

For $\eta > 0$, let $\nu = W\, dP/\eta$, viewed as a probability on tagged points (σ, y), $\sigma \in \{-1, 1\}$, $y > 0$. Write $c_+ = b$, $c_- = a$, $p_0 = 1 - w$, and

$$k_0(\sigma, y) = \frac{1}{c_\sigma^2 y}, \qquad k_2(\sigma, y) = \frac{1}{(y - c_\sigma)^2}, \qquad k_1(\sigma, y) = \frac{1}{y - c_\sigma}.$$

The contacts have zero ν mass. Singular expressions are integrated only off the contacts; the following identities prove integrability.

Lemma F.2. *With $g(q) = \varphi_Q(q)$ and $r_q(\sigma, y) = \exp(\mathrm{i}\sigma q\, \cdot)[0, c_\sigma, c_\sigma, y]$,*

$$\begin{aligned} &\varphi_P(q) - g(q) = \eta \int r_q\, d\nu, \\ &\eta \int k_0\, d\nu = p_0 - P\{0\} \leq p_0, \\ &\eta \int_\sigma k_2\, d\nu = u - c_\sigma P\{\sigma c_\sigma\} \leq u, \qquad \int_\sigma k_1\, d\nu = 0. \end{aligned} \tag{95}$$

The same ν occurs at every frequency, and $|r_q| \le |q|^3/6$. For $\eta = 0$, $P = Q$ and every characteristic-function difference is zero, without defining ν.

Proof. Hermite interpolation at $0, c, c$ has remainder $y(y-c)^2 \exp(\mathrm{i}\sigma q\,\cdot)[0, c, c, y]$. On each tag, the linear and quadratic terms have the same integrals against P and Q, by their matching first two tagged moments. Both interpolating polynomials have constant term 1. Their constant contributions agree only after summing the tags and adding the atom at zero, since both total masses are 1. Hermite–Genocchi gives the response bound. Multiplication of the three costs by $W = y(y-c)^2$ gives $(y-c)^2/c^2$, y, and $y(y-c)$, respectively. Their integrals give (95), including the omitted contact masses. Absolute integrability of k_1 follows from Cauchy and the k_2 bound. Lemma 9.2 handles $\eta = 0$. □

Fix a threshold H and finite coefficients $\kappa_j(H)$, defined below. For the bounded class $|X| \le R_* = 17/4$, all suprema below are over $\sigma \in \{-1, 1\}$ and $0 < y \le R_*$, off the contact. The linear response is

$$A_H(\sigma, y) = \Re \sum_{j=0}^{N-1} 4\kappa_j(H) g(q_j)^3 r_{q_j}(\sigma, y).$$

For any fixed $\lambda_0, \lambda_+, \lambda_- \ge 0$ and $\theta_+, \theta_- \in \mathbb{R}$, a bound

$$S_H \ge \sup_{\sigma, y} \{A_H - \lambda_0 k_0 - \lambda_\sigma k_2 - \theta_\sigma k_1\} \tag{96}$$

implies

$$\eta \int A_H \, d\nu \le \Gamma_H(\eta) := p_0\lambda_0 + u(\lambda_+ + \lambda_-) + S_H\eta. \tag{97}$$

Upper bounds for p_0, u may replace them. The two equality prices retain their signs. In particular, S_H and Γ_H need not be nonnegative. The prices are fixed before taking the supremum over the whole tagged domain. This preserves the cancellation between frequencies; separate absolute estimates for each r_q do not.

F.2.2 The fourth-power remainder

For complex $\psi = g + \delta$,

$$\psi^4 - g^4 = 4g^3\delta + \delta^2(\psi^2 + 2\psi g + 3g^2). \tag{98}$$

Suppose $|g| \le A \le 1$, $|\psi| \le 1$ and $|\delta| \le E$. Two complete bounds for the remainder are

$$E^2(p^2 + 2pA + 3A^2), \quad p = \min(1, A + E), \qquad v(4A^2 + 2 - v), \quad v = \min(E^2, 2A^2 + 1). \tag{99}$$

The first follows directly from (98). For the second, use

$$\psi^4 - g^4 - 4g^3\delta = 12\delta^2 \int_0^1 (1-t)(g + t\delta)^2 \, dt$$

and $|g + t\delta|^2 = (1-t)|g|^2 + t|\psi|^2 - t(1-t)|\delta|^2$.

A sharper two-disk bound is also finite algebra. Rotate $g = a \ge 0$ and put $\delta = x + \mathrm{i}y$, $s = |\delta|^2$. Then

$$|\psi^4 - g^4 - 4g^3\delta|^2 = s^2\{36a^4 + 4a^2 s + s^2 + 8ax(6a^2 + s) + 24a^2x^2\}.$$

For fixed a, s, the expression in braces is convex in x. Its maximum on the two disks occurs at a collinear endpoint or on $|\psi| = 1$. Put $r_+ = \min(E, 1)$, $m = \min(A, 1 - r_+)$, $r_- = \min(E, 1 + A)$. The collinear bounds are

$$r_+^2(6m^2 + 4mr_+ + r_+^2), \qquad r_-^2(6A^2 - 4Ar_- + r_-^2), \qquad \begin{cases} r_-^4, & r_- \le 1, \\ r_-^2(3r_-^2 - 8r_- + 6), & r_- \ge 1. \end{cases}$$

On the unit circle it remains to maximize

$$Q_A(s) = s^2\{18A^4 + 12A^2 + 6 - (12A^2 + 8)s + 3s^2\} \tag{100}$$

on $[(1 - A)^2, \min(E^2, (1 + A)^2)]$. Its endpoints and the sole interior local maximum

$$s_- = (3A^2 + 2 - A\sqrt{4 - 3A^2})/2$$

suffice. For $A = 0$ the interval is degenerate or empty. Taking the maximum of the collinear values and the square root of this last maximum gives the lens bound. To see that no $a < A$ is lost, note that the unit-circle expression is convex in a^2; its lower feasible endpoint is already collinear. The collinear expressions increase with their admissible radius after maximization over a. Thus these are whole-disk bounds, not sampled values. Their minimum with (99) is valid.

On a closed excess interval $[L, U]$, an affine radius $e_j(\eta) = A_j + B_j\eta$ is used only when it is nonnegative at both ends and bounds $|\delta_j|$ throughout. Freeze $p_j = \min(1, a_j + \max\{e_j(L), e_j(U)\})$ in the first bound of (99). This gives a quadratic in η with nonnegative leading coefficient. Alternatively, freezing E on the whole cell gives a constant disk or lens bound. A choice between complete bounds is allowed; a coefficientwise minimum is not.

F.2.3 A physical Prawitz comparison

We use the closed-CDF Prawitz majorant [10], in the normalization proved in Lemma F.1. To distinguish the factor from the kernel K used there, put

$$\widehat{K}(u) = 2K(u) = (1 - u) + \mathrm{i}R(u), \qquad R(u) = (1 - u)\cot(\pi u) + 1/\pi.$$

For $Z = (X_1 + \cdots + X_4)/2$,

$$F_Z(x) \le J_P(x) := \frac{1}{2} + \int_0^1 \Re\{\widehat{K}(u)e^{-6\mathrm{i}ux}\varphi_P(3u)^4\}\, du. \tag{101}$$

The real integrand is regular at zero; no convergence of the complex integral by itself is asserted. With $D = x - Z$ and $S(u) = uR(u)$ it is

$$F_x(u) = (1 - u)\mathbb{E}\cos(6Du) + S(u)\mathbb{E}[6D\,\mathrm{sinc}(6Du)].$$

The cotangent partial fractions, extended to the endpoints, give

$$0 \le S \le 1/3, \qquad |S'| < 2, \qquad |S''| < 5.$$

Here is a direct derivative check. Separate the first pole:

$$S = \frac{1}{\pi} - \frac{2}{\pi}(A + MG), \quad A = \frac{u^2}{1 + u}, \quad M = u^2(1 - u), \quad G = \sum_{j \ge 2} \frac{1}{j^2 - u^2}.$$

On $[0,1]$, $|A'| \le 3/4$, $|A''| \le 2$, $|M| \le 1/6$, $|M'| \le 1$ and $|M''| \le 4$. The derivative series converge uniformly, and

$$G \le 3/4, \qquad |G'| \le 10/27 < 1/2, \qquad |G''| \le 106/135 < 7/6.$$

The first sum telescopes. For the other two use $j^2 - u^2 \ge 3j^2/4$ and $\sum_{j\ge 2} j^{-k} \le 2^{-k} + \int_2^\infty t^{-k}\,dt$, with $k = 4, 6$. The product rule gives $|S'| < 19/18 < 2$ and $|S''| < 223/54 < 5$. For $u > 1/2$, put $v = 1 - u$ and use $v\cot(\pi v) \le 1/\pi$; for $u \le 1/2$ the cotangent is nonnegative. This proves $R \ge 0$. The positive partial-fraction terms then give $0 \le S \le 1/\pi < 1/3$. Since $\operatorname{sinc}(v) = \int_0^1 \cos(tv)\,dt$,

$$|F_x''(u)| \le 42\mathbb{E}|D| + 108\mathbb{E}D^2 + 24\mathbb{E}|D|^3. \tag{102}$$

These differentiations are justified by the displayed moments.

Now assume $|X| \le R_* = 17/4$ and $\rho < 46/25$. The fourth moment follows from these hypotheses; it is not an additional input. Write $\kappa = \mathbb{E}X^3$, $M_4 = \mathbb{E}X^4$. Lemma 9.2 gives the useful cell bounds

$$\begin{aligned} |\kappa| &\le \min\{\rho, |b-a| + \eta\}, \\ M_4 &\le \min\{R_*\rho, a^2 - ab + b^2 + (R_* + 2\max(a,b))\eta\}. \end{aligned} \tag{103}$$

For $x = H/2$, choose whole-cell bounds κ_* and $M_{4,*}$ from these formulas and put

$$D_2 = x^2 + 1, \qquad D_4 = x^4 + 6x^2 + 2|x|\kappa_* + (M_{4,*} + 9)/4.$$

Then $\mathbb{E}|D| \le \sqrt{D_2}$ and $\mathbb{E}|D|^3 \le \sqrt{D_2 D_4}$. Uniformly on $|x| \le 2$, the simpler bounds $\mathbb{E}D^2 \le 5$, $\mathbb{E}D^4 < 54$, $\mathbb{E}|D| < 9/4$, $\mathbb{E}|D|^3 < 33/2$ give $|F_x''| < 2061/2$.

Take $N = 256$, $u_j = (j + 1/2)/N$, $q_j = 3u_j$ and

$$\kappa_j(H) = \widehat{K}(u_j)e^{-\mathrm{i}q_j H}/N, \qquad C_H(Q) = 1 + 2\Re\sum_j \kappa_j(H)g(q_j)^4 - 2\Phi(H/2).$$

Composite midpoint quadrature gives one raw numerator reserve

$$Q_H = \min\left\{\frac{687}{8N^2}, \frac{42\sqrt{D_2} + 108D_2 + 24\sqrt{D_2 D_4}}{12N^2}\right\}. \tag{104}$$

It controls the integral of the original law P once. No separate reference, linear-transfer or remainder quadrature error is added.

The raw thresholds are $H_\ell = -4 + \ell/8$, $0 \le \ell \le 57$. To fill the intervals between them, bound the characteristic function on entire frequency cells, not just at the midpoints. If $a_j \ge |g(q_j)|$ and e_j bounds $|\varphi_P(q_j) - g(q_j)|$ on the whole excess cell, then

$$P_j = \min\{1, a_j + e_j + 3/(2N)\}$$

bounds $|\varphi_P(3u)|$ throughout $j/N \le u \le (j+1)/N$, since $\mathbb{E}|X| \le 1$. Two complete kernel bounds are

$$u^2|\widehat{K}(u)| \le u^2(1-u) + u/3, \qquad u^2|\widehat{K}(u)| \le u^2(1-u) + u(1-u)/2.$$

The second follows by retaining the first term of the cotangent partial fractions: $S(u) \le (1-u)(1+2u)/(\pi(1+u)) < (1-u)/2$. Let d_j be the smaller of the two exact integrals over the jth frequency cell. Their separate totals are $1/4$ and $1/6$. Thus, for the same whole geometry and excess cell,

$$|J_P''(x)| \le B = 36\sum_j d_j P_j^4.$$

Since $\sup|\Phi''| < 1/4$, the smooth physical function $H \mapsto 2\{J_P(H/2) - \Phi(H/2)\}$ needs the single interpolation reserve

$$E_H = \frac{h^2}{16}(B + 1/4), \qquad h = 1/8. \tag{105}$$

The elementary cube, mass and unit radii

$$\eta q_j^3/6, \qquad \max(a,b)^2 p_0 q_j^2, \qquad 1 + a_j$$

suffice for this curvature calculation. The mass bound follows from $|r_q| \le q^2/y$ and the first budget in (95). To verify the former inequality without dividing at a contact, put $f(y) = e^{\mathrm{i}\sigma q y}$. The divided-difference identity gives

$$y f[0,c,c,y] = f[c,c,y] - f[0,c,c].$$

Each second divided difference has modulus at most $q^2/2$ by Hermite–Genocchi, including repeated nodes. Hence, for $\eta > 0$,

$$|\varphi_P(q) - g(q)| \le \eta q^2 \int \frac{1}{y}\, d\nu \le q^2 \max(a,b)^2 p_0.$$

The zero-excess case has $\varphi_P = g$ and requires no division.

For $H \le -4$, Cantelli gives $F_Z(H/2) \le 1/5$. For $H \ge 25/8$, $\Phi(H/2) > 4/5$. The latter inequality follows, for example, by integrating the alternating Gaussian Taylor polynomial through degree six on $[0, 25/16]$ and using $\phi(0) > 19947/50000$. Consequently both threshold tails have positive normalized score at most $2/5$.

F.2.4 The low-excess certificate

On a geometry rectangle and a closed excess cell $[L,U]$, let M_H bound $C_H(Q) - Tr$ for the entire rectangle. Let Γ_H be obtained from (97), and let

$$R_2(\eta) = r_0 + r_1\eta + r_2\eta^2, \qquad r_2 \ge 0,$$

bound the sum of the fourth-power remainders weighted by $|\kappa_j(H)|$. Combining (98) at the same H gives

$$2\{F_{4,P}(H) - \Phi(H/2)\} - T\rho \le P_H(\eta) := M_H + Q_H + E_H + 2\Gamma_H(\eta) + 2R_2(\eta) - T\eta. \tag{106}$$

When used at grid points, E_H reserves interpolation to adjacent thresholds. The leading coefficient is nonnegative. Endpoint nonpositivity therefore proves the bound on the entire closed excess cell. Equivalently its degree-two Bernstein coefficients are

$$P_H(L), \quad \frac{P_H(L) + P_H(U)}{2} - \frac{p_2(U-L)^2}{2}, \quad P_H(U),$$

where p_2 is the quadratic coefficient. For a quantitative signed score, with $r_L \le r$ and $\rho < 46/25$, use

$$T + \max_{\eta = L,U} \left\{ \frac{P_H(\eta)}{46/25}, \frac{P_H(\eta)}{r_L + \eta} \right\}.$$

Both denominator faces are needed when the numerator is negative. The cap branch is convex; for the other branch the derivative numerator has derivative $P_H''(\eta)(r_L + \eta) \ge 0$, so an interior critical point can only be a minimum.

Here are the finite enclosures that produce these polynomials. The center is enclosed by signed Taylor expansion in (w,β) or in the contacts (a,b) along probability-preserving paths. For either pair of coordinates i,j,

$$\partial_i C_H = 8\Re\sum_j \kappa_j g^3 \partial_i g, \qquad |\partial_{ik} C_H| \le 8\sum_j |\kappa_j|(3D_iD_k + D_{ik}),$$

where D_i, D_{ik} bound the derivatives of the explicit three-atom characteristic function. The first-order terms retain their signs; the second-order error bounds the whole rectangle. The function $r(w,\beta)$ is jointly convex. Indeed, writing $r = w^{-1/2} f(\beta)$, $f = (1+\beta^2)/\sqrt{1-\beta^2}$, the Hessian determinant has the positive factor

$$3ff'' - (f')^2 = \frac{9 + 9\beta^2 + 15\beta^4 - \beta^6}{(1-\beta^2)^3}.$$

Subtracting its supporting plane therefore encloses $C_H - Tr$ without an absolute Hessian charge for r.

For the signed response, put $R_* = 17/4$, $y_k = k/64$, $0 \le k \le 272$, and use their nearest-node closed cells, clipped to $[0, R_*]$. Hermite–Genocchi gives

$$\begin{aligned} r_q(\sigma,y) &= \sum_{m=0}^{60} (\mathrm{i}\sigma)^{m+3} b_m h_m + E_q, & |E_q| &\le \frac{q^3}{6}\frac{(qR_*)^{61}}{61!}, \\ b_0 &= q^3/6, & b_m &= b_{m-1} qR_*/(m+3), \\ h_0 &= 1, & h_m &= (y/R_*)h_{m-1} + (m+1)(c/R_*)^m. \end{aligned} \tag{107}$$

The bound $|\partial_y r_q| \le q^4/24$ adds the half-cell reserve $q^4/(48\cdot 64)$. The signed frequency coefficients are summed before bounding their polynomial on a y cell. If g_0, c_0' are nominal values, $e_q \ge |g - g_0|$ is the geometry error, $p_q \ge |g|$ and $a_{0,q} \ge |g_0|$, then

$$|g^3 - g_0^3| \le e_q(p_q^2 + p_q a_{0,q} + a_{0,q}^2), \qquad |\partial_c r_q| \le q^4/12.$$

These give a whole-rectangle correction for $g^3 r_q - g_0^3 r_{q,0}$, with no second copy of the series or mesh reserve. Thus the priced supremum bounds actual references throughout the rectangle, not only its nominal center.

The prices used in the finite certificate are chosen from

$$\begin{aligned} \lambda_0 &\in \{0, 1/1000, 1/400, 1/200, 1/100, 1/50, 1/25, 2/25, 4/25, 8/25, 16/25\}, \\ \lambda_\pm &\in \{0, 1/2000, 1/1000, 1/400, 1/200, 1/100, 1/50\}, \\ \theta_\pm &\in \{0, \pm 1/200, \pm 1/100, \pm 1/50\}. \end{aligned}$$

For each fixed choice, all 273 cells on each of the two tags enter the supremum. When $\lambda_\sigma = 0$, only $\theta_\sigma = 0$ is used; otherwise the reciprocal contact rays would give an infinite supremum. On a cell $[y_L, y_U]$, with $c \in [c_L, c_U]$, $k_0 \ge 1/(c_U^2 y_U)$. The remaining cost is the concave quadratic $-\lambda_\sigma z^2 - \theta_\sigma z$ in $z = 1/(y-c)$. Its endpoints and admissible vertex suffice; when the cell crosses the contact both reciprocal rays are included. No division at the contact or at zero is used for the response series.

The same budgets yield whole-cell radius models by pricing $|r_q|$ or its signed coordinate projections. They are combined with the three elementary radii above and the complete two-disk remainder. An independent absolute branch uses

$$|\psi^4 - g^4| \le E(p^3 + p^2 A + pA^2 + A^3), \qquad p = \min(1, A + E),$$

or the complete linear-plus-remainder bound. It is evaluated as a whole alternative. In every branch, the actual reference modulus, the actual normalization $\rho = r + \eta$, one midpoint reserve and one physical threshold reserve are retained. The final maximum over H is taken only after assembling each complete row.

F.2.5 Closing the large excess

Let $p = P(X \geq 0)$ and $q = 1 - p$; zero belongs to the nonnegative class. Condition on exactly j nonnegative observations among four. For $Z = (X_1 + \cdots + X_4)/2$, the conditional moments are

$$\mu_j = \frac{u(j-4p)}{2pq}, \qquad V_j = \frac{1}{4}\left[j\left(\frac{v_+}{p} - \frac{u^2}{p^2}\right) + (4-j)\left(\frac{v_-}{q} - \frac{u^2}{q^2}\right)\right].$$

Closed-event Cantelli gives, for every x,

$$F_Z(x) \leq \sum_{j=0}^{4} \binom{4}{j} p^j q^{4-j} A_j(x), \qquad A_j(x) = \begin{cases} 1, & x \geq \mu_j, \\ V_j/(V_j + (\mu_j - x)^2), & x < \mu_j. \end{cases} \tag{108}$$

This includes variance zero and atoms at the threshold. The necessary sign-mass interval is

$$w(1-\beta)/2 \leq p \leq 1 - w(1+\beta)/2.$$

It is covered by closed cells. On a threshold cell $[x_L, x_U]$, bound $A_j(x_U)$ and subtract a lower bound for $\Phi(x_L)$. The five degree-four Bernstein coefficients bound the binomial mixture jointly; its masses are not maximized independently. Each retained sign-mass cell supplies a nonnegative raw numerator bound d_i and a correlated lower bound $r_i \leq r$, so its score is at most $d_i/(r_i + \eta)$.

If r_B is the whole-box lower bound for r, put

$$c_C^0 = \max\left\{0, \max_i (d_i/T - r_i), \frac{2}{5T} - r_B\right\}, \qquad c_C = 10^{-4}\lceil 10^4 c_C^0 \rceil + 10^{-4}.$$

This closes every $\eta \geq c_C$ by a strict bound below T, including the threshold tails. Independently, Shevtsova closes every $\rho \geq 979013/533000$:

$$\frac{3031}{10000}\left(1 + \frac{323}{500\rho}\right) \leq T.$$

Hence $c_S = \max\{0, 979013/533000 - r_B\}$ is another closed sufficient cutoff. Take $c_B = \min(c_C, c_S)$, selecting one complete route. The low-excess interval $[0, c_B]$ and the large-excess range $[c_B, \infty)$ include their common endpoint.

F.2.6 The finite cover and conclusion

Proposition F.3 (Bounded-law certificate implication)**.** *Assume all the terminal inequalities and domain covers in the following finite scheme have been verified. Then, for every standardized law P with*

$$|X| \leq 17/4, \quad \rho(P) < 46/25, \quad 625/2116 \leq w \leq 1, \quad |\beta| \leq 13/20,$$

and every real H,

$$\frac{2\{F_{4,P}(H) - \Phi(H/2)\}}{\rho(P)} \leq T.$$

The same assertion holds for the genuine reflected law $-P$.

Proof. Divide $[625/2116, 1]$ into 20 equal closed bands and $[0, 13/20]$ into 13 closed bands of width $1/20$. There are 260 rectangles. The minimum of $r^2 = (1+\beta^2)^2/[w(1-\beta^2)]$ is attained at the largest w and smallest $|\beta|$. A rectangle with $\min r^2 \geq (46/25)^2$ is empty for the strict physical cap. For every other rectangle construct its own large-excess cutoff. If $c_B = 0$, the large-excess bound covers all $\eta \geq 0$ directly. If $c_B > 0$, divide $[0, c_B]$ into 16 equal

closed intervals. At each interval check all 58 thresholds, their 57 joins, the two tails, and both actual reflected orientations, using the preceding inequalities. A parent is accepted by its complete bound or by complete bounds for all four children obtained by bisecting both coordinates. Each child has its own cutoff; no part of another child's domain or numerical conclusion is used in its place.

The certificate condition requires a disposition for every one of the 260 closed parent rectangles: a proved necessary-condition contradiction, a complete direct comparison, or complete comparisons on all four children. Acceptance requires the complete rational inequalities and their reconstruction, with no unresolved rectangle. All elementary operations are outward rational enclosures; square roots are checked by integer squaring. Trigonometric, Gaussian and cotangent values are enclosed by finite series with analytic remainders, using the arithmetic rules of Section F.2. The response uses (107), not divided quotients at contacts. The complete exact replay reconstructs this whole cover and its upper bounds; its execution and acceptance record are described in Appendix AB. A sampled value or an optimizer's status would not supply these inequalities.

For completeness, acceptance has a direct mathematical interpretation. Every actual canonical point belongs to a closed parent. In a nonempty parent or child, the large-excess inequalities cover $\eta \geq c_B$, while the low-excess inequalities cover all of $[0, c_B]$, including $\eta = 0$. The same unknown law supplies the single ν in every priced frequency response. Quadrature and physical interpolation extend the finite rows to every central threshold; the tails cover the rest. Thus the accepted rational inequalities prove the proposition for all actual laws, including bounded infinite supports and zero atoms. □

G A support criterion at a specified index

The support calculations throughout the finite range use one elementary criterion. Its inputs are a lower bound for the attained score, upper bounds for moments of the same marginal, and a predecessor allowance. The Gaussian calculation keeps the actual integer n throughout.

Lemma G.1 (Direct Gaussian support criterion)**.** *Let $n \geq 2$, and let (P, H) be an attained positive closed-event ratio maximizer with score C. Write $\rho = \mathbb{E}|X|^3$, $\beta = \mathbb{E}[X|X|]$ and $A = \sqrt{n-1}\Delta_{n-1}(P)/\rho$. Suppose known numbers c, r, b_-, b_+, S, e, q satisfy*

$$0 < c \leq C, \quad 1 \leq \rho \leq r, \quad -1 < b_- \leq \beta \leq b_+ < 1, \quad q \geq 0,$$

$$\sqrt{n/(n-1)} \leq S, \qquad n(\sqrt{n/(n-1)} - 1) - 1/2 \leq e, \qquad n(A - C)_+ \leq q.$$

Fix $R \geq 1$ and a support sign $\eta \in \{-1, 1\}$, and put $B_+ = b_+$, $B_- = -b_-$. Define

$$Q_\eta(v) = -v^3 + \tfrac{3}{2}rv^2 + 3B_\eta v + re, \qquad L_\eta(v) = \tfrac{2}{5}(v^2/2 + 1/2 + e) + cQ'_\eta(v).$$

Suppose that

$$Q_\eta(R), Q'_\eta(R), Q''_\eta(R), L_\eta(R), L'_\eta(R) < 0, \qquad 1/5 - 3c < 0. \tag{109}$$

Define the actual-index Gaussian expression

$$G_n(H, t) = n\sqrt{n}\{\Phi((H-t)/\sqrt{n-1}) - \Phi(H/\sqrt{n})\} + nt\phi(H/\sqrt{n}) + \tfrac{H}{2}(t^2-1)\phi(H/\sqrt{n}),$$

and suppose U_η satisfies

$$\sup_{H \in \mathbb{R}} \eta G_n(H, R) \leq U_\eta, \qquad U_\eta + cQ_\eta(R) + rSq < 0.$$

Then $\operatorname{supp} P$ *contains no point in the ray $\{\eta v : v \geq R\}$. The two signs can be checked separately, with different signed-moment bounds.*

Proof. Contact gives

$$0 \le G_n(H,t) + C\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} + \rho s_n n(A-C),$$

where $s_n = \sqrt{n/(n-1)}$ and $e_n = n(s_n - 1) - 1/2$. The last term is at most rSq. At $t = \eta v$ with $v \ge R$, the bracket is at most $Q_\eta(v)$. Since $Q_\eta''' = -6$, the first three gates make this polynomial strictly negative on the full ray. Thus multiplication by $C \ge c$ gives an upper bound $cQ_\eta(v)$.

The derivative is $n\phi(z) + Ht\phi(z) - ns_n\phi(s_n y)$, where $z = H/\sqrt{n}$ and $y = (H-t)/\sqrt{n}$. Taylor's theorem and $\|\phi''\|_\infty = \phi(0)$ bound $|n\phi(y) - n\phi(z) - Ht\phi(z)|$ by $\phi(0)t^2/2$. The identity $\partial_s[s\phi(sy)] = -\phi''(sy)$ bounds the remaining scaling error by $n\phi(0)(s_n - 1)$. This proves, for every $n \ge 2$, $|\partial_t G_n(H,t)| \le \phi(0)(t^2/2 + 1/2 + e)$. Therefore the derivative in v of $G_n(H,\eta v) + cQ_\eta(v) + rSq$ is at most $L_\eta(v)$. Its second derivative as a polynomial bound is $L_\eta'' = 2/5 - 6c < 0$; the last two endpoint gates imply $L_\eta(v) < 0$ on $[R,\infty)$. Finally $G_n(H,-R) = -G_n(-H,R)$, so the assumed U_η bounds $G_n(H,\eta R)$ uniformly in H. The strict endpoint inequality and monotonicity contradict contact on every point of the specified support ray. □

How the Gaussian-line certificate supplies U_+ and U_-.

Let $z = H/\sqrt{n}$ and $a = (H-R)/\sqrt{n-1}$. Direct differentiation gives

$$\partial_H G_n(H,R) = ns_n\phi(a) + \left[-n - RH + \tfrac{1}{2}(R^2-1)(1 - H^2/n)\right]\phi(z).$$

Choose an integer H_0 such that $H_0 \ge 5\sqrt{n}$ and $H_0 - R \ge 5\sqrt{n-1}$. An outward enclosure of the square roots makes this choice exact. On the whole real line,

$$|\partial_H^2 G_n(H,R)| \le M_{n,R} := \frac{\sqrt{n}}{4}\left(\frac{n}{n-1} + 1\right) + \frac{2}{5}R + \frac{2|R^2-1|}{5\sqrt{n}}.$$

On a closed threshold cell with midpoint h and half-width w, let $[g_-, g_+]$ and $[d_-, d_+]$ enclose $G_n(h,R)$ and $\partial_H G_n(h,R)$. Taylor's theorem then gives the two upper bounds

$$g_+ + \max(|d_-|,|d_+|)w + M_{n,R}w^2/2, \qquad -g_- + \max(|d_-|,|d_+|)w + M_{n,R}w^2/2.$$

They bound $G_n(H,R)$ and $-G_n(H,R)$ respectively throughout the cell. The certificate includes every consecutive cell from $-H_0$ to H_0, including their endpoints.

On the two exterior rays, both distribution-function arguments lie in the same normal tail, so

$$|G_n(H,R)| \le n\sqrt{n}\,\tau + \left(nR + \tfrac{1}{2}H_0|R^2-1|\right)\phi(H_0/\sqrt{n}), \quad |H| \ge H_0,$$

where $\tau = 3 \cdot 10^{-7} > \Phi(-5)$ is sufficient. Take the largest positive-cell bound and the exterior bound for U_+, and the largest negative-cell bound and the exterior bound for U_-. This proves a statement about the entire threshold line, not just about its grid points.

Exact arithmetic and the native input interface.

The implementation `direct_gaussian_support.py` uses integer-bracketed square roots and outward rational interval operations. Its density enclosure first divides the nonnegative exponential argument by a power of two until it is at most one, uses the alternating Taylor polynomial through degree23 with the degree24 upper remainder, and repeatedly squares with outward rounding. For a nonnegative normal argument at most five, its distribution-function enclosure integrates the first80 terms of the exponential series and adds the next-term error; floor and ceiling arithmetic encloses every term. The normal argument is rounded outward

to a grid of width 10^{-12}, and monotonicity of Φ supplies the interval endpoints. Outside this range it uses the preceding Mills bound. The density constant is enclosed by Machin's arctangent identity and integer square roots. These rules justify the midpoint value and derivative intervals used above.

The native function `target(n,K,R,q,...)` uses $c = 819/2000 + K/n$. The hypothesis $C \geq C_{\mathrm{E}} + K/n$ implies $C > c$. Its default third-moment cap comes from the two published affine Berry–Esseen estimates already used in the introduction:

$$r = \min\left\{\frac{0.3031 \cdot 0.646}{c - 0.3031}, \frac{0.3322 \cdot 0.429}{c - 0.3322}\right\}.$$

Both denominators are positive on its admitted domain. The signed-moment inequality gives the default symmetric interval for β. An explicitly supplied tighter ρ or signed-β interval must come from a previously established property of the same marginal. The program checks all the strict polynomial gates and the two full Gaussian-line bounds. Its return value proves the support conclusion of this lemma only after these analytic and same-marginal hypotheses have been supplied. In particular, it does not itself prove a score bound or transfer a result from one index to another.

H Support acquisition at the fifth index

The finite Fourier and count estimates require an explicit support radius. Compactness alone is insufficient: the radius enters their error bounds. Here we obtain the radii in the following table directly from contact at $n = 5$, uniformly in the raw threshold. Corollary H.2 gives the tighter radii used by the signed-moment sectors of the initial acquisitions.

Proposition H.1 (The fifth-index support table)**.** *Assume $C_4 \leq C_{\mathrm{E}}$ and $C_5 > C_{\mathrm{E}}$. Let (P, H) be a positive closed-event ratio maximizer supplied by Corollary 2.2. If $\rho(P)$ belongs to a row in the following table, then* $\operatorname{supp} P \subset (-R, R)$ *for that row's radius. The values of b are upper bounds for $|\mathbb{E}[X|X|]|$. The last column is a rational lower bound on the strict endpoint margin in the proof.*

ρ interval	*b*	*R*	*endpoint margin >*
$[1, 11/10]$	0.253257	23/10	0.251666
$[11/10, 23/20]$	0.307221	12/5	0.153351
$[23/20, 237/200]$	0.338920	5/2	0.174691
$[237/200, 6/5]$	0.351391	13/5	0.350224
$[6/5, 5/4]$	0.389174	13/5	0.020061
$[5/4, 7/5]$	0.478710	117/40	0.005870
$[7/5, 3/2]$	0.525521	25/8	0.008946
$[3/2, 8/5]$	0.565414	133/40	0.059409
$[8/5, 17/10]$	0.600001	7/2	0.033467
$[17/10, 7361/4000]$	0.641563	15/4	0.061370

Proof. Write $C = C_5$, $\beta = \mathbb{E}[X|X|]$, $s = \sqrt{5/4}$ and $e = 5(s-1) - 1/2$. The score of the same marginal at index four is $A = 2\Delta_4(P)/\rho \leq C_4 < C$. Hence the signed predecessor term $d_P = 5(A - C)$ is nonpositive. Define

$$\mathcal{H}(H, t) = 5\sqrt{5}\{\Phi((H-t)/2) - \Phi(H/\sqrt{5})\} + 5t\phi(H/\sqrt{5}) + \tfrac{H}{2}(t^2 - 1)\phi(H/\sqrt{5}).$$

Contact and the predecessor bound imply, at every support point t,

$$0 \leq \mathcal{H}(H, t) + C\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e\} + \rho s d_P. \tag{110}$$

To verify the scale term, the probability part contributes $5\rho(sA-C)$. The identity $5(sA-C) = sd_P + 5(s-1)C$ combines with the constant $-C\rho/2$ in the contact polynomial to give exactly $C\rho e$. Thus there is no approximation in (110).

For a row with upper moment endpoint r, put

$$c = 819/2000, \qquad E = 91/1000, \qquad Q(v) = -v^3 + \tfrac{3}{2}rv^2 + 3bv + rE,$$

$$L(v) = \tfrac{2}{5}(v^2/2 + 1/2 + E) + cQ'(v).$$

The elementary constants satisfy $c < C_{\mathrm{E}} < C$, $e < E$ and $s < 1119/1000$. The signed-moment inequality of Lemma 2.5, $(1+\beta^2)^2 \le \rho^2(1-\beta^2)$ implies $|\beta| \le b$, because direct rational substitution gives $(1+b^2)^2 > r^2(1-b^2)$ for each row. The exact arithmetic check gives

$$Q(R), Q'(R), Q''(R), L(R), L'(R) < 0, \qquad 1/5 - 3c < 0. \tag{111}$$

Since $Q''' = -6$, these gates imply $Q(v) < 0$ for all $v \ge R$. The bracket multiplying C in (110) is at most $Q(|t|)$, so on those rays its product with C is at most $cQ(|t|)$. The predecessor term is nonpositive.

The Gaussian envelope has the uniform derivative bound

$$|\partial_t \mathcal{H}(H,t)| \le \phi(0)(t^2/2 + 1/2 + e) < \tfrac{2}{5}(t^2/2 + 1/2 + E). \tag{112}$$

Indeed, with $z = H/\sqrt{5}$ and $y = (H-t)/\sqrt{5}$, $\partial_t \mathcal{H} = 5\phi(z) + Ht\phi(z) - 5s\phi(sy)$. Taylor's theorem and $\|\phi''\|_\infty = \phi(0)$ bound $|5\phi(y) - 5\phi(z) - Ht\phi(z)|$ by $\phi(0)t^2/2$. Also $\partial_s[s\phi(sy)] = -\phi''(sy)$, so the remaining error is at most $5\phi(0)(s-1)$. This proves (112) for every real H, t. For either sign η, the derivative in $v \ge R$ of $\mathcal{H}(H, \eta v) + cQ(v)$ is at most $L(v)$. Now $L'' = 2/5 - 6c < 0$; the last two gates in (111) make L strictly negative throughout the ray. It remains to prove a strict upper bound at $v = R$, on the whole threshold line.

For that purpose, Gaussian differentiation gives

$$|\partial_H^2 \mathcal{H}(H,R)| \le \frac{\sqrt{5}}{4}\left(\frac{5}{4} + 1\right) + \frac{2}{5}R + \frac{2|R^2-1|}{5\sqrt{5}} \le M_R := \frac{81}{64} + \frac{2}{5}R + \frac{|R^2-1|}{5}.$$

The first bound uses $\|\phi'\|_\infty = 1/\sqrt{2\pi e} < 1/4$ (the maximum of $|x|e^{-x^2/2}$ occurs at $|x| = 1$), $\|\phi''\|_\infty < 2/5$ and $\|\phi'''\|_\infty \le 2/\pi < 4/5$; the last follows either by differentiation or by the Gaussian Fourier integral. The second bound uses $2 < \sqrt{5} < 9/4$. For a twice differentiable function with $|f''| \le M$ on $[a,b]$, its deviation from the affine endpoint interpolant is at most $M(b-a)^2/8$. This follows by comparing the interpolation error with $\pm M(x-a)(b-x)/2$. Consequently, the values at

$$H = -16, -16 + 1/10, \ldots, 16$$

plus $M_R/800$ bound $|\mathcal{H}(H,R)|$ on every one of the 320 closed intervening cells.

The two exterior rays are covered separately. Since $16 \ge 5\sqrt{5}$ and $16 - R \ge 10$, Mills' bound gives, with $\tau = 10^{-6} > \Phi(-5)$,

$$|\mathcal{H}(H,R)| \le 5\sqrt{5}\,\tau + \left(5R + 8|R^2-1|\right)\phi(16/\sqrt{5}), \qquad |H| \ge 16.$$

Here $|H|\phi(H/\sqrt{5})$ decreases for $|H| \ge 16$. Finally $\mathcal{H}(H,-R) = -\mathcal{H}(-H,R)$, so the same absolute enclosure handles both support rays. The resulting whole-line bound B_R satisfies $B_R + cQ(R) < 0$; the table records lower bounds for $-cQ(R) - B_R$. Together with the ray monotonicity, this contradicts (110) whenever $|t| \ge R$. □

Exact evaluation of the finite table.

The check is a short independent rational program, rather than a sampled numerical maximum. It brackets π using the alternating arctangent series and $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, and brackets positive square roots by integer squaring on the grid 10^{-25}. For $0 \le x \le 26$, truncating e^{-x} after the even term 120 and adding the negative next term gives an enclosing interval. For $H^2 < 25v$, the formula

$$\Phi(H/\sqrt{v}) = \frac{1}{2} + \frac{\phi(0)}{\sqrt{v}} \sum_{k=0}^{\infty} \frac{(-1)^k H^{2k+1}}{(2v)^k k!(2k+1)}$$

is truncated after the even term 80, with the next term as remainder bound. The relevant tails of both alternating series decrease in absolute value. For $H^2 \ge 25v$, use $\Phi(-5) < 10^{-6}$ and reflection. All interval operations retain their signs. The check evaluates all 3,200 closed threshold cells in the ten rows, both exterior rays, every polynomial gate, the signed-moment enclosures and the constants. Insufficient support, an invalid moment bound and a threshold mesh too coarse to prove the assertion are rejected in separate controls. The full fractions are saved with the execution record; displayed decimals are exact rationals rounded down from the proved margins.

Corollary H.2 (Radii for the fifth-index Fourier and resource bounds)**.** *Under the assumptions of Proposition H.1, one has*

ρ *interval*	$\lvert\beta\rvert$ *upper bound*	R	*endpoint margin* $>$
$[1, 11/10]$	0.253257	11/5	0.045938
$[11/10, 23/20]$	0.307221	47/20	0.039564
$[23/20, 237/200]$	0.338920	49/20	0.050250
$[237/200, 6/5]$	0.351391	99/40	0.016341

In each row, $\operatorname{supp} P \subset (-R, R)$*. These radii hold on the whole signed-moment range and therefore on every narrower signed sector used by the resource calculation.*

Proof. Use exactly the polynomial gates, interpolation bound, Gaussian enclosures, and two infinite tails from Proposition H.1. The additional exact checks use the four displayed radii, on the same 320 closed threshold cells per row. All five ray gates and the strict endpoint inequalities pass. The lower margins shown in the table are rounded downward from the exact rational margins. The trial radii 21/10, 9/4, 12/5 and 12/5, respectively, fail the acceptance criterion. These failures are rejection controls, not counterexamples to a support theorem. □

I Support at the actual indices six through fourteen

The support calculation at index five extends to each subsequent small index, but its Gaussian expression must be evaluated at that index. We record these evaluations explicitly; no numerical maximum at index five is used at another index.

Proposition I.1 (Actual-index low-moment support)**.** *Let* $6 \le n \le 14$*, assume* $C_{n-1} \le C_{\mathrm{E}} < C_n$*, and let* (P, H) *be a positive closed-event ratio maximizer from Corollary 2.2. For each applicable row below,* $\rho(P)$ *in the stated closed interval implies* $\operatorname{supp} P \subset (-R, R)$*. The number* b *bounds* $|\beta| = |\mathbb{E}[X|X|]|$*; the final column is a strict rational lower bound for the endpoint margin, uniform over the indicated indices.*

indices	ρ *interval*	b	R	*margin* $>$
$6, \dots, 14$	$[1, 11/10]$	0.253257	23/10	0.259317
$6, \dots, 14$	$[11/10, 23/20]$	0.307221	12/5	0.161008
$6, \dots, 9$	$[23/20, 237/200]$	0.338920	5/2	0.182254
$6, \dots, 9$	$[23/20, 6/5]$	0.351391	13/5	0.357463

The overlapping last two rows serve different purposes: the smaller radius is used in the low-moment contact calculation, and the wider moment interval is used by the supported Fourier estimate.

Proof. Put $C = C_n$, $s_n = \sqrt{n/(n-1)}$, $e_n = n(s_n - 1) - 1/2$, and $A = \sqrt{n-1}\Delta_{n-1}(P)/\rho$. Then $d_P = n(A - C) \le 0$. Define

$$G_n(H,t) = n\sqrt{n}\left\{\Phi\left(\frac{H-t}{\sqrt{n-1}}\right) - \Phi\left(\frac{H}{\sqrt{n}}\right)\right\} + nt\phi\left(\frac{H}{\sqrt{n}}\right) + \frac{H}{2}(t^2-1)\phi\left(\frac{H}{\sqrt{n}}\right).$$

The contact inequality at the same marginal gives, at every support point t,

$$0 \le G_n(H,t) + C\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} + \rho s_n d_P. \tag{113}$$

Indeed $n(s_nA - C) = s_n d_P + n(s_n - 1)C$; together with the constant $-C\rho/2$ in the contact polynomial this gives the displayed $C\rho e_n$. Thus the immediate predecessor enters only through the nonpositive last term.

For a table row let r denote its upper moment endpoint and set

$$\begin{aligned} c &= 819/2000, \qquad E = 91/1000, \\ Q(v) &= -v^3 + \tfrac{3}{2}rv^2 + 3bv + rE, \\ L(v) &= \tfrac{2}{5}(v^2/2 + 1/2 + E) + cQ'(v). \end{aligned}$$

We have $c < C_{\mathrm{E}} < C$ and $e_n < E$. The latter inequality can be checked without a square root by $((E + 1/2 + n)/n)^2 > n/(n-1)$, for each of the nine indices. The signed-moment inequality in Lemma 2.5 and the strict rational comparison $(1+b^2)^2 > r^2(1-b^2)$ give $|\beta| \le b$. Every row satisfies

$$Q(R), Q'(R), Q''(R), L(R), L'(R) < 0, \qquad L'' = 2/5 - 6c < 0.$$

Hence $Q(v) < 0$ for $v \ge R$, and the polynomial term in (113) is at most $cQ(|t|)$.

The Gaussian derivative estimate used at index five holds with the actual index:

$$|\partial_t G_n(H,t)| \le \phi(0)\{t^2/2 + n(s_n - 1)\} < \tfrac{2}{5}(t^2/2 + 1/2 + E). \tag{114}$$

For completeness, with $z = H/\sqrt{n}$ and $y = (H-t)/\sqrt{n}$, the derivative is $n\phi(z) + Ht\phi(z) - ns_n\phi(s_n y)$. Taylor's theorem for ϕ bounds $|n\phi(y) - n\phi(z) - Ht\phi(z)|$ by $\phi(0)t^2/2$. The identity $\partial_s[s\phi(sy)] = -\phi''(sy)$ bounds the remaining scaling error by $n\phi(0)(s_n - 1)$. For either sign η, the derivative of $G_n(H, \eta v) + cQ(v)$ is therefore at most $L(v)$, which is negative on $[R, \infty)$. It remains to exclude $v = R$ for every real threshold H.

Differentiating twice in H gives the uniform bound

$$|\partial_H^2 G_n(H,R)| \le M_{n,R} := \frac{\sqrt{n}}{4}\left(\frac{n}{n-1} + 1\right) + \frac{2}{5}R + \frac{2|R^2-1|}{5\sqrt{n}}.$$

Here one uses $\|\phi'\|_\infty < 1/4$, $\|\phi''\|_\infty < 2/5$, and $\|\phi'''\|_\infty < 4/5$. Set $H_0 = 35/2$ for $6 \le n \le 9$, and $H_0 = 22$ for $10 \le n \le 14$. The rational interval evaluations at the nodes $-H_0, -H_0 + 1/10, \dots, H_0$, enlarged by $M_{n,R}/800$, cover every closed interval between consecutive nodes. This is the standard second-derivative interpolation error $Mh^2/8$, applied to both signs of G_n.

For both exterior rays, the exact squared comparisons $H_0^2 \ge 25n$ and $(H_0 - R)^2 \ge 25(n-1)$ put the two Gaussian arguments in the same tail beyond five standard deviations. Consequently, with $\tau = 10^{-6} > \Phi(-5)$,

$$|G_n(H,R)| \le n\sqrt{n}\,\tau + \left(nR + \frac{H_0}{2}|R^2-1|\right)\phi(0)e^{-H_0^2/(2n)}, \qquad |H| \ge H_0.$$

The difference of the two distribution functions is bounded by τ, since both lie in the same tail interval of length τ. Also $|H|\phi(H/\sqrt{n})$ decreases on these rays. The reflection identity $G_n(H,-R) = -G_n(-H,R)$ supplies the other support endpoint.

Let $B_{n,R}$ be the larger of the central interpolation bound and the exterior bound. Exact rational arithmetic proves $B_{n,R} + cQ(R) < 0$ in all 26 rows; the table gives lower bounds for $-cQ(R) - B_{n,R}$. Ray monotonicity now contradicts (113) whenever $|t| \geq R$. □

Reproduction of the table.

The independent programs `check_support6_14.py` and `check_support6_9_bc.py` use the rational Gaussian, exponential and square-root enclosures described after Proposition H.1. They check respectively 8,600 and 1,400 closed threshold cells, both exterior rays, and all moment and derivative gates. The actual integer n occurs in every Gaussian expression. The saved records retain the full rational margins; each displayed decimal is an exact rational rounded down. These support conclusions apply to violating ratio maximizers under the stated predecessor hypothesis. They do not assert that arbitrary laws with a bounded third moment have bounded support.

Proposition I.2 (Support for the complementary Fourier intervals)**.** *Under the same predecessor and ratio-maximizer hypotheses, each row below also gives* $\operatorname{supp} P \subset (-R, R)$ *on its whole stated moment interval. The signed-moment bound and endpoint-margin convention are those of Proposition I.1.*

indices	*ρ interval*	*b*	*R*	*margin >*
$6, \dots, 9$	$[6/5, 7/5]$	0.478710	3	0.281666
$6, \dots, 9$	$[7/5, 3/2]$	0.525521	16/5	0.316642
$6, \dots, 9$	$[3/2, 8/5]$	0.565414	17/5	0.392665
$6, \dots, 9$	$[8/5, 17/10]$	0.600001	18/5	0.516295
$6, \dots, 9$	$[17/10, 7361/4000]$	0.641563	19/5	0.278368
$10, \dots, 14$	$[23/20, 71/50]$	0.488731	3	0.139019

Proof. Use the same $c = 819/2000$, $E = 91/1000$, contact inequality and polynomials Q, L as in the preceding proof, substituting each listed (r, b, R). Every signed-moment and derivative gate holds strictly. For these larger support radii take

$$H_0 = 35/2 \quad (6 \leq n \leq 8), \qquad H_0 = 18 \quad (n = 9), \qquad H_0 = 22 \quad (10 \leq n \leq 14).$$

These choices satisfy both squared tail comparisons in that proof. The enlargement at $n = 9$ is needed for $R = 19/5$: $(18 - 19/5)^2 \geq 25(9 - 1)$, whereas $H_0 = 35/2$ would not satisfy this condition. Apply the same interpolation bound on the closed mesh of width $1/10$ and the same estimate on both exterior rays.

The program `check_support6_14_high.py` evaluates all 25 actual-index rows, comprising 9250 closed threshold cells. Its exact rational margins are strictly larger than the displayed rounded-down values. Thus $B_{n,R} + cQ(R) < 0$ throughout each row; ray monotonicity excludes every support point with $|t| \geq R$. These bounds supply the supported Fourier premises in the complementary-moment part of Section 11. □

J The common moment information in the Fourier bounds

The finite Fourier estimates use two kinds of information. Bounds valid for every standardized law cover the larger moment ranges. Smaller moment ranges also use the support bound already obtained from contact. The distinction is essential: a support hypothesis cannot be inferred from a successful numerical integral.

Let X be centered with $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, $\gamma = \mathbb{E}X^3$, and $f(u) = \mathbb{E}e^{iuX}$. In this section put

$$h = \mathbb{E}|X|, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- = h/2, \quad A = \mathbb{E}X_+^2, \quad B = \mathbb{E}X_-^2,$$
$$\zeta = 2AB/m, \qquad b(r) = \frac{\sqrt{r^2+8} - r}{2}.$$

Here $X_- = \max\{-X, 0\}$, so $A + B = 1$ and all of h, m, A, B are positive. The letter b is used for this root function to avoid confusion with the signed second moment $\beta = \mathbb{E}[X|X|]$.

Lemma J.1 (Anchored characteristic function)**.** *For $0 \le u \le \pi$,*

$$\cos u \le \Re f(u) \le \cos u + (\rho - 1)u^3/6, \qquad |\Im f(u)|^2 \le \frac{u^6}{36}(\rho - 1)(\rho + 5/3).$$

Proof. Write $\operatorname{sinc} t = \int_0^1 \cos(st)\, ds$. For real a, b,

$$|\operatorname{sinc} a - \operatorname{sinc} b| \le |a - b|/2, \qquad |\operatorname{sinc} a - \operatorname{sinc} b| \le |a^2 - b^2|/6.$$

The first follows from the Lipschitz bound for cosine. For the second, $v \mapsto \cos\sqrt{v}$ has derivative of absolute value at most $1/2$ on $v \ge 0$, including its limit at zero; integrate with respect to s. For $r \ge 0$, integration from 1 to r of the first estimate gives

$$\left|\cos(ur) - \cos u + \frac{u \sin u}{2}(r^2 - 1)\right| \le \frac{u^3}{12}(r-1)^2(2r+1).$$

Since $\mathbb{E}(|X| - 1)^2(2|X| + 1) = 2(\rho - 1)$, averaging proves the upper real bound. For the lower bound, when $0 < u \le \pi$ set $a = \sin u/(2u)$ and $J(v) = \cos v + av^2$. The function $\operatorname{sinc} v$ decreases on $[0, \pi]$, so J decreases up to u and increases from u to π. For $v \ge \pi$, $J(v) \ge -1 + a\pi^2 = J(\pi) \ge J(u)$. Evenness gives the same conclusion for negative v. Average $J(uX) \ge J(u)$ and use $\mathbb{E}X^2 = 1$.

The second sinc estimate, together with $\mathbb{E}X = 0$, gives $|\Im f(u)| \le u^3\mathbb{E}[|X|\,||X|^2 - 1|]/6$. Put $W(r) = (r-1)^2(2r+1)$. Cauchy's inequality applies to the factorization $r|r^2 - 1| = \sqrt{W(r)}\, r(r+1)/\sqrt{2r+1}$. The exact identity

$$r^3 + \frac{29r^2 + 1}{18} - \frac{2r^2(r+1)^2}{2r+1} = \frac{(r-1)^2(4r+1)}{18(2r+1)} \ge 0$$

bounds the second squared factor in expectation by $(\rho + 5/3)/2$; the first has expectation $2(\rho - 1)$. All expectations are finite under the stated third-moment hypothesis. The value at $u = 0$ is immediate. □

Lemma J.2 (One common moment surrogate)**.** *For an independent copy X', put $\eta = \mathbb{E}|X - X'|^3/2 - \rho$. Then*

$$\eta \le h \le 1, \qquad \zeta \ge h, \qquad \rho + \zeta \ge 2/h, \qquad \zeta \ge b(\rho), \qquad \gamma^2 + \zeta^2 \le \rho^2.$$

For every real u one also has $36|\Im f(u)|^2 \le u^6(\rho^2 - \zeta^2)$. If $|X| \le R$ almost surely, then additionally

$$\eta \le b(\rho) + \frac{(R - \sqrt{\rho^2 + 8})_+^2}{4R}.$$

In particular $R \le \sqrt{\rho^2 + 8}$ implies $\eta \le b(\rho)$.

Proof. The scalar inequality $|x-y|^3 \le (x-y)^2(|x|+|y|)$, independence and centering give $\mathbb{E}|X-X'|^3 \le 2\rho + 2h$; Cauchy gives $h \le 1$. Signwise Cauchy gives

$$A^2 \le m\mathbb{E}X_+^3, \quad B^2 \le m\mathbb{E}X_-^3, \quad \frac{m^2}{A} + \frac{m^2}{B} \le \mathbb{P}(X \ne 0) \le 1.$$

Thus $AB \ge m^2$, $\zeta \ge 2m = h$, and $\rho + \zeta \ge (A+B)^2/m = 2/h$. Multiplication yields $\zeta(\rho+\zeta) \ge 2$, proving $\zeta \ge b(\rho)$.

For completeness, the sine-circle bound uses an exact mixture, not an extreme-point assertion about convolution. Let μ_+ and μ_- be the restrictions of the law of $|X|$ to its positive and negative sign classes. On $a, b > 0$ define

$$d\nu(a,b) = \frac{a+b}{m}\, d\mu_+(a) d\mu_-(b).$$

Integrating the centered law with atoms $a, -b$ and probabilities $b/(a+b), a/(a+b)$ against ν recovers the nonzero part of the original law; the zero atom is added separately. For this two-point law the second sinc estimate implies

$$\frac{6|\Im f_{a,b}(u)|}{|u|^3} \le \frac{ab|a^2-b^2|}{a+b}.$$

Together with $e_{a,b} = 2a^2b^2/(a+b)$, the right side is one coordinate of a vector of length $ab(a^2+b^2)/(a+b)$. Minkowski's inequality in $\mathbb{R}^2$ therefore bounds the vector $(\zeta, 6\Im f(u)/u^3)$ by ρ. The third-moment assumption justifies all integrals. Letting $u \to 0$ proves the assertion for γ: Taylor's remainder is dominated by $|X|^3/6$, so dominated convergence applies after using $\mathbb{E}X = 0$.

For the final assertion the exact same-sign deficit is

$$h - \eta = \mathbb{E}\Big[\min(|X|,|X'|)(X-X')^2 \mathbf{1}_{\{XX'>0\}}\Big].$$

On a common sign class, $\min(a,b) \ge ab/R$. Expansion of the two independent class integrals gives

$$h - \eta \ge \frac{h\rho - 2(A^2+B^2)}{R} = \frac{h(\rho+\zeta) - 2}{R}.$$

If $h \le b(\rho)$, the already proved $\eta \le h$ suffices. Otherwise, write $b = b(\rho)$, use $\zeta \ge h$ and $b(\rho+b) = 2$ to obtain

$$\eta - b \le \frac{(h-b)(R-\rho-h-b)}{R} \le 0,$$

when $R \le \rho + 2b = \sqrt{\rho^2+8}$. For general R, the quadratic $y(R - \sqrt{\rho^2+8} - y)$ on $y \ge 0$ is at most $(R-\sqrt{\rho^2+8})_+^2/4$. This proves the stated remainder too. In particular radius three suffices for every $\rho \ge 1$. □

For a universal calculation put $H_* = \max\{h, b(\rho)\}$. Then $b(\rho) \le H_* \le 1$, $\eta \le H_*$, $\zeta \ge H_*$ and $\rho + \zeta \ge 2/H_*$. On a closed cell $\rho \in [\ell, r]$, $H_* \in [a,b]$, it is consequently safe to use

$$\eta \le b, \qquad \zeta \ge \max\{a, 2/b - r, b(r)\}.$$

Every cell uses both bounds from the same marginal law. A lower bound on H_* is not a lower bound on h; no later sign-probability argument may substitute it for one. If support $|X| \le R \le \sqrt{\ell^2+8}$ has independently been acquired, the last part of Lemma J.2 instead permits $\eta \le b(\ell)$ and $\zeta \ge b(r)$. For larger R its displayed nonnegative remainder must be retained.

J.1 Keeping the cubic compensation in the high frequencies

Suppose now that $|X| \le R$ and $\rho \le r$. We may use

$$M_4 = \min\left\{Rr, 1 + \frac{(R+1)^2}{R+1/2}(r-1)\right\},$$
$$M_5 = \min\left\{RM_4, 1 + \frac{R^3+2R^2+3R+3/2}{R+1/2}(r-1)\right\}$$

as upper bounds on $\mathbb{E}X^4$ and $\mathbb{E}|X|^5$ of the same law. To verify the less immediate branches, set $y = |X|$ and expand

$$\rho - 1 = \mathbb{E}(y-1)^2(y+1/2), \quad \mathbb{E}X^4 - 1 = \mathbb{E}(y-1)^2(y+1)^2,$$
$$\mathbb{E}|X|^5 - 1 = \mathbb{E}(y-1)^2(y^3+2y^2+3y+3/2).$$

The ratios of the last two factors to $y+1/2$ increase on $y \ge 0$: their derivatives are respectively $y(y+1)/(y+1/2)^2$ and $y(4y^2+7y+4)/(2(y+1/2)^2)$. The other branches follow from $y \le R$. These moment bounds require no bound on the product-gap defect.

Put $G = \min\{(r-1)(r+5/3), r^2 - Z^2\}$, where $0 < Z \le b(r)$. Assume that a proved nonnegative envelope $m(u)$ bounds $|f(u)|$ on the interval being integrated. Set

$$g = e^{-u^2/2}, \quad v = -i\gamma u^3/6, \quad b_u = \sqrt{G}\,u^3/6, \quad W = g\sqrt{1+b_u^2},$$

and, for $0 \le u \le \pi$,

$$A_u = \max\{|\cos u - g|, |\cos u + (r-1)u^3/6 - g|\},$$
$$B_u = \min\{M_4u^4/24, M_5u^5/120\} + b_u(1-g).$$

Taylor's theorem gives

$$|\Im f(u) + \gamma u^3/6| \le \min\{M_4u^4/24, M_5u^5/120\}.$$

The anchored real estimate therefore gives $|f - g(1+v)| \le \sqrt{A_u^2 + B_u^2}$. Telescoping the nth powers and then applying the integral Taylor remainder to $(1+sv)^n$, $0 \le s \le 1$, proves, for every integer $n \ge 2$,

$$|f(u)^n - g^n(1+nv)| \le \sqrt{A_u^2+B_u^2}\sum_{j=0}^{n-1} m(u)^j W^{n-1-j} + \binom{n}{2} b_u^2 g^n (1+b_u^2)^{(n-2)/2},$$
$$|f(u)^n - g^n(1+nv)| \le m(u)^n + g^n\sqrt{1+n^2b_u^2}.$$

One may take their pointwise minimum. These formulas use no quotient of $m(u) - W$, so coincident and zero values cause no exceptional cases.

On a closed frequency cell $[a,b] \subset [1/2, T]$, a rigorous upper bound for this minimum is multiplied by the exact weight $\int_a^b u^{-3}\,du = (a^{-2} - b^{-2})/2$. The infinite remainder is bounded without a periodicity assumption: for $T > 0$,

$$\int_T^\infty \frac{|f(u)^n - g(u)^n(1+nv(u))|}{u^3}\,du \le \frac{1}{2T^2} + \frac{e^{-nT^2/2}}{nT}\left(T^{-3} + \frac{n\sqrt{G}}{6}\right).$$

Indeed $|f| \le 1$, $\sqrt{1+x^2} \le 1 + |x|$, $u^{-3} \le T^{-3}$ on the normal part, and $\int_T^\infty e^{-nu^2/2}\,du \le e^{-nT^2/2}/(nT)$. This explains both the compensation and the complete tail in the native high-frequency certificates.

To preserve cancellation in the real residual on a closed frequency cell, put $d_0(u) = \cos u - e^{-u^2/2}$ and $d_1(u) = d_0(u) + (r-1)u^3/6$. Their derivatives are

$$d_0'(u) = -\sin u + ue^{-u^2/2}, \qquad d_1'(u) = d_0'(u) + (r-1)u^2/2.$$

If D_j encloses d_j' throughout a cell of width w and midpoint u_0, the fundamental theorem of calculus gives $d_j(u) \in d_j(u_0) + [-w/2, w/2]D_j$. Intersecting this with a direct interval evaluation remains valid. This avoids losing the cancellation in $\cos u - e^{-u^2/2}$ by bounding its two terms separately.

J.2 The compatible modulus profile

The scalar certificate uses $\kappa = 62/625$ and proves

$$\cos x \le 1 - x^2/2 + \kappa|x|^3 \qquad (x \in \mathbb{R}). \tag{115}$$

Here is the entire reduction to its finite check. For $x \ge 625/124$ the right side is at least one. On the remaining interval, the degree-sixteen Taylor polynomial P_{16} bounds cosine above: the sixteenth derivative of $\cos - P_{16}$ is $\cos - 1 \le 0$, and its lower derivatives vanish at zero. The degree-thirteen polynomial

$$Q(x) = \kappa + \sum_{j=2}^{8} \frac{(-1)^{j+1} x^{2j-3}}{(2j)!}$$

satisfies $x^3 Q(x) = 1 - x^2/2 + \kappa x^3 - P_{16}(x)$. On the eighty cells $[i/16, (i+1)/16]$, $0 \le i < 80$, and the last cell $[5, 625/124]$, the exact verifier checks the inverse Bernstein identity and all fourteen coefficients. The smallest coefficient is $2479189/65585520000 > 0$. Lemma E.2 proves positivity on every closed cell; evenness completes (115). The native checker rejects twenty-one malformed objects and also checks 324 exact evaluations of the power/Bernstein identity.

Put $a = 1/2 - 4\kappa = 129/1250$, $C = 1 - \cos 4$ and $A_0 = 16a/C$. The profile

$$q(z) = \begin{cases} 1/2 - \kappa z, & 0 \le z \le 4, \\ A_0(1 - \cos z)/z^2, & 4 \le z \le 2\pi, \\ 0, & z \ge 2\pi \end{cases}$$

is nonnegative, convex and nonincreasing on $[0, \infty)$, and satisfies $\cos z \le 1 - z^2 q(z)$. Indeed (115) gives $0 < A_0 \le 1$. The pieces agree at their endpoints. On the middle interval the first derivative is nonpositive and the second has the sign of

$$B_0(z) = 6 + (z^2 - 6)\cos z - 4z \sin z.$$

On $[4, 3\pi/2]$, $B_0'(z) = -2z\cos z + (2 - z^2)\sin z \ge 0$ and $B_0(4) \ge 4$: indeed $7\pi/6 \le 4 \le 3\pi/2$ implies $\sin 4 \le -1/2$, and $\cos 4 \ge -1$, so $6 + 10\cos 4 - 16\sin 4 \ge 4$. On $[3\pi/2, 2\pi]$, $B_0(z) \ge 6$. The derivative at 2π is zero. The derivative jump at 4 has the correct sign because $258\sin 4 + 119(1 - \cos 4) > 0$. Its exact lower bound is obtained by replacing sine by its degree-fifteen Taylor lower bound and cosine by P_{16}. The same scalar checker verifies this rational inequality and the needed $25/8 < \pi < 22/7$ by Machin's formula.

To apply the profile, weight the distribution of $X - X'$ by $(X - X')^2/2$. This is a probability measure whose first absolute moment is $\rho + \eta$. Jensen's inequality gives, for $u \ge 0$,

$$|f(u)|^2 = \mathbb{E}\cos(u(X - X')) \le 1 - 2u^2 q(u(\rho + \eta)).$$

If $\rho+\eta \le D$, monotonicity permits D on the right. Thus the linear part gives $1-u^2+2\kappa Du^3$; when $4 \le Du \le 2\pi$ the curved part gives $1-2A_0(1-\cos(Du))/D^2$. Take the minimum with one and with the coordinate-disk estimate from Lemmas J.1–J.2. The upper square of a real interval is the larger of the squares of its two endpoints, even when that interval crosses zero or lies below it.

J.3 From characteristic functions to the CDF

We use Prawitz's smoothing inequality in the form recorded in I. Shevtsova, *Annales Mathematicae et Informaticae* 39 (2012), Lemma 4.1, p. 263. For $n \ge 2$, $c > 0$, $0 < \tau \le 1$, put

$$K_2(t) = (1-t) + i\{(1-t)\cot(\pi t) + 1/\pi\}, \qquad 0 < t < 1.$$

With $T = c\sqrt{n}$, that inequality gives

$$\begin{aligned}\Delta_n \le{}& \int_0^\tau |K_2(t)||f(ct)^n - e^{-nc^2t^2/2}|\,dt + \int_\tau^1 |K_2(t)|\,|f(ct)|^n\,dt \\ &+ \int_0^\tau |K_2(t) - i/(\pi t)|e^{-nc^2t^2/2}\,dt + \frac{1}{\pi}\int_\tau^\infty e^{-nc^2t^2/2}\frac{dt}{t}.\end{aligned}$$

Our standardized iid laws satisfy its independence, centering and positive total-variance hypotheses. The cited convention uses a strict CDF. Continuity of Φ makes the supremum identical for closed CDFs: take one-sided limits at each evaluation point. Thus atoms require no smoothing of the original law.

For $g(u) = e^{-u^2/2}$ and envelopes $m \ge |f|$, $E_0 \ge |f-g|$, the first integrand uses $|f^n - g^n| \le \min\{E_0 \sum_{j=0}^{n-1} m^j g^{n-1-j}, m^n + g^n\}$. Every literal integer n is evaluated separately. The original 20–24 integral verifier covers $[\epsilon, \tau]$ and $[\tau, 1-\epsilon]$ by closed cells, where $\epsilon = 1/1024$; its numerical parameter domain is $0 < c < \pi$ and $\epsilon < \tau \le 3/10 < 1-\epsilon$. In particular $c\tau < 1$ licenses the stable low-frequency Taylor bounds. On a cell of width w and midpoint t_0, it uses either $w \sup F$ or $w\{F(t_0) + w^2 \max(0, \sup F'')/24\}$ for a fixed smooth majorant F. The latter follows by Taylor expansion and integration; a failed derivative-domain guard triggers the former bound on the entire cell. Switching branches at a sampled midpoint would not justify this rule.

For the omitted endpoint pieces, the respective upper bounds are

$$\begin{aligned}L_\epsilon &= n\left[\frac{rc^3}{6}\left(\frac{\epsilon^4}{4} + \frac{\epsilon^3}{3\pi}\right) + \frac{c^4}{8}\left(\frac{\epsilon^5}{5} + \frac{\epsilon^4}{4\pi}\right)\right], \\ N_\epsilon &= \epsilon + \pi\epsilon^2/2, \qquad H_\epsilon = \epsilon^2/2 + \pi\epsilon^3/3.\end{aligned}$$

They follow from $|K_2(t)| \le 1 + 1/(\pi t)$ near zero, cubic Taylor error for f, $e^{-u^2/2} - 1 + u^2/2 \le u^4/8$, and

$$0 \le 1/x - \cot x \le \frac{x}{3(1-x^2/6)} \le x \quad (0 < x \le 1).$$

For this last estimate use $\sin x - x\cos x = \int_0^x t\sin t\,dt \le x^3/3$ and $\sin x \ge x - x^3/6$. Apply the same identity with $1-t$ near one. Finally the infinite Gaussian term is at most $e^{-nc^2\tau^2/2}/(\pi nc^2\tau^2)$, by $\int_a^\infty e^{-v}v^{-1}\,dv \le e^{-a}/a$. For a strip $\rho \in [\ell, r]$, multiply the full four-term bound by $\sqrt{n}/\ell$, with outward enclosures. A complete closed strip cover and a strict rational comparison then establish the claimed CDF bound.

J.4 Sharper scalar bounds for the same integrals

For $a > 0$ write $E_1(a) = \int_a^\infty e^{-v} v^{-1}\, dv$. Integration by parts, followed by Jensen's inequality for the convex function $t \mapsto (a+t)^{-2}$ and an exponential random variable of mean one, gives

$$E_1(a) = \frac{e^{-a}}{a} - e^{-a} \int_0^\infty \frac{e^{-t}}{(a+t)^2}\, dt \le e^{-a} \left(\frac{1}{a} - \frac{1}{(a+1)^2} \right).$$

The last Gaussian term in Prawitz's inequality is $E_1(a)/(2\pi)$, where $a = nc^2\tau^2/2$. This improvement has no additional moment or support hypothesis.

Some small-index certificates require $\tau > 3/10$. Their low-frequency range is still controlled explicitly. Put $U = 6/5$. On $0 \le x \le U$, the alternating sine bounds and integration give

$$\sin x \ge x \left(1 - x^2/6 + x^4/120 - x^6/5040\right), \qquad 0 \le \sin x - x\cos x \le x^3/3 - x^5/30 + x^7/840.$$

The denominator below is at least $1 - U^2/6 - U^6/5040 > 0$, and the numerator factor is at least $1/3 - U^2/30 > 0$. Consequently

$$0 \le 1/x - \cot x \le L(x) := \frac{x(1/3 - x^2/30 + x^4/840)}{1 - x^2/6 + x^4/120 - x^6/5040}, \qquad L(0) = 0.$$

It follows that

$$|K_2(t) - i/(\pi t)| \le (1-t)\sqrt{1 + L(\pi t)^2} \quad (0 < t \le 19/50).$$

At the high endpoint put $s = 1 - t$. The identity $K_2(t) = s + i\, s(1/(\pi s) - \cot(\pi s))$ gives $|K_2(t)| \le s\sqrt{1 + L(\pi s)^2}$ for $t \ge 3/4$. The rational upper bound $\pi < 22/7$ implies $\pi(19/50) < 6/5$ and $\pi/4 < 6/5$. The numerical quadrature must therefore declare $\epsilon < \tau \le 19/50 < 1 - \epsilon$, $c < \pi$, and $c\tau \le 6/5$. It differentiates the displayed smooth rational majorant on whole cells; it does not differentiate an interval-selected minimum.

The cancellation series on this larger range also have explicit tails. Truncate the even series for cosine and $e^{-u^2/2}$ at degree twenty. For their difference, the error in each t derivative of order $j = 0, 1, 2$, with $u = ct \in [0, U]$, is at most

$$\frac{2(22)_j c^j u_{\max}^{22-j}}{2^{11} 11!}.$$

Indeed, for terms of Gaussian index $k \ge 11$, the ratios of consecutive absolute derivative terms are bounded respectively by

$$\frac{U^2}{2(k+1)}, \qquad \frac{U^2}{2k}, \qquad \frac{U^2(2k+1)}{2k(2k-1)}.$$

All decrease with k and are less than $1/8$ at $k = 11$. The cosine ratios are $U^2/\{(2k+2-j)(2k+1-j)\} < 1/8$. The first cosine tail is $2^{11}11!/22! < 1/4$ times the first Gaussian tail, for every j. Summing both geometric tails gives a factor less than $(1 + 1/4)/(1 - 1/8) = 10/7 < 2$. Uniform absolute convergence justifies these differentiations. For the removable function $2(\cos u - e^{-u^2/2})/u^3$, divide the series symbolically, so its value tail is bounded by $4u_{\max}^{19}/(2^{11}11!)$ and its value at zero is obtained directly from the series. No numerical division by a nearly zero frequency is needed.

K Fourier refinements at a small fixed index

At a small index, replacing a finite frequency integral by an integral over the whole half-line can lose much of the available margin. The following estimates retain the finite interval and the actual integer power. They use the moments of the same supported marginal throughout.

K.1 A finite-interval compensated estimate

Let $n > 2$ be an integer, $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $|X| \le R \le 6$, $\gamma = \mathbb{E}X^3$, $\mu_j = \mathbb{E}X^j$, and $f(t) = \mathbb{E}e^{itX}$. Assume $\mu_4 \le M \le 111/10$ and $|\gamma| \le g \le 37/20$. The coefficient and moment bounds proved in the formula atlas give

$$|a| \le A_*, \quad |z| \le Z, \quad \mu_6 \le M_6, \qquad a = (3-\mu_4)/24, \quad z = (10\gamma - \mu_5)/120.$$

Define

$$E = \frac{M_6}{720} + \frac{g^2}{72} + \frac{(1/2 + g^2/144)^3}{5}, \qquad B = A_* + Z/2 + E/4,$$

$$a_n = (n-1)/2, \qquad b_n = (n-2)(45-M)/96, \qquad I_j(A) = \int_0^{1/2} t^j e^{-At^2}\,dt.$$

All these quantities are nonnegative, and the two damping rates are positive. Put

$$L_0 = \sqrt{n}\left\{ nZI_2(a_n) + \frac{n(n-1)A_*g}{6} I_4(a_n) + \frac{n^3g^3}{1296} I_6(n/2) \right\},$$

$$L_H = \sqrt{n}\left\{ nA_*I_2(a_n) + \frac{n^2g^2}{72} I_4(n/2) \right\},$$

$$L_2 = n\left\{ nEI_3(a_n) + \frac{n(n-1)B^2}{2} I_5(b_n) \right\}.$$

For every real raw threshold H, the complete low-frequency estimate is

$$\int_0^{1/2} \frac{\left| \Im\left[e^{-itH}\{ f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6) \} \right] \right|}{t^3}\,dt \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n}. \tag{116}$$

To prove it, set $h(t) = e^{-t^2/2 - i\gamma t^3/6}$. Taylor's formula gives

$$f - h = -at^4 - izt^5 + r_6, \qquad |r_6| \le Et^6, \qquad |f-h| \le Bt^4 \quad (0 \le t \le 1/2).$$

Indeed, for $w = -t^2/2 - i\gamma t^3/6$, one has $|w| \le t^2(1/2 + g^2/144) < 1/6$. The quadratic exponential polynomial contributes at most $g^2t^6/72$ beyond degree five; its remaining error is at most $|w|^3 e^{|w|}/6 \le |w|^3/5$. The characteristic-function remainder is at most $M_6t^6/720$. Symmetrization and the global fourth-degree cosine upper bound give

$$|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45-M)t^2/48.$$

Thus $|f|$, $|h|$, and every point of their complex line segment are at most $e^{-(45-M)t^2/96}$. The integral Taylor remainder for the polynomial $z \mapsto z^n$ consequently yields

$$f^n - h^n = n(f-h)h^{n-1} + R_2, \qquad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8 e^{-b_nt^2}.$$

The r_6 and R_2 integrals give L_2/n. For the leading terms use

$$|\Im[e^{-itH}(-at^4 - izt^5)h^{n-1}]| \le e^{-a_nt^2}\{(Z + A_*|H|)t^5 + (n-1)A_*gt^7/6\}.$$

Finally, for $u = n\gamma t^3/6$, the rotated imaginary part of $e^{-nt^2/2}(e^{-iu} - 1 + iu)$ is bounded by $e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2)$. Integration proves (116), including its continuous endpoint at zero.

For the small indices the quantities $I_j(A)$ are enclosed directly. The degree-20 and degree-21 Taylor polynomials of e^{-x} are, respectively, upper and lower bounds on all $x \geq 0$, by the sign of the integral remainder. Their integrated values are the rational sums

$$\sum_{k=0}^{m} \frac{(-A)^k}{k!(j+2k+1)2^{j+2k+1}}, \qquad m = 20, 21.$$

The selected routines require $0 \leq A \leq 8$ and $0 \leq j \leq 16$; the rates above meet those gates for $5 \leq n \leq 9$. Every coefficient multiplying an integral is nonnegative. When two complete triples (L_0, L_H, L_2) are available, either may be selected by a rule independent of H. Selecting different coefficients from different triples is not justified by this argument.

K.2 Containing rectangles and exact finite powers

The supported fourth-moment bound also improves the real coordinate. If $\rho \in [\rho_l, \rho_u]$, $|X| \leq R$, and $\mu_4 \leq M_4$, then

$$\Re f(t) \leq 1 - t^2/2 + M_4 t^4/24.$$

If $R^2 t^2 \leq 30$, a second lower endpoint is

$$\Re f(t) \geq 1 - t^2/2 + \rho_l^2 t^4 (1 - R^2 t^2/30)/24.$$

These follow from the global fourth-degree upper and sixth-degree lower Taylor polynomials of cosine, $\mu_6 \leq R^2 \mu_4$, and $\mu_4 \geq \rho^2$ by Cauchy's inequality. The nonnegative coefficient gate is needed before substituting ρ_l^2. Intersect these real intervals with the anchored bounds of Lemma J.1.

Lemma K.1 (Two corners of a complex-power rectangle)**.** *Let $n \geq 2$ be an integer, $a \geq 0$, $0 < x_0 \leq x_1$ and $Y \geq 0$. If*

$$(n-2)Y \leq 3x_0/2, \qquad (2n-1)x_0^n \geq (n-1)a,$$

then for $x \in [x_0, x_1]$ and $|y| \leq Y$,

$$|(x+iy)^n - a|^2 \leq \max\{|(x_0+iY)^n - a|^2, |(x_1+iY)^n - a|^2\}.$$

Proof. Write $z = x + iy$, $r = |z|$ and $F(x,y) = |z^n - a|^2$. Since $|\arg z| \leq Y/x_0$, the first gate and $3/2 < \pi/2$ imply $\Re z^{n-2} \geq 0$. Direct differentiation gives

$$F_{yy} = 2nr^{2n-4}\{r^2 + 2(n-1)y^2\} + 2an(n-1)\Re z^{n-2} \geq 0,$$
$$F_{xx} = 2nr^{2n-4}\{r^2 + 2(n-1)x^2\} - 2an(n-1)\Re z^{n-2}.$$

Divide the latter by $2nr^{n-2} > 0$. Its first term is $r^{n-2}((2n-1)x^2 + y^2) \geq (2n-1)x_0^n$; the subtracted term is at most $a(n-1)$. Thus F is convex in each coordinate. Apply convexity first in y and then in x, and use $F(x,Y) = F(x,-Y)$. □

For Fourier smoothing take $a = e^{-nt^2/2}$ and any proved containing rectangle for $f(t)$. Both gates must hold throughout the frequency cell. If either fails, the geometric-sum estimate remains available. The maximum of the two pointwise corner bounds is enclosed before integration; the maximum of their separate integrals would not suffice. Complex powers are finite polynomial recurrences, so zero or equal bases create no division. A fourth-Taylor endpoint is kept fixed on each integration cell when differentiating its smooth majorant.

Cancellation can be retained when computing the real endpoints. For example, on $0 \le u \le 1$,

$$1 - u^2/2 + M_4 u^4/24 - e^{-u^2/2} = \frac{M_4 - 3}{24} u^4 + \sum_{k=3}^{10} \frac{(-1)^{k+1} u^{2k}}{2^k k!} + R(u).$$

For derivative order $j = 0, 1, 2$ with $u = ct$, the absolute remainder is at most $2(22)_j u_{\max}^{22-j} c^j / (2^{11} 11!)$; consecutive absolute tail terms have ratio below $1/2$ on this domain. Alternatively, at higher frequencies enclose each explicit smooth residual $d(u) = \cos u - e^{-u^2/2}$ or $d(u) + (\rho_u - 1)u^3/6$ by its midpoint value plus the half-width times an enclosure of its derivative on the entire cell. Intersecting two such valid enclosures preserves their common function. These operations differentiate scalar envelopes, not the law's CDF.

L A five-summand corner without a support assumption

The last small moment rectangle can be treated directly by conditioning on signs. This avoids contact and Fourier estimates in that rectangle. Let $\beta = \mathbb{E}[X|X|]$, $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$ and $a_\pm = (1 \pm \beta)/2$. Zero atoms belong to the nonpositive class. Signwise Cauchy gives

$$m \ge \frac{1 + \beta^2}{2\rho}, \qquad \frac{m^2}{a_+} \le p \le 1 - \frac{m^2}{a_-}, \qquad m^2 \le pq.$$

Set $h = pq/m$, $Y = p + hX$, $B = \mathbf{1}_{\{X>0\}}$, and $U = Y - B$. Both conditional means of U vanish, with variances

$$v_0 = p^2(qa_-/m^2 - 1), \qquad v_1 = q^2(pa_+/m^2 - 1).$$

Given $\sum_{i=1}^5 B_i = j$, the noises are independent with variance $V_j = (5 - j)v_0 + jv_1$. Write $b_j = \binom{5}{j} p^j q^{5-j}$ and define

$$T(V, d) = \begin{cases} 1, & d \le 0, \\ V/(V + d^2), & d > 0. \end{cases}$$

Conditional Cantelli and the exact normal coordinate yield

$$\frac{\sqrt{5}}{\rho}\{\mathbb{P}(S_5 \le t) - \Phi(t/\sqrt{5})\} \le \frac{\sqrt{5}}{\rho}\left\{\sum_{j=0}^{5} b_j T(V_j, j - z) - \Phi\left(\frac{(z - 5p)m}{\sqrt{5pq}}\right)\right\}, \qquad z = 5p + ht. \tag{117}$$

The value $T(0, 0) = 1$ is essential: a deterministic sum at the threshold belongs to the closed event.

Proposition L.1 (The positive corner)**.** *For every standardized iid marginal satisfying*

$$\frac{477}{400} \le \rho \le \frac{6}{5}, \qquad \frac{17}{50} \le \beta \le \frac{35139006743001}{10^{14}},$$

the positive closed-CDF score in (117) *is strictly below* $4097/10000 < C_{\mathrm{E}}$ *at every real threshold.*

Proof. Here are the enclosures defining the finite proof. On a closed β slab $[b_l, b_u]$ put $m_l = (1 + b_l^2)/(2(6/5))$. The moment inequalities enclose p in $[2m_l^2/(1 + b_u),\ 1 - 2m_l^2/(1 - b_l)]$. On a closed p, z box $[l, u] \times [a, b] \subset (0, 1/2) \times [0, 5]$ use

$$v_0 \le \max\{0, u^2((1 - l)(1 - b_l)/(2m_l^2) - 1)\},$$
$$v_1 \le \max\{0, (1 - l)^2(u(1 + b_u)/(2m_l^2) - 1)\}.$$

Each binomial mass is maximized at the projection of $j/5$ onto $[l,u]$, and each Cantelli factor is maximized at $z=b$ and the variance upper bound. Clip the sum of the six upper bounds at one. If $a-5u\ge 0$, the normal argument is at least $(a-5u)m_l/(\sqrt{5}\,u(1-u))$. Otherwise it is at least $(a-5u)/\sqrt{5u(1-u)}$: use $m\le\sqrt{pq}$ on the negative part and the fact that $(z-5p)/\sqrt{pq}$ decreases in p for $z\ge 0$, $p<1/2$. Its derivative has numerator $-[5p+z(1-2p)]/2$. Outward square roots and a rational Gaussian lower enclosure therefore bound the complete box. Take the nonnegative part of the CDF difference before multiplying by $\sqrt{5}/\rho_l$.

The twelve closed β slabs have endpoints

$$17/50,\quad 341/1000,\dots,351/1000,\quad 35139006743001/10^{14}.$$

Together with the five intervals $z\in[k,k+1]$, $0\le k<5$, these give sixty roots. The supplied binary subdivisions have 1,027 leaves. Exact evaluation on every closed leaf gives a maximum upper bound

$$204848338979/500000000000<4097/10000.$$

The independent evaluator uses a 100-term integrated Gaussian Taylor polynomial with its explicit remainder, rational Machin bounds and integer square-root enclosures. Complete root order, both children at every split, and all leaf bounds are checked; twenty-one negative controls reject missing cells, false bounds and malformed data.

Both omitted rays have direct bounds. Use the full rectangle's m_l and enclosing $[p_l,p_u]$, put $v_u=p_u(1-p_u)$ and $s_u=v_u^2/m_l^2-v_u$. On the feasible domain $m^2\le v$, v^2/m^2-v increases with v and decreases with m. Thus $V_j\le\max\{j/p_l,(5-j)/(1-p_u)\}s_u$. For $z\le 0$, discard the nonnegative Gaussian CDF and evaluate the Cantelli upper sum at zero, using binomial upper masses $\binom{5}{j}p_u^j(1-p_l)^{5-j}$. The resulting score is below $331/1000$. For $z\ge 5$, the CDF is at most one and normal Cantelli gives the score upper bound

$$\frac{\sqrt{5}}{\rho_l}\frac{1}{1+5m_l^2/p_u^2}<187/1000.$$

These exact rational comparisons and the sixty roots cover all real z. The proposition concerns the stated positive branch; it is not silently extended to the opposite β rectangle by reflection. □

M The finite-frequency sine bound used at indices 26–30

Let the positive and negative support magnitudes lie in $[a_+,R_+]$ and $[a_-,R_-]$, respectively. Assume

$$0<A\le a_+^2+a_-^2,\qquad B\ge R_+^2+R_-^2,\qquad M\ge\max(R_+,R_-).$$

Put $R(u)=1-Au^2(1-Bu^2/42)/20$. For $u\ge 0$, $Mu\le\pi$ and $Bu^2\le 42$,

$$|\Im f(u)|^2\le(\rho^2-\zeta^2)u^6R(u)^2/36. \tag{118}$$

In particular $R(u)\ge 1-21A/(40B)\ge 19/40$. Outside these frequency gates the ordinary sine-circle inequality remains available.

Here is a proof retaining the same marginal. Write $\operatorname{sinc}t=\int_0^1\cos(vt)\,dv$. For $u\sqrt{z}\le\pi$,

$$0\le-\frac{d}{dz}\operatorname{sinc}(u\sqrt{z})=\frac{u^2}{2}\int_0^1 v^2\operatorname{sinc}(uv\sqrt{z})\,dv\le\frac{u^2}{6}-\frac{u^4z}{60}+\frac{u^6z^2}{1680}.$$

The upper bound follows by integrating the global polynomial $\operatorname{sinc}t\le 1-t^2/6+t^4/120$; nonnegativity follows from $0\le uv\sqrt{z}\le\pi$. Integrating between a^2 and b^2 gives

$$|\operatorname{sinc}(au)-\operatorname{sinc}(bu)|\le\frac{|a^2-b^2|u^2}{6}\left(1-\frac{(a^2+b^2)u^2}{20}+\frac{(a^4+a^2b^2+b^4)u^4}{840}\right).$$

The last bracket is at most $R(u)$, since $a^4 + a^2b^2 + b^4 \le B(a^2 + b^2)$ and $1 - Bu^2/42 \ge 0$. At $a = b$ use the undivided inequality.

For the centered two-point law at $a, -b$, put

$$e_{a,b} = \frac{2a^2b^2}{a+b}, \quad r_{a,b} = \frac{ab(a^2+b^2)}{a+b}, \quad h_{a,b} = \frac{abu}{a+b}\{\operatorname{sinc}(au) - \operatorname{sinc}(bu)\}.$$

The identity $(2ab)^2 + (a^2 - b^2)^2 = (a^2 + b^2)^2$ gives $\|(e_{a,b}, 6h_{a,b}/(u^3R(u)))\|_2 \le r_{a,b}$. Integrate against the exact centered two-point mixture from the ordinary sine-circle proof. The three integrals are ζ, ρ and $\Im f(u)$, respectively. The triangle inequality proves (118). The case $u = 0$ is immediate.

For each of the 55 acquired rows ($26 \le n \le 30$, $32 \le j \le 42$), retain its closed probability interval $[p_l, p_u]$, an actual-noise bound $s \le s_u$, and an upper bound $d > 0$ on the inward sum $a_0 + a_1$. The interface requires $d < p_l$, $d \le 1 - 2p_u$ and

$$\frac{p_u^2 + (1 - p_u - d)^2}{p_u(1 - p_u) + s_u} \ge \frac{17}{10}, \qquad R_-^2 + R_+^2 \le \frac{1037}{100}, \qquad \max(R_-, R_+) \le \frac{13}{5}. \tag{119}$$

Indeed the minimum of $(p - a_0)^2 + (1 - p - a_1)^2$ under $a_0, a_1 \ge 0$, $a_0 + a_1 \le d \le 1 - 2p$ is $p^2 + (1-p-d)^2$: use the full inward sum and differentiate in a_0, obtaining $2(1-2p-d)+4a_0 \ge 0$. This numerator decreases in p on the retained interval, while $p(1 - p) + s$ increases. Thus (119) gives the required standardized inner support sum. It licenses (118) with $A = 17/10$, $B = 1037/100$, $M = 13/5$ on every row.

A direct resource-to-sine map at indices 26–30.

The three common constants in (119) follow directly from the acquired floor-25 resource. Write its original simultaneous intercepts as d_0, L, and its prices as b, g, h, so that

$$d_{\text{floor}} - b = gd_0 = hL > 0, \qquad b \ge 0.$$

These identities are checked exactly against the exported residual; L is the original joint intercept, even when another bound further clips the variance. The weighted resource implication gives

$$\frac{n}{25}(b + g\delta) + h\lambda_n \le d_{\text{floor}}, \qquad \frac{n}{25}\frac{\delta}{d_0} + \frac{\lambda_n}{L} \le 1.$$

The second inequality follows by subtracting $nb/25$, dividing by $d_{\text{floor}} - b$, and using $n \ge 25$. Therefore

$$\delta \le \delta_u := \min\left\{\frac{25d_0}{n}, \frac{25d_{\text{floor}}/n - b}{g}\right\}.$$

The bounded conditional supports give $s \le \mathcal{B}\delta$, where $\mathcal{B} = \max(O_0, O_1)\max(p_u, 1 - p_l)$ and $\lambda_n = ns$. Substitution in the joint plane yields

$$\lambda_n \le \left\{L^{-1} + (25\mathcal{B}d_0)^{-1}\right\}^{-1} =: \lambda_u.$$

This step uses one joint plane before optimizing. For all 55 rows, the choices $d = \delta_u$ and $s_u = \lambda_u/n$ satisfy the two preceding gates and

$$\frac{p_u^2 + (1 - p_u - \delta_u)^2}{p_u(1 - p_u) + \lambda_u/n} > \frac{17}{10}.$$

The smallest left side is greater than $1716/1000$ and occurs at $n = 26$ on P_{38}. The same rows satisfy $R_-^2 + R_+^2 \le 1037/100$ and $\max(R_-, R_+) \le 13/5$. The actual-index support proofs and

the full 100-panel resource cover are checked separately. Thus the finite-sine majorant applies on every surviving panel. Its integral uses only these common constants and the rational moment and frequency intervals below.

The symmetrization inputs admit a shorter derivation than the older special first-moment envelope. Since $(13/5)^2 < (11/10)^2 + 8$, the general bounded-support inequality for η gives $\eta \le b(\rho)$ throughout $[11/10, 6/5]$, where $b(r) = (\sqrt{r^2+8} - r)/2$. The universal inequality $\zeta \ge b(\rho)$ was proved separately. Since b decreases and $r + b(r)$ increases, on a strip $[r_l, r_u]$ we may use simultaneously

$$\rho + \eta \le r_u + b(r_u), \qquad \rho^2 - \zeta^2 \le r_u^2 - b(r_u)^2.$$

Rational brackets z_l, z_u for $b(r_u)$ are checked by $z_l^2 + r_u z_l \le 2 \le z_u^2 + r_u z_u$ with $0 < z_l \le z_u$. The integrator uses $r_u + z_u$ and $r_u^2 - z_l^2$ in their indicated upper-bound directions. The value z_u is not separately asserted to bound η at the lower end of the strip.

The selected 23 closed strips cover $[11/10, 6/5]$. All five actual indices are evaluated on each strip, giving 115 complete four-term Fourier inequalities. The optional fourth-Taylor real endpoint and Lemma K.1 are used only when their whole-cell gates hold. The remaining cells use a previously valid geometric-sum or direct interval bound.

The implementation sometimes integrates the maximum of the two corner majorants more accurately. If f_0, f_1 are twice continuously differentiable on $[t_0 - h, t_0 + h]$, enclose their midpoint values above by a_i, their midpoint derivatives by $[d_i^-, d_i^+]$, and both second derivatives above by $C \ge 0$. Taylor's theorem bounds the integral of their pointwise maximum by

$$\int_0^h \max_i (a_i + d_i^+ x)\, dx + \int_0^h \max_i (a_i - d_i^- x)\, dx + \frac{Ch^3}{3}.$$

Each affine maximum has at most one crossing, found exactly. This bounds the maximum before integration and assumes nothing about how often the original smooth curves cross.

N Moment constraints in finite count comparisons

The finite count calculation discards some parameter boxes before testing their probability bound. Each such rejection must follow from a necessary inequality for the same marginal. This section derives these inequalities. It does not assert that a point satisfying them is realizable by a law.

Use $hX = B - p + U$, where $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $v = pq$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$ and $h = v/m$. A positive central support gap has already been established. Write

$$U_0 \in [-O_0, a_0], \qquad U_1 \in [-a_1, O_1], \qquad \mathbb{E}U_i = 0, \quad s_i = \mathbb{E}U_i^2, \quad x = \mathbb{E}U_0^3, \quad y = \mathbb{E}U_1^3.$$

Here $a_0 < p$, $a_1 < q$ and $O_i, a_i \ge 0$. Set $s = qs_0 + ps_1$, $\lambda = ns$, and, when $s > 0$, $\alpha = ps_1/s \in [0, 1]$. Then

$$h^2 = v + s, \qquad s_0 = \frac{(1-\alpha)s}{q}, \qquad s_1 = \frac{\alpha s}{p}.$$

If $s = 0$, both conditional laws vanish and no division by s is used.

N.1 One pair of conditional third moments

For a centered $W \in [-L, U]$ of variance w, with $L, U > 0$, the nonnegative polynomials $(W + L)(W - w/L)^2$ and $(U - W)(W + w/U)^2$ give

$$w^2/L - Lw \le \mathbb{E}W^3 \le Uw - w^2/U. \tag{120}$$

At a zero endpoint, centering forces $W = 0$ almost surely. The two exact cubic identities are

$$\rho h^3 = v(1-2v) + 3v(s_0 + s_1) - qx + py, \tag{121}$$
$$\gamma h^3 = v(1-2p) + 3v(s_1 - s_0) + qx + py. \tag{122}$$

The differing sign of x is essential. The sign restrictions on the conditional supports justify the absolute cube in the first identity.

These identities restrict the single pair (x, y) to a rectangle and linear strips. For example, replacing p by the midpoint p_m in $-qx + py$ incurs error $(p - p_m)(x + y)$. On a closed cubic rectangle its absolute value is at most the probability half-width times $\max(|x_l + y_l|, |x_u + y_u|)$. In the signed-cubic identity the corresponding factor is $y - x$, bounded by $\max(|y_l - x_u|, |y_u - x_l|)$. All other products use directed interval endpoints. Thus these strips contain every actual common pair.

N.2 The physical defect as a further half-plane

Let $\sigma = \sqrt{10}$ and retain the physical gap defect $D = (3+\sigma)\rho - \gamma - 3J$. The endpoints of the central gap are $A = (p - a_0)/h$, $B = (q - a_1)/h$. The outer-gap inequality gives

$$D \geq \sigma\rho - \gamma - 6mAB + 6B(\mathbb{E}X_+^2 - mB) + 6A(\mathbb{E}X_-^2 - mA).$$

Substituting the count coordinates and the two cubic identities yields

$$\begin{aligned} Dh^3 \geq\ & B_\sigma(p) + 6vF(a_0, a_1) \\ & + qs_0\{3p(\sigma+3) - 6a_0\} + ps_1\{3q(\sigma+1) - 6a_1\} \\ & - (\sigma+1)qx + (\sigma-1)py, \\ B_\sigma(p) =\ & v\{(\sigma-1) + 2p - (2\sigma+6)v\}, \\ F(a_0, a_1) =\ & a_0 + a_1 - a_0^2 - a_1^2 - a_0a_1. \end{aligned} \tag{123}$$

The polynomial B_σ is the Bernoulli defect square. Replacing σ by a lower enclosure decreases the complete right side, because its coefficient is exactly $\rho h^3 \geq 0$. After that replacement, signed interval products and the explicit nonnegative-coefficient gates license the termwise lower bounds. If such a gate fails, this additional half-plane is omitted.

Substitution of (120) into (123) gives the stronger scalar inequality

$$\begin{aligned} Dh^3 \geq\ & B_\sigma + 6vF(a_0, a_1) \\ & + q\left\{[3p(\sigma+3) - (\sigma+7)a_0]s_0 + (\sigma+1)s_0^2/a_0\right\} \\ & + p\left\{[3q(\sigma+1) - (\sigma+5)a_1]s_1 + (\sigma-1)s_1^2/a_1\right\}. \end{aligned} \tag{124}$$

The terms with $a_i = 0$ are treated by $s_i = 0$, not by division. Also $s_i \leq O_i a_i$, by averaging the nonnegative quadratic $(U_i + O_i)(a_i - U_i)$, with reflection for class one.

On a rectangle clipped by $a_0 + a_1 \leq d < 1$, the function F is nonnegative and concave: its Hessian is $\left(\begin{smallmatrix} -2 & -1 \\ -1 & -2 \end{smallmatrix}\right)$. Its minimum therefore occurs at a vertex of the clipped polygon. There is no requirement $d < 1/2$ for this vertex argument. The older coordinate-monotonicity shortcut requires separate gates. For example, a lower bound for the a_0 derivative in (124) is

$$6v_l(1-2d) - q_u(\sigma_u + 7)(s_0)_u - q_u(\sigma_u + 1)O_0^2.$$

The reflected expression uses $\sigma + 5$ and $\sigma - 1$. Positivity of these gates implies $d < 1/2$ and licenses that shortcut only on its smaller domain. A bisection discards an interval only when its entire defect lower bound exceeds the already acquired upper bound.

N.3 Second moments and necessary noise floors

The exact second-moment relations are

$$\beta = \frac{v(1-2p) + s(2\alpha - 1)}{v+s}, \qquad m = \frac{v}{\sqrt{v+s}}. \tag{125}$$

The second expression increases with v and decreases with s. For $s > 0$, the first can be solved for α; the case $s = 0$ is tested directly. If an acquired upper bound β_u is below $t_l = 1 - 2p_u$, then

$$v_l(t_l - \beta_u) \le s(\beta_u + 1 - 2\alpha_l).$$

A nonpositive right coefficient excludes the box; a positive one gives a lower bound for s. If an acquired lower bound β_l exceeds $t_u = 1 - 2p_l$, use instead $v_l(\beta_l - t_u) \le s(2\alpha_u - 1 - \beta_l)$. These tests retain the same α as the conditional probabilities.

There is a further floor from the absolute third moment. The elementary bounds $-x \le O_0 s_0$, $y \le O_1 s_1$ in (121) give

$$\rho_l (v+s)^{3/2} \le v(1-2v) + Ms,$$

where M bounds $(3p+O_0)(1-\alpha)+(3q+O_1)\alpha$ at all four p, α corners. Convexity of $(v+s)^{3/2}$ implies

$$A(v) := \rho_l v^{3/2} - v + 2v^2 \le [M - \tfrac{3}{2}\rho_l\sqrt{v}]s.$$

If $A(v_l) > 0$, then $\rho_l\sqrt{v_l} > 1 - 2v_l$, so $A'(v_l) > 1/2 + v_l > 0$; moreover $A'' > 0$. Hence a positive lower bound for $A(v_l)$ and an upper bound for the right coefficient give a necessary noise floor on the whole box. A nonpositive upper coefficient proves infeasibility. Directed square roots preserve both conclusions.

Finally, the common inward bound $a_0 + a_1 \le d$, $d > 0$, together with (120), implies

$$\rho (v+s)^{3/2} \ge v(1-2v) + B_* s + H_* s^2/d, \tag{126}$$

where

$$B_* = 3p(1-\alpha) + 3q\alpha - d\max(\alpha, 1-\alpha), \qquad H_* = \left(\frac{1-\alpha}{\sqrt{q}} + \frac{\alpha}{\sqrt{p}}\right)^2.$$

For the quadratic term use

$$\frac{(1-\alpha)^2}{qa_0} + \frac{\alpha^2}{pa_1} \ge \frac{((1-\alpha)/\sqrt{q} + \alpha/\sqrt{p})^2}{a_0 + a_1}.$$

The zero-variance restrictions cover vanishing endpoints by a limit. The minimum of B_* is at the four p, α corners, because it is affine in p and concave in α. On a p interval $[p_l, p_u]$ a positive lower envelope for $\sqrt{H_*}$ is $(1-\alpha)/\sqrt{q_u} + \alpha/\sqrt{p_u}$, where $q_u = 1 - p_l$. This affine function is minimized at an α endpoint. The global upper Taylor bound

$$(v+s)^{3/2} \le v^{3/2} + \tfrac{3}{2}\sqrt{v}\, s + \tfrac{3}{8}s^2/\sqrt{v}$$

turns (126), with $\rho \le \rho_u$, into a necessary quadratic inequality in s. Its exact minimum on the closed variance interval is tested at both endpoints and any interior vertex. Strict positivity excludes the whole box. The implementation defers $d = 0$ to its deterministic branch.

N.4 An additional outgoing event and full phase coverage

For centered $W \in [-a_1, O_1]$ and $-a_1 \le b < O_1$,

$$(W + a_1)(W - b) \le \mathbf{1}_{\{W>b\}}(O_1 + a_1)(O_1 - b).$$

Thus $\mathbb{P}(W > b)$ is at least the nonnegative part of $(s_1 - a_1 b)/((O_1 + a_1)(O_1 - b))$. For $b < -a_1$ the strict event is certain; for $b \ge O_1$ use zero. If any of the k class-one variables exceeds $b = u + B + (k-1)a_1$, their sum exceeds $u + B$. Independence gives the lower probability $1 - (1 - r)^k$, where r is the preceding bound. The remaining centered sum, of variance V_0 and upper support $(n-k)a_0$, is at least $-B$ with probability at least

$$\max\left\{\frac{B^2}{V_0 + B^2}, \frac{B}{B + (n-k)a_0}\right\} \quad (B > 0).$$

These follow from Cantelli and Markov, respectively. At $B = 0$ use one if $V_0 = 0$ and zero otherwise. Multiplication of the two lower bounds is valid because the two conditional groups are independent. This supplies the optional single-event outgoing witness.

The acquired lower bound $m_l > 0$ also gives

$$\lambda = n(v^2/m^2 - v) \le n(v^2/m_l^2 - v), \qquad h \le v_u/m_l.$$

The quadratic on the right is convex, so its maximum on the v interval occurs at an endpoint. If $-H_- \le H \le H_+$, the gates

$$np_l - H_- h_u > -1/2, \qquad np_u + H_+ h_u < n + 1/2$$

ensure that all nearest counts belong to $\{0, \dots, n\}$. The original joint budget $d/d_0 + \lambda/\Lambda \le 1$ retains its own intercept Λ when a separate sharper noise cap clips the count domain. Replacing that intercept by the sharper cap would assert an additional joint inequality and is not licensed here.

N.5 A universal lower bound for each sign probability

Six probability panels at index six are excluded before the count calculation. The exclusion is the following elementary moment inequality; it does not require support, a central gap, or absence of a zero atom.

Lemma N.1 (Sign mass and third moment)**.** *If $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\rho = \mathbb{E}|X|^3 < \infty$, then*

$$\rho^2\mathbb{P}(X > 0), \quad \rho^2\mathbb{P}(X < 0) \;\ge\; \frac{8(7\sqrt{7} - 17)}{27} \;>\; \frac{11261}{25000}.$$

Proof. Write $p = \mathbb{P}(X > 0)$, $a = \mathbb{E}X_+^2$, and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. Centering and unit variance imply $0 < a < 1$. Cauchy's inequality, first with factors $X_+^{1/2}$ and $X_+^{3/2}$ and then with their negative-sign counterparts, gives

$$a^2 \le m\mathbb{E}X_+^3, \qquad (1-a)^2 \le m\mathbb{E}X_-^3.$$

Also $m^2 \le pa$. Consequently

$$p\rho^2 \ge f(a) := \frac{(a^2 + (1-a)^2)^2}{a}.$$

The derivative has the sign of $6a^2 - 2a - 1$, because

$$f'(a) = \frac{(2a^2 - 2a + 1)(6a^2 - 2a - 1)}{a^2}.$$

Thus the unique minimum on $(0,1)$ is at $a=(1+\sqrt{7})/6$, where $f(a)=8(7\sqrt{7}-17)/27$. The strict rational comparison follows from $2645751^2<7\cdot 10^{12}$ and $8(7\cdot 2645751/10^6-17)/27>11261/25000$. Apply the same argument to $-X$ for the other sign probability. □

The exact constant is attained. Take $p=(5-\sqrt{7})/6$, $q=1-p=(1+\sqrt{7})/6$, and $X=(B-p)/\sqrt{pq}$ for $B\sim\text{Bernoulli}(p)$. Then $\mathbb{E}X_+^2=q$ and $p\rho^2=(p^2+q^2)^2/q=f(q)$ equals the minimum above. The smaller rational constant is used only to simplify the finite checks.

For a closed probability panel $p\in[j/100,(j+1)/100]$ and $\rho\le r$, the necessary lower bound therefore excludes the entire panel whenever

$$\frac{j+1}{100}r^2<\frac{11261}{25000}.$$

This verifies precisely the six sign-mass dispositions used at index six:

ρ interval	j values	largest $(j+1)r^2/100$
$[1,11/10]$	$35,36$	$4477/10000$
$[11/10,23/20]$	$32,33$	$8993/20000$
$[23/20,237/200]$	$30,31$	$56169/125000$

Each last-column value is strictly less than $11261/25000$. The inequalities include both panel endpoints. They apply to the actual sign probability in the count representation, regardless of which discrepancy orientation supplied that representation.

N.6 Scalar consequences of the common inward budget

Some finite count domains use a stronger absolute-third-moment floor or noise ceiling before their five-dimensional subdivision. The following calculation supplies those scalar restrictions. It uses the same sign conditioning as Section N.

Reflect only for the scalar calculation, so that $0<p\le 1/2$, $q=1-p$, $v=pq$. This preserves ρ, total conditional variance s and total inward displacement. The count probability, its phase and its original orientation are not reflected. Suppose the inward endpoints obey $a_0+a_1\le d$ with $0<d\le p$. The conditional third-moment inequalities and Cauchy's inequality give

$$\rho\ge I(p,s,d):=\frac{v(1-2v)+(3p-d)s+s^2/(qd)}{(v+s)^{3/2}}. \tag{127}$$

Indeed, writing $b=qs_0$, $c=ps_1$ and $b+c=s$, the absolute-cubic numerator is at least

$$v(1-2v)+3pb+3qc-a_0b-a_1c+\frac{b^2}{qa_0}+\frac{c^2}{pa_1}.$$

Its linear part is at least $(3p-d)s$, while its quadratic part is at least $(b+c)^2/(q(a_0+a_1))\ge s^2/(qd)$. Vanishing inward endpoints force the corresponding centered variance to vanish; those terms are handled before division. If $d=0$, then $s=0$ directly.

The derivative has the exact numerator

$$(v+s)^{5/2}I_s=v(6p-3p^2-d-3/2)+\left\{\frac{2p}{d}-\frac{3p-d}{2}\right\}s+\frac{s^2}{2qd}. \tag{128}$$

Consequently, on $p\in[l,u]\subset(0,1/2]$, the two gates $d\le l$ and $f:=6l-3l^2-d-3/2\ge 0$ imply

$$\rho\ge\rho_B(p):=\frac{1-2p(1-p)}{\sqrt{p(1-p)}}\ge\rho_B(u).$$

The last function is decreasing on this half-interval. If a previously acquired ceiling is $\rho \le R$, a point with

$$R^2 p(1-p) - (1-2p(1-p))^2 < 0$$

is impossible. This polynomial increases on $0 < p \le 1/2$, because its derivative is $(1-2p)(R^2+4-8p(1-p)) \ge 0$. Exact bisection may therefore raise the closed lower probability endpoint to a known infeasible point; retaining that endpoint is a conservative enlargement.

When $f > 0$ and $s \le S$, formula (128) also gives

$$I_s \ge \frac{l(1-l)f}{[u(1-u)+S]^{5/2}}.$$

Integration from zero supplies a new variance ceiling. Directed square roots and outward division make every finite update valid separately.

A useful weaker floor survives failure of the monotonicity gate. Assume still $0 < d \le l$ and set

$$a = (3/2 - 6l + 3l^2 + d)_+, \qquad B = 2l/d - 3u/2 \ge 5/4.$$

The only possible decrease of I is bounded below by $-(va - Bs)_+ / v^{5/2}$. Integrating this negative triangle gives

$$\rho \ge \rho_B(u) - \frac{a^2}{2B\sqrt{l(1-l)}}. \tag{129}$$

In particular, failure of the monotonicity gate must not be replaced by the unsupported assertion $\rho \ge \rho_B(p)$.

There is a second variance contraction which works with either sign of f. The global Taylor upper bound for $(v+s)^{3/2}$, combined with (127), gives the necessary inequality

$$Gs^2 + fs \le \epsilon\,[u(1-u)+S]^{3/2}, \tag{130}$$

where

$$\epsilon = (R - \rho_B(u))_+, \qquad G = \frac{1}{(1-l)d} - \frac{3(1-2l(1-l))}{8l(1-l)} \ge \frac{5}{8l(1-l)} + \frac{3}{4} > 0.$$

To see this, subtract $\rho_B(p)(v+s)^{3/2}$ from the lower numerator in (127), use the quadratic Taylor upper bound on the subtracted term, and then use $p \ge l$ and $\rho_B(p) \ge \rho_B(u)$. The positive part in ϵ is necessary when $f < 0$. The larger real root of (130), enclosed from above, is a valid new upper bound for s, even for negative f.

Finally suppose the original joint resource has the stronger form

$$d_{\text{actual}} \le d_0(1 - s/S_0), \qquad S_0 = \Lambda/n.$$

To test a proposed upper endpoint $a \in (0, S_0)$, assume $s \ge a$. Then $d_{\text{actual}} \le d_0(1 - a/S_0)$, so the preceding quadratic calculation can be repeated with this smaller inward bound. If its resulting upper bound is strictly less than a, the whole region $s \ge a$ is impossible. Otherwise the probe proves no lower bound on s. A finite sequence of such hypothetical tests is sound without any convergence assertion. Throughout it, the original S_0 remains in the joint budget; a separately improved noise ceiling clips that budget and does not replace its intercept.

O Finite acquisition for one fixed marginal

The count argument needs more than an upper bound for the defect. Its input includes the sign probability, the central gap, the two conditional variances, and bounds for several signed

moments. This appendix explains how the finite calculation acquires them together. The sample size stays fixed throughout each application.

Suppose $n \geq 5$ is a strict first failure, and take the attained positive closed-event ratio maximizer (P, H). Then $C = C_n > C_{\mathrm{E}}$, and its same-law predecessor loss is zero. First acquire support by Lemma G.1 or one of its separately proved specializations. This step uses the contact inequality and the Gaussian CDF, without higher moments. Only afterwards use

$$\rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad \beta = \mathbb{E}[X|X|], \quad m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad \mu_j = \mathbb{E}X^j.$$

The signed-moment and Fourier formulas in Appendix T then apply to this supported marginal. At small indices their numerical integrals use the literal value of n and, where needed, (116). No later-index approximation is substituted.

O.1 What a state asserts

A state is a finite list of simultaneous inequalities for this one law: closed intervals for ρ, β, γ, m, signed support radii, a score ceiling, and any already established raw, defect, or threshold bounds. Once a central gap is known, it also contains intervals for $p = \mathbb{P}(X > 0)$ and its two endpoints. The inequalities are necessary conditions. Their feasible set may contain points that do not represent any probability law, which is harmless for an upper bound.

The following rules preserve the actual parameter point. Intersect a coordinate interval with a newly proved necessary interval; replace an upper bound by the smaller of two valid upper bounds; or subdivide a closed interval into two closed children sharing their endpoint. A negative upper bound for a nonnegative quantity excludes a cell. A failed denominator, Taylor-domain, or positivity gate does not exclude the law: that proposed bound is unavailable, and the cell must receive another valid bound or remain unresolved.

O.2 Moment contractions with the correct sign radii

Write $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. The inequalities

$$m^2 \leq a_+ a_-, \qquad a_\pm^2 \leq m b_\pm, \qquad m \geq \frac{1+\beta^2}{2\rho}$$

retain the same m on both signs. They also hold when there is a zero atom. Thus solving them on a closed moment box is a valid contraction, but choosing a different mean separately for different tests is not an acquisition of one common state.

Here is the version of the moment matrices that permits different positive and negative support radii. On either sign write $Y = |X|$, $0 \leq Y \leq r$, and let p_s, m, a, b, c_4, c_5 be its unnormalized moments of orders zero through five. Positivity of the weights $r - Y$, $Y(r-Y)$, and $Y^2(r-Y)$ gives

$$\begin{pmatrix} rp_s - m & rm - a \\ rm - a & ra - b \end{pmatrix}, \quad \begin{pmatrix} rm - a & ra - b \\ ra - b & rb - c_4 \end{pmatrix}, \quad \begin{pmatrix} ra - b & rb - c_4 \\ rb - c_4 & rc_4 - c_5 \end{pmatrix} \succeq 0.$$

For positive denominators their determinant inequalities imply

$$\begin{aligned} p_s &\geq m/r + (rm - a)^2/[r(ra - b)], \\ c_4 &\leq rb - (ra - b)^2/(rm - a), \\ c_5 &\leq rc_4 - (rb - c_4)^2/(ra - b). \end{aligned}$$

A zero diagonal entry forces the corresponding off-diagonal entry to vanish. One may check that condition or discard the nonnegative correction, but one cannot divide by zero. In interval form a correction is bounded below by the square of a positive-part lower numerator divided by an upper positive denominator. The sign radius r is used only for its own class. Summing the first inequality over the two signs and using $p_+ + p_- \le 1$ can improve the common upper mean.

One further bound used at small indices strengthens the signed constant coefficient. Set $\delta = \gamma - \rho\beta$ and $Y = |X|$. Then

$$V := \mathbb{E}\{Y(Y^2 - \rho Y)^2\} = \mathbb{E}|X|^5 - 2\rho\mu_4 + \rho^3, \qquad \delta = \mathbb{E}\{Y \operatorname{sgn}(X)(Y^2 - \rho Y)\}.$$

Here $m > 0$ by centering and unit variance. Cauchy's inequality for the nonnegative measure $Y\,dP$ gives $\delta^2 \le 2mV$. Consequently

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\mathbb{E}|X|^5 \le \frac{\rho^3}{4} - \frac{(\gamma - \rho\beta)^2}{8m}. \tag{131}$$

This deduction can be combined with the two other deductions in Appendix T by taking their maximum. Their sum would spend the same nonnegative quantity more than once.

O.3 The first raw bound comes before defect feedback

Let $e = C - C_{\mathrm{E}} > 0$, $w = H^2/(2n)$ and $\phi_0 = \phi(0)$. The exact conserved identity, with nonnegative contact loss L, is

$$ne\rho + nC_{\mathrm{E}}\rho(1 - e^{-w}) + n\phi_0 e^{-w} D/6 + n^{3/2}L = \mathcal{R}.$$

Initially use only $D \ge 0$, acquired support and moments, and complete LOW and HIGH estimates. The first full cover must prove a raw upper bound $\mathcal{R} \le r < 2n/5$. In particular, it must not assume the defect bound that it proposes to export.

Since $C\rho > C_{\mathrm{E}} > 2/5$, rearrangement now gives

$$\phi_0 e^{-w}(\gamma + 3J)/6 = C\rho + \sqrt{n}L - \mathcal{R}/n > 0.$$

This is precisely the sign license needed for removing the Gaussian weight. Lemma R.1, used here at $N = n$, therefore proves

$$ne\rho + n\phi_0 D/6 + n^{3/2}L \le \mathcal{R}, \qquad e^{-w} \ge 1 - 5r/(2n), \qquad D \le 15000r/(997n).$$

Every later refinement may use these inequalities, always from a previous completed pass. The optional additional phase curvature is (138); it is included once in the budget used to update D. The raw remainder and that smaller budget have different meanings.

The signed moment in the actual identity remains γ, even if an auxiliary Fourier estimate is evaluated after reflection. In particular, when $\gamma \ge g_0$ and $\phi \in [\phi_l, \phi_u]$, the valid curvature lower bound is

$$C\rho + \tfrac{1}{2}\min\{\phi_l g_0, \phi_u g_0\}.$$

Section T.7 proves the coherent corner rule for this expression. Complete score and moment corners are taken before maximizing over H or completing the square. If two complete Fourier estimates are available, their resulting upper bounds may be minimized; mixing their separate coefficients is not justified.

O.4 From the acquired gap to the count input

With $\sigma = \sqrt{10}$, the bound $D < \sigma\rho - \gamma$ excludes zero from the support, because otherwise $J \le \rho$. Both signs have positive probability, so a genuine central gap $(-A, B)$ follows. The signed endpoint, central-integral and outer-gap bounds of Appendix T can now refine the state. In particular, the complete central rectangle is bounded by its tensor Bernstein coefficients, using the lower mean for a negative coefficient and the upper mean for a positive coefficient.

The joint inequality (58) then acquires the triangle (59). Its remaining defect budget is shared by inward displacement and conditional variance. All refinements of conditional moments price their entire nonnegative linear cost under this single budget, as in Lemma R.2. This supplies the actual common-law hypotheses for the count calculation and the optional filters in Appendix N.

The induction is finite. At each complete pass, every actual parameter point belongs to an initial cell and remains in a child or in a cell with a proved exclusion. A declared grid with sizes $(N_\rho, N_\beta, N_\gamma, N_m)$ must account for exactly $N_\rho N_\beta N_\gamma N_m$ cells; skipping an excluded block must account for its full multiplicity. An export to the probability panel $[j/100, (j+1)/100]$ takes maxima of upper bounds and minima of lower bounds over every active cell touching that panel. In particular, an endpoint at $j/100$ belongs to both adjacent panels. This is why the export range is

$$\lceil 100 p_l \rceil - 1 \ \le j \le \ \lfloor 100 p_u \rfloor,$$

intersected with $\{0, \dots, 99\}$. An empty intersection or a strict necessary-inequality contradiction can close a cell; an incomplete calculation cannot.

The resulting proof is an induction on finite enclosures, not a convergence claim about an iterative search. Its last obligations are concrete: the complete covers must pass, each resource must consume its proved initializer, and each count contract must be implied by the resource for the same moment and probability panel. These are the joins checked by the accompanying reproduction programs.

O.5 The actual-index acquisitions at 35 through 39

These five steps use the actual index in every INITIAL and resource bound. They therefore need no transfer from a smaller evaluation index. For $35 \le n \le 38$, Lemma 8.4 supplies $C_n < 41/100 + 4177/(13824n)$. This is below all four native INITIAL ceilings in Section 12 and also below $423493/10^6$, the smaller ceiling used by the local calculations at 35. The sharp predecessor gives contact allowance zero; the weaker allowances $19/100$ and $63/340$ retained in the index-35 support and resource programs are consequently valid. At 39, use the separately proved positive-allowance step to obtain the smaller ceiling $41449/10^5$.

The global INITIAL covers have respectively $10, 16, 12, 15, 12$ finite refinement passes, followed by a full 100-panel export. Each pass retains one law's earlier bounds, proves support before using its higher moments, and rounds upper bounds upwards and lower bounds downwards. The global support bounds are $39/10$ at 35 and $19/5$ at 36–39. At 35, the separate support calculation actually gives $96/25 < 39/10$. Local refinements use separately checked smaller radii.

At 35, four short INITIAL refinements cover the intervals $[.29, .34]$, $[.34, .39]$, $[.39, .45]$ and $[.45, .57]$; they use respectively $4, 3, 3, 2$ passes. Bounds shared by a whole interval are the weakest bounds over its still possible panels: the maximum of their upper bounds and the minimum of their lower bounds. These are intersected with the already acquired bounds for that interval. Thus no moment from one law is assigned to another. The resulting count interfaces cover $P_{31}, \dots, P_{48}$. Three further alternating INITIAL/resource calculations cover $P_{16}, \dots, P_{28}$: they exclude $P_{16}, \dots, P_{22}$ and produce count interfaces on $P_{23}, \dots, P_{28}$. The

first short interval also supplies the two interfaces on P_{29}, P_{30}. Finally the global source, supplemented by the last short interval on $P_{49}, \dots, P_{56}$, excludes every panel in $P_0, \dots, P_{15}$ and $P_{49}, \dots, P_{99}$. The union consists of 74 excluded panels and 26 count panels, with every probability endpoint included.

At 36, a two-pass localized INITIAL calculation on $[.30, .42]$ replaces the global source only on $P_{30}, \dots, P_{41}$; its proved one-sided support radii are 3 and $37/10$. All other panels retain their global source. At 37–39, the global exports are used directly. The full resource traversals leave respectively $P_{24}, \dots, P_{48}$, $P_{24}, \dots, P_{47}$, $P_{25}, \dots, P_{47}$ and $P_{26}, \dots, P_{47}$. Every other panel reaches an explicit moment, support or budget contradiction during the finite traversal.

At 35 the phase budget also retains the actual excess. Since $C > C_{\mathrm{E}}$ and $C_{\mathrm{E}} < 20487/50000$, its lower charge is $35 \max\{0, C - 20487/50000\}\rho$. For each fixed phase, the resulting upper budget is affine in C on either side of this breakpoint, and affine in ρ for fixed C. Its maximum on the score–moment rectangle therefore occurs among the six corners obtained from the two score endpoints, the breakpoint, and the two moment endpoints. Subtracting this charge before the phase comparison is justified by the same conserved identity used in Appendix O.6.

Each surviving row supplies upper bounds for the gap, noise, amplitudes and the two signed phases, together with one simultaneous affine defect lower bound. The count contract weakens each of these bounds in the valid direction. For the phase envelopes, every acquired corner polynomial is bounded above on the whole signed interval by at least one count-certificate corner polynomial. The comparison checks the endpoints and, when it lies inside the interval and the difference is convex, its vertex. Taking maxima therefore preserves the inequality. These comparisons connect the freshly acquired same-law rows to the $26+25+24+23+22 = 120$ actual-index count forests.

O.6 Why the floor calculations retain their force at later indices

The reuse of a finite resource calculation needs a statement about each analytic bound, rather than monotonicity of the optimal constants C_n. Let N be its fixed evaluation index and $n \geq N$ the actual index. Fix one contrary maximizing marginal and all of its acquired moments, endpoints and signed phase constraints. Throughout a finite refinement sequence retain

$$\mathcal{R}_n \leq r, \qquad nD \leq Nd. \tag{132}$$

Here $\mathcal{R}_n$ is the complete raw Fourier remainder, not one selected summand. The smaller inequality $D \leq d$ is available for index-free moment and gap estimates, but does not replace the second inequality in (132).

The index-free operations are intersections of necessary moment intervals, signwise Cauchy bounds for $\mu = \mathbb{E}X_+$, central-gap bounds, and physical endpoint prices. For example,

$$\mu^2 \leq p_u(1+\beta_u)/2, \qquad \mu^2 \leq (1-p_l)(1-\beta_l)/2, \qquad \mu^2 \leq \max_{p \in [p_l, p_u]} p(1-p)$$

do not depend on n. If a proved gap inequality gives $6\mu AB \geq P > 0$, then $(A+B)^2 \geq 2P/(3\mu_u)$, by $4AB \leq (A+B)^2$. Such an improvement requires the already acquired gap and positive mean. It cannot be used to manufacture a missing initial gap.

For the six-piece high-frequency window bound of Lemma S.5, keep its admissible event parameters and window fixed. Its nonconstant index factors are positive sums and products of

$$\sqrt{n}e^{-cn} \quad (2Nc > 1), \qquad n^{-1}, \qquad e^{-knb^2}/\sqrt{n},$$

and of the decreasing function

$$r(x) = \frac{x+2}{x^2+4x+2}, \qquad r'(x) = -\frac{x^2+4x+6}{(x^2+4x+2)^2} < 0.$$

Thus the ordinary first piece, peak-core correction, peak tails and zero-peak tail decrease. The normal comparator has the form $\sqrt{n}e^{-n/8}\{2r(n/8) + g_c/3\}$, and decreases for $N \geq 5$. The improved first piece has an additional factor $r(n\ell/4)$ with $\ell > 0$, and uses its own positive damping and $2Nc > 1$ gates. For the optional improved zero piece, the derivative sign of $f(x) = \sqrt{x}e^{-x}r(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4.$$

This is $-11/128$ at $x = 1/4$ and is strictly decreasing for $x > 0$. Hence the optional bound decreases whenever the native gate $Nkb^2 \geq 1/4$ holds. All these statements retain the original window conditions $0 < a \leq 1$, $m_e > 0$, $0 < d_0 \leq 12$, $1 \leq M \leq 111/10$, $0 < w \leq 1/4$, and positive sine slope. A minimum of complete valid window bounds stays valid; one need not assume that the minimizing window remains the same at every index. Rational numerical upper bounds are taken at the fixed floor N.

The low-frequency envelopes have index factors

$$(n/(n-1))^{3/2}, \quad (n/(n-1))^{3/2}/(n-1), \quad a_n^{-j}, \quad b_n^{-j}, \quad n^{-1/2},$$

where $a_n = (n-1)/(2n)$, $b_n = (45-M)(1-2/n)/96$ and $j > 0$. The first two decrease and a_n, b_n increase positively on the retained moment domain. Sums and products with nonnegative fixed numerators therefore decrease. This applies to the full remainders $e/(2a_n^2) + B^2/(2b_n^3)$ and $L_2/\sqrt{n}$. Every paired-moment eligibility condition remains necessary; the index comparison cannot repair a failed moment-rectangle hypothesis.

The phase estimate supplies

$$\phi(H/\sqrt{n}) \in [\phi_{0,l}(1 - 5r/(2N)), 2/5]$$

under its raw-remainder and positivity gates. The actual lower endpoint increases as n increases. Similarly, for $r, D_l \geq 0$,

$$r/n - \phi_{0,l}(1 - 5r/(2n))D_l/6$$

increases with $1/n$, so evaluation at N is an upper bound. Put $g_0 = (79/25)\rho - 3 - d$ and $t = (d - 4/25)_+$. The general phase curvature has derivative $(g_0 + t)/2 \geq 0$ with respect to the lower density: if $d \leq 4/25$, use $g_0 \geq 4/25 - d$; otherwise $g_0 + t = (79/25)(\rho - 1)$. The signed paired curvature is

$$C\rho + \tfrac{1}{2}\min\{\phi_l g_0, (2/5)g_0\}.$$

It increases with ϕ_l for $g_0 \geq 0$ and is constant in ϕ_l for $g_0 < 0$. This proves the required direction with both signs retained. The positive additions and all native curvature gates are checked before using the phase maximum.

For the sharp contrary score, $C > C_{\mathrm{E}}$ and $C_{\mathrm{E}} < 40974/10^5$ give

$$n(C - C_{\mathrm{E}})\rho \geq N \max\{0, C - 40974/10^5\}\rho_l.$$

The breakpoint $C = 40974/10^5$ is therefore included in the score cover. On either side the charge is affine in C; the same coherent (C, ρ) corner argument bounds the remaining phase expression. A charged phase budget B with $nD \leq B/\kappa$, $\kappa = 997/15000$, exports $d = B/(\kappa N)$. A negative budget excludes the law before division. Maximum over a complete

cover, upward rounding and minimum with a previous valid entry all preserve (132). Induction over the finite saved passes proves the invariant without a limiting argument. A positive K/n calculation is separate: its $K\rho$ charge must be retained and is not covered by this sharp-target simplification.

Finally, if the physical affine resource is

$$D \geq b + gd_{\mathrm{in}} + (N/n)h_N\lambda_n, \qquad \lambda_n = ns, \quad b, g, h_N \geq 0,$$

then (132) gives

$$(n/N)(b + gd_{\mathrm{in}}) + h_N\lambda_n \leq d, \qquad b + gd_{\mathrm{in}} + h_N\lambda_n \leq d.$$

The original joint intercepts therefore bound the actual noise. An equivalent directed contract records $D \leq Nd/n$, noise price Nh_N/n, and actual phase normalization $15000/(997n)$. The actual index remains in every binomial mass and count probability. The present applications are $N = 25$, $26 \leq n \leq 30$, and $N = 31$, $32 \leq n \leq 34$, with a separate support calculation for every actual index and every support state used by the refinements.

Extracting a phase bound from the whole charged envelope.

The two local B1 refinements at floor 31 also tighten the signed phase intervals. Suppose that, for either sign and $h = |H| \geq 0$, the same acquired law satisfies

$$0 \leq D \leq c_n \max_j\{A_j + F_j h - Q_j h^2\}, \qquad c_n > 0, \quad Q_j > 0 \quad \text{for every } j.$$

The maximum is over the complete charged corner family; its members cannot be selected independently of the law. At least one corner polynomial is nonnegative at the actual h. Its discriminant $\Delta_j = F_j^2 + 4Q_jA_j$ is therefore nonnegative, and solving that quadratic inequality gives

$$h \leq \frac{F_j + \sqrt{\Delta_j}}{2Q_j}.$$

Choose a nonnegative rational enclosure u_j with $u_j^2 \geq \Delta_j$ whenever $\Delta_j \geq 0$, and set

$$r_j = \begin{cases} 0, & \Delta_j < 0, \\ \max\{0, (F_j + u_j)/(2Q_j)\}, & \Delta_j \geq 0. \end{cases}$$

A negative discriminant cannot supply the nonnegative corner, so $h \leq \max_j r_j$. Intersect this bound with the previously acquired bound for the same sign. The construction is valid at a double root and at $h = 0$. Its checker requires every $Q_j > 0$; it has no unchecked linear or constant branch.

The fixed floor-31 normalization is $15000/(997 \cdot 31)$. Under the weighted transfer, the actual normalization becomes $c_n = 15000/(997n)$, which stays positive. The implication from $D \geq 0$ to the nonnegativity of at least one charged corner is therefore unchanged. Separate maxima of the roots of the minus and plus families give the two signed phase caps used by the actual-index count contracts.

Transporting the physical variance refinement.

One resource step uses the bounded conditional supports to obtain $s \leq B\delta$, where s is the rescaled conditional variance, $\delta = d_{\mathrm{in}}$, and

$$B = \max(O_0, O_1)\max(p_u, 1 - p_l) > 0.$$

Here O_0, O_1 are outward support offsets in the same conditional coordinates. If the acquired joint plane is $\delta/d_0 + \lambda_n/L \le 1$, with $\lambda_n = ns$, then

$$\lambda_n \le \frac{1}{L^{-1} + (nBd_0)^{-1}}.$$

The physical variance parameter is $\theta_n = \lambda_n/(np(1-p) + \lambda_n)$. Writing

$$v_l = \min_{p \in [p_l, p_u]} p(1-p) > 0$$

gives

$$\theta_n \le \frac{1}{1 + nv_l/L + v_l/(Bd_0)} \le \frac{1}{1 + Nv_l/L + v_l/(Bd_0)} \quad (n \ge N).$$

The last quantity is precisely the floor calculation's variance refinement. Downward rounding of the support scale enlarges O_0, O_1; upward rounding of this last bound and intersection with the previous physical θ bound preserve its direction. This transports the bound for θ, rather than treating the floor's diagnostic λ_N as an actual-index bound for λ_n. The original joint intercept L remains the one used in the affine plane. For the selected floor-25 first-band resource there are 193 such saved refinements, each checked at every actual index 26–30.

P The fixed-index positive allowance at 39

The acquisition used by the sharp argument at 39 requires the intermediate cap $C_{39} < .41449$. It is sufficient first to exclude

$$C_{39} \ge C_{\mathrm{E}} + K/39, \qquad K = 741/4000,$$

because $.40974 + K/39 = .41449$ and $C_{\mathrm{E}} < .40974$. This is a separate fixed-index comparison. Its inputs are $C_{39} < .4226$, $39(C_{39} - C_{\mathrm{E}}) < .50$, predecessor loss at most .33, and both deleted score bounds at most .4253. Sharp predecessors and the one-step estimate supply these inputs. Indeed $\sqrt{39/38} < 51/50$ and $\sqrt{38} > 6$ give

$$39(C_{39} - C_{\mathrm{E}}) < \frac{39}{38}\left(\frac{41}{200} + \frac{17}{250} + \frac{17}{1250}\right) = \frac{55887}{190000} < .3,$$
$$C_{39} < .41 + .3/39 < .4226.$$

Under the contrary hypothesis the predecessor loss is zero, and the two sharp deleted bounds are below .4253. No positive-allowance result at another index is used.

The support polynomial in Section Q.3 is evaluated with

$$(N, L, K, q, R) = (39, 39, 741/4000, .33, 3.84).$$

Its derivation applies at this index, including the scale estimate valid for $n \ge 36$. Subsequent radius contractions use the same fixed-index polynomial. Central bounded-threshold contractions use the positive-error series with excess allowance .50 and loss .33. The initializer covers all signed moment cells and all 100 probability panels. It retains the lower score $C \ge .4095 + K/39$ and the charge $K\rho_l$, because $39(C - C_{\mathrm{E}})\rho \ge K\rho_l$. This charge is part of the actual resource budget and is kept through the coherent score corners and the later refinements.

The confinement proof of Section U applies with its actual parameters $N = 39$, $J_0 = 37$ and deletion factor strictly larger than $\sqrt{\max\{(39/38)^3, 39/37\}}$. The acquired initial strip is $(-w, 1+w)$ with $w \in \{.55, .575\}$; the conditional cubic coefficient is 30 on the left and 24

centrally. In the proof replace .55 by w in the cubic bound, interval endpoints, width gates and upper contradiction $L_g - wU_+ > \mathcal{E}$. The gate

$$c_\rho^2[l(1-l)]^3 > [3(1-l)+w]^2, \qquad [l,h] \subset (0,1/2),$$

licenses the cubic bound. Together with $d_0 < w$, the strip and the inward endpoints give $|U_i| < w$. All positive slopes, inverse denominators and signed budget gates are checked at the actual index. After each contraction the newly proved amplitude is the next strip. Thus this is the same confinement proof with explicit parameters, without extending the fixed-index statement to another integer.

The local comparison then consumes 78 probability rows covering $[151/500, 229/500] = [.302, .458]$, with all the probability and moment complements excluded by the preceding acquisition. Its 624 signed profile cells retain both signs of H and four noise bands. The whole-count part has 90 signed rows. Its closed phase cover is: nearest-count comparison on $(0, .05]$ and $[.05, .08]$ at every phase; nearest count for $|u| \geq .19$ and whole count for $|u| \leq .19$ on $[.08, .10]$; the corresponding boundary .27 on $[.10, .12]$; and whole count at every phase above .12. Each band is intersected with its own acquired noise ceiling.

Here the target is $C < C_B + K/39$, so the scalar tests differ from the zero-allowance tests in Section V. All of them follow from the same identity (200). Write $b = \sqrt{39}\pi_k$ and retain the verified two-sided bound $m \leq b \leq 1$. A nearest-count bracket D may now have either sign; consequently

$$bD \geq \min\{mD, D\}.$$

If the repair is at most $r\lambda/39$, with $r \geq 0$, a noise band $[a, b_\lambda]$ is excluded by the intermediate-noise test

$$\min\{mD, D\} - rb_\lambda/39 + K/39 > 0.$$

Indeed the identity gives $\rho_B(C - C_B) < K/39$. The elementary inequality $\rho_B \geq 1$ follows from $(1-2v)^2 - v = (1-v)(1-4v) \geq 0$, $0 < v \leq 1/4$. Thus $C < C_B + K/39 \leq C_{\mathrm{E}} + K/39$.

At arbitrarily small positive noise the rescaled quartic and tail estimates give the same inequality per unit λ. The tiny-noise test adds the allowance

$$\frac{K}{39\max\{.05, \Lambda\}}$$

to that per-noise reserve. Multiplying by $0 < \lambda \leq \min\{.05, \Lambda\}$ again gives a bound strictly below $K/39$. No positive lower noise cutoff is inserted.

For the whole-count test, write

$$S(\lambda) = \max\{mB(\lambda), B(\lambda)\} + I(\lambda)/39.$$

Both the bracket and the inverse correction may have either sign; the first maximum uses $m \leq b \leq 1$, while the division by 39 uses the actual integer. Hence $\rho_B(C - C_B) \leq \lambda S(\lambda)$. The saved gate on $[a, b_\lambda]$ is

$$K\rho_{B,l}/39 - b_\lambda S(\lambda) > 0,$$

where $\rho_{B,l}$ is a positive lower bound for ρ_B. If $S < 0$ the conclusion is immediate; otherwise it follows by $\lambda \leq b_\lambda$ and $\rho_B \geq \rho_{B,l}$. The formula is convex in positive λ, since its only nonaffine terms are positive multiples of $(\alpha\lambda + \beta)^{-1}$ and $1/\lambda$, and maxima of convex functions preserve convexity. Both endpoint gates therefore cover the whole noise cell.

Finally, zero noise makes the marginal exactly Bernoulli and gives $C \leq C_{\mathrm{E}} < C_{\mathrm{E}} + K/39$. The weak all-probability Bernoulli theorem suffices; no finite binomial slack table is needed here. Once the nine original arithmetic actions and their full source interfaces are verified, these strict comparisons exclude the hypothesized positive allowance and supply the cap required by the sharp39 initializer. The positive-allowance and sharp39 checks remain distinct numerical obligations.

How to read these estimates

The main proof uses one attained ratio maximizer and the contact inequality at coefficient $C = C_n$. The first section bounds its support before any higher-moment calculation. The resource and Fourier sections then acquire simultaneous information about that same law. The local comparison section uses only the acquired information and the actual iid binomial count. The final refinements retain a joint resource, correct the sign of the cubic term, and use the signed Gaussian phase. They provide the analytic licenses for the finite-index implementation. Numerical domain coverage and arithmetic acceptance remain separate from these analytic proofs.

The contact interface used below is precisely the normalized fixed-target contact inequality at $c = C_n$, supplied by the main manuscript's ratio attainment corollary. It is stated explicitly before its use. The Bernoulli theorem and the finite moment caps are the only external inequality inputs to this selected local analytic argument.

Q One support estimate, uniform in the threshold

Q.1 The scale correction

Write $\phi_0 = \phi(0)$. For $n > 1$ put $s_n = \sqrt{n/(n-1)}$ and $e_n = n(s_n - 1) - 1/2$.

Lemma Q.1. *For every integer $n \geq 36$, one has $0 < e_n < .384/n$.*

Proof. Put $x = 1/n$ and $c = .384 = 48/125$. Direct expansion gives

$$(1-x)(1+x/2+cx^2)^2 - 1 = x^2\{(2c-3/4) - (c+1/4)x + (c^2-c)x^2 - c^2x^3\}.$$

The bracket is decreasing for $x \geq 0$, since $0 < c < 1$. At $x = 1/36$ its exact rational value is $823/4050000 > 0$. Squaring positive quantities gives $s_n < 1 + x/2 + cx^2$ for $0 < x \leq 1/36$. Similarly $(1-x)(1+x/2)^2 - 1 = -3x^2/4 - x^3/4 < 0$, which gives the lower bound. □

Q.2 A signed Gaussian envelope

Define, for every real H, t and $n > 1$,

$$\mathcal{H}_n(H,t) = n^{3/2}\{\Phi((H-t)/\sqrt{n-1}) - \Phi(H/\sqrt{n})\} + nt\phi(H/\sqrt{n}) + \tfrac{H}{2}(t^2-1)\phi(H/\sqrt{n}).$$

Write $v = n-1$, $s = \sqrt{n/v}$, $u = |t|$, and

$$A = \frac{u^3 - 3u}{6}, \quad B = \frac{u^4 - 6u^2 + 3}{24\sqrt{v}}, \quad D = \frac{u^5 - 15u}{120v}.$$

Lemma Q.2 (Fifth-order identity with a uniform finite remainder)**.** *If $u > \sqrt{3}$, then, with $z = H/\sqrt{v}$,*

$$\frac{\mathcal{H}_n(H, \pm u)}{s^3} = \pm A(1-z^2)\phi(z) + B\phi^{(3)}(z) \mp D\phi^{(4)}(z) + R,$$

where the sign of the remainder need not be fixed and

$$|R| \leq E_v^{(5)}(u) := \frac{51u^6}{14400v^{3/2}} + \frac{51(3u^2-2)}{960v^{3/2}} + \frac{u}{8v^2}.$$

This statement is uniform over every real H.

Proof. Let $F_w(x) = \Phi(x/\sqrt{w})$, with primes denoting x derivatives. The heat identity $\partial_w F_w = F_w''/2$ and ordinary Taylor's theorem give, with unlabelled derivatives at (v, H),

$$\begin{aligned} F_{v+1} &= F_v + F_v''/2 + F_v^{(4)}/8 + R_0, & |R_0| &\le \sup_{w\in[v,v+1]} \|F_w^{(6)}\|_\infty/48, \\ F_{v+1}' &= F_v' + F_v'''/2 + F_v^{(5)}/8 + R_1, & |R_1| &\le \sup_{w\in[v,v+1]} \|F_w^{(7)}\|_\infty/48, \\ F_{v+1}'' &= F_v'' + F_v^{(4)}/2 + R_2, & |R_2| &\le \sup_{w\in[v,v+1]} \|F_w^{(6)}\|_\infty/8. \end{aligned}$$

Expand $F_v(H-t)$ through degree five, with remainder at most $u^6\|F_v^{(6)}\|_\infty/720$. In

$$\frac{\mathcal{H}_n}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF_{v+1}'(H) - \frac{t^2-1}{2}F_{v+1}''(H),$$

the first three derivative coefficients cancel. The next three are

$$\left(\frac{t}{2} - \frac{t^3}{6}\right)F_v''' + \frac{t^4 - 6t^2 + 3}{24}F_v^{(4)} + \left(\frac{t}{8} - \frac{t^5}{120}\right)F_v^{(5)}.$$

The remaining absolute value is bounded by

$$\left\{\frac{u^6}{720} + \frac{1}{48} + \frac{u^2-1}{16}\right\}\sup_{w\in[v,v+1]}\|F_w^{(6)}\|_\infty + \frac{u}{48}\sup_{w\in[v,v+1]}\|F_w^{(7)}\|_\infty.$$

Fourier inversion of the Gaussian, differentiated under its absolutely integrable Fourier transform, gives

$$\|\phi^{(k)}\|_\infty \le \frac{1}{\pi}\int_0^\infty y^k e^{-y^2/2}\,dy.$$

Integration by parts yields $\|\phi^{(5)}\|_\infty \le 8/\pi < 2.55$ and $\|\phi^{(6)}\|_\infty \le 15\phi_0 < 6$. Since $F_w^{(j)} = w^{-j/2}\phi^{(j-1)}$, the displayed remainder becomes $s^3E_v^{(5)}(u)$ after multiplication by $n^{3/2}$. The three retained derivative coefficients give the identity. □

Lemma Q.3 (A lower tangent bound for the fourth Gaussian derivative)**.** *For every real z,*

$$\phi^{(4)}(z) \ge \phi_0(3 - 15z^2/2).$$

Proof. The same Gaussian Fourier integral gives $\phi^{(4)}(z) = \pi^{-1}\int_0^\infty y^4 e^{-y^2/2}\cos(yz)\,dy$. Use the globally valid inequality $\cos w \ge 1 - w^2/2$ inside this integral. Its fourth and sixth Gaussian moments are $3\phi_0$ and $15\phi_0$, respectively. □

Lemma Q.4 (Uniform signed fifth-order envelope)**.** *Suppose $u > \sqrt{3}$ and $A, B, D \ge 0$. If $W = A - 15D/2 > 0$, put*

$$T = A - 3D + \frac{9B^2}{4W}.$$

If $T \ge A/2 + 2B + 3D$, then, for every real H and either sign,

$$\mathcal{H}_n(H, \pm u) \le s^3\{\phi_0 T + E_v^{(5)}(u)\}.$$

Proof. For positive $t = u$ and $|z| \le 1$, divide the retained terms by ϕ_0 and use the preceding tangent bound to obtain

$$A - 3D - (A - 15D/2)z^2 + 3B|z| \le A - 3D + 9B^2/(4W).$$

Here $(1-z^2)\phi(z) \le (1-z^2)\phi_0$ and $|\phi^{(3)}(z)| \le 3|z|\phi_0$ on this interval. For $|z| \ge 1$ the first term is nonpositive, and the Fourier bounds $\|\phi^{(3)}\|_\infty < 2\phi_0$ and $\|\phi^{(4)}\|_\infty \le 3\phi_0$ give $2B+3D$. For negative $t = -u$, the first term has global upper bound $2e^{-3/2}A\phi_0 < A\phi_0/2$, and the other terms give $(2B+3D)\phi_0$. The hypothesis on T covers both exterior cases. The inequality $2e^{-3/2} < 1/2$ follows from four positive terms of the exponential series at $3/2$. Finally add the uniform remainder. □

Lemma Q.5 (A global Gaussian remainder bound)**.** *For every integer $n \ge 2$ and all real H, t,*

$$|\mathcal{H}_n(H,t)| < \frac{|t|^3}{15} + \frac{|t|}{5} + \frac{1}{4} < \frac{|t|^3}{15} + \frac{|t|}{5} + \frac{1}{2}.$$

Proof. Write $F_w(x) = \Phi(x/\sqrt{w})$, $v = n-1$ and $y = H - t$. The identity defining $\mathcal{H}_n$ is

$$n^{-3/2}\mathcal{H}_n(H,t) = F_v(y) - F_n(H) + tF_n'(H) - \frac{t^2-1}{2}F_n''(H).$$

Taylor expansion of $F_n(y)$ at H through degree two has remainder at most $\phi(0)|t|^3/(6n^{3/2})$, since $\|\phi''\|_\infty = \phi(0)$. After this expansion the remaining correction is

$$\frac{1}{2}\{F_n''(H) - F_n''(y)\} + \frac{1}{2}\int_v^n \{F_n''(y) - F_w''(y)\}\, dw.$$

Here we used $\partial_w F_w = F_w''/2$. The first term is at most $\phi(0)|t|/(2n^{3/2})$ in absolute value. Applying the heat identity once more bounds the second term by

$$\frac{1}{4}\int_v^n \int_w^n |F_r^{(4)}(y)|\, dr\, dw \le \frac{\|\phi'''\|_\infty}{8v^2} \le \frac{1}{4\pi v^2}.$$

The last inequality follows by differentiating the absolutely convergent Gaussian Fourier integral, which gives $\|\phi'''\|_\infty \le 2/\pi$. Multiply by $n^{3/2}$ and use $\phi(0) < 2/5$. The function $x^{3/2}/(x-1)^2$ decreases for $x > 1$, because its logarithmic derivative is $3/(2x) - 2/(x-1) = -(x+3)/(2x(x-1)) < 0$. Hence, for $n \ge 2$,

$$\frac{n^{3/2}}{4\pi(n-1)^2} \le \frac{\sqrt{2}}{2\pi} < \frac{1}{4},$$

using $\sqrt{2} < 3/2$ and $\pi > 3$. This proves the stronger bound and therefore the exact ray majorant used by the original checkers. □

Q.3 Finite bands and the exact support polynomial

For integers $36 \le N \le L$, use the simultaneous endpoint bounds

$$B_+ = \frac{u^4 - 6u^2 + 3}{24\sqrt{N-1}}, \quad D_+ = \frac{u^5 - 15u}{120(N-1)}, \quad D_- = \frac{u^5 - 15u}{120(L-1)}.$$

For an infinite index range, set $D_- = 0$ instead. Put

$$W_* = A - 15D_+/2, \qquad T_* = A - 3D_- + \frac{9B_+^2}{4W_*}.$$

The proof of Lemma Q.4 applies uniformly on the whole band if $W_* > 0$ and $T_* \ge A/2 + 2B_+ + 3D_+$. In particular the favorable term is frozen at the upper index endpoint, and the adverse terms at the lower one. No monotonicity of a signed coefficient is assumed.

Let (P, H) be an attained positive closed maximizing pair, standardized to mean zero and variance one. Write $C = C_n \geq C_E$, $\rho = \mathbb{E}|X|^3$, $\beta = \mathbb{E}[X|X|]$, and $d = n(C_{n-1} - C_n)_+$. The finite contact/predecessor bound is

$$\rho I(t) \leq \mathcal{H}_n(H, t) + C\{-|t|^3 + 3\rho t^2/2 + 3\beta t + \rho e_n\} + \rho s_n d,$$

with $I = 0$ on the entire closed support. Suppose also, for a finite band, that $C_n - C_E \geq K/n$, where $K > 0$. The finite moment bounds of [15, p. 125], namely $C\rho \leq .3031(\rho + .646)$ and $C\rho \leq .3322(\rho + .429)$, give

$$c = .4095 + K/L < C_n, \qquad \rho < r = \min\left\{\frac{.3031(.646)}{c - .3031}, \frac{.3322(.429)}{c - .3322}\right\}.$$

For the unrestricted nonnegative case put $K = 0$, $c = .4095$. The sign-moment inequality $\rho \geq (1 + \beta^2)/\sqrt{1 - \beta^2}$ supplies $|\beta| < b$ whenever $(1 + b^2)^2 > r^2(1 - b^2)$ and $0 < b < 1$. All these statements concern the same current law. The lower endpoint cannot replace L in K/L.

When a preceding resource enclosure supplies sharper actual bounds $\rho \leq r$ and $|\beta| \leq b$, these may instead be used in the same polynomials. Their validity is the only extra premise.

Use rational $\ell^2 \leq N - 1$ and $S^2 > N/(N - 1)$. For $u = |t|$ define

$$Q(u) = -u^3 + 3ru^2/2 + 3bu + .384r/N.$$

First check $Q < 0$, before replacing C by its lower bound c. Replace $1/\sqrt{N - 1}$ by $1/\ell$ in the adverse remainder and in the term $2B_+$ of the domination gate. Write E_* for that rational remainder. Strict negative contact for every n in the band follows from the polynomial inequality

$$\frac{2}{5}S^3\left\{(A - 3D_-)W_* + \frac{(u^4 - 6u^2 + 3)^2}{256(N - 1)}\right\} + S^3E_*W_* + \{cQ + rSq\}W_* < 0.$$

The second polynomial gate is $W_* > 0$. The third is

$$\{A/2 - 3D_- - 3D_+ - 2B_+\}W_* + \frac{(u^4 - 6u^2 + 3)^2}{256(N - 1)} > 0.$$

This is the cleared-denominator version of the domination hypothesis. The bound on T_* is positive, so the replacement $\phi_0 < .4$ and $s_n < S$ both have their correct direction.

The exact kernel tests these polynomials on all adjacent closed intervals of length .05 from R to 4.5. Positive Bernstein coefficients prove their signs on every point of each cell, including endpoints. On the rays $u \geq 4.5$ use Lemma Q.5, which gives the same bound $|\mathcal{H}_n(H, t)| \leq u^3/15 + u/5 + 1/2$. The resulting cubic and its first three derivatives are strictly negative at 4.5, as is the coefficient Q before replacing C by c. This excludes the whole rays. Thus, if $d \leq q$, actual support is in $(-R, R)$, including exclusion of arbitrarily small exterior atoms and exterior accumulation points.

Q.4 The support rows used in this proof

For the all-index local argument, take $K = 0$, $R = 3.84$, $q = .19$ on

$$[45, 49],\ [50, 59],\ [60, 89],\ [90, 134],\ [135, 199],\ [200, \infty).$$

Only indices at least 47 are used. The last row uses $D_- = 0$; there is no upper index restriction. The exact checker reconstructs all closed u cells and the two exterior rays. Missing an index band or the infinite row is an acceptance failure.

Q.5 One bounded-threshold series

Put $c_k = (-1)^k/[2^k k!(2k+1)]$, and let $(x)_j$ denote the rising factorial. Define the following rational polynomials:

$$P_j(H,t) = \sum_{k=0}^{j+1} \frac{c_k (k+1/2)_{j+1-k}}{(j+1-k)!}(H-t)^{2k+1} - c_{j+1}H^{2j+3} + \frac{(-1)^{j+1}tH^{2j+2}}{2^{j+1}(j+1)!} + \frac{(-1)^j H^{2j+1}(t^2-1)}{2^{j+1}j!}.$$

In particular,

$$P_0 = (t^3-3t)/6, \qquad P_1 = \frac{3t-t^3}{4}H^2 + \frac{t^4-6t^2+3}{8}H + \frac{-t^5+10t^3-15t}{40}.$$

Fix $h > 0$, $E(x) = (193/71)^{\lceil x \rceil}$, and

$$B_r = \frac{7(h+r)}{6}E((h+r)^2/6) + \left(h + r + \frac{h(r^2+1)}{8}\right)E(h^2/8).$$

Lemma Q.6. *For every $n \geq 4$, $|H| \leq h$ and $|t| \leq r$,*

$$\left| \mathcal{H}_n/\phi(0) - \sum_{j=0}^{5} P_j/n^j \right| \leq 4^7 B_r/n^6.$$

Proof. Put $z = 1/n$. Integration of the Gaussian exponential series gives $\mathcal{H}_n/\phi(0) = S(z)/z$, where

$$S(z) = \sum_{k\geq 0} c_k\{(H-t)^{2k+1}(1-z)^{-k-1/2} - H^{2k+1}\}z^k + te^{-H^2z/2} + \tfrac{1}{2}zH(t^2-1)e^{-H^2z/2}.$$

Its constant coefficient vanishes, and expansion of the binomial factor gives $[z^{j+1}]S = P_j$. At $z_0 = 1/4$, the sum obtained by replacing all coefficients by their absolute-value majorants is at most B_r: use $(1-z_0)^{-1/2} < 7/6$, $1/(2k+1) \leq 1$, and $e < 193/71$. The latter bound follows, for example, by summing the exponential series through degree six and bounding the rest by a geometric series with ratio $1/8$. All products and rearrangements are therefore absolutely convergent. The discarded terms have degrees at least seven in S, so their sum at $0 \leq z \leq z_0$ is at most $(z/z_0)^7 B_r$. Divide by z to get the result. This proves a finite uniform bound, including every threshold in the stated interval and every $n \geq 4$. □

Q.6 A common contact test for radius and strip refinement

Suppose $n \geq 47$, $0 \leq C - C_E \leq .7/n$, $\delta_n \leq .19$ and $|X| \leq R_0$. A previously acquired row gives

$$\rho \leq r, \quad 0 \leq b_l \leq \beta \leq b_u < 1, \quad |H| \leq h.$$

Define $Q(t) = -|t|^3 + 3\rho t^2/2 + 3\beta t$ and $L_P(t) = C_E Q(t) + \phi(0)(t^3-3t)/6$. For $t > 0$, upper polynomials for $L_P(t), Q(t)$ are

$$(1/15 - .4095)t^3 + 1.5(.40974)rt^2 + (3(.40974)b_u - .3988/2)t, \quad -t^3 + 1.5rt^2 + 3b_u t.$$

For $t < 0$, write $u = |t|$ and use

$$(-.3988/6 - .4095)u^3 + 1.5(.40974)ru^2 + (.2 - 3(.4095)b_l)u, \quad -u^3 + 1.5ru^2 - 3b_l u.$$

These coefficient choices are valid because $b_l \ge 0$ is checked first. On any closed cell of one sign, let L_J be an upper bound for the first polynomial and let $Q_J \ge 0$ bound the positive part of the second. For $j = 1, 2, 3$, take m_j to be the positive part of the largest tensor Bernstein coefficient of P_j on $[-h, h] \times J$. For $j = 4, 5$, the absolute monomial bound $\sum_{a,b} |[H^a t^b] P_j| h^a R_J^b$ suffices, where $R_J = \max_{t \in J} |t|$. The support contact is then bounded above by

$$L_J + \frac{.7 Q_J}{47} + .4 \left(\sum_{j=1}^{5} \frac{m_j}{47^j} + \frac{4^7 B_{R_J}}{47^6} \right) + r \left(.19 S + \frac{.384(.425)}{47} \right), \qquad S > \sqrt{47/46}. \tag{133}$$

Indeed substitute the finite series in contact, split $C = C_E + (C - C_E)$, and use $C < .40974 + .7/47 < .425$. Every error retained in (133) is nonnegative and decreases with n. A strictly negative rational result therefore excludes the entire cell for every integer $n \ge 47$, including atoms and accumulation points.

For a coordinate strip, conditional centering gives

$$X = \ell (Y - p), \quad \ell = \sqrt{(1 - \theta)/(p(1 - p))}, \quad \beta = (1 - \theta)(1 - 2p) + \kappa, \quad |\kappa| \le \theta.$$

If $p \in [p_l, p_u] \subset (0, 1/2)$ and $\theta \le \Theta$, put $\ell_l \le \sqrt{(1 - \Theta)/(p_u(1 - p_u))}$. The forbidden strip $Y \notin (-.55, 1.55)$ maps into

$$X \le -\ell_l (p_l + .55), \qquad X \ge \ell_l (1.55 - p_u).$$

Apply (133) between these thresholds and $\pm R_0$. An interval beyond the known radius is empty; an equal endpoint is a singleton that is still checked. If only an upper β bound is given, use $b_l = (1 - \Theta)(1 - 2p_u) - \Theta$, after checking its nonnegativity.

The selected consequences used in the resource schedule are:

input rows	prior radius	conclusion	closed cells
10 central preparation rows	3.84	$\lvert X \rvert < 2.5$	1080
45 central rows	2.5	$-.55 < Y < 1.55$	1431
38 left rows	2.7	$-.55 < Y < 1.55$	3073

The respective largest contact bounds are below $-.084$, $-.097$ and $-.01658$. These implications acquire no row by themselves: each is used only after its input row is proved by the preceding resource step. The left radii are applications of the all-threshold support polynomial, with $(N, L, K, q) = (47, \infty, 0, .19)$ and the actual moment caps

$$(r, b, R) = (1.390, .430, 3), \qquad (1.2563, .3926, 2.7).$$

Thus the same heat proof supplies the initial radius, both left contractions, and the five finite-base rows. The bounded-threshold series supplies the central contraction and both coordinate strips.

R Entry as a resource enclosure

The entry calculation has one job: turn an attained extremizer into a short list of simultaneous hypotheses for the local comparison. Its many successive tables are applications of one enclosure principle. A table is a cover of possible laws; a refinement intersects that cover with newly proved inequalities. There is no assertion that an iteration converges, and no improved bound is used to justify itself.

Throughout, X has mean zero and variance one, $n \ge N$, and H is the raw sum threshold. Put

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad M_j = \mathbb{E}|X|^j,$$
$$C = C_n, \quad e = C - C_E \ge 0, \qquad C_E = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The input is an attained positive closed maximum, together with the contact inequality stated below, an actual support bound, and an independently established score cap. The global proof supplies these inputs. A conditional upper bound on moments is never used to acquire the support needed to define those moments.

R.1 The conserved quantity

Write $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \ge t}]$. It is nonnegative, $\int \tau = 1$, and $2\int |t|\tau(t)\,dt = \rho$. On each bounded gap $g = (a, b)$ of the closed support it has the constant value τ_g. Define

$$J = \sum_g \tau_g (b-a)^2, \qquad J_1 = \sum_g \tau_g \int_a^b (t-a)t\,dt,$$
$$J_2 = \sum_g \tau_g \int_a^b (t-a)t|t|\,dt, \qquad D = (3+\sigma)\rho - \gamma - 3J, \qquad \sigma = \sqrt{10},$$
$$L = \sum_g \tau_g \int_{[a,b)} (t-a)\,\nu(dt), \qquad \nu = \mathcal{L}(H - S_{n-1}).$$

The half-open convention includes the lower endpoint with weight zero and excludes the upper endpoint. In particular it treats atoms without an implicit midpoint convention. Bounded support makes all sums absolutely convergent. The gap calculation below proves $D \ge 0$; clearly $L \ge 0$.

Put $\varphi = \phi(H/\sqrt{n})$ and

$$K_1 = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma,$$
$$K_q = M_5/4 - \rho M_4/2 - \tfrac{3}{2}\beta\gamma + \rho/2, \qquad U = J_1 + (M_4 - 3)/6.$$

The exact remainder identity is

$$ne\rho + nC_E\rho(1 - e^{-w}) + n\phi(0)e^{-w}D/6 + n^{3/2}L = \mathcal{R}, \qquad w = H^2/(2n), \tag{134}$$

where

$$\mathcal{R} = C\{3\beta J/2 + 3\rho J_1 - 3J_2 - K_q\} + H(CK_1 + \varphi U) - H^2(C\rho + \gamma\varphi/2) + E_n. \tag{135}$$

The complete Fourier estimate supplies a *coherent* pair $|E_n| \le K_F + s_F|H|$: the constant from one estimate is not combined with the slope from another unless both were separately proved. Section S derives (134) and the whole Fourier bound from the contact interface.

Lemma R.1 (Removing the Gaussian weight)**.** *Suppose a first pass proves $\mathcal{R} < r < Nc_0$, where $C\rho \ge c_0$. Every later valid upper bound $\mathcal{R} \le r'$ then gives*

$$ne\rho + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}, \quad e^{-w} \ge 1 - \frac{5r'}{2n}, \quad nD \le \frac{15000r'}{997}. \tag{136}$$

The last two bounds use $C_E\rho > 2/5$ and $\phi(0) > 997/2500$.

Proof. Rearrange the exact identity before changing its Gaussian weight: $\varphi(\gamma+3J)/6 = C\rho+\sqrt{n}L-\mathcal{R}/n>0$. Thus $C_E\rho-\phi(0)D/6>0$. Replacing e^{-w} by one in (134) lowers its left side, proving the first assertion. The original nonnegative phase term proves $1-e^{-w}\le r'/(nC_E\rho)<5r'/(2n)$. The defect bound follows from the first assertion. The initial sign license is proved before any feedback is used. □

The same identity also gives a useful score update. If this cell forces $D\ge d_l\ge 0$ and a preceding pass gives raw ceiling r, then

$$C\le .40974+\frac{r/N-.3988(1-5r/(2N))d_l/6}{\rho_l}.$$

A negative numerator excludes the cell because $e\ge 0$. Otherwise its numerator increases with $1/n$, and division by the positive lower ρ_l is outward. Thus the update is valid for every $n\ge N$ and uses only the previous raw bound.

For later local comparisons we retain the sharper phase resource. If previous bounds give $\mathcal{R}\le r$, $nD\le Nd$, then

$$ne\rho+\kappa nD+n^{3/2}L\le \mathcal{B}(H),\qquad \kappa=997/15000, \tag{137}$$

where an upper bound for $\mathcal{B}$ is obtained from the raw quadratic by increasing its curvature by

$$q_{\mathrm{extra}}(\rho)=\frac{1-5r/(2N)}{2}\left(.4095\rho-\frac{d}{15}\right)>0. \tag{138}$$

Indeed the subtracted exact phase term is $n(1-e^{-w})(C_E\rho-\phi(0)D/6)$. Use $1-e^{-w}\ge we^{-w}$, $D\le d$, $\phi(0)<.4$, and the previous density floor. The output field `Qbudget` includes (138); it must not be charged a second time.

R.2 One positive-measure rule

Suppose zero lies in the gap $(-A,B)$, $A,B>0$. Let

$$p=\mathbb{P}(X>0),\quad q=1-p,\quad m=\mathbb{E}X_+=\mathbb{E}X_-,\quad V=\mathbb{E}[X\mid \mathbf{1}_{X>0}],$$
$$W=X-V,\quad \theta_\pm=\mathbb{E}[W^2;\pm X>0],\quad \theta=\theta_++\theta_-.$$

Then $m^2=(1-\theta)pq$, and the conditional means have magnitudes $\bar{A}=m/q$, $\bar{B}=m/p$. Set $\Delta_A=\bar{A}-A\ge 0$, $\Delta_B=\bar{B}-B\ge 0$. We also use the unnormalised endpoint displacements

$$M_+=\mathbb{E}[(X-B)\mathbf{1}_{X>0}]=p\Delta_B,\qquad M_-=\mathbb{E}[(-X-A)\mathbf{1}_{X<0}]=q\Delta_A.$$

Lemma R.2 (Positive-measure pricing)**.** *For $x=|X|$, $a=|V|$, and a function h constant on each sign class,*

$$\mathbb{E}[h(|X|^j-|V|^j)]=\mathbb{E}\left[hW^2\sum_{k=0}^{j-2}(k+1)a^kx^{j-2-k}\right],\quad j\ge 2. \tag{139}$$

Also, for $d\mu_0=(|X|-1)^2(|X|+1/2)\,dP$,

$$\mu_0(\mathbb{R})=\rho-1,$$
$$M_4-1=\int\frac{(x+1)^2}{x+1/2}\,d\mu_0,$$
$$M_5-1=\int\frac{x^3+2x^2+3x+3/2}{x+1/2}\,d\mu_0,$$
$$K_1+3\rho\beta=\int\frac{\operatorname{sgn}(X)x(x+2-3\rho/2)}{x+1/2}\,d\mu_0.$$

For any nonnegative resources z_i satisfying $\sum a_i z_i \le r$, $a_i > 0$, a signed linear cost obeys

$$\sum b_i z_i \le r \max\{0, b_1/a_1, \dots, b_k/a_k\}. \tag{140}$$

Proof. Divide $x^j - a^j - ja^{j-1}(x-a)$ by $(x-a)^2$; the linear term has zero conditional mean. The radial identities follow by multiplying out and using $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. Finally each b_i is at most a_i times the displayed maximum, and all z_i are nonnegative. □

This one rule accounts for the moment, signed Taylor, endpoint movement, outer-gap, and confinement-error estimates. In particular, no two costs may each spend the full residual and then be added.

R.3 The simultaneous gap budget

Let $p_* = (4-\sigma)/2$ and $T_+ = \mathbb{E}X_+^3 - mB^2$, $T_- = \mathbb{E}X_-^3 - mA^2$. If $J_o = J - m(A+B)^2$, signwise integration gives

$$D = m(\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2 + (\sigma-1)T_+ + (\sigma+1)T_- + 3(T_+ + T_- - J_o). \tag{141}$$

Moreover, writing $R_+ = \mathbb{E}[X_+(X_+ - B)]$ and similarly R_-,

$$T_+ + T_- - J_o \ge 2BR_+ + 2AR_- \ge 0.$$

If zero belongs to the support, every gap has one sign and $J \le \rho$, so $D \ge \sigma\rho - \gamma \ge (\sigma-1)\rho$. Otherwise (141) proves $D \ge 0$. Consequently $D < (\sigma-1)\rho$ licenses the actual central gap. We also have $J \le \rho + 2m \le \rho + 1$; if $D < 2$, then $T_+ + T_- \le D/2$ and $A + B \ge \sqrt{2-D}$.

To verify the gap estimate, on a positive outer gap (u, v), $(v-u)^2 \le 2\int_u^v (t-B)\,dt$. Summing with weights τ_g and applying Tonelli gives $J_o^+ \le T_+ - 2BR_+$; reflect for the other sign. Expanding the definition of D gives (141). The inequality $AB \le 1$ follows by integrating $(X+A)(X-B) \ge 0$, and $m \le 1/2$ follows from $\mathbb{E}|X| \le 1$.

Conditional expansion about $\bar{A}, \bar{B}$ now gives the stronger single inequality

$$\begin{aligned} D \ge{}& \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p-p_*)^2 + 6m\{(A+B)(\Delta_A+\Delta_B) + \Delta_A\Delta_B\} \\ &+ 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-. \end{aligned} \tag{142}$$

For example $\mathbb{E}X_+^3 = m\bar{B}^2 + \mathbb{E}[(X+2\bar{B})(X-\bar{B})^2; X > 0]$ and $R_+ = \theta_+ + m(\bar{B} - B)$. The cubic residual is at least $3B\theta_+$. The deterministic shift left after substituting both signs is $6m\{\bar{A}\bar{B} - AB + B\Delta_B + A\Delta_A\}$, equal to the second term above. The remaining deterministic expression, using $m^2 = (1-\theta)pq$, is exactly the displayed binary square.

R.4 A state, its update, and its exported resource

A state for a closed probability interval $I = [l, u]$ records

$$|X| \le R, \quad C \in [c_0, c_1], \quad \mathcal{R} \le r, \quad nD \le Nd, \quad \theta \le \Theta < 1, \quad A \ge A_l > 0, \quad B \ge B_l > 0,$$

together with simultaneous signed-moment and coefficient intervals. The distinction between $nD \le Nd$ and $D \le d$ is essential. Let $s_+ = \max_I p(1-p)$, $s_- = \min_I p(1-p)$ and

$$b_I = \frac{(6+2\sigma)(1-\Theta)^{3/2}}{\sqrt{s_+}}\,\mathrm{dist}(p_*, I)^2, \qquad r_I = d - b_I,$$

$$K_+ = 3(\sigma+1)B_l, \quad K_- = 3(\sigma+3)A_l, \quad K = \min(K_+, K_-), \quad m_l = \sqrt{(1-\Theta)s_-}.$$

If $r_I < 0$, exclude the entire closed cell. Otherwise (142) gives, simultaneously,

$$6m(A_l + B_l)(\Delta_A + \Delta_B) + K_+\theta_+ + K_-\theta_- \le r_I. \tag{143}$$

Thus the actual law satisfies the intersected bounds

$$\begin{aligned} \theta &\le \min\{\Theta, r_I/K\}, \\ A &\ge \max\left\{A_l, \sqrt{(1-\Theta)l/(1-l)} - \frac{r_I}{6m_l(A_l+B_l)}\right\}, \\ B &\ge \max\left\{B_l, \sqrt{(1-\Theta)(1-u)/u} - \frac{r_I}{6m_l(A_l+B_l)}\right\}. \end{aligned} \tag{144}$$

Each right side is evaluated from established bounds; sequential improvements may then be used by the next expression. A nonpositive endpoint estimate fails this update, rather than excluding a law. Initialization uses the same formulas with the valid lower sum $A + B \ge \sqrt{2-d}$.

In Bernoulli coordinates set $Y = B_0 + U$, $\mathbb{E}[U \mid B_0] = 0$, $B_0 \sim \text{Bernoulli}(p)$, $X = \ell(Y - p)$ and $d_Y = (\Delta_A + \Delta_B)/(\bar{A} + \bar{B})$. Then $\ell^2 = (1-\theta)/(pq)$ and $\lambda = n\mathbb{E}U^2 = npq\theta/(1-\theta)$. The *weighted* versions of (143) prove

$$n\theta \le \frac{Nr_I}{K}, \quad nd_Y \le \frac{Nr_I}{6(1-\Theta)(A_l+B_l)}, \quad \lambda \le \frac{Ns_+r_I}{(1-\Theta)K}. \tag{145}$$

Indeed $nD \le Nd$ whereas $nb_I \ge Nb_I$. These exports are derived from the weighted resource; multiplying an unweighted variance ceiling by N would not prove them.

For $j = 3, 4, 5$ set

$$L_3(v) = R + 2v, \quad L_4(v) = R^2 + 2Rv + 3v^2, \quad L_5(v) = R^3 + 2R^2v + 3Rv^2 + 4v^3.$$

Put $v_+ = \sqrt{(1-l)/l}$, $v_- = \sqrt{u/(1-u)}$ and $k_j = [1 - (1-\Theta)^{j/2}]/\Theta$, with $k_j = j/2$ at $\Theta = 0$. For $u < 1/2$, valid Bernoulli moment ceilings are, with $s = l(1-l)$,

$$B_3^* = (1-2s)/\sqrt{s}, \quad B_4^* = 1/s - 3, \quad B_5^* = (1 - 4s + 2s^2)/s^{3/2}.$$

They decrease towards $p = 1/2$. Convexity places $(1-\theta)^{j/2}$ below its endpoint chord on $[0, \Theta]$; positive-measure pricing gives

$$M_j \le B_j^* + r_I \max\left\{0, \frac{L_j(v_+) - k_jB_j^*}{K_+}, \frac{L_j(v_-) - k_jB_j^*}{K_-}\right\}. \tag{146}$$

The same rule, using $T_+ \le 2mv_+\Delta_B + L_3(v_+)\theta_+$ and its reflection, bounds

$$W_o := (\sigma-1)T_+ + (\sigma+1)T_- \le r_I \max\left\{\begin{array}{c} \frac{(\sigma-1)v_+}{3(A_l+B_l)}, \frac{(\sigma+1)v_-}{3(A_l+B_l)}, \\ \frac{(\sigma-1)L_3(v_+)}{K_+}, \frac{(\sigma+1)L_3(v_-)}{K_-} \end{array}\right\}. \tag{147}$$

Intersect every new bound with the previous one. The versions which merge θ_+ and θ_- are obtained by replacing both costs by their maximum and both prices by their minimum; they need no new proof.

R.5 The enclosure theorem

On each cell, the formula atlas bounds the four coefficients in (135). The result, for each actual sign $s = \operatorname{sgn} H$, is a finite family of quadratics

$$\mathcal{R} \le \max_j (A_j + F_{j,s}|H| - Q_j^{\mathrm{raw}} H^2), \qquad Q_j^{\mathrm{raw}} > 0. \tag{148}$$

The maxima retain all score–moment corners. In the initial signed cover, intervals contain both signs of β, γ; no orientation is discarded. In the probability cover, the central integral bound is a maximum of affine functions of ρ, hence convex. The other ρ-dependence is affine or the maximum of convex quadratics. For fixed H, the score dependence is convex. Endpoints are therefore taken *before* completing the square.

Theorem R.3 (Finite resource enclosure)**.** *Consider a finite ordered schedule of the following operations:*

1. *Split a cell into an explicitly covered family of closed cells.*
2. *Intersect its enclosures with an inequality proved in the atlas.*
3. *Exclude a cell by an empty necessary interval or a strictly negative nonnegative resource.*
4. *Apply a separately proved support or confinement implication, checking all its hypotheses against this cell's already acquired data.*
5. *Export a raw, weighted, or signed phase bound by* (148)*,* (137)*, and* (145)*.*

If exact outward arithmetic checks every domain, corner and cover, every output contains all attaining laws satisfying the input. The conclusion holds for all integers $n \ge N$ wherever the formula's monotonicity gates are checked. It includes zero conditional variance, singleton intersections and arbitrarily small atoms.

Proof. Induct over the schedule. A split preserves the union, intersections use necessary inequalities, and an excluded cell contains no feasible law. Each callback consumes only earlier hypotheses for that same law. For a scalar raw output, use $fh - qh^2 \le (f_+)^2/(4q)$ for $h \ge 0$, $q > 0$; using f^2 is a valid weaker bound. For a resource output first add (138) to each curvature. A negative maximum excludes the cell; a nonnegative one may be divided by κN to give its next defect parameter. The phase bound follows from the positive root of $A + fh - qh^2 \ge 0$, replacing A by A_+ if necessary. A strict output is exported only when the certificate retains a strict upper gap, such as its strict Fourier or target enclosure; closed-domain coverage does not change strictness. All constants are frozen before evaluating corners. Thus no iteration step relies on an unproved fixed point. The stated index uniformity is that of the displayed finite formulas, not a fixed-law limit. □

For the Fourier cancellation estimate only, reflect (X, H) if needed: $I_0(-X, -H) = I_0(X, H)$. Both odd coefficients change sign, and $\int e^{-x^2/2}|zx^2 - cx^4|\,dx$ is unchanged by $(z, c) \mapsto (-z, -c)$. Consequently a lower endpoint used there bounds $|\gamma|$, not necessarily the original signed γ. The actual contact, gap and phase data keep their original orientation. A useful additional signed lower bound is

$$\gamma \ge (1-\Theta)^{3/2} \frac{1-2u}{\sqrt{u(1-u)}} - \left(R + 2\sqrt{\frac{u}{1-u}}\right)\Theta. \tag{149}$$

To prove it expand $\mathbb{E}X^3 = \mathbb{E}V^3 + \mathbb{E}[W^2(X+2V)]$; the positive residual is nonnegative and the negative residual has the displayed lower bound. The binary term decreases in both $p < 1/2$ and θ. Thus $\max(0, \text{any signed lower bound})$ is a valid lower bound for $|\gamma|$ inside this auxiliary Fourier estimate.

R.6 Support, confinement, and a short acyclic schedule

Two different support implications are useful. Gaussian contact excludes a point by a strict negative upper bound on its contact expression; closed cells cover the finite exterior and a derivative-sign certificate covers any remaining ray. The shared Gaussian support theorem supplies this callback, including the bounded-threshold radius and coordinate-strip contracts; it is a prior theorem, not a test on moments alone. Noise confinement proves $|U| \le a$ on an already acquired row. It then implies, without a small-noise limit,

$$|X| \le R_* := \frac{\max\{u + a, 1 - l + a\}}{\sqrt{\min_I p(1-p)}}. \tag{150}$$

The denominator uses $\mathbb{E}U^2 \ge 0$, so zero noise is included. The Fourier/moment formulas remain valid for $R \ge 1$; the paired Pearson choice additionally checks $R > 2g$. In particular there is no intrinsic lower radius 2.3: the radial ratios in Lemma R.2 increase on all $x \ge 0$, and the other formulas use their explicit domains.

Here is the selected proof schedule. Each row names *an application* of Theorem R.3, not a different analytic theorem.

Node	Input and operation	Output consumed later
I	$N = 47$, $R = 3.84$, $C \le .4247$; 13 signed passes plus export	100 closed probability cells; 35 survive; $\mathcal{R} < 1.885$, $nD < 47(.590)$
C	Central preparation, contact radius 2.5, probability refinement	45 cells $[.38, .47]$, width .002; canonical simultaneous input
L	Complete complementary cover, contact radius 3, refinement	39 cells $[.302, .38]$; all other complementary cells excluded
C1	First confinement from C; radius formula; enclosure	First improved central table
C2	Antichain confinement from C; intersect C1; enclosure	Selected central table: 45 cells, $[.38, .47]$
S	Separate sign-resource pricing on C2 and L	One left cell excluded; 38 left cells remain
L2	Left contact radius 2.7 from S; enclosure	Selected left table: 38 cells, $[.304, .38]$

Every shared endpoint belongs to its adjacent closed cells. The confinement producing C2 consumes C, not C2. The selected central local comparison uses C2; it does not need the optional further central refinement in S. The final cover is exactly $[.304, .47]$.

Corollary R.4 (Uniform acquired entry)**.** *Let an attained positive maximum have $n \ge 47$, $C_E \le C \le .4247$, $0 \le n(C - C_E) \le .7$, decrement at most .19, and both deleted score caps at most .4253. Given the shared contact support and strip implications, the exact schedule above places its law in one of the 38 selected left rows or 45 selected central rows, with every resource in that row holding simultaneously. This assertion has no upper bound on n.*

Proof. The shared all-threshold support theorem supplies radius 3.84. Apply the finite enclosure theorem in the displayed order, retaining the weighted bounds at every edge. The initialization's stored upper index 98 occurs only through $.4095 + K/98$; here $K = 0$, so it imposes no upper-index restriction. All subsequent index-dependent inequalities are checked for every $n \ge 47$ by their stated monotonicity gates. The accepted closed covers exclude the complementary cells and give the stated 83 rows. □

The local-comparison interface.

For each final row supply $p \in I$, $\theta \le \Theta$, $\lambda \le \Lambda$, the actual support/amplitude, $d_Y \le d_0$, both signed phase limits, and all four coherent quadratics bounding (137). On a phase/noise

subcell, let T_H be the maximum of each concave quadratic at its clipped vertex, then the maximum over corners and the minimum with the scalar bound. If a proved moment inequality is

$$D \geq \alpha d_Y + b\theta + P, \qquad \alpha > 0, \quad \ell \geq \ell_-, \quad \lambda \geq L_0,$$

then the exact strict resource supplied to the comparison is

$$nd_Y < G := \frac{T_H/\kappa - NP - b\ell_-^2 L_0}{\alpha}. \tag{151}$$

This uses $n\theta = \ell^2\lambda$ and $nP \geq NP$. If $G \leq 0$, the entire closed subcell is excluded. Formula (151) is the same resource as (137), not an additional allowance.

Validation contract.

The exact supplement must pin the input and output hashes of every node, reconstruct all closed covers, and recompute its inequalities. Every square-root enclosure is checked by squaring nonnegative endpoints; every division has a positive-denominator gate. Exponentials and the finite Gaussian integral use the stated rational series enclosures. The first raw pass checks $r < .4N$ before exporting a defect; later passes consume their predecessors' raw and weighted bounds. Excluded cells retain an explicit mathematical reason and count towards coverage; an empty final row is not a negative remainder bound. Reordered, duplicated, missing, malformed and falsely strengthened objects reject. The accompanying source map identifies the exact implementation of each formula and each independently accepted callback. This consolidates the proof; it does not replace the arithmetic replays by a hash check.

S The minimal contact–Fourier foundation

The only contact input needed here is the following explicit interface. At the attained positive maximum, with $F = \mathbb{P}(S_n \leq H)$, $z = H/\sqrt{n}$ and $\nu = \mathcal{L}(H - S_{n-1})$, one has $\nu([y,\infty)) \leq \mathfrak{q}(y)$ for every real y, with equality on the full closed support, where

$$\begin{aligned} n\mathfrak{q}(y) = nF + \tfrac{1}{2}\{z\phi(z) + C\rho/\sqrt{n}\} - (\sqrt{n}\phi(z) + 3C\beta/\sqrt{n})y \\ - \tfrac{1}{2}\{z\phi(z) + 3C\rho/\sqrt{n}\}y^2 + C|y|^3/\sqrt{n}. \end{aligned} \tag{152}$$

This contact statement is the normalized main-manuscript contact inequality at $c = C_n$. Attainment is supplied by its ratio-attainment corollary. Everything that follows in this section is derived from it. We abbreviate $M_4 = \mu_4$, $M_5 = \eta_5$ when comparing with the implementation.

S.1 Compensated smooth inversion

Write $Z_n = S_n/\sqrt{n}$, $f(t) = \mathbb{E}e^{itX}$, and let N be standard normal. The smooth payoff $q_z(x) = (x - z)|x - z|$ gives the exact error

$$\epsilon_n(z) = \mathbb{E}q_z(Z_n) - \mathbb{E}q_z(N) - \frac{2\gamma\phi(z)}{3\sqrt{n}}.$$

Lemma S.1. *For every real H and $n \geq 1$,*

$$\epsilon_n(H/\sqrt{n}) = -\frac{4}{\pi n}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt, \qquad V_n(t) = f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6). \tag{153}$$

The integral is absolutely convergent. With the split at $t = 1/2$,

$$I_{\mathrm{low}} = \int_0^{1/2} \frac{|\Im(e^{-itH} V_n(t))|}{t^3}\, dt, \qquad I_{\mathrm{high}} = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt, \tag{154}$$

one has $|\epsilon_n(z)| \le 4(I_{\mathrm{low}} + I_{\mathrm{high}})/(\pi n)$.

Proof. The scalar identities

$$\int_0^{\infty} \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|, \qquad \int_0^{\infty} \frac{|ty - \sin(ty)|}{t^3}\, dt = \frac{\pi}{4} y^2$$

follow by scaling and integration by parts. One elementary evaluation of the constant uses $J_a(b) = \int_0^{\infty} e^{-at}(1 - \cos bt)t^{-2}dt$: $J_a''(b) = a/(a^2 + b^2)$, $J_a(0) = J_a'(0) = 0$, so $J_a(1) = \int_0^1 \arctan(b/a)db \to \pi/2$. Monotone convergence removes $a > 0$. For a finite signed measure μ with zero mass and first moment and finite absolute second moment, absolute Fubini applied to the *compensated* scalar integrand therefore gives

$$\int (s - H)|s - H|\, d\mu(s) = -\frac{4}{\pi} \int_0^{\infty} t^{-3} \Im(e^{-itH} \widehat{\mu}(t))\, dt. \tag{155}$$

Apply this to the law of S_n minus the pushforward, by $x \mapsto \sqrt{n}x$, of the signed density $\phi(x)[1 + \gamma(x^3 - 3x)/(6\sqrt{n})]$. Gaussian integration by parts gives its transform $e^{-nt^2/2}(1 - in\gamma t^3/6)$, total mass one and first moment zero. Since $(x^3 - 3x)\phi = -\phi'''$, integration twice by parts and then across z gives $-\int q_z \phi''' = 4\phi(z)$. This proves the formula and its scale factor. The high integral is finite from $|f| \le 1$ and Gaussian decay; the low integral is finite by the absolute Fubini argument. □

The payoff is smooth enough for compensated inversion. No discontinuous CDF is inverted, so no atom midpoint convention is introduced.

S.2 Derivation of the conserved identity

Proof. Compute $\mathcal{A} = \int \tau(y)[\mathfrak{q}(y) - \nu([y, \infty))]dy$ in two ways. Put $r = -\mathfrak{q}'$. Contact at both endpoints of a gap gives

$$\nu([a, b)) = \int_a^b r(t)dt, \quad \int_a^b [\mathfrak{q}(y) - \nu([y, \infty))]dy = \int_a^b (t - a)r(t)dt - \int_{[a,b)} (t - a)d\nu(t).$$

The integrand vanishes on the support and $\tau = 0$ on the exterior rays. Summing, and differentiating (152), gives

$$\mathcal{A} = \frac{\phi(z)J}{2\sqrt{n}} + \frac{G}{n^{3/2}} - L, \qquad G = \frac{3}{2} C\beta J + (H\phi(z) + 3C\rho)J_1 - 3CJ_2. \tag{156}$$

All sums and integrals are absolutely convergent on the compact convex hull of the support, since $|J_1| \le R_0 J/2$, $|J_2| \le R_0^2 J/2$ when $|X| \le R_0$.

For the second computation let X^* have density τ, independent of S_{n-1}. Signwise Fubini gives $\mathbb{E}g'(X^*) = \mathbb{E}[Xg(X)]$ for continuously differentiable g. In particular, $\mathbb{E}X^* = \gamma/2$, $\mathbb{E}(X^*)^2 = \mathbb{E}X^4/3$ and $\mathbb{E}|X^*|^3 = \mathbb{E}|X|^5/4$. Applying the identity to each summand shows that $(S_{n-1} + X^*)/\sqrt{n}$ has density $\tau_{Z_n}(t) = \mathbb{E}[Z_n \mathbf{1}_{Z_n \ge t}]$. Its CDF at z is $T_z = \mathbb{E}[(Z_n^2 - zZ_n)\mathbf{1}_{Z_n \le z}]$, by integrating that density; the boundary integrand is zero. Substituting the three moments into (152) therefore gives

$$\mathcal{A} = F - \frac{\gamma\phi(z)}{2\sqrt{n}} + \frac{H\phi(z)(3 - \mathbb{E}X^4)/6 + CK_q}{n^{3/2}} - T_z. \tag{157}$$

Multiply contact equality by X and integrate. By independence and symmetry, $n\mathbb{E}[X\nu([X,\infty))]$ equals $\mathbb{E}[S_n\mathbf{1}_{S_n\le H}]$. Expanding the other side yields, for $M_z=\mathbb{E}[Z_n\mathbf{1}_{Z_n\le z}]$,

$$M_z+\phi(z)=-\frac{\gamma z\phi(z)}{2\sqrt{n}}+\frac{CK_1}{n}. \tag{158}$$

Finally $q_z(x)=(x-z)^2-2(x-z)^2\mathbf{1}_{x\le z}$, so Gaussian integration gives

$$T_z-\Phi(z)=-\tfrac{1}{2}\{\mathbb{E}q_z(Z_n)-\mathbb{E}q_z(N)\}+z(M_z+\phi(z))-z^2(F-\Phi(z)). \tag{159}$$

Insert (158) and the definition of ϵ_n into (157). With

$$P_0=H^2(C\rho+\gamma\phi(z)/2)-H\{CK_1+\phi(z)(\mathbb{E}X^4-3)/6\}+CK_q,$$

the result is $\sqrt{n}\mathcal{A}=C\rho-\gamma\phi(z)/6+P_0/n+\sqrt{n}\epsilon_n(z)/2$. Equating this with (156) proves

$$C\rho+\sqrt{n}L=\frac{\phi(z)}{6}(\gamma+3J)+\frac{G-P_0}{n}-\frac{\sqrt{n}}{2}\epsilon_n(z). \tag{160}$$

Now substitute $\gamma+3J=(3+\sqrt{10})\rho-D$ and $\phi(z)=\phi(0)e^{-w}$, and expand $G-P_0$. □

S.3 One complete low-frequency transfer

Put $f(t)=\mathbb{E}e^{itX}$, $h(t)=e^{-t^2/2-i\gamma t^3/6}$, $e_n(t)=e^{-nt^2/2}(1-in\gamma t^3/6)$, and

$$I_0(H)=\int_0^{1/2}|\Im(e^{-itH}(f^n-e_n))|\frac{dt}{t^3},\qquad I_\infty=\int_{1/2}^{\infty}|f^n-e_n|\frac{dt}{t^3}.$$

The inversion bound (154) charges $2\sqrt{n}(I_0+I_\infty)/\pi$; both integrals below are complete.

Lemma S.2 (Signed Gaussian transfer)**.** *Suppose $n\ge N>2$, $\mu_4\le M\le 11.1$, $|\gamma|\le g\le 1.85$, $\mu_6\le M_6$, and the actual coefficients*

$$a=(3-\mu_4)/24,\quad z=(10\gamma-\mathbb{E}X^5)/120,\quad c=a\gamma/6$$

satisfy $|a|\le a_*$, $|z|\le Z$. Put

$$E_6=M_6/720+g^2/72+(1/2+g^2/144)^3/5,\quad B_6=a_*+Z/2+E_6/4. \tag{161}$$

Let $a_0=(N-1)/(2N)$, $0<b_0\le(45-M)(1-2/N)/96$, and $\lambda_N=\sqrt{\pi/2}(N/(N-1))^{3/2}$. Then, for every real H,

$$I_0(H)\le(L_0+L_H|H|)/\sqrt{n}+L_2/n,\qquad L_2=E_6/(2a_0^2)+B_6^2/(2b_0^3). \tag{162}$$

Two admissible choices, each used as a complete triple, are as follows. The triangle choice is

$$L_0=c_0Z+c_1a_*g+251g^3/17280,\quad L_H=c_0a_*+251g^2/4800, \tag{163}$$

where $c_0\ge\lambda_N$, $c_1\ge\lambda_N/2$. For the cancellation choice, take any rectangle containing the actual (z,c), $I_u\ge\sqrt{\pi/2}$ and $F_N\ge(N/(N-1))^{3/2}$. Define

$$U(z,c)=\begin{cases}I_u(|z|+3|c|), & zc\le 0,\\ I_u\big||z|-3|c|\big|+6\sqrt{|zc|}e^{-|z|/(2|c|)}, & zc>0,\end{cases}$$
$$J_u=\max_{\text{corners}}U(z,c).$$

Then

$$L_0=F_NJ_u+\frac{F_NI_u}{N-1}(35a_*g^3/48+35Zg^2/24)+251g^3/17280,$$
$$L_H=F_NI_u\{a_*+5Zg/[2(N-1)]\}+251g^2/4800. \tag{164}$$

Neither the transfer nor its cancellation choice assumes signs for a,z,c or γ.

Proof. Taylor with integral remainder gives $f - h = -at^4 - izt^5 + r_6$, $|r_6| \le E_6 t^6$ for $0 \le t \le 1/2$. Indeed for $w = -t^2/2 - i\gamma t^3/6$, $|w| \le t^2(1/2 + g^2/144) < 1/6$; the terms through degree five in $1 + w + w^2/2$ are exact, its degree-six term costs $g^2 t^6/72$, and the remaining exponential series costs $|w|^3 e^{|w|}/6 \le |w|^3/5$. Add the true sixth-moment remainder.

Four integrations of $1 - \cos u \ge 0$ give $\cos u \le 1 - u^2/2 + u^4/24$. Symmetrization, using $\mathbb{E}(X - X')^4 = 2\mu_4 + 6$, therefore gives $|f(t)|, |h(t)| \le e^{-(45-M)t^2/96}$ on $[0, 1/2]$. Every point on their joining segment has the same bound, so polynomial Taylor expansion yields

$$f^n - h^n = n(f - h)h^{n-1} + R_2, \qquad |R_2| \le \tfrac{1}{2} n^2 B_6^2 t^8 e^{-b_0 n t^2}.$$

The r_6 and R_2 integrals are at most

$$\frac{1}{n}\{E_6/(2a_0^2) + B_6^2/(2b_0^3)\} = L_2/n. \tag{165}$$

For the remaining comparator put $u = n\gamma t^3/6$. Its imaginary part is bounded by $e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2)$, so its divided integral is at most

$$n^{-1/2}(251g^3/17280 + 251g^2|H|/4800). \tag{166}$$

Here the Gaussian moments follow from integration by parts, $\int_0^\infty x^{2j} e^{-x^2/2} dx = (2j-1)!!\sqrt{\pi/2}$, and $\sqrt{2\pi} < 251/100$.

It remains to price the leading quartic and fifth terms. Set $b = n - 1$, $v = b\gamma t^3/6$. The triangle estimate uses $|\sin(Ht + v)| \le |H|t + bgt^3/6$, $|\cos| \le 1$; the Gaussian moments of orders two and four give respectively λ_N and $\lambda_N/2$. This proves (163). For cancellation, after division by t^3, rotation by Ht bounds the leading imaginary part by

$$|zt^2 - atv| + a_* t|\sin v - v| + Zt^2|\cos v - 1| + |H|t\{a_* t + Zt^2|\sin v|\}.$$

Multiply by $ne^{-bt^2/2}$ and use the global sine/cosine remainders. The first integral equals $nJ(z, c)/b^{3/2}$, where $J(z, c) = \int_0^\infty e^{-x^2/2}|zx^2 - cx^4|dx$. This norm is convex and $J(-z, -c) = J(z, c)$, so its rectangle maximum is at a corner. For $zc \le 0$ the integral is $\sqrt{\pi/2}(|z| + 3|c|)$. For $z, c > 0$, split at $r = \sqrt{z/c}$; integration by parts gives

$$J = \sqrt{\pi/2}(z - 3c) + 6cre^{-r^2/2} + 2(3c - z)\int_r^\infty e^{-x^2/2} dx.$$

Discard the last term when nonpositive, and otherwise bound its integral by $\sqrt{\pi/2}$. This proves $J \le U$; simultaneous sign reflection covers the remaining quadrant. The other four phase terms use moments of orders two, six, eight and ten and give the coefficients $1, 5/2, 35/24, 35/48$ in (164). Finally $n/(n-1)$, $1/(n-1)$, a_0^{-1} and the allowable damping inverses decrease with n, proving the all-index bounds at N. □

Lemma S.3 (Retaining the signed Gaussian tail)**.** *For $r > 0$, put*

$$\mathcal{M}(r) = e^{r^2/2}\int_r^\infty e^{-t^2/2}\,dt, \qquad \mathcal{M}_k(r) = \cfrac{1}{r + \cfrac{1}{r + \cfrac{2}{\ddots + \cfrac{k}{r}}}} \quad (k \ge 1), \qquad \mathcal{M}_0(r) = 1/r.$$

Odd truncations are lower bounds for $\mathcal{M}$; even truncations are upper bounds. For $z, c > 0$, set $x = z/c$ and

$$d_{k+1} = 1, \qquad d_j = 1 + \frac{j}{x d_{j+1}} \quad (j = k, \dots, 1), \qquad m_k(x) = d_1^{-1}.$$

Then the cancellation norm in Lemma S.2 satisfies

$$J(z,c) \le \sqrt{\pi/2}(z-3c) + \sqrt{zc}\,e^{-x/2}\left(6 + \frac{2(3-x)}{x} m_k(x)\right), \tag{167}$$

where $k = 11$ if $x \ge 3$ and $k = 12$ if $x < 3$. The factor in parentheses is positive. One may take the minimum with the previous bound $U(z,c)$ at each corner of the coefficient rectangle.

Proof. The positive integrals $I_j(r) = \int_0^\infty t^j e^{-rt-t^2/2}\,dt$ satisfy $I_1 = 1 - rI_0$ and $I_{j+1} = jI_{j-1} - rI_j$ by integration by parts. Thus, writing $q_j = I_j/I_{j-1} > 0$,

$$I_0 = \frac{1}{r+q_1}, \qquad q_j = \frac{j}{r+q_{j+1}}.$$

Replacing the final positive q_{k+1} by zero reverses the inequality at each substitution. This proves the alternating finite bounds, since translation gives $I_0 = \mathcal{M}(r)$. No convergence of an infinite continued fraction is required. The displayed rational recursion gives $m_k(x) = r\mathcal{M}_k(r)$ when $r = \sqrt{x}$.

Splitting the norm at $r = \sqrt{z/c}$, as in the preceding proof, gives the exact identity

$$J(z,c) = \sqrt{\pi/2}(z-3c) + \sqrt{zc}\,e^{-x/2}\left(6 + \frac{2(3-x)}{x} r\mathcal{M}(r)\right).$$

The multiplier of $\mathcal{M}$ is nonpositive when $x \ge 3$ and positive when $x < 3$. Substitute the corresponding finite lower or upper bound. Positivity of the remaining factor is immediate for $x < 3$; for $x \ge 3$, use $0 < m_{11}(x) < 1$ to bound it below by $6 - 2(x-3)/x > 4$. Finally the exact norm J, rather than either upper-bound formula, is convex, so its containing rectangle is still exhausted by its four corners. □

At every positive coefficient corner use the displayed finite Mills bound. A negative Gaussian coefficient uses a lower enclosure for $\sqrt{\pi/2}$; a nonnegative coefficient uses an upper one. The remaining factor is positive, so outward upper square-root and exponential enclosures are valid. A simultaneous reflection of both odd coefficients leaves the norm unchanged.

The two coefficient choices in the certificates.

All coefficients come from the actual marginal. For $Q = X^2 - \gamma X - 1$, multiplication gives

$$\begin{aligned} \mathbb{E}Q^2 &= \mu_4 - 1 - \gamma^2, & \mathbb{E}X^5 &= \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X+2\gamma)], \\ \mu_6 &= 1 + 3\gamma^2 + \gamma^4 + \mathbb{E}[Q^2\{(X+\gamma)^2 + 2\gamma^2 + 2\}]. \end{aligned} \tag{168}$$

For any radius $R_0 \ge |X|$ with $R_0 \ge 2g$, the two signs of $\mathbb{E}X^5 - 10\gamma$ are bounded, at $t = |\gamma|$, by

$$\begin{aligned} P_{R_0}(t) &= R_0(M-1) + (10-2M)t - R_0t^2 + t^3, \\ Q_{R_0}(t) &= R_0(M-1) + (2M-10)t - R_0t^2 - t^3. \end{aligned}$$

The gate $R_0 - 2g \ge 0$ permits substituting M for μ_4 in both signs. The triangle choice uses the valid enclosing radius $R_0 = 6$, $a_* = \max(2, M-3)/24$, $M_6 = RE_5$, $b_0 = (45-M)(1-2/N)/96$ and

$$120Z = \min\{E_5 + 10g, Z_0\},$$

$$Z_0 = \begin{cases} P_6(g), & M \le 5, P_6'(g) \ge 0, \\ 6(M-1) + (10-2M)^2/[4(6-g)], & M \le 5, P_6'(g) < 0, \\ 6(M-1) + (2M-10)^2/24, & M > 5. \end{cases} \tag{169}$$

If $M \le 5$, $P_6 \ge Q_6$ and $P_6'' < 0$ on $[0,g]$; use the increasing endpoint or $t^3 \le gt^2$ and complete the square. If $M > 5$, $P_6 \le 6(M-1)$ and the upper quadratic for Q_6 gives the third row. One may also use the actual radius by setting $r_0 = \max(R, 2g) > g$ and

$$Z_r = \begin{cases} r_0(M-1) + (10-2M)t_* - (r_0-g)t_*^2, & M \le 5, \\ & t_* = \min\{g, (10-2M)/[2(r_0-g)]\}, \\ r_0(M-1) + (2M-10)t_* - r_0 t_*^2, & M > 5, \\ & t_* = \min\{g, (2M-10)/(2r_0)\}. \end{cases}$$

Indeed for $M \le 5$, $P_{r_0} \ge Q_{r_0}$ and $t^3 \le gt^2$; for $M > 5$, $P_{r_0} \le r_0(M-1)$ and dropping $-t^3$ bounds Q_{r_0}. Each displayed quadratic is maximized at its clipped vertex. Thus the selected implementation takes $120Z = \min(E_5 + 10g, Z_0, Z_r)$. It uses $c_0 \ge \sqrt{\pi/(16a_0^3)} = \lambda_N$ and $c_1 = c_0/2$. The weaker choice $c_1 \ge \sqrt{\pi/(256a_0^5)} = \lambda_N N/[2(N-1)]$ is also permitted by the same transfer lemma.

For the paired choice the already proved cell has $0 \le \gamma_l \le |\gamma| \le g \le 1$, $M \le 2$, and $R > 2g$. After reflecting only this auxiliary Fourier integral if necessary, set

$$\begin{gathered} a_* = (2-\gamma_l^2)/24, \qquad b_0 = (43/96)(1-2/N), \\ M_6 = \min\{RE_5, 1+3g^2+g^4 + (R^2+2Rg+3g^2+2)(M-1-\gamma_l^2)\}, \\ t_* = \min\{g, (10-2M)/[2(R-g)]\}, \\ 120Z = \min\{E_5 + 10g, R(M-1) + (10-2M)t_* - (R-g)t_*^2\}, \\ z_l = \max\{-Z, [8\gamma_l - \gamma_l^3 - (R+2\gamma_l)(M-1-\gamma_l^2)]/120\}, \quad z_u = Z, \\ c_l = (3-M)\gamma_l/144, \qquad c_u = (2-\gamma_l^2)g/144. \end{gathered}$$

Check $M - 1 - \gamma_l^2 \ge 0$ and that the rectangle is ordered. Here $P_R \ge Q_R$, and $P_R(t) \le R(M-1) + (10-2M)t - (R-g)t^2$, whose maximum is at t_*. The numerator of the lower z bound increases in $\gamma \ge 0$ because its derivative is $10 - 2M + 2R\gamma + 3\gamma^2 > 0$. Pearson's first and third identities give a_* and M_6. If $3g^2 \le 2$, the increasing function $2\gamma - \gamma^3$ further licenses $c_u \le (2g - g^3)/144$. Thus this positive rectangle is one specialization of the signed transfer, with exactly the coefficients used by the fixed acceptance programs.

S.4 The complete high-frequency integral

For $b, \kappa > 0$ and $n > 0$, put

$$T_n = \sqrt{n} \int_b^\infty e^{-n\kappa t^2} t^{-3}\, dt, \qquad x = n\kappa b^2.$$

Lemma S.4 (HIGH_LAGUERRE2)**.** *For every $x > 0$,*

$$T_n \le \frac{\sqrt{n} e^{-x}}{2b^2} \frac{x+2}{x^2+4x+2}.$$

If $N\kappa b^2 \ge 1/4$, the right side decreases with $n \ge N$.

Proof. Substitute $t^2 = b^2(1+s)$ and use positive Fubini to obtain

$$T_n = \frac{\sqrt{n} e^{-x}}{2b^2} \int_0^\infty \frac{e^{-xs}}{(1+s)^2}\, ds = \frac{\sqrt{n} e^{-x}}{2b^2} \int_0^\infty \frac{u}{x+u} e^{-u}\, du.$$

Put $P(u) = u^2 - 4u + 2$ and $D(x) = x^2 + 4x + 2 = P(-x) > 0$. Exact division gives

$$\frac{1}{x+u} - \frac{Q_x(u)}{D(x)^2} = \frac{P(u)^2}{(x+u)D(x)^2} \ge 0,$$

where

$$Q_x(u) = -u^3 + (x+8)u^2 - (x^2+8x+20)u + x^3 + 8x^2 + 20x + 16.$$

Integration uses only $\int_0^\infty u^j e^{-u} du = j!$, proved by integration by parts. It gives $\int Q_x(u)e^{-u}du = (x+3)D(x)$ and hence

$$\int_0^\infty \frac{u}{x+u} e^{-u} du = 1 - x\int_0^\infty \frac{e^{-u}}{x+u} du \le 1 - \frac{x(x+3)}{D(x)} = \frac{x+2}{D(x)}.$$

For the all-index statement, differentiate $\sqrt{x}e^{-x}(x+2)/D(x)$. After multiplication by positive factors, its derivative has numerator $4 - 10x - 22x^2 - 13x^3 - 2x^4$. This polynomial is negative at $x = 1/4$ and strictly decreasing for $x > 0$. Thus it is negative on the entire required half-line. □

Let $EX = 0$, $EX^2 = 1$, $EX^4 \le M$, and write $\gamma = EX^3$, $f(t) = Ee^{itX}$. For independent copies suppose an actual event E satisfies

$$P(E) \ge a > 0, \qquad |X - X'| \ge d_0 > 0 \text{ on } E, \qquad E[(X-X')^2 \mid E] \le M_E.$$

The event is measured under the product of the same marginal law. Define the Taylor comparator $e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6)$ and

$$I_\infty = \int_{1/2}^\infty |f(t)^n - e_n(t)| t^{-3}\, dt, \qquad L(x) = \frac{x+2}{x^2+4x+2}.$$

The sign of the imaginary term is immaterial for its absolute-value bound. Use $b = 9/10$, $0 < r \le 1/4$,

$$h = \frac{333}{106} - \frac{(22/7)^3 r^2}{6} > 0, \quad \kappa = a(23/48)^2 d_0^2,$$
$$S_3 = \sum_{j=1}^{20} j^{-3} + \frac{1}{800}, \qquad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + \frac{1}{4(20-r)^4}.$$

Set

$$c = \frac{1 - \max_{u\in\{1/4,81/100\}}[1 - u + (M+3)u^2/12]}{2},$$
$$c_0 = \frac{1}{8} - \frac{M+3}{384}, \quad s = \frac{1 - (M+3)53/600}{2}.$$

The usual damping domain requires $c > 0$.

Lemma S.5 (HIGH_WINDOW50)**.** *For every integer $n \ge 5$, the complete upper bound is*

$$\sqrt{n} I_\infty \le A_n + \frac{M_E}{4\pi^{3/2} h\sqrt{a}} \left(S_3 + \frac{3S_5}{nah^2} \right) + \frac{M_E\sqrt{n}}{2\pi^2} e^{-nah^2 r^2}$$
$$+ Z_n + \frac{\sqrt{n}}{2b^2} e^{-na(23/48)^2} + \sqrt{n} e^{-n/8} \left(2L(n/8) + \frac{\sqrt{M-1}}{3} \right),$$

where always $A_n = 2\sqrt{n}e^{-nc}$ is permitted, and, if $s > 0$, it can be replaced by the smaller of itself and

$$2\sqrt{n} e^{-nc_0} L(ns/4).$$

The zero-cell term Z_n may be taken as the smaller of

$$\frac{e^{-n\kappa b^2}}{2b^4\kappa\sqrt{n}}, \qquad \frac{\sqrt{n}e^{-n\kappa b^2}}{2b^2} L(n\kappa b^2).$$

Each term may be evaluated at a common floor N for all $n \ge N$ provided

$$2Nc' > 1 \quad (c' \in \{c, ah^2r^2, a(23/48)^2, 1/8\}).$$

For the new first term also require $2Nc_0 > 1$; for the second zero-cell option require $N\kappa b^2 \ge 1/4$. If either extra condition is unavailable, retain the corresponding old term. Any fixed collection of permitted windows may be evaluated, and the minimum of its complete six-term bounds remains valid.

Proof. We give the entire frequency partition to make clear what changes. On $[1/2, b]$, symmetrization and $\cos u \le 1 - u^2/2 + u^4/24$ yield

$$|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12, \qquad |f(t)|^n \le e^{-nc(t^2)}, \quad c(u) = u/2 - (M+3)u^2/24.$$

Bounding $c(t^2)$ below by c and using $\int_{1/2}^{b} t^{-3}dt < 2$ gives the old first term. Concavity supplies the more precise chord bound

$$c(u) \ge c_0 + s(u - 1/4), \qquad 1/4 \le u \le 81/100,$$

because the difference is exactly $(M+3)(u-1/4)(81/100-u)/24$. If $s > 0$, extend only the resulting positive exponential majorant to the half-line. The preceding rational Gaussian-tail lemma, with lower endpoint $1/2$, gives

$$\sqrt{n}\int_{1/2}^{b} e^{-nc(t^2)}t^{-3}dt \le e^{-n(c_0-s/4)}\sqrt{n}\int_{1/2}^{\infty} e^{-nst^2}t^{-3}dt \le 2\sqrt{n}e^{-nc_0}L(ns/4).$$

For $t \ge b$, conditioning on the actual event and convex Jensen give

$$|f(t)|^n \le e^{-nE\sin^2(t(X-X')/2)} \le E[e^{-na\sin^2(td/2)} \mid E], \qquad d = |X - X'|.$$

Positive Tonelli allows a separate partition for every $d > 0$. Put $\ell = 2\pi/d$. The zero cell is $[0, \ell/2]$; the other cells are $\ell[j - 1/2, j + 1/2]$, $j \ge 1$. In each nonzero cell take the core $t = \ell(j + v)$, $|v| \le r$. These cores are disjoint even for $r \le 1/2$. For our smaller range $r \le 1/4$, the elementary sine bound gives $|\sin(\pi v)| \ge h|v|$ throughout a core, and $|\sin(\pi v)| \ge hr$ throughout its wings. The latter uses that $\sin(\pi|v|)$ increases on $[0, 1/2]$.

Since $j - r > 0$, Taylor's formula on the two sides gives

$$\frac{(j-v)^{-3} + (j+v)^{-3}}{2} \le j^{-3} + 6(j-r)^{-5}v^2.$$

Integrate this majorant against $e^{-nah^2v^2}$ on the full real line. The result is $\sqrt{\pi}[j^{-3}/\sqrt{nah^2} + 3(j-r)^{-5}/(nah^2)^{3/2}]$. The Jacobian is $t^{-3}dt = d^2(4\pi^2)^{-1}(j+v)^{-3}dv$. Summing and averaging d^2 gives the second term. The two series tails are bounded by their decreasing integrals after 20, producing exactly S_3, S_5. The total wing weight is at most $\int_{\ell/2}^{\infty} t^{-3}dt = d^2/(2\pi^2)$, giving the third term. Extending cores or wings below b only adds a positive integral.

In the zero cell write $u = td/2 \le \pi/2$. For $u \le 1/2$, use $\sin u \ge (23/24)u$ and the rational Gaussian-tail lemma with rate κ; its two choices give Z_n. For $u \ge 1/2$, use $\sin u \ge 23/48$ and total weight $1/(2b^2)$, giving the fifth term. Shared endpoints have zero integration measure, including when the conditional distance distribution has atoms.

Finally keep the comparator on the entire interval $[1/2, \infty)$. Its absolute-value bound splits its integral into

$$\sqrt{n}\int_{1/2}^{\infty} e^{-nt^2/2}t^{-3}dt + \frac{n^{3/2}|\gamma|}{6}\int_{1/2}^{\infty} e^{-nt^2/2}dt.$$

The first part is at most $2\sqrt{n}e^{-n/8}L(n/8)$. For the second, $t \ge 1/2$ implies

$$\int_{1/2}^{\infty} e^{-nt^2/2}dt \le 2\int_{1/2}^{\infty} te^{-nt^2/2}dt = 2e^{-n/8}/n.$$

Pearson's elementary identity $E[(X^2 - 1 - \gamma X)^2] = EX^4 - 1 - \gamma^2 \ge 0$ gives $|\gamma| \le \sqrt{M-1}$ and proves the sixth term. Any separately proved upper bound $|\gamma| \le g$ may replace $\sqrt{M-1}$ in that term. The selected implementation uses $g = \min(1.85, \sqrt{M-1})$.

For uniformity, $\sqrt{n}e^{-nc'}$ decreases when $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ always decrease. The derivative of $L(x)$ is $-(x^2+4x+6)/(x^2+4x+2)^2 < 0$. The new first term is thus decreasing under its stated gate, and the comparator decreases for every $n \ge 5$. The earlier tail lemma proves the separate $N\kappa b^2 \ge 1/4$ gate for its new zero-cell option. These comparisons concern the bounding expressions, so they are uniform even when the marginal law varies with n. □

Events, not artificial kernels.

The entry uses only the following actual choices, with the direct rate from the same fourth moment in each case:

event	a	d_0	M_E
$(X-X')^2 \ge 9/25$	$(2-9/25)^2/(2M+6)$	$3/5$	$2/a$
$XX' < 0, D \le d \le .8$	$1/(2r_*^2)$	$\sqrt{2-d}$	$4r_*^2$
$XX' < 0, p \in [l,u]$	$2\min(l(1-l), u(1-u))$	$\max(A_l + B_l, \sqrt{2-d})$	$2/a$

The first row follows from Cauchy using $\mathbb{E}[(X-X')^2; E] \ge 2 - d_0^2$ and $\mathbb{E}(X-X')^4 = 2\mu_4 + 6$. The second is Lemma T.1; all its moment and $\alpha < 3$ gates are checked at use. In the third, $D \le d < 2$ ensures a genuine central gap, $P(XX' < 0) = 2pq$, and restriction of the squared distance gives $M_E = 2/a$. Taking a minimum of *complete* high-frequency bounds is legitimate. No tail is dropped and the third construction also allows a cell crossing $1/2$.

T Coefficient atlas for the enclosure

These formulas are inequalities for one actual law. They are the two coordinate descriptions of the same enclosure: signed moments before a probability gap is located, and conditional moments afterwards. A bound is used only with its stated gates, and every refinement is an intersection.

T.1 Signed moments with a common first moment

Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Cauchy gives $m^2 \le a_+a_-$ and $b_\pm m \ge a_\pm^2$. Consequently

$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}, \qquad m \ge \max\left\{\frac{1+\beta^2}{2\rho}, \frac{a_+^2}{b_+}, \frac{a_-^2}{b_-}\right\}.$$

These statements also hold with a zero atom. Projection of $X|X|$ onto the span of X and $|X|$ gives

$$\mu_4 \ge \rho^2 + \frac{(\gamma - \rho\beta)^2}{1-\beta^2}.$$

For completeness, this follows by minimizing the nonnegative expectation $E(X|X| - uX - v|X|)^2$ over u, v; its Gram matrix has diagonal entries one and off-diagonal entry β, and is positive definite.

Two support-localizing moment matrices give the additional upper bounds as follows. If $D_\pm = Ra_\pm - b_\pm > 0$, then

$$2m + \sum_{\pm} \frac{(Rm - a_\pm)^2}{D_\pm} \le R, \qquad \mu_4 \le R\rho - \sum_{\pm} \frac{D_\pm^2}{Rm - a_\pm}.$$

Indeed the first follows by summing the determinant inequalities for

$$E[(R - |X|)(1, |X|)^{\mathsf{T}}(1, |X|); \pm X > 0].$$

The second follows from the corresponding matrices with an additional factor $|X|$. One more factor $|X|$ yields, sign by sign,

$$E[X_\pm^5] \le RE[X_\pm^4] - \frac{(Rb_\pm - E[X_\pm^4])^2}{D_\pm}.$$

Zero denominators are treated by dropping their nonnegative correction; they never imply exclusion without a separate contradictory numerator. The common m is retained throughout.

A second useful upper bound for m follows by integrating $(x - a)^2(R - x) \ge 0$ for $x = |X| \in [0, R]$. For every fixed $a > 0$,

$$2m \le \frac{R - \rho + 2a + Ra^2}{a(2R + a)}.$$

The cover uses only the eleven rational choices $a = 2/10, \ldots, 12/10$.

Write $\delta = \gamma - \rho\beta$. Cauchy and a square completion on each sign give the signed constant bound

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \le \frac{\rho^3}{4} - \frac{(\rho + \gamma)(\delta_+)^2}{8(1 + \beta)^2} - \frac{(\rho - \gamma)((-\delta)_+)^2}{8(1 - \beta)^2}.$$

Indeed on either sign, write $a = \mathbb{E}X_s^2$, $b = \mathbb{E}X_s^3$, $c = \mathbb{E}X_s^4$, $f = \mathbb{E}X_s^5$. Cauchy gives $c \ge b^2/a$, $f \ge c^2/b$, and $\rho c/2 - f/4 \le \rho^2 b/4 - (c - \rho b)^2/(4b)$. Distance to the half-line $c \ge b^2/a$ proves the stated penalty on summing both signs. Independently, $\mu_4^2 \le \rho\eta_5$ and the projection bound above give

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \le \frac{\rho^3}{4} - \frac{(\gamma - \rho\beta)^4}{4\rho(1 - \beta^2)^2}.$$

The larger of the two nonnegative deductions is retained.

T.2 An actual central gap without assuming $D < 2$

If zero belongs to the support, the signwise gap calculation gives $J \le \rho$. Thus $D \ge \sqrt{10}\rho - \gamma \ge (\sqrt{10} - 1)\rho$. Whenever $D < (\sqrt{10} - 1)\rho$, there is therefore a central gap $(-A, B)$, $A, B > 0$, with no zero atom. Put $p = P(X > 0)$, $q = 1 - p$, $T_+ = b_+ - mB^2$ and $T_- = b_- - mA^2$. The exact gap decomposition is

$$D = mQ + (\sigma - 1)T_+ + (\sigma + 1)T_- + 3(T - J_o), \qquad Q = (\sqrt{\sigma + 1}A - \sqrt{\sigma - 1}B)^2, \quad \sigma = \sqrt{10}.$$

All terms are nonnegative. Hence $T \le D/(\sigma - 1)$ and

$$(A + B)^2 \ge A^2 + B^2 \ge \frac{\rho - D/(\sigma - 1)}{m}.$$

Moreover $A \le a_-/m$, $B \le a_+/m$ and $J \le \rho + 2m$. The last bound gives the compulsory defect

$$D \ge \sigma\rho - \gamma - 6m.$$

The inequalities $T - J_o \geq 2BR_+ + 2AR_-$ give the useful additional endpoint constraint

$$D \geq (\sigma - 1)b_+ + 6a_+B - (\sigma + 5)mB^2.$$

If $D < (\sigma - 1)b_+$, this downward quadratic forces B to be at least its positive root. The reflected constraint has a_-, b_-, A and the coefficients $\sigma + 1, \sigma + 7$. In a closed moment box, positive coefficients are rounded down, negative coefficients up, and the lower square-root endpoint is used in the resulting root bound.

The actual opposite-sign event has probability $2pq$. The first two moment constraints give lower bounds for both p and q. For independent copies one has exactly

$$E[(X - X')^2 \mid XX' < 0] = \frac{1 + \beta(1 - 2p) + 4m^2}{2pq} = \frac{1 + 4m^2 + \beta}{2p} + \frac{1 + 4m^2 - \beta}{2q}.$$

For an upper bound, use the upper m endpoint and both endpoints of p and β. The expression is convex in p and affine in β. This improves the former $2/P(XX' < 0)$ bound inside the complete HIGH estimate. Both copies still have the same actual marginal.

T.3 Preserving central and outer integral costs

Let

$$P_1(A, B) = B^3/3 + AB^2/2 - A^3/6, \quad P_2(A, B) = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

Then $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A + B)(\rho - B)$,

$$G(A, B; \rho) \leq G(A, \rho; \rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

The last expression increases with A because its derivative is $(\rho - A)^2(\rho + 2A)/6$, and is convex in ρ because its second ρ derivative is $\rho^2 + \rho A$. Its closed-rectangle maximum is therefore bounded using the upper A endpoint and both ρ endpoints.

Positive outer gaps satisfy $(t-a)(\rho-t) \leq (\rho-B)_+^2/4$. On negative outer gaps the positive part is bounded by $(R - \rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \leq \frac{3}{8}\{(\rho - B)_+^2 T_+ + (R - \rho)^2 T_-\}.$$

The available weighted outer budget is $D - mQ$, not the entire D. The cover intersects this budget with the individual bounds $T_+ \leq b_{+,u} - m_l B_l^2$ and $T_- \leq b_{-,u} - m_l A_l^2$.

For the signed phase term $U = J_1 + (\mu_4 - 3)/6$, retain

$$-RT_-/2 \leq J_{1,o} \leq RT_+/2.$$

The central minimum of P_1 occurs at B_l and an A endpoint; its maximum occurs at B_u and A clipped to the interval at $A = B_u$. A nonnegative polynomial bound is multiplied by m_u for an upper bound and m_l for a lower bound; a negative one uses the opposite endpoint. For example, the binary endpoints $A = 3/2$, $B = 2/3$ make this maximum negative; the sign switch is necessary even when $D < 2$. The fourth-moment sign intervals also sharpen $K_1 = E[X|X|^3] - 3\beta - 3\rho\gamma/2$. The complete K_1, U, ϕ rectangles are used in the signed expression $|CK_1 + \phi U|$, without replacing it by two separate absolute values.

A complete coarse bound before a central gap is licensed.

Always $-\mu_4/6 \le J_1 \le \mu_4/2$ and $-3J_2 - \eta_5/4 \le 0$. On a gap of length $h = b - a$, these follow from

$$\int_a^b (t-a)t\,dt + \tfrac{1}{2}\int_a^b t^2 dt = hb^2/2, \quad \tfrac{3}{2}\int_a^b t^2 dt - \int_a^b (t-a)t\,dt = h(2a+b)^2/6,$$

and, on a negative gap $(-v, -u)$, from $\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3$. For $1 \le r \le \rho$ and actual support radius $R \le 6$, set

$$\Gamma(R,r) \ge \max_{r \le v \le R} \left\{ -\frac{v^2}{4} + \frac{(R+r)v}{3} - \frac{Rr}{2} + \frac{r^3(2R-r)}{12v^2} \right\}_+ .$$

Then

$$3(\rho J_1 - J_2) \le \frac{3}{2}\Gamma(R,r)(\rho - \gamma) + \frac{3}{8}\rho^2 J.$$

On negative gaps extract $x = |t|$ and discard the nonpositive part $x < \rho$. Replacing ρ by r and the outer endpoint by R gives $\int_r^v (R-x)x(x-r)\,dx$. Dividing this integral by v^2 gives exactly the displayed expression. Signwise Tonelli now charges $\Gamma\mathbb{E}X_-^3 = \Gamma(\rho-\gamma)/2$. On positive gaps $(t-a)(\rho - t) \le \rho^2/4$, giving $\rho^2 J/8$. The implementation uses 64 closed v-intervals: on each, maximize the concave quadratic at its clipped vertex $2(R+r)/3$ and the positive reciprocal term at the left endpoint. This proves its interval bound. The alternative bound $\Gamma \le 5/2$ follows at $r = 1, R = 6$ by $3v^4 - 28v^3 + 66v^2 - 11 > 0$ on $[1, 6]$: the endpoint values are $30, 205$, and the interior critical values are $(85 \mp 35\sqrt{5})/2 > 0$. These monotone enclosing parameters cover all $r \ge 1, R \le 6$.

T.4 Moment and gap coefficients for a whole cell

The following coefficient rules are collected here to make the finite calculation transparent. Every rule is an inequality for an actual law; there is no extremal-law assumption inside a cell. Write $v = 1 - \theta$. For $p \in [l, u] \subset (0, 1)$ set

$$s_0 = \min\{l(1-l), u(1-u)\}, \quad v_* = \max\{\sqrt{u/(1-u)}, \sqrt{(1-l)/l}\},$$
$$r_B = (1-2s_0)/\sqrt{s_0}, \quad g_B = \max_{t=l,u} |1-2t|/\sqrt{t(1-t)}, \quad b_B = \max_{t=l,u} |1-2t|,$$
$$L_3 = R + 2v_*, \quad L_4 = R^2 + 2Rv_* + 3v_*^2,$$
$$L_5 = R^3 + 2R^2 v_* + 3Rv_*^2 + 4v_*^3, \quad e_B = r_B \frac{1 - 4s_0 + 2s_0^2}{s_0(1-2s_0)}.$$

The polynomials $T_3 = x + 2a$, $T_4 = x^2 + 2ax + 3a^2$ and $T_5 = x^3 + 2ax^2 + 3a^2 x + 4a^3$ are nonnegative. Their maxima at $x = R$, $a = v_*$, in Lemma R.2, give, if $\theta \le \Theta < 1$,

$$\begin{aligned} \rho &\le r_B + (L_3 - r_B)\Theta, & |\gamma| &\le g_B + (L_3 - g_B)\Theta, \\ |\beta| &\le b_B + (1 - b_B)\Theta, & \mu_4 &\le (1-\Theta)^2(1/s_0 - 3) + L_4\Theta, \\ \eta_5 &\le (1-\Theta)^2 e_B + L_5\Theta. \end{aligned} \tag{170}$$

Replacing θ by Θ uses the explicitly checked inequalities $L_3 > r_B, g_B$, $L_4 > 2(1/s_0 - 3)$ and $L_5 > 2e_B$. The exact binary factors are $v^{3/2}, v, v^2, v^{5/2}$; the upper bounds use $v^{3/2} \le v$ and $v^{5/2} \le v^2$. Thus cells crossing $1/2$ and zero conditional variances are included. The same cubic expansion, in the later domain $p \in [.37, .47]$, $A \ge .75$, $B \ge 1.05$, gives

$$\rho \ge v^{3/2}\rho_B(p) + 3\min(A, B)\theta \ge \rho_B(p) + .45\theta, \qquad \rho_B(p) = \frac{1 - 2pq}{\sqrt{pq}}. \tag{171}$$

Indeed $\rho_B < 1.11$ there and $v^{3/2} \geq 1 - 3\theta/2$. Also $\beta \geq 1 - 2p - 2(1-p)\theta$.

For the signed coefficient $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$, define

$$K(p, v) = (1-2p)\{(3v^3/2 - v^2)/(pq) - 3v^3 + v^2 + 3v\}.$$

Direct substitution gives $K_1(V) = -K(p, v)$, with the coefficient 3 unchanged. For $v \geq 2/3$, K decreases in p: both coefficients in braces are nonnegative and $(1-2p)/(pq)$ decreases. Its derivative in v has the sign of $1 - 2p$, since the derivative of the braces is at least $9v^2 - 6v + 3 > 0$. Hence both variance endpoints are retained. The error is

$$|K_1(X) - K_1(V)| \leq \Theta\{L_4 + 3 + \tfrac{3}{2}L_3(R_\rho + g_B)\} \tag{172}$$

when $\rho \leq R_\rho$. This follows from the conditional expansions and $\rho\gamma - \rho_V\gamma_V = \rho(\gamma - \gamma_V) + \gamma_V(\rho - \rho_V)$.

Here are the remaining signed charges. They also explain the two alternative bounds retained in the initial, coarse moment cover:

$$-\mu_4/6 \leq J_1 \leq \mu_4/2, \qquad -3J_2 - \eta_5/4 \leq 0, \tag{173}$$

$$\rho J_1 - J_2 \leq \tfrac{5}{4}(\rho - \gamma) + \tfrac{1}{8}\rho^2(\rho + 1), \tag{174}$$

$$-75D/104 \leq J_{1,o} \leq 3D/2. \tag{175}$$

To prove the first line, a gap of length $h = b - a$ satisfies

$$\int_a^b (t-a)t\,dt + \tfrac{1}{2}\int_a^b t^2 dt = hb^2/2, \quad \tfrac{3}{2}\int_a^b t^2 dt - \int_a^b (t-a)t\,dt = h(2a+b)^2/6.$$

Sum with weight τ_g. On a negative piece $(-v, -u)$, $\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3 \geq 0$; Tonelli proves the J_2 assertion. For (174), the positive part is at most $\rho^2 J/8$. The negative part is at most $\mathbb{E}[V\int_1^V (6-x)x(x-1)dx; V \geq 1]$, $V = X_-$, hence at most $(5/2)\mathbb{E}X_-^3$. The last comparison is equivalent to $3V^4 - 28V^3 + 66V^2 - 11 > 0$ on $[1, 6]$: the endpoint values are $30, 205$, and at the possible interior critical points $(7 \pm \sqrt{5})/2$ its value is $(85 \mp 35\sqrt{5})/2 > 0$. For (175), use $J_{1,o} \geq -3T_-$, $J_{1,o} \leq 3J_o \leq 3T$ and (141).

The central integrals are $J_{j,c} = mP_j(A, B)$, with

$$P_1 = B^3/3 + AB^2/2 - A^3/6, \qquad P_2 = B^4/4 + AB^3/3 - A^4/12.$$

They increase in B; their A derivative changes from positive to negative at $A = B$. A rectangle minimum is therefore at B_l and an A endpoint. Multiply a negative minimum by an upper bound for m, and a nonnegative minimum by a lower bound. For the upper central charge, maximizing at $(\bar{a}, \bar{b})$ if $p \leq 1/2$ gives $J_{1,c} \leq v^2(2-3p)/(6pq)$, a decreasing function of p on that interval. If $p \geq 1/2$, maximize first at $A = B$ to get $J_{1,c} \leq (2/3)v^2q^2/p \leq 1/3$. Thus every cell may use

$$J_{1,c} \leq \max\{1/3, (2-3l)/(6l(1-l))\}. \tag{176}$$

For $|X| \leq R$, $R \geq 1$, define

$$k_4(R) = \frac{(R+1)^2}{R+1/2}, \qquad k_5(R) = \frac{R^3 + 2R^2 + 3R + 3/2}{R+1/2}.$$

Then

$$\mu_4 \leq \min\{R\rho, 1 + k_4(R)(\rho - 1)\}, \quad \eta_5 \leq \min\{R\mu_4, 1 + k_5(R)(\rho - 1)\}, \quad \mu_6 \leq R\eta_5. \tag{177}$$

Writing $u = |X|$ in (R.2), the first two constants are the endpoint values of the ratios $u + 3/2 + 1/[4(u + 1/2)]$ and $u^2 + 3u/2 + 9/4 + 3/[8(u + 1/2)]$, whose derivatives are nonnegative. The other bounds are pointwise. The following coarse cancellation estimates continue to use the valid radius-six majorants: The last identity in (R.2) gives

$$|K_1 + 3\rho\beta| \le (96 - 18\rho)(\rho - 1)/13. \tag{178}$$

Indeed $u(u + 2 - 3r/2)/(u + 1/2)$ is convex on $[0, 6]$, its maximum is $(96 - 18r)/13$, and its negative part is at most $31/40$, smaller than that maximum for $1 \le r \le 37/20$. The following bounds are used only for $|X| \le 6$ and $1 \le \rho \le 37/20 < 2$, with the signed moment hypotheses below. The other bounds needed on this domain are

$$\begin{aligned} &|K_1| \le 6\rho - 3\rho^2/2 + 3|\beta|, \\ &K_1 \le M + 3|\beta| - 3\rho g_-/2, && -K_1 \le 3\rho + 3|\beta| + (3\rho/2 - 3)g_-, \\ &-K_q \le (\rho/2 - 3/8)M - \rho/2 + 1/8 + 3|\beta|g/2, && -K_q \le 8\rho^3/27 + (3|\beta|/2 - 1/2)\rho, \end{aligned} \tag{179}$$

where $\mu_4 \le M$, $|\gamma| \le g$, $\gamma \ge g_-$, and $K_q = \eta_5/4 - \rho\mu_4/2 - 3\beta\gamma/2 + \rho/2$. For the first two lines use $|\mathbb{E}[X|X|^3] - 3\gamma| \le 3\rho$, and note the negative coefficients of γ. For the last line use $\eta_5 \ge \mu_4^{3/2} \ge 3\mu_4/2 - 1/2$, or maximize $\rho y^2/2 - y^3/4$ over $y \ge 0$.

Compatible weaker resource constants.

The implementation also permits the weaker binary charge $b_I = (616/25)(1 - \Theta)^2 \operatorname{dist}(p_*, I)^2$. It follows from $2(6 + 2\sigma) > 616/25$ and $(1 - \theta)^{3/2} \ge (1 - \theta)^2$. With $K = \min(K_+, K_-)$ and $v_* = \max(v_+, v_-)$, the merged charge (143) gives

$$T_+ + T_- \le T_* = (d - b_I) \max\{v_*/[3(A_l + B_l)], (R + 2v_*)/K\}, \qquad d_o = \min\{d, (\sigma + 1)T_*\}.$$

Thus $W_o \le d_o$ and, for $R \le 6$,

$$-75d_o/104 \le J_{1,o} \le 3d_o/2, \qquad 3(\rho J_{1,o} - J_{2,o}) \le \frac{3}{8} \max\{\rho^2/(\sigma - 1), (R - \rho)^2/(\sigma + 1)\} d_o.$$

These are precisely the merged-cost versions of the sign-specific resource rule; they do not introduce another available budget.

T.4.1 Conditional lower moments improve two signed charges

Let $s_u = \max_{p \in [l,u]} pq$ and $\theta \le \Theta$. The unsigned cases of (139) give

$$\mu_4 \ge M_l := \max\{1, (1 - \Theta)^2(1/s_u - 3)\}, \tag{180}$$

$$\eta_5 \ge E_l := (1 - \Theta)^{5/2} \frac{1 - 4s_u + 2s_u^2}{s_u^{3/2}}. \tag{181}$$

Both binary moment functions decrease in $s \in (0, 1/4]$. For the fifth this follows because the derivative has the sign of $s^2 + 2s - 3/2 < 0$; the fourth is immediate. The fourth-moment lower bound sharpens the lower endpoint of $J_1 + (\mu_4 - 3)/6$; its explicit use appears in the cell map below.

The constant signed charge is

$$-K_q = \frac{\rho\mu_4}{2} - \frac{\eta_5}{4} + \frac{3}{2}\beta\gamma - \frac{\rho}{2}.$$

The old coherent estimate used $\eta_5 \geq 3\mu_4/2 - 1/2$ and $\mu_4 \leq M$. A second estimate uses (181). Their difference is independent of ρ. Consequently the improved bound is exactly the old one minus

$$\frac{1}{4}\max\{0, E_l - 3M/2 + 1/2\}. \tag{182}$$

This is a constant shift on the cell and does not spoil convexity. It does not assert the generally invalid implication $\eta_5 \geq 3M/2 - 1/2$ from an upper bound on μ_4.

T.4.2 Signed conditional Taylor cancellation

Let $K_1 = \mathbb{E}[X|X|^3] - 3\beta - \frac{3}{2}\rho\gamma$. On a sign class put $x = |X|$, $a = |V|$, $W = X - V$ and $\varepsilon = 1$ on the positive class, -1 on the negative class. In the actual central-gap domain there is no atom at zero. With $w_3 = x + 2a$, $w_4 = x^2 + 2ax + 3a^2$ and $\gamma_V = \mathbb{E}V^3$, exact conditional expansion gives

$$K_1(X) - K_1(V) = \mathbb{E}\Big[W^2\left\{\varepsilon(w_4 - 3 - \tfrac{3}{2}\rho w_3) - \tfrac{3}{2}\gamma_V w_3\right\}\Big]. \tag{183}$$

Apply (139) to $j = 2, 3, 4$, with $h = 1$ or $h = \varepsilon$, and use $\rho\gamma - \rho_V\gamma_V = \rho(\gamma - \gamma_V) + \gamma_V(\rho - \rho_V)$.

The positive coefficient in (183) is $F(x, a; t) = x^2 + 2ax + 3a^2 - 3 - \frac{3}{2}t(x + 2a)$ with $t = \rho + \gamma_V$; the negative coefficient is $-F$ with $t = \rho - \gamma_V$. Their ranges are bounded on complete rectangles of actual conditional means and support magnitudes. The positive-support magnitude is at most R. The same formula permits separate actual bounds on the two signed support magnitudes. On the late domain $p < 1/2$, bounds for γ_V are

$$\sqrt{(1-\Theta)^3(1-2u)^2/[u(1-u)]} \leq \gamma_V \leq (1-2l)/\sqrt{l(1-l)}.$$

The positive mean lies in $[\sqrt{(1-\Theta)(1-u)/u}, \sqrt{(1-l)/l}]$, and the negative magnitude mean lies in $[\sqrt{(1-\Theta)l/(1-l)}, \sqrt{u/(1-u)}]$. Use the lower bound $\rho \geq \rho_B(u)$ only after checking $3\min(A_l, B_l) \geq (3/2)\sup_I \rho_B$. Indeed conditional expansion and $(1-\theta)^{3/2} \geq 1 - 3\theta/2$ then imply $\rho \geq (1-\theta)^{3/2}\rho_B(p) + 3\min(A, B)\theta \geq \rho_B(p)$. Otherwise retain the universal lower bound one.

Since $x + 2a > 0$, F decreases with t. For fixed t its Hessian is $\left(\begin{smallmatrix} 2 & 2 \\ 2 & 6 \end{smallmatrix}\right) > 0$. Its rectangle maximum is at a corner. Its minimum is among the four corners, the minimizer on each edge, and the stationary point $(x, a) = (3t/8, 3t/8)$ if that lies in the rectangle. On fixed-x edges use $a = t/2 - x/3$, clipped to the edge; on fixed-a edges use $x = 3t/4 - a$, also clipped. This is a complete finite extremum list. Multiplication by $0 \leq \theta \leq \Theta$ uses $\Theta\min(0, L)$ and $\Theta\max(0, U)$ for the lower and upper coefficient bounds. It intersects this new interval with the retained absolute-error interval; neither need be discarded.

T.5 Generic event and complete central rectangle

Lemma T.1 (Generic small-defect event)**.** *Assume $|X| \leq R \leq 6$ and $\rho \leq 37/20$. Let $0 \leq d < 2$ and $D \leq d$. Define*

$$r_* = \min\{37/20, (3+d)/(\sigma - 1)\}, \qquad W = \sqrt{37d/10},$$
$$c = \sqrt{\sigma + 1}, \qquad e = \sqrt{\sigma - 1}, \qquad C_0 = c + e,$$

$$h_* = \frac{C_0}{6}\sqrt{12 + W^2} + \frac{W}{3C_0}, \qquad L_* = \max\left\{h_*, \frac{(2c - e)h_* + 3W}{C_0}\right\}.$$

Then $\rho \leq r_*$, $A + B \leq h_*$ and $|2B - A| \leq L_*$. If $\alpha = L_*/6 \leq 3$, then

$$|J_1| \leq \alpha(\rho + 1) + (3 - \alpha)d/2, \quad \mu_4 \leq h_*\rho + Rd/2, \quad \eta_5 \leq h_*^2\rho + R^2 d/2.$$

For independent copies, the actual event $E = \{XX' < 0\}$ obeys

$$P(E) \ge a := 1/(2r_*^2), \quad |X - X'| \ge d_0 := \sqrt{2-d} \text{ on } E, \quad \mathbb{E}[(X-X')^2 \mid E] \le 2/a.$$

Proof. The inequality $J \le \rho + 1$ gives $D \ge (\sigma - 1)\rho - 3$. Equation (141) gives $w^2 \le 2\rho D \le 37d/10$ for $w = cA - eB$. Since $ce = 3$,

$$A + B = \frac{C_0}{6}\sqrt{w^2 + 12AB} - \frac{w}{3C_0}, \qquad 2B - A = \frac{(2c-e)(A+B) - 3w}{C_0}.$$

Use $AB \le 1$, $|w| \le W$; also $2B - A \ge -(A+B)$. Now $J_{1,c} = J_c(2B-A)/6$, $|J_{1,o}| \le 3J_o$ and $0 \le J_o \le T \le d/2$ imply $|J_1| \le \alpha J + (3-\alpha)J_o$.

For either inner endpoint b_* and $b_* \le x \le R$,

$$x^4 - h_* x^3 \le Rx(x^2 - b_*^2), \qquad x^5 - h_*^2 x^3 \le R^2 x(x^2 - b_*^2).$$

Indeed $h_* \ge b_*$, and after factoring $x - b_*$ the remaining comparisons follow from $x \le R$. Integrate and use $T \le d/2$. Finally $m^2 \le pq$ and $m \ge 1/(2\rho)$ give $2pq \ge a$; the conditional squared distance is bounded by its unconditional expectation 2, divided by the event probability. □

The central part of $\rho J_1 - J_2$ is exactly $mG(A, B; \rho)$, where $G = \rho P_1 - P_2$. Let a proved probability cell give $A \in [A_l, A_u]$, $B \in [B_l, B_u]$, $m \in [m_l, m_u]$. Here

$$A_u = \sqrt{u/(1-u)}, \quad B_u = \sqrt{(1-l)/l}, \quad m_l = \sqrt{(1-\Theta)s_0}, \quad m_u = \sqrt{\max_{p \in [l,u]} pq}.$$

The closed-box Bernstein identity.

For a polynomial $P(x) = \sum_{k=0}^{d} t_k x^k$,

$$P(x) = \sum_{i=0}^{d} b_i \binom{d}{i} x^i (1-x)^{d-i}, \qquad b_i = \sum_{k \le i} t_k \binom{i}{k} \Big/ \binom{d}{k}.$$

Indeed $\binom{d}{i}\binom{i}{k} = \binom{d}{k}\binom{d-k}{i-k}$ and the binomial theorem recover each monomial x^k. The basis terms are nonnegative and sum to one on the closed unit interval. Tensor products prove the two-variable version below, including all boundary points.

Lemma T.2 (Complete central rectangle)**.** *Write $G(A_l + (A_u - A_l)x, B_l + (B_u - B_l)y; \rho) = \sum t_{kh} x^k y^h$. Define the 25 rational coefficients*

$$b_{ij}(\rho) = \sum_{\substack{k \le i \\ h \le j}} t_{kh} \frac{\binom{i}{k}}{\binom{4}{k}} \frac{\binom{j}{h}}{\binom{4}{h}}, \qquad 0 \le i, j \le 4,$$

and

$$\mathcal{G}(\rho) = \max_{i,j} \{m_l b_{ij}(\rho),\, m_u b_{ij}(\rho)\}.$$

Then $\rho J_{1,c} - J_{2,c} \le \mathcal{G}(\rho)$. Moreover $\mathcal{G}$ is convex in ρ.

Proof. Apply the Bernstein identity (T.5) separately to the two coordinates after the affine substitution. The Bernstein basis is nonnegative and sums to one on the closed square, so G is a convex combination of its 25 coefficients. For each coefficient retain both m endpoints, or equivalently use m_u when it is nonnegative and m_l otherwise. This gives a uniform bound on the entire three-dimensional rectangle, including boundaries. Each endpoint-multiplied coefficient is affine in ρ; their maximum is convex. No interior stationary point is omitted. □

T.6 One interval-feedback theorem

The signed identity from the preceding section is

$$\mathcal{R} = C\{3\beta J/2 + 3\rho J_1 - 3J_2 - K_q\} + H(CK_1 + \varphi U) - H^2(C\rho + \gamma\varphi/2) + E_n, \tag{184}$$

where $U = J_1 + (\mu_4 - 3)/6$, $\varphi = \phi(H/\sqrt{n})$. For any one coherent low-frequency triple and complete high-frequency value K_H at floor N,

$$|E_n| \le K_F + s_F|H|, \quad K_F = c_\pi(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = c_\pi L_H, \quad c_\pi \ge 2/\pi. \tag{185}$$

The implementation uses $16/25$ before the signed phase and $212/333$ afterward; both are valid since $\pi > 333/106$.

Lemma T.3 (Continuous cells and feedback)**.** *Suppose bounds for one actual cell give*

$$r_0 \le \rho \le r_1, \quad 0 < c_0 \le C \le c_1, \quad |\beta| \le b, \quad |\gamma| \le g, \quad \mu_4 \le M, \quad D \le d,$$

$K_1 \in [k_l, k_u]$, $U \in [u_l, u_u]$, *and* $0 \le \varphi_- \le \varphi \le 2/5$. *Freeze the shared outer charge* d_o, *lower fifth moment* E_l, *upper* $s_u = \max_{p\in[l,u]} p(1-p)$, *and central envelope* $\mathcal{G}$ *from the preceding bounds. Define*

$$\begin{aligned} A(r) &= \tfrac{3}{2}b(r + 2\sqrt{s_u}) + 3\mathcal{G}(r) + \tfrac{3}{8}\max\{r^2/(\sigma-1), (R-r)^2/(\sigma+1)\}d_o \\ &\quad + (r/2 - 3/8)M - r/2 + 1/8 + \tfrac{3}{2}bg - \tfrac{1}{4}\max\{0, E_l - 3M/2 + 1/2\}, \\ B(c) &= \max_{k\in\{k_l,k_u\},\, u\in\{u_l,u_u\},\, \psi\in\{\varphi_-,2/5\}} |ck + \psi u|, \\ Q(c,r) &= cr + \tfrac{1}{2}\varphi_-(3.16r - 3 - d) - \tfrac{1}{2}(2/5 - \varphi_-)(d - .16)_+. \end{aligned}$$

If all four corner values of Q *are positive, then, for every real* H,

$$\begin{aligned} \mathcal{R} &\le \max_{(c,r)\in\{c_0,c_1\}\times\{r_0,r_1\}} \{K_F + cA(r) + (B(c) + s_F)|H| - Q(c,r)H^2\} \\ &\le \max_{c,r \text{ corners}} \{K_F + cA(r) + (B(c) + s_F)^2/(4Q(c,r))\}. \end{aligned} \tag{186}$$

If T *is a strict upper bound for the final maximum in* (186)*, the weight-removal step in Lemma R.1 gives*

$$D < \frac{15000T}{997N}, \quad \varphi > \frac{997}{2500}\left(1 - \frac{5T}{2N}\right), \quad C < C_{\mathrm{E}} + V/N \quad \textit{if also } \mathcal{R}/\rho < V. \tag{187}$$

Only positive density floors are used. These statements remain valid at $H = 0$, *on closed cell edges, and for zero conditional variance.*

Proof. The constant coefficient follows from the shared outer budget, Lemma T.2, and (179), using

$$J \le \rho + 2ABm \le \rho + 2(1-\theta)^{3/2}\sqrt{pq} \le \rho + 2\sqrt{s_u}.$$

This follows from $A \le \bar{a}$, $B \le \bar{b}$, $m^2 = (1-\theta)pq$; it is used only in the charge $3|\beta|J/2$. Since $\gamma \ge 3.16\rho - 3 - d$, its negative part is at most $(d - .16)_+$. Using the upper density only for that negative part gives $C\rho + \gamma\varphi/2 \ge Q$. Absolute value is separately convex in each coordinate of $ck + \psi u$, proving the formula for B. For fixed r and $|H|$, the resulting upper bound is convex in c. Take those endpoints *before* completing the square. At either endpoint $c > 0$, it is convex in r, since $\mathcal{G}$ and the maximum of the two outer quadratics are convex, every other term of A is affine, and Q is affine in r. This proves the first line of (186); $fh - qh^2 \le f^2/(4q)$ proves the second. Bilinearity of Q proves its positivity throughout the rectangle from its corners.

□

T.7 Signed phase curvature in the resource schedule

The auxiliary Fourier orientation does not change the signed third moment in the actual transfer identity. The paired phase estimate must therefore retain its sign. On a fixed resource cell put

$$L = \frac{997}{2500}\left(1 - \frac{5r}{2N}\right), \qquad g_0(\rho) = 3.16\rho - 3 - d, \qquad U = .4,$$

where r, d are the preceding raw and defect upper bounds, and the checker requires $0 \le L \le U$. The actual density lies in $[L, U]$ and the actual signed moment satisfies $\gamma \ge g_0(\rho)$. Consequently

$$C\rho + \tfrac{1}{2}\gamma\phi \ge q_{\mathrm{s}}(C, \rho) := C\rho + \tfrac{1}{2}\min\{Lg_0(\rho), Ug_0(\rho)\}. \tag{188}$$

This follows by first multiplying the lower bound on γ by the nonnegative actual density, and then minimizing over its two endpoints. The simpler expression $C\rho + Lg_0(\rho)/2$ is licensed if $g_0 \ge 0$, or if nonnegativity of the actual γ has been separately proved. It is not licensed merely by reflecting the auxiliary Fourier integral.

The corrected price preserves the four-corner reduction. For fixed C, q_{s} is concave in ρ, being affine plus the minimum of two affine functions. For fixed ρ it is affine in C. In the phase envelope $A + f|H| - q_{\mathrm{s}}H^2$, the remaining terms are separately convex in C, ρ, by the moment and coefficient bounds already proved. Thus the entire envelope is separately convex, and two successive one-dimensional endpoint bounds place its maximum at the four corners. The same argument applies after adding the affine conserved-resource curvature e_ρ and subtracting a barrier price $K\rho$. If the four corrected divisors are positive, concavity makes them positive throughout the rectangle, and square completion gives the valid raw and resource bounds

$$\max_{C,\rho\ mcorners}\left(A + \frac{f^2}{4q_{\mathrm{s}}}\right), \qquad \max_{C,\rho\ mcorners}\left(A + \frac{f^2}{4(q_{\mathrm{s}} + e_\rho)} - K\rho\right).$$

Any threshold-root export likewise uses the corrected corner curvature.

For completeness, nonnegativity in the later acquired domains can be checked without replaying their iterations. Write $B = \mathbf{1}_{\{X>0\}}$, $V = \mathbb{E}(X \mid B)$, $W = X - V$, $\theta = \mathbb{E}W^2$, and $p = \mathbb{P}(X > 0)$. Conditional centering and $W \ge -R - V$ imply

$$\gamma = \mathbb{E}V^3 + 3\mathbb{E}(VW^2) + \mathbb{E}W^3 \ge \mathbb{E}V^3 + 2\mathbb{E}(VW^2) - R\theta.$$

For $p \in [l, u] \subset (0, 1/2)$ and $\theta \le \Theta < 1$ this gives

$$\gamma \ge \frac{(1-\Theta)^{3/2}(1-2u)}{\sqrt{u(1-u)}} - \left(R + 2\sqrt{\frac{u}{1-u}}\right)\Theta. \tag{189}$$

Here the binary cubic decreases in $p < 1/2$, and the right side decreases as its variance bound increases. These are bounds on the same actual law throughout every later call. The fixed acquired input domains license (189) before those calls are made. The exact domain check covers 302 input rows: 45 central-input rows, 45 left-input rows, 45 first-confinement feedback rows, 45 Sperner feedback rows, 84 signed-resource rows, and 38 final left-radius rows. With the common already acquired radius $R = 3.84$, each right side is greater than .096. Thus all later paired calls have an actual nonnegative signed third moment, independently of any auxiliary Fourier orientation.

There is also a convenient cell-constant correction. Since $\rho \ge 1$, $(-g_0(\rho))_+ \le (d - .16)_+$, so the weaker price

$$C\rho + \tfrac{1}{2}Lg_0(\rho) - \tfrac{1}{2}(U - L)(d - .16)_+$$

is valid. This is the correction selected for the additional index–46 calculation. Both forms keep the raw remainder separate from the barrier-subtracted resource budget.

The broad preliminary and complementary resource passes are replayed with (188); their subsequent consumers retain the same fixed target tables only when the complete corrected replay verifies every exported resource. A diagnostic equality of two corner maxima alone does not replace this check, since threshold bounds also consume the phase coefficients. Later uses of the simpler formula require their own explicit nonnegative signed-moment or small-defect license.

T.8 The unsigned first-stage cover is the same enclosure

The first finite barrier uses a cover of ρ alone before introducing the signed cover. Here are its exact coefficient choices, so that the first barrier needs no separate inherited entry theorem. Fix a closed cell $\rho \in [l,u] \subset [1,1.85]$, actual radius $1 \le R \le 6$, and write $k_j(R)$ for the radial ratios of Lemma R.2. Set

$$M = \min\{11.1, Ru, 1 + k_4(R)(u-1)\}, \quad g = \min\{1.85, u, \sqrt{M-1}\},$$
$$E_5 = \min\{RM, 1 + k_5(R)(u-1)\}, \qquad b = \sqrt{[u\sqrt{u^2+8} - u^2 - 2]/2}.$$

The last expression solves $\rho \ge (1+\beta^2)/\sqrt{1-\beta^2}$, proved by $\rho m \ge a_+^2 + a_-^2$ and $m^2 \le a_+ a_-$. If a previous pass licenses $D \le d < 2$, the generic-event lemma also permits the ceilings $h_* u + Rd/2$ and $h_*^2 u + R^2 d/2$ for M, E_5; all dependent quantities are recomputed after taking these minima. Use the triangle LOW triple and the minimum of complete permitted HIGH values, with the same moment cap. Put

$$K_F = \tfrac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \qquad s_F = \tfrac{16}{25} L_H.$$

The radial identity bounds $|K_1|$ by

$$k_{\mathrm{rad}} = 3ub + \frac{R(R + 2 - 3l/2)}{R + 1/2}(u-1),$$

provided the displayed radial coefficient is at least $31/40$. Indeed its positive maximum is at R and its negative part is at most $31/40$ on $[0,R]$ for $1 \le \rho \le 1.85$; convexity of $x(x+2-3\rho/2)/(x+1/2)$ proves this directly. Two additional valid bounds are

$$k_{\mathrm{old}} = 6u - 3u^2/2 + 3b, \qquad k_R = Ru/2 + \max_{t=l,u} |R/2 - 3t/2| g + 3b.$$

For k_R, bound the positive and negative fourth moments separately by $R(\rho \pm \gamma)/2$. Before defect feedback use $k = \min(k_{\mathrm{rad}}, k_{\mathrm{old}}, k_R)$. After feedback one may replace k_{old} by $\max(k_+, k_-)$, where

$$g_l = 3.16l - 3 - d, \quad g_u = 3.16u - 3 - d,$$
$$k_+ = M + 3b - \tfrac{3}{2} \min_{r=l,u;, v=g_l,g_u} rv,$$
$$k_- = 3u + 3b + \max_{r=l,u;, v=g_l,g_u} (3r/2 - 3)v.$$

These substitutions bound the lower affine graph for the actual γ, not an independently chosen skewness.

Put $\Delta = 2u$ initially and $\Delta = (1-3.16)l + 3 + d$ after feedback. The constant coefficient of the raw identity is bounded by

$$A_* = \min\{\tfrac{3}{2}b(2u+1) + \tfrac{3}{2}\Gamma(R,l)\Delta + \tfrac{3}{8}u^2(u+1) + \tfrac{8}{27}u^3 - l/2,$$
$$\tfrac{3}{2}b(2u+1) + 2uM - l/2\}.$$

This is the coarse gap bound above together with the two bounds for $-K_q$. The density coefficient obeys

$$v_H = \min\{.4j + .4\max(2, M-3)/6, .4\max(1/2, 2M/3 - 1/2)\},$$

where $j = 3(u+1)$ initially; the generic-gap choice is $j = \alpha(u+1) + (3-\alpha)d/2$, with its checked $\alpha < 3$. For a score endpoint enclosure $C \in [c_-, c_+]$, the resulting raw bound is

$$K_F + c_+ A_* + \frac{(c_+ k + v_H + s_F)^2}{4q_*}, \qquad q_* > 0, \quad k \ge 0, \quad A_* > 0, \tag{190}$$

where initially $q_* = c_- l - g/5$. After feedback use the larger of that quantity and

$$c_- l + \varphi_-(3.16l - 3 - d)/2 - (2/5 - \varphi_-)(d - .16)_+/2.$$

The score endpoints are the lower score floor and the upper allowed score cap; if the latter depends on ρ, use its values at u, l for c_-, c_+, respectively. Convexity in the actual score precedes the completion of the square, and these outward endpoint intervals cover it. For the phase resource add (138) before maximizing. A barrier $e \ge K/n$ subtracts Kl from the nonnegative resource. The first raw target must pass the license of Lemma R.1 first.

Optional signed improvement inside an unsigned cell.

If $g_0 = \max(-g, \sigma_l l - 3 - d) > g$, exclude the cell. When $g_0 \ge 0$, the sign-moment constraint forces $(1-\beta)^3 \le (1+\beta)(u - g_0)^2$. This inequality is monotone on $(-1, 1)$, so exact bisection yields a lower $b_0 \le \beta$; if it fails even at b, the cell is empty. For $x \in [g_0, g]$, set

$$f_-(x) = (l+x)^2/[2(1+b)] - R(u-x)/2 - 3b - 3ux/2,$$
$$f_+(x) = R(u+x)/2 - (l-x)_+^2/[2(1-b_0)] - 3b_0 - 3lx/2.$$

The signwise fourth-moment Cauchy bounds prove $\min f_- \le K_1 \le \max f_+$. The first minimum is at its clipped vertex $3u(1+b)/2 - l - R(1+b)/2$. The second maximum is among its endpoints, the clipped kink l, and the clipped vertex $l + (R/2 - 3l/2)(1 - b_0)$. Intersect with $[-k, k]$ and replace the linear term in (190) by the smaller full rectangle bound

$$\max_{c=c_-,c_+;,k=k_l,k_u;,\psi=\varphi_-,.4;,v=-1/2,\,2M/3-1/2} |ck + \psi v| + s_F.$$

All sixteen corners are retained. If $b_0 > 0$, the opposite-sign event also has probability at least $(1 + b_0^2)^2/(2u^2)$, from $m \ge (1+\beta^2)/(2\rho)$ and $m^2 \le pq$. It may improve the complete HIGH value. These are intersections with necessary conditions; the cover still examines every ρ-cell.

Moving from a small defect to probability coordinates.

For $0 < d \le .7$, start with

$$A_l = \frac{1.47\sqrt{2-d} - \sqrt{3.7d}}{3.52}, \qquad B_l = \frac{2.04\sqrt{2-d} - \sqrt{3.7d}}{3.52}.$$

They follow from $A + B \ge \sqrt{2-d}$ and $|\sqrt{\sigma+1}A - \sqrt{\sigma-1}B| \le \sqrt{3.7d}$; their positivity is checked, rather than assumed at every d. The pointwise inequality $x(x+b) \ge (29/5)b(x-b)$ for $x > b > 0$, applied to the two residual third moments, gives $D \ge (29/5)\min((\sigma+1)A, (\sigma-1)B)\theta$. Start $\Theta = 1$ and perform the declared finite number of updates:

$$\Theta \leftarrow \min\{\Theta, d/[(29/5)\min((104/25)A_l, (54/25)B_l)]\},$$
$$\epsilon = \sqrt{d/[(616/25)(1-\Theta)^2]}, \qquad [l, u] = [(4-\sigma_u)/2 - \epsilon, (4-\sigma_l)/2 + \epsilon],$$
$$A_l \leftarrow \max\{A_l, \sqrt{(1-\Theta)l/(1-l)} - d/[2(1-u)(179/25)A_l^2]\},$$
$$B_l \leftarrow \max\{B_l, \sqrt{(1-\Theta)(1-u)/u} - d/[2l(129/25)B_l^2]\}.$$

The last two bounds use $D \ge 2(\sigma+2)B^2M_+ + 2(\sigma+4)A^2M_-$ and $\Delta_A = M_-/q$, $\Delta_B = M_+/p$. Every division, interval endpoint and variance domain is gated. Subdivide this entire probability interval, then use the common cell update and coefficient atlas. Thus the unsigned first barrier, the signed initialization and the final probability refinements are three data choices for the same finite-enclosure theorem.

U One confinement lemma with two concentration choices

A proved amplitude is used as a support update, never as a premise of its own proof. The following derivation applies to every acquired row, with the same formulas for the central and left probability ranges.

U.1 Statement and hypotheses

Let $n \ge N = 47$, $Y = B + U$, $\mathbb{P}(B = 1) = p < 1/2$ and $\mathbb{E}(U \mid B) = 0$. Put $q = 1 - p$, $s = \mathbb{E}U^2$, $\lambda = ns$ and $X = \ell(Y - p)$, where $\mathbb{E}X^2 = 1$. Thus $\ell^2 = (1-\theta)/(pq)$ and $\theta = \ell^2 s$. The conditional label-zero support is at most $a \ge 0$ and the label-one support is at least $b \le 1$, where a, b are support points and $b - a = 1 - d$. In particular $0 \le a \le d$, $1 - d \le b \le 1$. Assume the initial strip $\operatorname{supp} Y \subset (-.55, 1.55)$. Write $\rho = \mathbb{E}|X|^3$, $\beta = \mathbb{E}[X|X|]$ and $\epsilon = \mathbb{E}|U|$.

Select an acquired closed probability panel $[l, h]$. Its fields give upper bounds $\Theta, d_0, \rho_0, \beta_+, T, \Lambda$, a lower bound β_-, and separate bounds H_+, H_- for the two signs of H. Assume

$$\theta \le \Theta, \quad d \le d_0, \quad \rho \le \rho_0, \quad \beta_- \le \beta \le \beta_+, \quad 0 \le \lambda \le \Lambda, \quad .4 < C \le C_0 = .4253.$$

For the positive sign $0 \le H \le H_+$; for the negative sign $-H_- \le H \le 0$. Let $x = np + H/\ell$ and $\nu = \mathcal{L}(x - \sum_{i<n} Y_i)$. Assume a contact majorant Q for $g(y) = \nu([y, \infty))$, with equality at each support point of Y, and derivative

$$\sqrt{n}(-Q'(y)) = \ell\phi(H/\sqrt{n}) + \frac{\ell}{n}\{3C\beta + (H\phi(H/\sqrt{n}) + 3C\rho)t - 3Ct|t|\}, \quad t = \ell(y - p). \tag{191}$$

Both actual deleted convolutions satisfy, separately,

$$Q_j(w) \le \frac{\ell\phi(0)w + 2C_0\rho}{\sqrt{j}}, \qquad j = n-1, n-2, \tag{192}$$

for every interval of width $w \ge 0$.

Here is the precise common resource assumption. Put $\kappa = 997/15000 < \phi(0)/6$, $J = \int_{[a,b)}(y - a)\nu(dy)$, and note that the central gap contributes $m\ell J = (1-\theta)J$ to L. Thus $L \ge (1-\theta)J$; retain $e \ge 0$ and

$$ne\rho + n\phi(0)D/6 + n^{3/2}L < \mathcal{B}(H), \qquad \mathcal{B}(H) = \min\{T, \max_j(A_j + F_{j,\pm}|H| - Q_jH^2)\}. \tag{193}$$

The four quadratics are the table's `quadratics` rows, using `Qbudget`, not `Qraw`. The Gaussian phase contribution has already been included in `Qbudget`.

For each panel let E, F_ℓ, m_0 be respectively strict lower, upper, and lower rational bounds for

$$\sqrt{(1-\Theta)/(h(1-h))}, \qquad 1/\sqrt{l(1-l)}, \qquad \sqrt{(1-\Theta)l(1-l)}.$$

The bounds use the grid 10^{-6}, with an extra unit outward. Define

$$c_+ = 2(5.16)E^2(1 - h - d_0)^2, \qquad c_- = 2(7.16)E^2(l - d_0)^2,$$
$$b_* = \min\{3(4.16)E(1 - h - d_0), 3(6.16)E(l - d_0)\},$$
$$P = m_0E^2(3.5\,\operatorname{dist}([l, h], [.41886, .41887]) - 2.041d_0)_+^2.$$

The simultaneous moment inequality has the following usable form:

$$D \geq c_+ M_+ + c_- M_- + b_* \theta + P, \quad d = (M_+/p + M_-/q)/\ell, \quad \epsilon \leq 2(M_+ + M_-)/\ell, \quad M_\pm \geq 0. \tag{194}$$

To derive it, the endpoint expansion of (141) gives $D \geq m(\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2 + 2(\sigma+2)B^2 M_+ + 2(\sigma+4)A^2 M_- + 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-$. Insert $A \geq E(l - d_0)$, $B \geq E(1 - h - d_0)$ and the rational lower coefficients displayed above. For the square, write $A = \ell(p - a)$, $B = \ell(1 - p - (1 - b))$, with $a + (1 - b) = d \leq d_0$. Its unshifted coefficient is $(\sqrt{\sigma+1} + \sqrt{\sigma-1})(p - p_*)$; $\sqrt{\sigma+1} + \sqrt{\sigma-1} > 3.5$ and $\sqrt{\sigma+1} < 2.041$ prove the stated lower square P. For completeness, on the positive class put $Z = X - B \geq 0$. Then $T_+ = 2B^2 M_+ + 3B\mathbb{E}[Z^2; X > 0] + \mathbb{E}[Z^3; X > 0]$ and $R_+ = BM_+ + \mathbb{E}[Z^2; X > 0]$. Consequently $(\sigma - 1)T_+ + 6BR_+ \geq 2(\sigma+2)B^2 M_+ + 3(\sigma+1)B\theta_+$, since $\mathbb{E}[Z^2; X > 0] \geq \theta_+$. The reflected expansion gives the negative-class bound. The centered absolute deviation on each class is at most twice its mean distance from the inner endpoint, proving the bound on ϵ. All quantities refer to the same marginal.

U.2 Concentration from an antichain

Lemma U.1 (Strict-gap concentration)**.** *Suppose $Y = U_0 + B(1 + U_1 - U_0)$, where B is Bernoulli(p) independent of the pair (U_0, U_1), and $1 + U_1 - U_0 \geq c > 0$ almost surely. For independent copies $Y_1, \ldots, Y_j$ and any interval I of diameter strictly less than c,*

$$\mathbb{P}\left(\sum_{i=1}^{j} Y_i \in I\right) \leq \max_{0 \leq k \leq j} \binom{j}{k} p^k (1-p)^{j-k}.$$

The same bound holds after adding any independent random variable.

Proof. Condition on all noise pairs. The sum is a fixed constant plus $\sum B_i w_i$, with every $w_i \geq c$. The family $\mathcal{F}$ of subsets whose weighted sums lie in I is an antichain: distinct comparable subsets have sums differing by at least c, larger than the diameter. A uniformly random permutation of $\{1, \ldots, j\}$ defines the maximal chain of its initial subsets. The chain contains a fixed k-subset with probability $1/\binom{j}{k}$, and meets an antichain at most once. Hence

$$\sum_{S \in \mathcal{F}} \binom{j}{|S|}^{-1} \leq 1.$$

Multiplying each summand by its binomial level mass proves the bound conditionally. Integration removes the conditioning. Conditioning on an additional independent variable proves the last assertion. This argument includes all atom collisions and any open or closed interval endpoints. □

For the actual marginal we may couple the two conditional noises independently and then sample an independent label. Since $U_0 \leq a$ and $U_1 \geq b - 1$, the step is at least $b - a = 1 - d$. Strict diameter is necessary: a Bernoulli law on $\{0, c\}$ assigns mass one to $[0, c]$, exceeding its largest atom when $0 < p < 1$.

A rational cap for every remaining index.

Put $J_0 = N - 2 = 45$ and define

$$r_L = l - (1 - l)/J_0, \qquad r_R = h + h/J_0, \qquad v_* = \min\{r_L(1 - r_L), r_R(1 - r_R)\}.$$

The checker verifies $0 < r_L \le r_R < 1$. Adjacent binomial-mass ratios show that any mode k at index $j \ge J_0$ satisfies $(j+1)p - 1 \le k \le (j+1)p$, hence $r = k/j \in [r_L, r_R]$. In particular $1 \le k \le j-1$. Finite Stirling bounds give

$$\sqrt{j}\binom{j}{k}p^k(1-p)^{j-k} \le \frac{\exp(1/(12j) - jD(r\|p))}{\sqrt{2\pi r(1-r)}} < \frac{.4}{\sqrt{v_*}(1 - 1/(12J_0))}.$$

Here $D(r\|p) \ge 0$ follows from Jensen's inequality for the logarithm; $\exp x \le (1-x)^{-1}$ for $0 \le x < 1$ follows term by term from the power series. The finite Stirling inequalities used are $\sqrt{2\pi m}(m/e)^m < m! < \sqrt{2\pi m}(m/e)^m \exp(1/(12m))$. Choose the strict 10^{-6} grid upper square root M of $.16/[v_*(1 - 1/(12J_0))^2]$. Thus, whenever $w < 1 - d$,

$$Q_j(w) < M/\sqrt{j}, \qquad j \ge n - 2 \ge J_0. \tag{195}$$

This covers the $n-1$ full summands of ν and the $n-2$ full summands remaining after one mark or deletion. An independent remaining conditional noise preserves the cap. We do not replace the latter index by $n-1$.

Either the score-derived cap or the strict-gap cap may be selected on an interval where its hypotheses hold. The first confinement uses the score cap; later central and left schedules use the strict-gap cap. For the left panels replace the conditional cubic coefficient 24 below by 28; in general any c_ρ satisfying $c_\rho^2[l(1-l)]^3 > [3(1-l) + .55]^2$ is valid.

U.3 Signed deletion, including atoms

Let $m = n - 1$, $Z = x - \sum_{i=1}^m Y_i$, $\mu = x - mp$ and let $K = \sum B_i$. For a fixed removed label put $T_b(I) = \mathbb{P}(x - \sum_{i<m} Y_i - b - U_b \in I)$. Exchangeability and $mp - K = Z - \mu + \sum U_i$ give the exact identities

$$\nu(I) - T_1(I) = \frac{\mathbb{E}[(Z - \mu + \sum U_i)1_I(Z)]}{mp}, \qquad T_0(I) - \nu(I) = \frac{\mathbb{E}[(Z - \mu + \sum U_i)1_I(Z)]}{mq}.$$

There is no absolute value around the sum of noises. For a full removed pair (Y_i, U_i), condition on the pair. The remaining sum gives a function $f(Y_i) \in [0, Q_{m-1}(w)]$. Since $\mathbb{E}U_i = 0$,

$$\mathbb{E}[U_i f(Y_i)] \le Q_{m-1}(w)\mathbb{E}(U_i)_+ = \tfrac{1}{2}\epsilon Q_{m-1}(w).$$

This also works when one conditional law is deterministic.

Let independent conditional noises satisfy $\Delta = U_1 - U_0 \ge -d$. For $\tau > 0$ put $I^+ = I + 1 + [-d, \tau]$. On $\Delta \le \tau$, the point for T_1 in I maps to the point for T_0 in I^+. On the complement, conditioning on both noises gives error at most $Q_{m-1}(w)\mathbb{P}(\Delta > \tau)$. Moreover

$$\mathbb{P}(\Delta > \tau) \le \min\left\{\frac{s}{s + p\tau^2}, \frac{d}{d+\tau}\right\}. \tag{196}$$

Indeed $\mathbb{E}\Delta = 0$, $\operatorname{Var}\Delta \le s/p$, and $\Delta + d \ge 0$ has mean d. Cantelli and Markov prove the two bounds. The exceptional event is strict, so $\Delta = \tau$ belongs to the good event.

If I, I^+ are disjoint, combine the two signed noise terms before bounding them. For the same remaining-sum law their conditional probabilities f_I, f_J satisfy

$$0 \le f_I \le Q_I, \quad 0 \le f_J \le Q_J, \quad f_I + f_J \le Q_{\text{hull}}.$$

Since $p < q$,

$$\frac{f_I}{p} + \frac{f_J}{q} \le \min\left\{\frac{Q_I}{p} + \frac{Q_J}{q}, \frac{Q_{\text{hull}}}{q} + \left(\frac{1}{p} - \frac{1}{q}\right)Q_I\right\}.$$

Multiplication by $\mathbb{E}U_+ = \epsilon/2$ proves the combined bound. Closed, open and half-open intervals are all covered by containment in the corresponding closed interval. No density or atom separation is used.

U.4 Slope bounds and the two exterior intervals

Split $|H|$ into 16 closed equal cells $[h_l, h_h]$ and λ into 24 closed equal cells $[s_l, s_h]$. On each box put

$$\rho_* = \min\{\rho_0, \rho_B(l) + 24 s_h/N\},$$
$$b_- = \max\{\beta_-, 1 - 2h - 2(1-h)\Theta\}, \qquad b_+ = \min\{\beta_+, 1 - 2l + \Theta\}.$$

Here $\rho_B(p) = (1 - 2p(1-p))/\sqrt{p(1-p)}$. Write $s_i = \mathbb{E}U_i^2$. Conditional centering and the two sign supports give

$$\rho = \ell^3\{pq(1-2pq) + 3pq(s_0+s_1) + p\mathbb{E}U_1^3 - q\mathbb{E}U_0^3\}.$$

The initial strip and $d_0 < .55$ imply $|U_i| < .55$. Thus $\rho \le \rho_B + \ell^3(3q + .55)s \le \rho_B + 24s$; the last coefficient inequality is checked as $24^2[l(1-l)]^3 > [3(1-l) + .55]^2$. Similarly $\beta = (1-\theta)(1-2p) + \ell^2(ps_1 - qs_0)$, and $|\ell^2(ps_1 - qs_0)| \le \theta$. These identities prove the stated outward bounds on ρ and β. Set $f = .4F_\ell$, $\varphi_- = .39886(1 - h_h^2/(2N)) > 0$. The exact upper bound for $H\phi(H/\sqrt{n})$ used below is

$$v_+ = \begin{cases} .4h_h & H \ge 0, \\ -\varphi_- h_l & H \le 0. \end{cases}$$

For all $t \le 0$, completing the square in (191) gives

$$\mathcal{M}_- = F_\ell((v_+ + 3C_0\rho_*)^2/(12(.4)) - 3(.4)b_-)_+, \qquad L_- = E\varphi_- - \mathcal{M}_-/N.$$

A positive bound $B_* > \sqrt{(1-l)/l}$ gives valid central-gap lower and positive-half-line upper slope bounds

$$\mathcal{M}_+ = F_\ell(.4h_h B_* + 3C_0 B_*(B_* - 1) - 1.2b_-)_+,$$
$$L_g = E\varphi_- - \max(\mathcal{M}_-, \mathcal{M}_+)/N,$$
$$U_+ = f + \frac{F_\ell}{N}\left\{3C_0 b_+ + \frac{(.4h_h + 3C_0\rho_*)^2}{12(.4)}\right\}.$$

They follow by completing a square on the positive half-line and using $t(t-1) \le B_*(B_* - 1)$ for $1 \le t \le B_*$; on $0 \le t \le 1$ that term has the favorable sign. Negative values are covered by $\mathcal{M}_-$. All divisions and positivity requirements are explicit gates.

The sharper lower-ray slope.

Take $\tau_- = 5r_0/12$, $\eta = 10^{-5}$ and $I = [a - r, a - d - \tau_- - \eta]$, where $r \ge r_0$. Since $a \le d$, the entire interval has $t \le t_* = -E(l + \tau_- + \eta) < 0$. Write $B(t) = b_- + \rho_* t + t^2$ and $B_*^{\min} = B(\min(t_*, -\rho_*/2))$. For the actual C and $t \le t_*$,

$$3C(\beta + \rho t + t^2) + H\phi t \ge 1.2B(t) + v_+ t + 3(C_0 - .4)\min(0, B_*^{\min}).$$

The inequality uses $t < 0$ and separates the possible negative part of $B(t)$ before varying C. Minimize the right side at the clipped vertex $\min(t_*, -(v_+ + 1.2\rho_*)/2.4)$ and call its value q_*. Then

$$L = E\varphi_- - F_\ell \max(0, -q_*)/N$$

is a lower slope on the whole interval, for every $n \ge N$. Positive corrections are discarded. Therefore contact gives

$$\sqrt{n}\nu(I) \ge L(r - d - \tau_- - \eta).$$

This one slope replaces the previous separate bounds on the long and short parts of the interval.

Lower interval geometry.

Its translate is $I^+ = [a - r + 1 - d, b - \eta]$, of width $r - \eta$. The initial strip implies $I^+ \subset [a + g, b)$, $g = 1 - .55 - 2d_0 > 0$, hence $\nu(I^+) \le J/g$. The intervals are disjoint throughout $r \le .55 + d_0$, since $\inf I^+ - \sup I = 1 - r + \tau_- + \eta > 0$. At $r = r_0$, concentration and directed distance bounds are

$$\begin{aligned} Q_I &= f(r_0 - \tau_- - \eta) + 2C_0\rho_*, & Q_J &= f(r_0 - \eta) + 2C_0\rho_*, \\ Q_{\text{hull}} &= f(1 + r_0 - \eta) + 2C_0\rho_*, \\ D_I &= (v_H - l - \tau_- - \eta)_+, & D_J &= (v_H + 1 - l - \eta)_+, \end{aligned}$$

where $v_H = -h_l/F_\ell$ for $H \ge 0$ and $v_H = h_h/E$ for $H \le 0$. These signs follow from $\mu = p + H/\ell$.

Upper interval geometry.

For a support point $y \ge 1 + r_0$, take $\tau_+ = 7r_0/15$, $I = [a, a + \tau_+]$ and $I^+ = [b, a + 1 + 2\tau_+]$. The strict gates $d_0 + 2\tau_+ < r_0$ and $1 - d_0 - \tau_+ > 0$ give $I^+ \subset [b, y)$ and disjointness. Contact, including the atom at b, and the leakage definition give

$$\sqrt{n}\nu(I) \ge L_g(1 - d) - \sqrt{n}J/\tau_+, \qquad \sqrt{n}\nu(I^+) \le U_+(.55 + d).$$

Use

$$\begin{aligned} Q_I &= f\tau_+ + 2C_0\rho_*, & Q_J &= f(2\tau_+ + d_0) + 2C_0\rho_*, \\ Q_{\text{hull}} &= f(1 + 2\tau_+) + 2C_0\rho_*, \\ D_I &= (d_0 + \tau_+ + h_h/E - l)_+, & D_J &= (1 + d_0 + 2\tau_+ + h_h/E - l)_+. \end{aligned}$$

The upper side deliberately keeps the absolute phase bound.

The strict-gap alternative.

If $1 - .55 - 2d_0 > 0$ and the displayed lower intervals have width below $1 - d$ throughout $r \in [r_0, .55 + d_0]$, take $Q_I = Q_J = M$. For the upper intervals the corresponding gate is $1 - 2d_0 - 2\tau_+ > 0$; then again $Q_I = Q_J = M$. Use the same directed distances. Their concentration caps are now independent of r, and use $a_e = F_d M\{1/l + 1/(1 - h)\}/2$ in the next formulas. The lower-ray gate reduces to $L > 0$: its contact side increases with r, while the entire error is constant on the acquired finite strip. For the score-derived cap retain the explicit affine slope gate below. No assertion outside that strip is needed.

U.5 One budget, two error bounds

Let F_d be the strict grid upper bound for $\sqrt{\max\{(47/46)^3, 47/45\}}$. It simultaneously covers the two concentration factors in (192) after multiplication by $\sqrt{n}$, uniformly for $n \ge 47$. For either pair of intervals set

$$\begin{aligned} a_e &= \frac{F_d}{2} \min\left\{ \frac{Q_I}{l} + \frac{Q_J}{1 - h}, \frac{Q_{\text{hull}}}{1 - h} + \left(\frac{1}{l} - \frac{1}{1 - l} \right) Q_I \right\}, \\ A &= F_d \left\{ Q_I D_I/l + Q_J D_J/(1 - h) \right\}, \qquad B = F_d Q_I. \end{aligned}$$

The probability denominators are outward because $1/p - 1/(1 - p)$ is decreasing on $(0, 1/2)$. Use $a_d = L$, $a_J = 1/g$ on the lower side, and $a_d = L_g + U_+$, $a_J = 1/\tau_+$ on the upper side. For the phase cell, maximize each concave budget quadratic at its vertex clipped to $[h_l, h_h]$, take their maximum, then the minimum with T; call this upper bound T_*. Put

$$R = T_*/\kappa - b_* E^2 s_l, \quad \alpha = \min(c_+ lE, c_-(1 - h)E), \quad G = (R - NP)/\alpha.$$

If $G \le 0$, the strict common budget excludes the entire closed box. Otherwise $nd < G$ and

$$B_\tau = \min\{s_h/(s_h + Nl\tau^2), G/(N\tau + G)\}$$

bounds (196). With

$$z = \max\left\{\frac{a_d/l + 2a_e}{Ec_+}, \frac{a_d/(1-h) + 2a_e}{Ec_-}, \frac{\kappa a_J}{1-\Theta}\right\},$$

one valid total error bound is

$$\mathcal{E}_0 = (zR + A)/N - zP + BB_\tau. \tag{197}$$

This follows by multiplying the single budget by z/κ. The inverse-index coefficient $zR + A$ is checked positive.

A stronger bound spends the same resource on the pair event. Let $R_0 = R - NP > 0$ and define the three pairs

$$(v_+, w_+) = \left(\frac{a_d/l + 2a_e}{Ec_+}\frac{R_0}{N}, \frac{R_0}{Ec_+ l}\right),$$

$$(v_-, w_-) = \left(\frac{a_d/(1-h) + 2a_e}{Ec_-}\frac{R_0}{N}, \frac{R_0}{Ec_-(1-h)}\right), \qquad (v_J, w_J) = \left(\frac{\kappa a_J}{1-\Theta}\frac{R_0}{N}, 0\right).$$

They are upper cost/displacement coordinates at the vertices of the simplex with resources nc_+M_+, nc_-M_- and $n^{3/2}L/\kappa$. For any $t \ge 0$,

$$\frac{D}{N\tau + D} \le \frac{N\tau}{(N\tau + t)^2}D + \frac{t^2}{(N\tau + t)^2} \qquad (D \ge 0).$$

The difference is $N\tau(D-t)^2/[(N\tau + D)(N\tau + t)^2]$. Consequently a further valid error bound is

$$\mathcal{E}_t = \frac{A}{N} + \max_{i \in \{+,-,J\}}\left(v_i + B\frac{N\tau}{(N\tau + t)^2}w_i\right) + B\frac{t^2}{(N\tau + t)^2}.$$

The checker takes the minimum of (197) and these eleven bounds with $t = jG/10$, $j = 0, \ldots, 10$. Each is individually valid; no numerical minimization or derivative of their minimum is assumed.

The lower contradiction is the strict gate $Lr_0 - L(\tau_- + \eta) > \mathcal{E}$. To extend it to every $r \ge r_0$, bound increments directly in the original affine-in-width signed deletion bound for the same law. The budget gives $n\epsilon < E_* = 2T/(\kappa E \min(c_+, c_-))$. The sufficient slope gate is

$$L > \frac{F_d f}{N}\left\{D_I/l + D_J/(1-h) + \frac{E_*}{2}(1/l + 1/(1-h))\right\} + F_d f B_\tau.$$

Using the uncoupled increment is safe even when the hull bound was used at r_0. The upper contradiction is $L_g - .55U_+ > \mathcal{E}$. Thus no support point lies below $a - r_0$ or above $1 + r_0$. Since $a \ge 0$, $b \ge 1 - d_0$ and $d_0 < r_0$, the label-zero and label-one supports both satisfy $|U| < r_0$.

The selected schedules.

The ordinary, central-antichain, and left-antichain certificates substitute their recorded r_0 into these same gates. Noise-dependent amplitudes are the same theorem with a positive lower noise endpoint in $R = T_*/\kappa - b_* E^2 s_l$. In the central rows the additional amplitudes are $.35, .30, .275$ on $[.05, .08], [.08, .10], [.10, \Lambda]$; on the left they are $.375, .325, .30$. Intersect each interval with its own row's Λ and each amplitude with that row's global amplitude. The fixed acceptance checks both signs of H, all closed phase/noise cells, clipped endpoints, and strict budget exclusions. No proof of LOW, MID, or a binomial score comparison is used by this confinement lemma.

V One local comparison theorem

A law concentrated near two points has a natural integer count and a small centered error. We compare it with its own Bernoulli count. The same exact identity handles every noise regime. There are two useful estimates of its right side: discard all but the nearest outgoing count, or retain all counts through a random cutoff. The estimates share their normalization, moments, and tail bounds; only the final scalar witness changes.

V.1 The identity and the common numerical test

Let $0 < p \le 1/2$, $q = 1 - p$, $v = pq$, and let

$$Y = B + U, \qquad \Pr(B = 1) = p, \qquad \mathbb{E}(U \mid B) = 0.$$

The two actual conditional laws U_0, U_1 obey

$$|U_j| \le A, \quad U_0 \le a_0, \quad U_1 \ge -a_1, \quad a_0, a_1 \ge 0, \quad d = a_0 + a_1 \le d_0 < \min(p, q), \quad d_0 \le A. \tag{198}$$

Write $s_j = \mathbb{E}U_j^2$, $s = qs_0 + ps_1$, $\lambda = ns$, $\ell = (v + s)^{-1/2}$, $\theta = s/(v + s)$ and $\rho = \mathbb{E}|\ell(Y - p)|^3$. At the actual threshold choose

$$x = np + H/\ell = k + u, \qquad k \in \{1, \dots, n - 1\}, \quad |u| \le \tfrac{1}{2}, \qquad \delta = k - np = H/\ell - u.$$

Either nearest integer is allowed at a tie. Put

$$\pi_j = \binom{n}{j} p^j q^{n-j}, \quad F_B(j) = \Pr(\mathrm{Bin}(n, p) \le j), \qquad \rho_B = (1 - 2v)/\sqrt{v},$$

with $\pi_j = 0$ outside $[0, n]$. Let W_j be an independent sum of j copies of U_1 and $n - j$ copies of U_0. The signed scores are

$$C = \frac{\sqrt{n}}{\rho}\{\Pr(\textstyle\sum Y_i \le x) - \Phi(H/\sqrt{n})\}, \qquad C_B = \tfrac{\sqrt{n}}{\rho_B}\{F_B(k) - \Phi(\delta/\sqrt{nv})\}.$$

Throughout this section $C \ge c_* > 0$; the selected application uses $c_* = .4$. Define

$$\begin{aligned} f(u) &= \frac{\Phi((\delta + u)/\sqrt{n(v + s)}) - \Phi(\delta/\sqrt{n(v + s)})}{\pi_k}, \\ Q &= \Pr(\textstyle\sum Y_i \le k + u) - F_B(k), \\ \mathcal{R} &= \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v + s)})\} + C(\rho_B - \rho). \end{aligned} \tag{199}$$

Direct subtraction gives the exact identity

$$\rho_B(C - C_B) = \sqrt{n}\pi_k\{Q/\pi_k - f(u)\} + \mathcal{R}. \tag{200}$$

In particular, the phase u, the sign of H, and both conditional laws refer to the same marginal throughout.

Theorem V.1 (Parameterized local comparison)**.** *Fix $n \ge N$, $\lambda > 0$ and a closed parameter cell on which* (198) *holds. Suppose $\sqrt{n}\pi_k \ge m > 0$. Either of the following witnesses proves $\rho_B(C - C_B) < -\varepsilon\lambda$.*

1. Nearest-count witness. *Valid bounds O, T, R_0, κ satisfy*

$$\Pr(W_k > u) + f(u) \ge O, \quad \sum_{i \ge 1} \Pr(W_{k+i} \le u - i) \le T, \quad \pi_{k+i}/\pi_k \le R_0, \quad \mathcal{R} \le \kappa\lambda/n.$$

Only legal counts occur in the sum; adding positive upper bounds for other counts is allowed. If $D = O - R_0 T > 0$ and

$$mD - \kappa\lambda/n > \varepsilon\lambda, \tag{201}$$

the conclusion follows.

2. Whole-count witness. *Bounds $B(\lambda) < 0$ and $I(\lambda) \ge 0$ satisfy*

$$\rho_B(C - C_B) \le \sqrt{n}\pi_k \lambda B(\lambda) + \lambda I(\lambda)/n.$$

It is enough that

$$mB(\lambda) + I(\lambda)/N < -\varepsilon. \tag{202}$$

The witnesses may differ from cell to cell. A finite closed cover and a separate treatment of $\lambda = 0$ therefore suffice for a complete local comparison.

Proof. Conditioning on the binomial count gives, with all atom conventions intact,

$$Q = -\sum_{i=0}^{k} \pi_{k-i} \Pr(W_{k-i} > u + i) + \sum_{i=1}^{n-k} \pi_{k+i} \Pr(W_{k+i} \le u - i). \tag{203}$$

Dropping the outgoing terms with $i > 0$ gives $Q/\pi_k - f(u) \le -D$. Use (200) and $D > 0$. For the second witness, use $B < 0$ before replacing $\sqrt{n}\pi_k$ by m, and $I \ge 0$ before replacing n by N. These sign requirements are part of the test, not conventions. □

V.2 Normalization and the repair, once for every witness

Here is an explicit finite recipe for the constants required above. Let $p \in [p_-, p_+] \subset (0, 1/2)$, $n \ge N$, $\lambda \le \Lambda$, $\theta \le \Theta$, and $|\delta| \le \Delta$, where $\Delta \ge 1$ also includes every binomial mode. Choose $0 < r_- < p_- - \Delta/N$ and $p_+ + \Delta/N < r_+ < 1/2$. Let $v_* > 0$ be a lower bound for $p(1-p)$. Choose positive E, L, R bounding, on $[r_-, r_+]$,

$$|D'''(r\|p)| = \frac{1 - 2r}{r^2(1-r)^2}, \qquad |(\log(r(1-r)))'| = \frac{1-2r}{r(1-r)}, \qquad \frac{1}{12r(1-r)},$$

respectively, where $D(r\|p) = r\log(r/p) + (1-r)\log((1-r)/q)$. Then, with $Z(t) = Lt/2 + R + Et^3/(6N)$,

$$\sqrt{n}\pi_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+\zeta}, \qquad n|\zeta| < Z(|\delta|). \tag{204}$$

For completeness, the finite Stirling formula used here is

$$\log(j!) = (j + \tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

If $a_j = \log(j!) - (j + 1/2)\log j + j$ and $z = (2j+1)^{-1}$, the integrated geometric series gives $0 < a_j - a_{j+1} = \sum_{i \ge 1} z^{2i}/(2i+1) < 1/[12j(j+1)]$. Summing gives the remainder relative

to $\lim a_j$. The integration-by-parts recursion for $\int_0^{\pi/2}\sin^j t\,dt$ and the inequalities between three successive integrals give Wallis's product; its finite product equals $e^{4a_j-2a_{2j}}j/[2(2j+1)]$ and tends to $\pi/2$. Thus $\lim a_j=\frac12\log(2\pi)$. Applying this formula to $n!,k!,(n-k)!$ and Taylor-expanding D and the logarithmic prefactor proves (204); the factorial error is at most $1/[12nr(1-r)]$.

Consequently a valid mass lower bound is any positive

$$m<\frac{.39886}{\sqrt{p_+(1-p_+)}}\left[1-\frac{\Delta^2/(2v_*)+Z(\Delta)}{N}\right]. \tag{205}$$

The same formula at a mode proves $\sqrt{n}\max_j\pi_j<1$ whenever $v_*[1-Z(\Delta)/N]^2>.4^2$ and $Z(\Delta)<N$. The rational bounds $.39886<\phi(0)<.4$ follow, for example, by the alternating arctangent series in Machin's identity.

Differentiating (199) gives

$$\log f'=-\tfrac12\log(1+s/v)+\delta^2/(2nv)-(\delta+u)^2/(2n(v+s))-\zeta. \tag{206}$$

Thus $n|\log f'|\le B_0\lambda+P(D,a)$ on $|\delta|\le D\le\Delta$, $|u|\le a\le 1/2$, where

$$B_0>\frac{1}{2v_*}+\frac{\Delta^2}{2Nv_*^2},\qquad P(D,a)=\frac{2Da+a^2}{2v_*}+Z(D).$$

Choose $0<b<1$ above $[B_0\Lambda+P(\Delta,1/2)]/N$ and put $F=(1-b)^{-1}$. The elementary estimates $e^z\le(1-z)^{-1}$ and $1-e^{-z}\le z$ imply

$$|f'-1|\le(B_1\lambda+P_1)/n,\qquad B_1=FB_0,\quad P_1=FP(D,a). \tag{207}$$

This supplies both a derivative interval $l_f\le f'\le h_f$ and the finer phase-error bound $|u-f(u)|\le a(B_1\lambda+P_1)/n$.

The exact count variance is

$$t_j=\mathbb{E}W_j^2=\lambda+(j-np)(s_1-s_0),\qquad |s_1-s_0|\le s/p\le c_vs,\quad c_v\ge 1/p_-. \tag{208}$$

It gives $\omega_-\lambda\le t_k\le\omega_+\lambda$ with $\omega_\pm=1\pm c_vD/N$. All uses below require the checked condition $\omega_->0$. Also $t_{k+i}\le iV_*\lambda$ for $V_*=1+c_v(\Delta+1)/N$. The binomial recursion gives, for every legal $i\ge1$,

$$\frac{\pi_{k+i}}{\pi_k}\le R_0,\qquad R_0>1+\frac{p_++\Delta-1}{Nv_*[1-(\Delta-1)/(Np_-)]}, \tag{209}$$

provided the denominator is positive and $p_++\Delta<2$. Indeed the first ratio has this bound and all subsequent rising factors are at most one. No count outside $[0,n]$ is used in this recursion.

Finally (198) fixes the signs inside the absolute cubic. Expansion and centering give

$$\mathbb{E}|Y-p|^3=v(1-2v)+3pq(s_0+s_1)+p\mathbb{E}U_1^3-q\mathbb{E}U_0^3\ge v(1-2v)+(3p-d)s.$$

The inequality $(1+z)^{3/2}\le1+3z/2+3z^2/8$ for $z\ge0$ yields

$$(v+s)^{3/2}(\rho-\rho_B)\ge s\left(6p-3p^2-\tfrac32-d-\frac{3(1-2v)s}{8v}\right). \tag{210}$$

Let $Q_*>0$ be a lower bound for the bracket and $m_c\le c_*/[p_+(1-p_+)+\Lambda/N]^{3/2}$. The normal variance correction is nonpositive for $\delta\le0$; for $\delta>0$ its derivative in s is at most $.4\delta/(2v^{3/2})$. Therefore

$$\mathcal{R}\le(P_*D_+-m_cQ_*)\lambda/n,\qquad P_*\ge\frac{.4}{2v_*^{3/2}},\quad D_+\ge\delta_+. \tag{211}$$

One may take $D_+=D$, but retaining the positive part is often decisive. All bounds concern $v+s$, which can exceed $1/4$.

V.3 Two elementary tools for probabilities

Lemma V.2 (A shifted quartic)**.** *Suppose W is a sum of independent centered variables bounded in absolute value by A, and $t = \mathbb{E}W^2$. Let $K_0 \geq (9 + 6\sqrt{3})/4$ and $a > 0$. For $r \geq 0$ and $\sigma \in \{-1, 1\}$,*

$$\Pr(W + \sigma r > 0) \geq \frac{(3a^2 - A^2)t - 3t^2 - 4Art - 6r^2t - r^4 + 3a^2r^2 + 2\sigma a^3 r}{K_0 a^4}. \tag{212}$$

Proof. The polynomial $h_a(z) = (-z^4 + 3a^2z^2 + 2a^3z)/(K_0a^4)$ is below $\mathbf{1}_{\{z>0\}}$: it is nonpositive for $z \leq 0$, and its maximum for $z > 0$ is at $z/a = (1+\sqrt{3})/2$ and is at most one. Independence gives $|\mathbb{E}W^3| \leq At$ and $\mathbb{E}W^4 \leq A^2t + 3t^2$. Expand $\mathbb{E}h_a(W + \sigma r)$ and apply these two inequalities. The value at the threshold is zero, so atoms require no limiting argument. □

Lemma V.3 (One common tail bound)**.** *If independent centered $X_i \leq A$ have total variance at most V, then, for $b > 0$, their sum has upper tail bounded by the minimum of one and*

$$\frac{V}{V + b^2}, \qquad \mathcal{B}(A, V, b) = e^{b/A}\left(\frac{V}{V + Ab}\right)^{b/A+V/A^2}, \qquad \frac{\mathbb{E}(P_\mu - j)_+^3}{(b/A + \mu - j)^3}, \tag{213}$$
$$\mu = V/A^2, \qquad 0 \leq j < b/A + \mu.$$

where P_μ is Poisson. The same holds for a lower tail when $|X_i| \leq A$. All bounds apply to closed events, hence also to strict events.

Proof. Cantelli's bound follows by applying Markov to $(\sum X_i + V/b)^2$. Taylor's integral remainder gives $\mathbb{E}e^{zX_i} \leq \exp\{\mathbb{E}X_i^2(e^{zA} - 1 - zA)/A^2\}$ for $z \geq 0$. Multiply and use $z = A^{-1}\log(1 + Ab/V)$ to get $\mathcal{B}$. The zero-variance case is immediate.

For the third bound, smooth $f(x) = (x - t)_+^3$ by a nonnegative smooth compactly supported density. Then $f'', f''', f'''' \geq 0$. For $\mathcal{A}f(x) = f(x + A) - f(x) - Af'(x)$, the Taylor quotient

$$\frac{f(x + y) - f(x) - yf'(x)}{y^2} = \int_0^1 (1 - r)f''(x + ry)\, dr$$

is nondecreasing in y. A centered $X \leq A$ of variance v consequently satisfies $\mathbb{E}f(x + X) \leq f(x) + (v/A^2)\mathcal{A}f(x)$. For $\mathcal{P}_z f(x) = \mathbb{E}f(x + A(P_z - z))$, $\partial_z \mathcal{P}_z f = \mathcal{P}_z \mathcal{A} f$ and

$$\partial_z^2 \mathcal{P}_z f = \mathcal{P}_z \mathcal{A}^2 f \geq 0, \qquad \mathcal{A}^2 f(x) = \int_0^A \int_0^A (A - r)(A - w)f''''(x + r + w)\, dr\, dw.$$

Thus $\mathcal{P}_{v/A^2} f \geq f + (v/A^2)\mathcal{A}f$. Replace the independent variables successively; translations and expectations preserve the derivative signs and Poisson rates add. Increasing the total variance to V is permitted by Jensen after adding independent centered Poisson noise. Removing the smoothing is justified by bounded actual upper supports and finite Poisson third moments. Markov at $t = A(j - \mu)$ proves the displayed bound. □

These bounds have finite rational evaluations. For example,

$$\mathbb{E}(P_\mu - j)_+^3 = \mu^3 + 3(1 - j)\mu^2 + (1 - 3j + 3j^2)\mu - j^3 \tag{214}$$
$$+ e^{-\mu} \sum_{h=0}^{\lfloor j \rfloor} \frac{\mu^h (j - h)^3}{h!}.$$

Replacing $e^{-\mu}$ by $(\sum_{h=0}^{24} \mu^h/h!)^{-1}$ enlarges the nonnegative correction. Alternatively sum the positive Poisson series through 32; the remaining tail is at most its first term divided by $1 - \mu(34 - j)^3/[34(33 - j)^3]$, provided $j < 33$ and that ratio is below one. For Bennett,

put $x = Ab/V$ and $L = [(b/A + \mu)\log(1 + x) - b/A]$. The positive series $\log(1 + x) = 2\sum_{h\geq 0} z^{2h+1}/(2h+1)$, $z = x/(2+x)$, gives lower bounds for L; any positive lower exponential series gives an upper bound for e^{-L}. Rounding a positive lower L down before exponentiation is also valid. These are choices of rational witnesses, not optimization claims.

V.4 The nearest-count witness, including vanishing noise

For $u = -r \leq 0$, add $-h_f r$ to (212) with $\sigma = 1$; for $u = r \geq 0$, add $l_f r$ with $\sigma = -1$. Both bound $\Pr(W_k > u) + f(u)$ from below. For a cell $t \in [t_-, t_+]$, $r \in [r_-, r_+]$, this expression is concave in t (second derivative $-6/(K_0 a^4)$), so only t_-, t_+ are needed. At each endpoint, bound a monomial with positive coefficient at r_- and one with negative coefficient at r_+. Any rational $a > 0$ is a valid witness. Other valid choices are

$$\frac{r_-^2}{t_+ + r_-^2} - h_f r_+ \quad (u \leq 0), \qquad l_f r_- \quad (u \geq 0). \tag{215}$$

They follow from Cantelli and nonnegativity, respectively. Their maximum or one selected witness is valid without an optimality proof.

For incoming counts use $t_{k+i} \leq iV_*\lambda$. On $u \in [-r_+, -r_-]$ the first distance is $1 + r_-$ and every later distance is at least i. On $u \in [r_-, r_+]$ they are $1 - r_+$ and $i(1 - r_+/2)$ for $i \geq 2$. Let b, b_o be these first and later distances. On $\lambda \in [L, U]$, let T_1 be any bound from Lemma V.3 at (A, V_*U, b). Since $\mathcal{B}(A, iV, ib) = \mathcal{B}(A, V, b)^i$, all later counts total at most $B^2/(1 - B)$ for $B = \mathcal{B}(A, V_*U, b_o) < 1$. Thus

$$D = O - R_0\{T_1 + B^2/(1 - B)\} > 0, \qquad mD - \kappa U/N > \varepsilon U \tag{216}$$

proves the entire cell when $\kappa \geq 0$. The selected MID normalizers explicitly verify this sign. This is the common central and left MID proof.

A cell touching $\lambda = 0$ requires estimates proportional to λ; a positive lower endpoint may not be substituted. The following rescaling provides them. Put $r = zt$, $t_* = \omega_+ U$ and $d_a = K_0 a^4$. The shifted quartic plus br, divided by t, becomes

$$F_{\sigma,b}(z, t) = \frac{3a^2 - A^2 - 3t}{d_a} + \left(\frac{2\sigma}{K_0 a} + b - \frac{4At}{d_a}\right) z + \frac{3a^2 t - 6t^2}{d_a} z^2 - \frac{t^3 z^4}{d_a}. \tag{217}$$

It extends to $t = 0$ and is concave in t. On $0 \leq z \leq z_0$, replacing z^4 by $z_0^2 z^2$ gives a lower quadratic at $t = t_*$; at $t = 0$ it is linear. Endpoints and a clipped quadratic vertex therefore give an exact lower bound $J(\sigma, b, z_0)$ for the whole rectangle.

For negative phase choose $z_-, c_- > 0$ and verify

$$J(1, -h_f, z_-) > c_-, \quad \frac{z_-^2}{1 + z_-^2 t_*} - h_f z_- > c_-, \quad \frac{2z_-}{(1 + z_-^2 t_*)^2} > h_f, \tag{218}$$
$$\frac{1}{1 + 4t_*} - \frac{h_f}{2} > c_- t_*, \qquad z_- t_* < 1/2.$$

The quartic handles $r \leq z_- t$. Beyond that point use $r^2/(t + r^2) - h_f r$. Its derivative at $r = z_- t$ is positive by the third gate; its second derivative has just one sign change, so any later critical point is a maximum. Its minimum is at $z_- t$ or $1/2$, covered by the other gates. Hence the own count exceeds $c_- t$. For positive phase choose $z_+, c_+ > 0$ with

$$J(-1, l_f, z_+) > c_+, \qquad l_f z_+ > c_+, \qquad z_+ t_* < 1/2. \tag{219}$$

The quartic handles $r \leq z_+ t$ and the normal term handles the rest. The own count exceeds $c_+ t$ and also exceeds $l_f r$.

For incoming tails proportional to λ, fix a Poisson anchor j. A positive series term of index h divided by $\lambda(b/A+\mu-j)^3$ has logarithmic derivative, multiplied by λ,

$$h-1-\mu-\frac{3\mu}{b/A+\mu-j},\qquad \mu=V_*\lambda/A^2.$$

If $b/A-j>0$ and this is nonnegative at $\lambda=U$ for the first positive index, it is nonnegative for every term and all $0<\lambda\le U$. Therefore the Poisson bound divided by λ is largest at U. Likewise $B^2/[\lambda(1-B)]$ is increasing whenever $\lambda B'/B\ge 1/2$. Differentiating the Bennett formula gives

$$\lambda B'/B = b/A-\mu\log(1+b/(A\mu));$$

The function $g(\mu)=b/A-\mu\log(1+b/(A\mu))$ decreases because $\log(1+x)>x/(1+x)$ for $x>0$. Moreover the logarithmic derivative of $B^2/[\lambda(1-B)]$, multiplied by λ, is $2g-1+gB/(1-B)\ge 0$ when $g\ge 1/2$. Thus checking the gate at U proves the whole interval. A rational upper bound for the logarithm verifies that gate. On cells using the fixed positive bound $l_f r_-$ instead, unscaled endpoint tails suffice: after a positive bracket is checked, divide it by U to obtain a lower bound per λ. These arguments, followed by (201), prove TINY uniformly down to arbitrary positive noise, including a deterministic conditional component.

V.5 The whole-count witness

Assume $c=1-d_0>1/2$. Couple all $2n$ conditional variables independently and set $W_j=\sum_{i\le j}U_{1,i}+\sum_{i>j}U_{0,i}$, extending it constantly outside $[0,n]$. Every increment of $j+W_j$ is at least c. Define

$$K=\max\{j\in\mathbb{Z}: j+W_j\le k+u\},\qquad R=K-k.$$

An independent binomial selector has the correct actual count laws, so

$$\Pr(\textstyle\sum Y_i\le k+u)=\mathbb{E}F_B(k+R). \tag{220}$$

Let $\chi=(n+1)p-k<2$ and $m_B=\max_j\pi_j$. For every integer r,

$$F_B(k+r)-F_B(k)-r\pi_k\le\frac{m_B}{(n+1)v}\left[(\chi-1)_+r_+ +\chi_+\binom{r_-}{2}+\binom{r_-}{3}\right]. \tag{221}$$

Indeed, the positive part of $\pi_{k+i}-\pi_{k+i-1}$ is at most $m_B(\chi-i)_+/((n+1)v)$, by $\pi_j-\pi_{j-1}=\pi_j((n+1)p-j)/(p(n-j+1))$. Double summation gives (221): for $r>0$ only $i=1$ can contribute; for $r=-h<0$ use $(\chi+i)_+\le\chi_++i$ and sum the multiplicities $h-1-i$. Zero extension makes the same calculation valid outside the legal count range.

Write $t=\mathbb{E}W_k^2$. Crossing monotonicity and Cauchy–Schwarz give

$$\mathbb{E}R_+\le\frac{t}{(c-u)^2},\quad \mathbb{E}\binom{R_-}{2}\le\frac{t}{[c+\min(u,c/3)]^2},\quad \mathbb{E}\binom{R_-}{3}\le\frac{t\sqrt{A^2+3t}}{(2c-1/2)^3}. \tag{222}$$

For the first, $R=i\ge 1$ implies $-W_k\ge ci-u$ and $(ci-u)^2\ge i(c-u)^2$. For $R=-h\le -2$, the next count is outside the closed event, so $W_k>c(h-1)+u$. Put $j=h-1$. The function $j(j+1)/(2(cj+u)^2)$ decreases for $j\ge 1$ if $u\le c/3$; for larger u reduce to that endpoint. Also $(j^3-j)/(6(cj-1/2)^3)$ decreases for $j\ge 2$, since its derivative has the sign of $2cj-(3j^2-1)/2<0$. These give the last two bounds using $\mathbb{E}(W_k)_+^3\le\sqrt{\mathbb{E}W_k^2\mathbb{E}W_k^4}$.

The marked crossing.

Let $D_i = U_{1,i} - U_{0,i}$, $\vartheta = \mathbb{E}(D_i)_+ = \mathbb{E}(D_i)_-$, $0 < \eta \le 1$, and $R_f = \mathbb{E}\{u + \eta - W_k\}$. The two deleted laws are

$$A_0 \overset{d}{=} kU_1 + (n-k-1)U_0, \qquad A_1 \overset{d}{=} (k-1)U_1 + (n-k)U_0.$$

Then

$$\mathbb{E}R \le u + \eta - R_f + J, \tag{223}$$
$$J = \frac{\vartheta}{\eta} \sum_{i \ge 1} \{\Pr(A_0 \le u + d_0 - ci) + \Pr(A_1 > u + \eta + i - A)\}.$$

Here are the pathwise details. Put $z = u - W_k$ and mark

$$Z_f = \sum_{i=1}^{n-k} (D_{k+i})_- \mathbf{1}_{\{W_k \le u - ci\}}, \quad Z_b = \sum_{i=1}^{k} (D_{k-i+1})_+ \mathbf{1}_{\{W_k > u + \eta + i\}}.$$

If $R > z + \eta$, the crossing activates the marks through R: for $R > 0$ use $z \ge cR$, while for $R < 0$ use $z < R - \eta$. In either case $R \le z + Z_f + Z_b$. Hence if $h = R - \lfloor z + \eta \rfloor > 0$, $Z_f + Z_b > h - 1 + \eta \ge \eta h$. Thus $R \le \lfloor z + \eta \rfloor + (Z_f + Z_b)/\eta$ on every path. On an active forward mark $U_0 > U_1 \ge -d_0$; deleting its pair implies $A_0 \le u + d_0 - ci$. On a backward mark delete $U_1 \le A$ to obtain the strict event for A_1. The remaining sum is independent of the entire pair, so the expectations factor. Extending to all $i \ge 1$ adds only nonnegative bounds. Finally $|R| \le 1 + (|W_k| + |u|)/c$ proves integrability and permits expectations. No coupling has replaced either deleted marginal.

A finite bound for every mark.

Fix $\lambda \in [L, U]$, $L > 0$, and $t \le \omega_+ \lambda$; put $T_L = \omega_+ L$, $T_U = \omega_+ U$. On a phase subcell $u \in [v_l, v_h]$, for $i = 1, 2$ let $b_1 = ci - d_0 - v_h > 0$ and $b_2 = i - A + v_l > 0$. Lemma V.3 gives a bound $\omega_+ \lambda K_i^{\mathrm{ind}}$, where

$$K_i^{\mathrm{ind}} = \sum_{j=1}^{2} \min \left\{ \frac{1}{b_j^2 + T_L}, \frac{\overline{P}(A, T_U, b_j)}{T_L} \right\}.$$

The denominator T_L is a lower bound for the chosen upper envelope $\omega_+ \lambda$, not for any actual variance. Alternatively couple $A_0 = Z + U_1$, $A_1 = Z + U_0$ with $Z \overset{d}{=} (k-1)U_1 + (n-k-1)U_0$. The two events project into $Z \le -\alpha$ and $Z > \beta$, where $\alpha = ci - 2d_0 - v_h > 0$, $\beta = i - A - d_0 + v_l > 0$. For any $z \in (-\alpha, \beta)$ their sum is at most $(\omega_+ \lambda + z^2)/\min(z + \alpha, \beta - z)^2$. Consequently a second kernel is

$$K_i^{\mathrm{com}} = \frac{1 + z^2/T_L}{\min(z + \alpha, \beta - z)^2}, \qquad z = \max\{-T_L/\beta, \min((\beta - \alpha)/2, T_L/\alpha)\}.$$

This center is inside the interval; differentiating the two quadratic ratios on either side of the midpoint proves its optimality, although any interior rational center suffices. Take the smaller kernel on each phase subcell and the maximum over their closed cover. Denote it K_i. For $i \ge 3$ use fourth moments and the convex midpoint integral for $(cx - b)^{-4}$. On $u \in [u_l, u_h]$ this gives

$$K = K_1 + K_2 + \frac{A^2 + 3T_U}{3c} \left\{ \frac{1}{((5/2)c - A - u_h)^3} + \frac{1}{((5/2)c - A + u_l)^3} \right\}, \qquad J \le \frac{\vartheta}{\eta} \omega_+ \lambda K. \tag{224}$$

All denominators must be positive. Indeed, the tail fourth moment is at most $A^2t + 3t^2$, and integration from $5/2$ to infinity covers every remaining integer, including the harmless added illegal counts.

If the same-law resource supplies $nd < G$ and $G^2 < c_v NL$, then

$$\vartheta \le \frac{c_v G\lambda}{c_v n\lambda + G^2}. \tag{225}$$

To prove this, for centered $X \ge -b$, $\mathbb{E}X^2 \le V$ and $b^2 \le V$, the quadratic $b(X - V/b)^2/(b + V/b)^2$ majorizes X_- on $[-b, \infty)$. Its expectation is at most $bV/(V + b^2)$. Apply this to $D_i \ge -d$, using $\mathrm{Var}(D_i) = s_0 + s_1 \le c_v\lambda/n$, and monotonicity in $b \le \sqrt{V}$. The case $d = 0$ is immediate. With $\eta = z\lambda$, $z > 0$, $zU \le 1$, (224) becomes

$$\frac{J}{\lambda} \le \frac{\omega_+ K c_v G}{z(c_v N\lambda + G^2)}. \tag{226}$$

The fractional-part gain.

If each centered summand has support span at most $b < 1/2$ and the sum has variance t, then, for any real w and $0 < r_* \le 2\sin^2(\pi b)/b^2$,

$$\mathbb{E}\{w - W\} \ge \frac{5}{23}(1 - e^{-r_* t}). \tag{227}$$

Indeed $1 - \cos(2\pi x) \le (23/5)x$ for $0 \le x \le 1$. A finite proof of this sole scalar inequality is recorded below. For an independent copy Z' of one summand, concavity of sine on $[0, \pi b]$ gives

$$|\mathbb{E}e^{2\pi i Z}|^2 = 1 - 2\mathbb{E}\sin^2 \pi(Z - Z') \le e^{-4\sin^2(\pi b)\,\mathrm{Var}(Z)/b^2}.$$

Multiply square roots over the independent summands and use periodicity. This argument includes zero variances and integer collisions. Here the conditional spans are at most $A + d_0$, so this is the b to use.

For a reproducible scalar proof, $x \ge 10/23$ follows from $1 - \cos \le 2$. On $[0, 10/23]$ use the degree-ten alternating upper polynomial for $1 - \cos(2\pi x)$, with directed coefficients:

$$Q(x) = \sum_{j=1}^{5} \frac{(-1)^{j+1}(2\pi_j x)^{2j}}{(2j)!}, \qquad \pi_j = \begin{cases} 3.141592653589794, & j \text{ odd}, \\ 3.141592653589793, & j \text{ even}. \end{cases}$$

Put $P(x) = 23/5 - Q(x)/x$. On the eight intervals with left endpoint $a_j = 5j/92$ and width $w = 5/92$, its degree-nine Bernstein coefficients are

$$b_{jk} = \sum_{i=0}^{k} \frac{\binom{k}{i}}{\binom{9}{i}} \sum_{h=i}^{9} [x^h]P(x) \binom{h}{i} a_j^{h-i} w^i, \qquad 0 \le j < 8, \quad 0 \le k \le 9.$$

The 80 rational coefficients are positive; the retained scalar acceptor reconstructs them. Expansion in powers of the unit-interval coordinate proves the formula; Bernstein weights are nonnegative and sum to one. The alternating remainder and the stated rational bounds on π follow from the same elementary series above. This completely specifies the finite scalar obligation. A rational rate follows by replacing $\sin(\pi b)$ with $x - x^3/6 + x^5/120 - x^7/5040$, $x = 3.14159b$, on the checked alternating-series domain.

Set $r = r_*\omega_-$, require $rU < 3$, and define

$$g = \frac{5}{23}\left(\frac{r^3L^2}{6} - \frac{r^4L^3}{24} + \frac{r^5L^4}{120} - \frac{r^6L^5}{720}\right). \tag{228}$$

The sixth-order Taylor lower bound for $1 - e^{-r\lambda}$ gives $R_f/\lambda \ge 5r/23 - 5r^2\lambda/46 + g$. The discarded higher-order part is increasing as a function of λ for $r\lambda < 3$: its derivative has the sign of $48 - 18x + (24/5)x^2 - x^3 > 0$ on $[0, 3]$. For instance this polynomial has positive degree-three Bernstein coefficients $48, 30, 132/5, 51/5$ there. Thus freezing it at L is valid.

One final expression.

On a signed phase cell find an interval $[x_l, x_h]$ containing H/ℓ by its four endpoint quotients, using positive bounds for ℓ. Let

$$D = \max(|x_l - u_h|, |x_h - u_l|), \quad D_+ = \max(0, x_h - u_l), \quad a = \max(|u_l|, |u_h|), \quad b_\chi = p_+ - x_l.$$

The same formula allows coarser upper bounds D, D_+, b_χ, as used by the central certificate. Check $D \le \Delta$ and $b_\chi + u_h < 2$. With $\omega_\pm = 1 \pm c_v D/N$, the curvature coefficient is

$$V_0 = \frac{\omega_+}{v_*}\left[\sup_{u_l \le u \le u_h}\left\{\frac{(b_\chi + u - 1)_+}{(c-u)^2} + \frac{(b_\chi + u)_+}{[c + \min(u, c/3)]^2}\right\} + \frac{\sqrt{A^2 + 3T_U}}{(2c - 1/2)^3}\right]. \tag{229}$$

On a rational phase subcell, both numerators are bounded at the right endpoint; the first denominator is smallest there, and the second is smallest at the left endpoint. This proves the supremum bound, including both kinks. Equations (220)–(222), $\sqrt{n} m_B \le 1$, and (200) now give the whole-count witness with

$$B(\lambda) = z + \frac{\omega_+ K c_v G}{z(c_v N \lambda + G^2)} - \frac{5r}{23} + \frac{5r^2\lambda}{46} - g, \tag{230}$$
$$I(\lambda) = V_0 + aB_1 + \frac{aP_1}{\lambda} + P_* D_+ - m_c Q_*.$$

We used $|u - f(u)| \le a(B_1\lambda + P_1)/n$ and $\sqrt{n}\pi_k \le 1$. Every constant is fixed before λ varies. Both B and $mB + I/N$ are convex on $\lambda > 0$; I is decreasing. Thus their two endpoint gates $B < 0$, $I > 0$, $mB + I/N < -.001$ prove the entire closed noise cell. This proves all three LOW variants by exactly the same argument.

V.6 The selected finite cover and the zero-noise endpoint

The retained tables contain 45 central probability intervals covering $[.38, .47]$ and 38 left intervals covering $[.304, .38]$. Each row supplies its own Θ, signed H bounds, d_0, Λ, and global amplitude; the confinement profiles may improve the amplitude with λ. All these are hypotheses of the present section, acquired earlier without using its conclusion. The shared resource has the form

$$nD_{\mathrm{mom}} < \frac{15000}{997} T_H, \qquad D_{\mathrm{mom}} \ge \alpha d + \beta\theta + P.$$

Consequently a noise cell $[L, U]$ has

$$nd < G = \{(15000/997)T_H - NP - \beta \ell_-^2 L\}/\alpha.$$

Here T_H is the maximum of the acquired concave quadratic budget on the closed signed H cell, further bounded by its scalar ceiling. It already contains the phase charge. If $G \le 0$, the entire cell is impossible and receives no score test; the exclusion is strict even at $G = 0$.

For positive noise the complete cover is the following. Each interval is intersected with the row ceiling Λ; shared endpoints are included.

Noise	Central phase	Left phase	Witness
$(0, .05]$	all	all	nearest count (scaled)
$[.05, .06]$	$\lvert u\rvert \ge .09$	$\lvert u\rvert \ge .09$	nearest count
$[.06, .08]$	$\lvert u\rvert \ge .13$	$\lvert u\rvert \ge .14$	nearest count
$[.08, .10]$	$\lvert u\rvert \ge .18$	$\lvert u\rvert \ge .19$	nearest count
$[.10, .12]$	$\lvert u\rvert \ge .29$	$u \le -.29$	nearest count
$[.10, .12]$	—	$u \in [.29, .5]$	whole count
$[.05, \Lambda]$	remaining phase	remaining phase	whole count

Beyond .12 the last row covers the full phase interval. The three LOW objects are one central cover, one left cover with phase slices, and one extra positive left cover. Their distinction is only the closed domain and the chosen outward constants. Signed H cells meet at zero; nearest count ties meet at $u = \pm 1/2$. The fixed coverage checker also checks every row-ceiling singleton and every entirely excluded LOW row.

The common TINY parameters are

$$N = 47, \quad p \in [.304, .47], \quad |H| \le 1.393, \quad \Theta = .0154, \quad d_0 = .0181, \quad A = .425, \quad \Lambda = .05.$$

For its scaled own witness use $a = .4784$, $K_0 = 9/(4 \cdot .4641)$, $(z_-, c_-) = (1.66, .514)$ and $(z_+, c_+) = (1.2, 1.036)$. Its six phase cells are $[-.5, 0], [0, .1], \dots, [.4, .5]$. The first two use scaled Poisson anchors $1.25, 1$, respectively; the other four use the fixed positive normal bound. The exact normalizers and gates specified in the accompanying certificate map verify these constants. The central MID amplitudes are $.35, .35, .30, .275$ in its four rows. The left ones are $.375, .375, .325, .30$, each further reduced by the row's global amplitude. No interpolation of the tables is used.

All six accepted comparisons give the same strict conclusion

$$\rho_B(C - C_B) < -.001\lambda.$$

Their proofs have now reduced to Theorem V.1, the shared normalization, and the two explicitly derived witnesses. The arithmetic objects still have their original domains and targets; no certificate branch has been silently deleted or relabeled.

If $\lambda = 0$, conditional centering and $p, q > 0$ imply $U_0 = U_1 = 0$ almost surely. The law is exactly Bernoulli. For an attained positive closed maximum its threshold is a binomial atom: lowering a noninteger threshold to the preceding atom leaves the CDF unchanged and strictly improves the normal term, while thresholds below the first atom have nonpositive score. Thus $u = 0$ at such a maximum and $C = C_B$. Schulz's all-probability Bernoulli theorem [12, Theorem 1] gives $C_B \le C_E$ for every $0 < p < 1$ and in particular for all the indices $n \ge 47$ used here. The same external theorem closes the positive-noise conclusion $C < C_B \le C_E$. At zero noise $C = C_B \le C_E$ already contradicts the strict contrary hypothesis $C > C_E$ used by the main first-failure argument; no finite Bernoulli reserve is needed. No estimate in this section divides by zero or passes from fixed-law convergence to uniformity in varying laws.

W Noise and inward displacement share one budget

The local comparison near a two-point law uses both a variance bound and a bound on the displacement of its inner support endpoints. These two bounds come from the same inequality. Keeping their joint constraint shrinks the allowed displacement as the variance increases. We give an elementary, self-contained proof, including the gap identity on which it rests. The final section explains its use in the exact local comparisons; the numerical acquisition of the input bounds is a separate obligation.

W.1 The idea

Suppose a proof has established, for nonnegative quantities d and λ, that

$$\frac{d}{d_*} + \frac{\lambda}{\Lambda} \le 1.$$

Replacing this by $d \le d_*$ and $\lambda \le \Lambda$ loses information. For example, if $\lambda \ge \Lambda/2$, the original inequality gives $d \le d_*/2$. The local characteristic-function and tail bounds improve when d decreases, so this elementary observation is useful precisely where the noise is largest.

Here d measures how far the actual inner endpoints lie inside the conditional means, and λ is the total conditional noise variance in a sum of n independent copies. The point is to establish the joint inequality for one and the same marginal distribution.

W.2 Definitions and the gap identity

Let X be bounded, $\mathbb{E}X = 0$, and $\mathbb{E}X^2 = 1$. Assume that its closed support has a gap $(-A, B)$ containing zero, with $A, B > 0$ and endpoints in the support. Write

$$\sigma = \sqrt{10}, \quad \rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad m = \mathbb{E}X_+ = \mathbb{E}X_-.$$

Thus $0 < p, q < 1$ and $m > 0$. For every bounded open gap $g = (a, b)$ of the closed support, let τ_g be the constant value on that gap of $\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X \geq t\}}]$. Set

$$J = \sum_g \tau_g (b - a)^2, \qquad D = (3 + \sigma)\rho - \gamma - 3J.$$

All summands in J are nonnegative. The bounds below also establish convergence of its outer-gap part. Endpoint atoms cause no ambiguity: τ_g denotes the value in the open gap.

Define

$$T_+ = \mathbb{E}X_+^3 - mB^2, \quad T_- = \mathbb{E}X_-^3 - mA^2, \qquad R_+ = \mathbb{E}X_+^2 - Bm, \quad R_- = \mathbb{E}X_-^2 - Am.$$

Each is nonnegative because the nonzero positive and negative magnitudes are at least B and A, respectively. Put $J_o = J - m(A + B)^2$.

Lemma W.1 (Gap remainder)**.** *The outer gaps satisfy*

$$J_o \leq T_+ + T_- - 2BR_+ - 2AR_-.$$

Consequently

$$D = m\big(\sqrt{\sigma + 1}A - \sqrt{\sigma - 1}B\big)^2 + (\sigma - 1)T_+ + (\sigma + 1)T_- \\ + 3(T_+ + T_- - J_o). \tag{231}$$

In particular $D \geq 0$.

Proof. On a positive outer gap (u, v), where $u \geq B$,

$$(v - u)^2 \leq 2 \int_u^v (t - B)\, dt.$$

Multiply by τ_g and sum. Tonelli's theorem, applied to nonnegative integrands, gives

$$J_o^+ \leq 2 \int_B^\infty \tau(t)(t - B)\, dt = \mathbb{E}[X_+(X_+ - B)^2] = T_+ - 2BR_+.$$

Reflection proves the corresponding negative bound. Their sum is the first assertion. To check the identity, substitute $\rho = m(A^2 + B^2) + T_+ + T_-$, $\gamma = m(B^2 - A^2) + T_+ - T_-$, and $J = m(A + B)^2 + J_o$. The cross coefficient in the square is $-6AB$ because $(\sigma + 1)(\sigma - 1) = 9$. Nonnegativity follows from the first assertion and $R_\pm, T_\pm \geq 0$. □

W.3 The simultaneous budget

Let $V = \mathbb{E}[X \mid \mathbf{1}_{\{X>0\}}]$, and let $W = X - V$. Define

$$\theta_+ = \mathbb{E}[W^2; X > 0], \qquad \theta_- = \mathbb{E}[W^2; X < 0], \qquad \theta = \theta_+ + \theta_-.$$

The two conditional means are $\bar{B} = m/p$ and $-\bar{A} = -m/q$. Therefore

$$m^2 = (1-\theta)pq, \qquad \Delta_A = \bar{A} - A \geq 0, \qquad \Delta_B = \bar{B} - B \geq 0.$$

In particular $\theta < 1$. The following inequality retains three distinct nonnegative costs: departure of p from its distinguished value, inward displacement, and conditional variance.

Theorem W.2 (One-law gap budget)**.** *With $p_* = (4 - \sqrt{10})/2$,*

$$\begin{aligned} D \geq \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p-p_*)^2 + 6m\{(A+B)(\Delta_A+\Delta_B) + \Delta_A\Delta_B\} \\ + 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-. \end{aligned} \tag{232}$$

Proof. Conditional expansion of the cubic gives

$$\mathbb{E}X_+^3 = m\bar{B}^2 + \mathbb{E}[(X+2\bar{B})(X-\bar{B})^2; X>0] \geq m\bar{B}^2 + 3B\theta_+.$$

The reflected inequality has $\bar{A}, A, \theta_-$ in place of $\bar{B}, B, \theta_+$. Also $R_+ = \theta_+ + m\Delta_B$ and $R_- = \theta_- + m\Delta_A$. Substitute these statements in Lemma W.1. The variance coefficients become $3(\sigma+1)B$ and $3(\sigma+3)A$. The remaining deterministic expression is

$$\begin{aligned} m\{(\sigma+1)\bar{A}^2 + (\sigma-1)\bar{B}^2 - 6\bar{A}\bar{B}\} \\ + 6m\{\bar{A}\bar{B} - AB + B\Delta_B + A\Delta_A\}. \end{aligned}$$

The second brace equals $(A+B)(\Delta_A+\Delta_B) + \Delta_A\Delta_B$. For the first, substitute $\bar{A} = m/q$, $\bar{B} = m/p$ and use

$$(\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2, \qquad \frac{m^3}{p^2q^2} = \frac{(1-\theta)^{3/2}}{\sqrt{pq}}.$$

This proves every term in (232). □

W.4 A triangle of allowed noise and displacement

Introduce Bernoulli coordinates

$$X = \ell(B_0 + U - p), \quad B_0 = \mathbf{1}_{\{X>0\}}, \quad \ell = \bar{A} + \bar{B} = \frac{m}{pq}, \quad \mathbb{E}[U \mid B_0] = 0.$$

Then

$$\ell^2 = \frac{1-\theta}{pq}, \quad m\ell = 1-\theta, \qquad d = \frac{\Delta_A + \Delta_B}{\ell}, \quad \lambda = n\mathbb{E}U^2 = \frac{npq\theta}{1-\theta}.$$

Theorem W.3 (Joint inward/noise constraint)**.** *Suppose $n \geq N > 0$, and bounds for this same law give*

$$nD \leq Nd_D, \quad \theta \leq \Theta < 1, \quad pq \leq v_*, \quad A + B \geq S_* > 0,$$

and

$$D \geq b + 6mS_*(\Delta_A + \Delta_B) + K\theta, \qquad b \geq 0, \quad K > 0.$$

Set $r = d_D - b$. If $r > 0$, define

$$d_* = \frac{r}{6(1-\Theta)S_*}, \qquad \Lambda = \frac{Nv_*r}{K(1-\Theta)}.$$

Then

$$\boxed{\frac{d}{d_*} + \frac{\lambda}{\Lambda} \le 1.} \tag{233}$$

If $r < 0$, the premises are impossible. If $r = 0$, they force $d = \lambda = 0$.

Proof. Multiply the assumed joint lower bound for D by n. Since all coefficients being weakened are nonnegative,

$$\begin{aligned} Nd_D \ge nD &\ge nb + 6n(1-\theta)S_*d + K\frac{(1-\theta)\lambda}{pq} \\ &\ge Nb + 6N(1-\Theta)S_*d + \frac{K(1-\Theta)}{v_*}\lambda. \end{aligned}$$

Subtract Nb. For $r > 0$, division by Nr proves (233). For $r \le 0$, positivity of both coefficients gives the stated alternatives without dividing by zero. □

The hypotheses are direct consequences of Theorem W.2. For example, on $p \in I = [l, u]$, take positive lower bounds A_*, B_*, put $S_* = A_* + B_*$ and

$$v_* = \max_{p \in I} p(1-p), \qquad K = \min\{3(\sigma+1)B_*, 3(\sigma+3)A_*\},$$

and use

$$b = \frac{(6+2\sigma)(1-\Theta)^{3/2}}{\sqrt{v_*}} \operatorname{dist}(p_*, I)^2.$$

Drop the nonnegative product $6m\Delta_A\Delta_B$. The remaining bound is the hypothesis of Theorem W.3.

Corollary W.4 (Closed noise bands and outward rounding)**.** *If $0 \le L \le \lambda \le U \le \Lambda$, then*

$$d \le d_*(1 - L/\Lambda).$$

Both intercepts in (233) *may first be replaced by larger positive numbers. The same conclusion then holds using those enlarged intercepts and the enlarged noise ceiling.*

Proof. Rearrange (233) and use $\lambda \ge L$. Increasing either denominator weakens its nonnegative summand, so it preserves the joint inequality. This also proves the rounding assertion. □

W.5 Use in the computer-assisted proof

The acquired tables store a weighted defect bound $nD \le Nd_D$ and the two intercepts d_*, Λ. Those are simultaneous statements about the actual extremizing law. An unweighted bound $D \le d_D$ alone would not justify the n-uniform noise ceiling.

On each closed noise interval, the local programs recompute the inward gap ceiling from the corollary. This improves the available support separation, the conditional cubic correction, the Poisson and common background tail distances, and the characteristic-function decay rate. The proof uses the resulting smaller allowed set. It does not add two different estimates that each spend the entire budget.

In the current exact central comparison the amplitude ceilings on the noise bands are

λ	$[.05, .08]$	$[.08, .10]$	$[.10, .12]$	$[.12, \Lambda]$
$\lvert U\rvert$ ceiling	.375	.325	.275	.25.

Each is intersected with an independently acquired first amplitude bound. A separate support/concentration argument proves these amplitude profiles before the final local score comparison uses them. The exact central profile replay checks 248 signed configurations and 210,130 strict rational gates, with 14 malformed or false-object controls.

The analytic statement proved here is Theorem W.3. It does not, by itself, prove attainment, a Fourier remainder, a numerical cutoff, or acquisition of any numerical resource table. Those edges remain explicit in the project dependency map. The proof uses no floating-point positivity claim and no limiting assertion about varying marginal laws.

X The sign of the cubic correction

The cubic correction may have either sign, so its multiplier must be bounded in the corresponding direction. The following argument uses the stated support, moment and score bounds, supplied in the finite application at $42 \le n \le 48$. Every conditional moment concerns the same iid marginal.

X.1 Conditional centering and the exact cubic bound

Let B be Bernoulli with parameter $0 < p < 1/2$, put $q = 1 - p$ and $v = pq$, and write $Y = B + U$, with $\mathbb{E}(U \mid B) = 0$. Let U_0, U_1 denote the conditional laws of U, and assume

$$U_0 \le a_0 < p, \qquad U_1 \ge -a_1 > -(1-p), \qquad a_0, a_1 \ge 0, \qquad d = a_0 + a_1.$$

Put $s_j = \mathbb{E}U_j^2$, $s = qs_0 + ps_1$, $\lambda = ns$, and

$$X = \frac{Y - p}{\sqrt{v+s}}, \quad \rho = \mathbb{E}|X|^3, \quad \rho_B = \frac{1-2v}{\sqrt{v}}.$$

Finite third moments are assumed. The support conditions fix the sign of $Y - p$ in each class, which is necessary for the following expansion.

Lemma X.1 (No positivity assumption on the cubic bracket)**.** *With the preceding notation,*

$$(v+s)^{3/2}(\rho - \rho_B) \ge sQ, \qquad Q = 6p - 3p^2 - \frac{3}{2} - d - \frac{3(1-2v)s}{8v}. \tag{234}$$

Proof. Expanding within each sign class and using conditional centering gives

$$\mathbb{E}|Y-p|^3 = v(1-2v) + 3pq(s_0+s_1) + p\mathbb{E}U_1^3 - q\mathbb{E}U_0^3.$$

The one-sided support bounds imply $\mathbb{E}U_1^3 \ge -a_1 s_1$ and $\mathbb{E}U_0^3 \le a_0 s_0$. Moreover $3pq(s_0+s_1) \ge 3p(qs_0 + ps_1) = 3ps$, because $q \ge p$. Consequently

$$\mathbb{E}|Y-p|^3 \ge v(1-2v) + (3p-d)s.$$

For $z \ge 0$, Taylor's formula gives $(1+z)^{3/2} \le 1 + 3z/2 + 3z^2/8$: the third derivative is negative. Subtract

$$\rho_B(v+s)^{3/2} = v(1-2v)(1+s/v)^{3/2} \le v(1-2v) + \tfrac{3}{2}(1-2v)s + \frac{3(1-2v)s^2}{8v}$$

from the preceding moment lower bound and use $3p - \frac{3}{2}(1-2v) = 6p - 3p^2 - \frac{3}{2}$. □

X.2 The coefficient must be bounded in the correct direction

Suppose $n \ge N$, $p \in [p_-, p_+] \subset (0, 1/2)$, $0 \le d \le d_0$, and $0 \le \lambda \le \Lambda$. Choose $0 < v_* \le p_-(1-p_-)$ and put

$$V^* = p_+(1-p_+) + \Lambda/N, \qquad Q_* = 6p_- - 3p_-^2 - \frac{3}{2} - d_0 - \frac{3(1-2v_*)\Lambda}{8v_*N}.$$

Then $Q \ge Q_*$: the first term increases with p, and $(1-2v)/v$ decreases with v. In particular, Q_* may be negative. Already with $d = s = 0$ its sign changes at $p = 1 - 1/\sqrt{2}$.

Let C be the score multiplying the moment correction, and assume $0 < c_* \le C \le c^*$. Choose positive multipliers satisfying

$$0 < m_- \le \frac{c_*}{(V^*)^{3/2}}, \qquad m_+ \ge \frac{c^*}{v_*^{3/2}},$$

and define the signed gain

$$G_* := \min\{m_- Q_*, m_+ Q_*\}. \tag{235}$$

Proposition X.2 (Signed gain and nonnegative repair)**.** *Let Φ denote the standard normal distribution function. For an actual count displacement δ, set*

$$\mathcal{R} = \sqrt{n}\left\{\Phi\left(\frac{\delta}{\sqrt{nv}}\right) - \Phi\left(\frac{\delta}{\sqrt{n(v+s)}}\right)\right\} + C(\rho_B - \rho).$$

If $D_+ \ge \max(\delta, 0)$ and $P_* \ge .4/(2v_*^{3/2})$, then

$$\mathcal{R} \le (P_* D_+ - G_*)\frac{\lambda}{n} \le \kappa\frac{\lambda}{N}, \qquad \kappa = \max\{0, P_* D_+ - G_*\}. \tag{236}$$

Proof. Write $A = C/(v+s)^{3/2}$. The hypotheses give $0 < m_- \le A \le m_+$. By Lemma X.1, $C(\rho - \rho_B) \ge sAQ \ge sAQ_*$. If $Q_* \ge 0$, $AQ_* \ge m_- Q_*$; if $Q_* < 0$, $AQ_* \ge m_+ Q_*$. This proves $C(\rho - \rho_B) \ge sG_*$ in both cases. One must use the *upper* multiplier in the negative case.

The normal part is nonpositive when $\delta \le 0$. For $\delta > 0$, differentiation with respect to an intermediate variance $t \in [0, s]$ gives

$$\frac{d}{dt}\left[-\sqrt{n}\,\Phi\left(\frac{\delta}{\sqrt{n(v+t)}}\right)\right] = \frac{\delta}{2(v+t)^{3/2}}\phi\left(\frac{\delta}{\sqrt{n(v+t)}}\right) \le \frac{.4\delta}{2v_*^{3/2}} \le P_* D_+.$$

Integrate over $[0, s]$, and combine with the moment bound. Finally $s = \lambda/n$ and $\kappa \ge 0$, so $(P_* D_+ - G_*)\lambda/n \le \kappa\lambda/n \le \kappa\lambda/N$. The clamp at zero is needed before replacing n by its lower bound. □

X.3 Exact checks and the finite application

The new LEFT42 normalization uses

$$N = 42, \qquad c_* = .4, \qquad c^* = .4226.$$

These are hypotheses supplied by the surrounding finite-index proof. The rational normalization checker verifies multiplier inequalities by

$$(m_-)^2(V^*)^3 < c_*^2, \qquad (m_+)^2 v_*^3 > (c^*)^2,$$

together with positivity of both multipliers. Squaring without those positivity checks would accept a false negative coefficient. The normal multiplier is checked similarly using $4P_*^2v_*^3 > .4^2$ and $P_* > 0$.

The checker recomputes the minimum in (235) from the same input row, then verifies a rational upper enclosure of the nonnegative coefficient in (236). Altering the sign, the score cap, or the acquired probability/noise row is not a licensed change of hypotheses. Exact arithmetic verifies these finite inequalities; Lemma X.1 and Proposition X.2 explain why they imply the probabilistic correction.

Y Keeping phase and resource information in the local comparison

We refine the local comparison of Section V by keeping the phase and the joint noise/displacement constraint throughout each estimate. The nearest-count and whole-count identities, quartic minorant and Poisson comparison were proved there; the conditional support inputs come from Section U. Each refinement below states which of those acquired bounds it uses.

Y.1 Notation and the information that must be retained

Let $Y = B + U$, where $\mathbb{P}(B = 1) = p$, $\mathbb{E}(U \mid B) = 0$, $v = p(1-p)$, $s = \mathbb{E}U^2$, $\lambda = ns$, and $\ell = (v+s)^{-1/2}$. The actual threshold has the form

$$x = np + H/\ell = k + u, \qquad |u| \le \tfrac{1}{2}, \qquad \delta = k - np = H/\ell - u.$$

The integer k and the value δ stay fixed when integrating a function of a new variable t below. Write $\pi_k = \mathbb{P}(\mathrm{Bin}(n,p) = k)$ and

$$f(t) = \frac{\Phi((\delta+t)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

The conditional errors satisfy $|U_j| \le A$, $U_0 \le a_0$, $U_1 \ge -a_1$, with $a_0, a_1 \ge 0$ and $d = a_0 + a_1$. All quantities in an application refer to the same iid marginal.

Lemma Y.1 (A joint budget on a noise interval)**.** *Suppose the acquired same-law resource inequality is*

$$\frac{d}{d_*} + \frac{\lambda}{\Lambda} \le 1, \qquad d_* > 0, \quad \Lambda > 0.$$

On a closed noise interval $L \le \lambda \le U \le \Lambda$ one may use

$$d \le d_L := d_*(1 - L/\Lambda).$$

Every conditional support, cubic-moment, tail-distance and characteristic function bound whose hypothesis is $d \le d_0$ may therefore be applied with $d_0 = d_L$ on that interval.

Proof. Rearrange the displayed resource inequality and use $\lambda \ge L$. The substitution is valid separately on every noise interval. Neither the pair of inequalities $d \le d_*$, $\lambda \le \Lambda$ nor independently maximized numerical coordinates imply the joint resource inequality. Its acquisition is a separate obligation. □

For example, the whole-count kernel uses $c = 1 - d_0$. Its moment coefficients are recomputed as

$$\alpha = 6m_0\ell_-^2(1-d_0), \qquad \beta = \min\{3(4.16)\ell_-(1-p_+-d_0), 3(6.16)\ell_-(p_- - d_0)\},$$

where m_0, ℓ_- are the previously justified moment and scale lower bounds. The ordinary cubic correction is recomputed with the same d_0. Likewise, a noise amplitude bound A gives a conditional support span at most $A + d_0$; the characteristic-function rate is recomputed from that span. These changes preserve the original identities and their sign conditions.

Y.2 A binomial mass bound on the actual phase cell

Suppose the finite Stirling argument has supplied positive constants E, L_0, R on a fixed segment containing every k/n under consideration, with

$$\sqrt{n}\,\pi_k = \frac{\phi(0)}{\sqrt{v}}\exp\{-\delta^2/(2nv)+\zeta\}, \qquad n|\zeta| \le Z_N(|\delta|), \qquad Z_N(t) = \frac{L_0 t}{2}+R+\frac{Et^3}{6N}, \quad n \ge N.$$

The constants may have been computed on a much wider global phase range. Let $v_- \le v \le v_+$ and let $q_+ \ge \sqrt{v_+}$.

Lemma Y.2 (Mass localization)**.** *On any smaller closed cell where $|\delta| \le D$,*

$$\sqrt{n}\,\pi_k \ge \frac{.39886}{q_+}\left[1 - \frac{D^2/(2v_-) + Z_N(D)}{N}\right].$$

The maximum of this expression and any previously proved mass lower bound is also a valid mass lower bound. There is no requirement $D \ge 1$.

Proof. Set $A_* = D^2/(2v_-) + Z_N(D) \ge 0$. The logarithmic formula gives $\sqrt{n}\pi_k \ge (.39886/q_+)e^{-A_*/n}$. Now $e^{-z} \ge 1 - z$ and $n \ge N$. If the bracket is negative the displayed inequality still holds, and the previously positive bound can be retained. The global Stirling segment, rather than the smaller value D, continues to justify E, L_0, R. □

Y.3 A sharper elementary harmonic constant

Lemma Y.3. *For every $x \ge 0$,*

$$1 - \cos(2\pi x) \le \frac{4553}{1000}x.$$

Proof. Put $K = 4553/1000$. For $x \ge 2/K$ the assertion follows from $1 - \cos \le 2$. On $[0, 2/K]$ take the degree-fourteen alternating upper Taylor polynomial $Q(x)$ for $1 - \cos(2\pi x)$. Its remainder is nonpositive: the first omitted term is negative, and the remaining term magnitudes decrease, since $(4\pi/K)^2 < 16 \cdot 15$.

Use the directed endpoints

$$\pi_- = 3.141592653589793 < \pi < 3.141592653589794 = \pi_+.$$

In each positive coefficient of Q replace π by π_+; in each negative coefficient replace it by π_-. This only increases Q. The exact checker expands $P(x) = K - Q(x)/x$, a polynomial of degree thirteen, on each of the64 intervals $[2j/(64K), 2(j+1)/(64K)]$. If $P(x) = \sum_{h=0}^{13} p_h x^h$, its Bernstein coefficients on $[a, a+w]$ are

$$b_k = \sum_{i=0}^{k} \frac{\binom{k}{i}}{\binom{13}{i}} \sum_{h=i}^{13} p_h \binom{h}{i} a^{h-i} w^i, \qquad k = 0, \dots, 13.$$

All896 rational coefficients are positive; the smallest is greater than .00012312. Thus $P > 0$ on the full interval. The value at zero follows by continuity (or directly).

For completeness the stated π endpoints are checked using alternating series for $\arctan(1/5)$ (32 terms) and $\arctan(1/239)$ (12 terms). The tangent addition formula gives $\tan(4\arctan(1/5) - \arctan(1/239)) = 1$. The angle lies between zero and $4/5 < \pi/2$, so it equals $\pi/4$. The positivity follows already from $4(1/5 - 1/(3 \cdot 5^3)) - 1/239 > 0$; the upper bound follows from $\arctan z < z$. This proves Machin's identity on the correct branch. The full rational recipe is `check_harmonic4553.py`; no sampled positivity is used. □

If the original characteristic-function argument gives $|\mathbb{E}e^{2\pi iW}| \le e^{-rt}$, Lemma Y.3 upgrades its fractional part estimate from $(5/23)(1 - e^{-rt})$ to

$$\frac{1000}{4553}(1 - e^{-rt}).$$

Consequently every occurrence of $5/23$ and $5/46$ in that harmonic gain is replaced by $1000/4553$ and $500/4553$, respectively. The existing degree-six Taylor gain and its monotonicity argument remain unchanged.

Y.4 The Gaussian derivative knows the sign of the threshold

Lemma Y.4 (Positive phase, with a fixed actual count)**.** *Assume $u \in [a, b] \subset [0, 1/2]$, $H/\ell \le h_*$, $|\delta| \le D$, $v \ge v_- > 0$, $\lambda \le U$, and the Stirling remainder bound above. Set*

$$G = \max\left(0, \max_{z \in [a,b]} (2h_* z - z^2)\right), \qquad \eta = \frac{U + G}{2v_-} + Z_N(D).$$

Then $f(u) \ge u(1 - \eta/N)$ for every $n \ge N$. The inner maximum is attained at the projection of h_* onto $[a, b]$. If $H \le 0$, one may take $h_* = 0$, hence $G = 0$.

Proof. For a fixed actual law and threshold, differentiate while keeping $\delta = k - np$ fixed:

$$\log f'(t) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta + t)^2}{2n(v + s)} - \zeta.$$

For $0 \le t \le u$, use $\log(1 + s/v) \le s/v$ and $v + s \ge v$ to obtain

$$\log f'(t) \ge -\frac{\lambda + 2\delta t + t^2}{2nv} - \zeta.$$

The convex function $t \mapsto 2\delta t + t^2$ attains its maximum on $[0, u]$ at an endpoint. Since the actual relation is $\delta = H/\ell - u$,

$$2\delta t + t^2 \le \max(0, 2(H/\ell)u - u^2) \le G.$$

Thus $\log f'(t) \ge -\eta/n$, and $f'(t) \ge e^{-\eta/n} \ge 1 - \eta/N$. Integrate from zero to the actual u. This argument does not change δ when t varies and does not reflect the distribution. □

In the code the signed H interval supplies the bound D. If $H \ge 0$, one can use $h_* = H_+/\ell_-$ and $D = \max(b, h_* - a)$; if $H \le 0$, use $D = H_-/\ell_- + b$. The new lower derivative constant is maximized with the older valid constant. Every resulting bound is checked positive.

The normal variance and cubic-moment repair is localized separately. On a phase cell, use its actual upper bound for δ_+, rather than $|\delta|$. If the resulting coefficient is κ, replace it by $\max(0, \kappa)$ before using $\kappa/n \le \kappa/N$. That sign condition is essential for an all-index conclusion.

Y.5 Coverage and a conservative cache

The selected resource table has 47 probability cells covering $[.38, .474]$. Zero noise is handled by the Bernoulli theorem. TINY covers all phases for $0 < \lambda \le \min(.05, \Lambda)$. Above .05, the LOW phase radii are $.09, .14, .19, .29, .5$ on noise bands separated by $.06, .08, .10, .12$. MID covers the complementary positive and negative phase intervals through .12. All intervals are closed at shared boundaries and all actual-event conventions are inherited from the exact count identity.

The LOW implementation caches a curvature bound at fixed probability, H and phase intervals. The cache key in the selected version omits c. This is conservative here: noise bands are traversed in increasing order, so Lemma Y.1 makes $c = 1 - d_L$ increase. Each denominator in the cached terms $(c - u)^{-2}$ and $(c + \min(u, c/3))^{-2}$ increases with c on the checked positive domain. The numerators are nonnegative and independent of c. Thus an older cache entry remains an upper bound. Other c-dependent quantities are recomputed in each noise band.

Each table is applied after acquisition has supplied its joint resource inequality and amplitude profile for the same marginal. The complete composition checks these inputs as well as the closed table coverage.

Z The signed Gaussian correction in nearest-count comparison

Let $n \ge N$, $v = p(1-p) > 0$, $s \ge 0$, $\lambda = ns$, and let the actual nearest count and phase satisfy

$$\delta = k - np = h - u, \qquad f(t) = \frac{\Phi((\delta + t)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

The actual k and δ are held fixed when t varies. Assume the proved finite Stirling formula on this count range is

$$\sqrt{n}\,\pi_k = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + \zeta\}, \qquad n|\zeta| \le Z_N(D), \quad |\delta| \le D.$$

Write $v \ge v_- > 0$ and $Z = Z_N(D) \ge 0$.

Lemma Z.1 (Negative phase)**.** *Suppose $u = -r$, $0 \le a \le r \le b$, and $h \le h_*$. Define*

$$t_* = \max(0, \min(b, h_* + b)), \qquad G_- = 2(h_* + b)t_* - t_*^2, \qquad \eta_- = G_-/(2v_-) + Z.$$

If $D^2 \le Nv_-$ and $\eta_- < N$, then

$$f'(-t) \le \frac{1}{1 - \eta_-/N} \quad (0 \le t \le r), \qquad u - f(u) \le \frac{b\eta_-}{n(1 - \eta_-/N)}.$$

Proof. Direct differentiation and the Stirling formula give

$$\log f'(-t) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2 s}{2nv(v+s)} + \frac{2\delta t - t^2}{2n(v+s)} - \zeta.$$

Since $\log(1+x) \ge x/(1+x)$ for $x \ge 0$, the first two terms are at most

$$\frac{s}{2(v+s)}\left(\frac{\delta^2}{nv} - 1\right) \le 0.$$

This cancellation removes the variance term from the upper derivative bound. Because $\delta = h + r$, $r \le b$ and $0 \le t \le r \le b$,

$$2\delta t - t^2 \le 2(h_* + b)t - t^2 \le G_-.$$

The last maximum is the concave quadratic maximum on $[0, b]$, achieved at the displayed projection t_*. It is nonnegative because $t = 0$ is available. Therefore $\log f'(-t) \le \eta_-/n$. Using $e^x \le (1 - x)^{-1}$ for $0 \le x < 1$ gives

$$f'(-t) - 1 \le \frac{\eta_-/n}{1 - \eta_-/n} \le \frac{\eta_-}{n(1 - \eta_-/N)}.$$

Integrate $u - f(u) = \int_0^r (f'(-t) - 1)\, dt$. □

Equivalently, $G_- = b^2 + 2h_*b$ for $h_* \ge 0$, and $G_- = \max(0, h_* + b)^2$ for $h_* < 0$. This sharper vertex is used in the selected LEFT42 LOW implementation.

Lemma Z.2 (Positive phase)**.** *Suppose $u \in [a, b] \subset [0, 1/2]$, $h \le h_*$, and define*

$$G_+ = \max\left(0, \max_{z \in [a,b]} (2h_*z - z^2)\right).$$

Then

$$u - f(u) \le \frac{b}{n}\left(\frac{\lambda}{2v_-} + \frac{G_+}{2v_-} + Z\right).$$

The inner maximum is at the projection of h_* onto $[a, b]$.

Proof. The exact derivative identity and $\log(1 + s/v) \le s/v$ give

$$\log f'(t) \ge -\frac{\lambda + 2\delta t + t^2}{2nv} - \zeta.$$

For fixed actual u, the function $2\delta t + t^2$ is convex on $[0, u]$. Its maximum is at an endpoint, and $\delta = h - u$ gives

$$2\delta t + t^2 \le \max(0, 2hu - u^2) \le G_+.$$

Use $e^{-x} \ge 1 - x$ and integrate from zero to u. □

Whole-count insertion and interval coverage.

The whole-count identity is split as $Q/\pi_k - f(u) = (Q/\pi_k - u) + (u - f(u))$. Thus one needs an upper bound on the signed correction $u - f(u)$; an absolute-value estimate is unnecessary. After division by λ, the new coefficients replacing the old Gaussian correction are

$$\begin{aligned} u \ge 0: &\quad \frac{b}{2v_-} + \frac{b\{G_+/(2v_-) + Z\}}{\lambda}, \\ u \le 0: &\quad \frac{b\eta_-}{(1 - \eta_-/N)\lambda}. \end{aligned}$$

On each fixed phase cell these have the form $A + B/\lambda$ with $A, B \ge 0$. They therefore preserve the original convex endpoint check on each closed noise interval. Every phase cell is confined to one half-line; zero is a shared closed endpoint. The count displacement bound D comes from the same actual $h - u$ cell, and the Stirling constants continue to come from the already proved global count segment.

The other all-index steps are unchanged: the event bracket is checked negative before substituting a binomial mass lower bound, and the entire inverse-n correction is checked

positive before replacing n by N. The inverse correction is $I(\lambda) = K + A + B/\lambda$, hence decreases with λ; positivity at both endpoints (in particular the upper endpoint) licenses the all-index step throughout the interval. The final score is convex in λ: its event fraction is a positive constant divided by an affine positive function, its B/λ coefficient is nonnegative, and its remaining terms are affine or constant. These are independent gates in the exact kernel. The formulas apply to both signs of h without reflecting the law or changing the count during integration. They also supply an improved upper derivative for the negative-phase MID inequalities.

Full-phase MID below the whole-count threshold.

For $\lambda \in [.05, .08]$, the selected shifted-quartic MID comparisons are checked on every closed phase interval of length .001 in $[0, 1/2]$, for both phase signs. The shifted-quartic polynomial and nearest-count identity are defined at phase zero, so the two closed covers include $u = 0$ directly. This permits the whole-count LOW comparison to start at $\lambda = .08$. The four MID bands use phase starts $0, 0, .19, .27$ at noise bands $[.05, .06]$, $[.06, .08]$, $[.08, .10]$, and $[.10, .12]$, respectively.

AA A larger domain for paired Fourier cancellation

The signed Gaussian transfer lemma in the cutoff49 formula atlas already allows $\mu_4 \le 11.1$ and $|\gamma| \le 1.85$. Its specialized positive coefficient rectangle was used only for $\mu_4 \le 2$, $|\gamma| \le 1$. The following extension changes that specialization. Bounded support and all numerical resources remain hypotheses to be acquired separately.

Proposition AA.1 (The actual absolute third moment)**.** *Suppose $0 < l \le p \le u < 1/2$, $0 \le \theta \le \Theta < 1$, and the actual conditional cubic identity gives*

$$\rho = \mathbb{E}|X|^3 \ge (1-\theta)^{3/2}\rho_B(p) + 3a\theta, \qquad a \ge 0, \quad \rho_B(p) = \frac{1 - 2p(1-p)}{\sqrt{p(1-p)}}.$$

Here a can be the minimum of the two acquired lower support endpoints. With outward bounds $r_- \le \rho_B(u)$ and $r_+ \ge \rho_B(l)$, a valid lower bound is

$$r_* := \max\{1, r_- - (3r_+/2 - 3a)_+\Theta\} \le \rho.$$

Proof. The function ρ_B decreases on $(0, 1/2)$ and convexity gives $(1-\theta)^{3/2} \ge 1 - 3\theta/2$. Therefore

$$\rho \ge \rho_B(p) - (3\rho_B(p)/2 - 3a)\theta \ge r_- - (3r_+/2 - 3a)_+\Theta.$$

The bound $\rho \ge 1$ follows from $\mathbb{E}X^2 = 1$ and monotonicity of the absolute moment norms. No separate requirement $3a \ge 3r_+/2$ is needed. □

Proposition AA.2 (Positive cancellation rectangle beyond $M = 2$)**.** *Let $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $|X| \le R$ almost surely, and let*

$$1 \le \mu_4 \le M < 3, \qquad 0 \le \gamma_l \le |\gamma| \le g, \qquad g^2 < 2, \quad R > 2g, \quad M - 1 - \gamma_l^2 \ge 0.$$

The paired formulas of the formula atlas remain valid, with exactly one change to their damping constant:

$$b_0 = \frac{45 - M}{96}(1 - 2/N), \qquad N > 2.$$

In particular, after simultaneous reflection of both odd Fourier coefficients, the valid rectangle for $a_0 = (3-\mu_4)/24$, $z = (10\gamma - \mathbb{E}X^5)/120$, $c = a_0\gamma/6$ *is*

$$a_* = (2-\gamma_l^2)/24, \qquad c_l = (3-M)\gamma_l/144, \quad c_u = (2-\gamma_l^2)g/144,$$
$$t_* = \min\{g, (10-2M)/(2(R-g))\},$$
$$120Z = \min\{E_5 + 10g, R(M-1) + (10-2M)t_* - (R-g)t_*^2\},$$
$$z_l = \max\{-Z, [8\gamma_l - \gamma_l^3 - (R+2\gamma_l)(M-1-\gamma_l^2)]/120\}, \quad z_u = Z,$$

where $E_5 \geq \mathbb{E}|X|^5$. *If* $3g^2 \leq 2$, c_u *may also be decreased to* $\min(c_u, (2g-g^3)/144)$. *The same sixth-moment bound remains valid:*

$$M_6 = \min\{RE_5, 1 + 3g^2 + g^4 + (R^2 + 2Rg + 3g^2 + 2)(M - 1 - \gamma_l^2)\}.$$

Proof. Pearson's identity gives $\mu_4 \geq 1 + \gamma^2$, so $0 \leq a_0 \leq a_*$. Both stated bounds on c follow by multiplying nonnegative quantities. The optional refinement uses the monotonicity of $2t - t^3$ on $[0, g]$ when $3g^2 \leq 2$.

Set $Q = X^2 - \gamma X - 1$. The exact identities

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \qquad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)]$$

give the upper absolute fifth-coefficient bound by maximizing

$$R(M-1) + (10-2M)t - (R-g)t^2, \qquad 0 \leq t \leq g.$$

The larger of the two Pearson sign bounds is still this one: $M < 3 < 5$, and $R > 2g$ permits replacing μ_4 by its upper bound. The quadratic has its maximum at the displayed clipped vertex. The numerator in the lower bound for z has derivative $10 - 2M + 2Rt + 3t^2 > 0$, so its lower endpoint is obtained at γ_l. The sixth-moment identity

$$\mu_6 = 1 + 3\gamma^2 + \gamma^4 + \mathbb{E}[Q^2\{(X+\gamma)^2 + 2\gamma^2 + 2\}]$$

gives the stated M_6.

The signed Gaussian transfer is applicable because $M < 3 < 11.1$ and $g < \sqrt{2} < 1.85$. Its norm is convex in (z, c), so the rectangle is checked at its four corners; all Gaussian and Mills estimates remain unchanged. Its damping hypothesis is $b_0 \leq (45 - M)(1 - 2/N)/96$. The older value $43(1-2/N)/96$ is unavailable when $M > 2$ and must be replaced as stated. □

A necessary implementation seam.

The same new r_* must be used in the four Fourier corners, the signed phase-curvature correction, all conserved budgets, and exported quadratic resources. The latter correction remains

$$q(C, \rho) = C\rho + \tfrac{1}{2}\min\{\phi_{\text{low}} g_0, .4g_0\}, \qquad g_0 = 3.16\rho - 3 - d.$$

Recomputing the old ρ_B lower endpoint in any of these locations would invalidate the rectangle association. The new adapter returns r_* as part of its paired result and the corrected callback uses that value.

The same lower bound improves signed fourth-moment errors

Let $V = \mathbb{E}(X \mid \text{sign class})$ and $W = X - V$. In the actual central-gap domain there is no zero atom. Write $\varepsilon = \operatorname{sgn} X$, $x = |X|$, $a = |V|$, $\beta = \mathbb{E}[X|X|]$, and

$$K_1(X) = \mathbb{E}[X|X|^3] - 3\beta - \tfrac{3}{2}\rho\gamma.$$

Conditional Taylor expansion gives the exact identity

$$K_1(X) - K_1(V) = \mathbb{E}W^2 \left\{ \varepsilon(w_4 - 3 - \tfrac{3}{2}\rho w_3) - \tfrac{3}{2}\gamma_V w_3 \right\},$$

where $w_3 = x + 2a$, $w_4 = x^2 + 2ax + 3a^2$ and $\gamma_V = \mathbb{E}V^3$. For the product term use $\rho\gamma - \rho_V\gamma_V = \rho(\gamma - \gamma_V) + \gamma_V(\rho - \rho_V)$. Every other term follows by expanding the signed or unsigned second, third and fourth power about the conditional mean; the linear remainder has conditional expectation zero.

Define

$$F(x, a; t) = x^2 + 2ax + 3a^2 - 3 - \tfrac{3}{2}t(x + 2a).$$

The positive-class coefficient is $F(x, a; \rho+\gamma_V)$; the negative-class coefficient is $-F(x, a; \rho-\gamma_V)$. For $x, a > 0$, F decreases with t. Thus any proved lower bound $r_* \le \rho$ may be used in the upper endpoint for the positive coefficient and lower endpoint for the negative coefficient. The proof does not need the stronger assertion $\rho \ge \rho_B$.

On each actual conditional mean/support rectangle, F is convex with Hessian $\left(\begin{smallmatrix} 2 & 2 \\ 2 & 6 \end{smallmatrix}\right)$. Its maximum is at a corner. Its minimum is attained at a corner, at an edge stationary point, or at $(x, a) = (3t/8, 3t/8)$ if that point is inside. The edge points are $a = t/2 - x/3$ at fixed x and $x = 3t/4 - a$ at fixed a, clipped to their respective edges. These formulas allow negative t as well.

If L and U bound the two signed coefficients together, then

$$\Theta \min(0, L) \le K_1(X) - K_1(V) \le \Theta \max(0, U),$$

since $\mathbb{E}W^2 \le \Theta$. The enclosing p and conditional-mean intervals are the same ones used by the formula atlas. Its outer condition $1 - \Theta > 2/3$, which licenses the separate binary K monotonicity calculation, is retained. Intersect this signed bound with the existing unsigned estimate and reject an empty intersection.

The implementation in `paired_k_extension` uses the proved r_* lower endpoint in this signed rectangle calculation. It retains the stated moment domain, the outer condition $1 - \Theta > 2/3$, and all intersection checks. The acquisition step must supply these premises before the improved fourth-moment bound is used.

AB Certificates and independent reproduction

The analytical reductions and the finite checks together prove the bounds used in Theorem 16.1. This appendix describes the full proof's checking procedure and then gives the retained compact entry calculation in greater detail.

AB.1 How the full proof is divided into checks

The finite computations have two obligations. A numerical checker proves an inequality on its stated parameter region. The continuous reduction and its interface checks must also show that the region contains every law still possible at that stage of the argument. Both are needed: an accepted polynomial or integral bound cannot be used with a support radius, moment cap, or predecessor loss belonging to a different law.

The continuous arguments in the preceding sections supply attainment, contact, moment identities, Fourier inversion, and the reductions to these regions. The programs then evaluate the resulting finite inequalities and check their complete closed covers. Their output therefore remains modular: a numerical result may correctly say that it is conditional on contact and a sharp predecessor even when the paper has already proved those hypotheses. Such wording is not an additional assumption in the assembled theorem.

Part of the proof	What its complete numerical check establishes
Indices 1–3	The scalar and supported tensor-majorant inequalities, including the quantitative three-summand bound used at 4.
Index 4	All 260 acquired geometry classes and all five HIGH requests per class; the physical comparison is checked on each whole parent region.
Index 5	The initializer and resource families, 295 count forests, the separate signed corner, and the complementary Fourier regions.
Indices 6–39	Actual-index support and Fourier bounds, each finite acquisition and resource step, and all resulting count domains. The positive allowance at 39 is checked before applying the sharp comparison at 39.
Indices 40–42	Both probability branches of the weighted floor-40 calculation, with actual-index binomial probabilities and complete signed phase coverage.
The two local exclusions	The finite 42–48 family and the uniform family from 47, including noise arbitrarily close to zero. Uniformity in the index follows from the analytical bounds, not from testing finitely many large indices.
Strictness from 700	The retained compact entry and scalar checks, applied after the weak all-index bound supplies the sharp predecessor.

The public COUNT batch contains 1243 forests: 295 for 5, 284 for 6–9, 364 for 10–19, and 300 for 26–39. Other finite families have their own checks. These counts describe proof objects, not a set of sampled probability laws. A forest is accepted only after its whole root region, every subdivision and terminal inequality have been verified.

Inputs, discovery, and acceptance.

The reproduction package distinguishes static inputs from execution outputs. Static inputs include mathematical source files, explicit plans, and untrusted candidate certificates. A historical result appearing in a source archive is not accepted as a new execution. Where an original checker uses an older parent object as numerical input, the interface reader matches that object to a complete current parent calculation before using its conclusion. Hashes identify the object being compared; the original arithmetic checker and the domain argument establish its meaning.

The four-summand replay illustrates why finding a certificate and verifying one are different tasks. Its read-only mode first recomputes the complete mathematical object, then compares every nonresource field with the supplied candidate. The candidate's discovery process need not be trusted. The execution record identifies this complete replay separately from the interrupted producer; it does not invent a missing exit observation or claim two completed fresh passes.

Arithmetic and runtime.

The Python acceptance programs use integers and rational arithmetic. The three-summand C++ checker uses Boost.Multiprecision's unbounded integer type, `cpp_int`; its Python companion uses integers, exact fractions, and integer square roots. The Gaussian bounds are rational enclosures. This differs from using a large floating-point precision and treating the printed digits as exact. Some retained candidate-planning wrappers require NumPy, including the own-score-cap planner; complete exact acceptance follows that planning step. An optimizer's floating-point output is not an acceptance certificate.

Campaign records retain commands, runtime and input identities, top-level stdout and stderr, and actual return codes. Native traces also identify the compiled three-summand kernel and its 111 invocations; successful nested compiler and kernel streams are not all archived separately. Interrupted outer supervisors retain their missing observations explicitly. The tested C++ build used Apple clang and Boost 1.90 headers already installed on the host.

The current process supervisors are explicitly documented macOS adaptations; the numerical sources and acceptance targets are unchanged. A larger supervisor memory allowance is recorded as a different resource policy, not as a stronger mathematical result. Assertions must remain enabled: `-O`, `-OO`, and `PYTHONOPTIMIZE` are not permitted for Python acceptance.

Independent reproduction.

Start with a fresh physical workspace and verify the package's source manifest. Restore the foundation, five-summand and finite static inputs using their exact archive and manifest hashes. The finite restorer places the old scalar outputs in an explicitly named archive directory; the scalar preparer copies only the three source files to the canonical calculation directory. Its unchanged checker then constructs both outputs anew. The public COUNT candidates are installed separately from the hash-pinned archive described in `COUNT_CANDIDATES.md`.

The top-level reproduction index supplies the commands, their working directories, source pins, output names and prerequisite executions. Its component maps are the foundation `FULL_COMMANDS_V2.json`, finite `FULL_COMMANDS_V3.json`, scalar `SCALAR_KAPPA_FRESH_COMMANDS_V2.json`, five-summand `N5_FRESH_V3_DAG.json`, and bridge `EXECUTION_DAG.json`. The positive-39 and compact-entry commands are listed separately in the same index. Resolve workspace placeholders before changing the working directory. A new campaign uses one interpreter consistently for each producer and its reader; preserved older observations retain their actual original runtime identities.

Run the complete acceptance commands and their final interface readers. A preliminary file, a zero exit from a source-preparation command, or a printed success word is not a substitute for these readers. The supplied negative controls check omitted, duplicated, reordered, wrongly scoped and corrupted objects. They test rejection behavior; the mathematical proofs of coverage and soundness remain in the text. The source package, preserved execution evidence and instructions for a fresh execution are labelled separately so that a reader can audit one without mistaking it for another.

AB.2 The retained compact-entry proof object

The retained format and target are, respectively,

`FIXED_C_COARSE_ENTRY_V7`
`INITIAL42_FIXED_TARGET_COARSE_V7`.

The JSON object stores four lists of 24 moment-cell records and all 100 probability-panel records. A live cell records its support radius, remainder and defect ceilings, and active/excluded descendant counts; a previously excluded cell is marked accordingly. It does not store 11264 individual leaves. The checker reconstructs the subdivisions, exclusions, bounds, and panel incidences. The soundness argument is Proposition 5.3.

For example, the first signed pass records a checkpoint for cell index 0. The checker reconstructs its moment interval as $[1 + 3361i/96000, 1 + 3361(i+1)/96000]$ at index i; the JSON stores i, not these two endpoints. Its descendants and their enclosures must still be recomputed.

Quantity for ρ-cell 0	Value and provenance
Moment interval (reconstructed)	$[1, 99361/96000]$
Support radius (stored)	$5/2$
Remainder ceiling	$27071971/50000000 = 0.54143942$
Defect ceiling d	$18233099/100000000 = 0.18233099$
Active / excluded descendants	128 / 0

The last numerical bound means $nD \le 42d = 7.65790158$ for every actual $n \ge 42$ in the cell, rather than only for a law at index 42.

The compact-entry paths in the rest of this appendix are relative to the repository root. The retained source and data can be read at the immutable baseline commit linked in Section 17; they are unchanged by this revision.

Certificate
`mathematics/fixed-target-v7/2026-10-27/entry-proof.json`

Checker
`mathematics/fixed-target-v7/2026-10-27/accept_entry.py`

Source-isolated replay
`mathematics/fixed-target-v8/2026-10-28/isolated_entry_replay.py`

Selected exporter and exploratory producer
`mathematics/fixed-target-v7/2026-10-27/export_entry.py`, `mathematics/fixed-target-v7/2026-10-27/signed_probe.py`

Two-pass ancestor
`mathematics/fixed-target-v6/2026-10-27/signed-two-pass-check.py` and its pinned dependencies.

The exploratory producer considers candidate subdivisions, support radii, and passes. The exporter selects the first four stages of a six-stage discovery trace and rounds ceilings outward. Acceptance reconstructs the two coarse and four signed passes; it does not rely on a discovery success label. The selected path uses integer and rational arithmetic. Generation and checking share mathematical kernels, so this is not a claim of independent implementations.

AB.3 The acceptance procedure

The acceptance program carries out the following steps in a fixed order.

1. Check the format and target names, exact record fields, canonical rational strings, four ordered lists of 24 cells, and all 100 ordered closed panels, including their numerical target domains. Check source hashes before importing the numerical kernels.

2. Recompute the two coarse passes. Their 24 endpoint pairs must agree with $[1 + 3361i/96000, 1 + 3361(i + 1)/96000]$.

3. For each live cell in each signed pass, recompute the support radius and the $1 \times 8 \times 8 \times 2$ subdivision in (ρ, β, γ, m). Evaluate the exclusions and complete upper bounds on these descendants using the previous pass's ceilings.

4. Require the active and excluded counts to agree with the stored counts and to sum to 128. For a surviving cell, require each stored ceiling to enclose the recomputed value and to be no larger than the previous ceiling. A cell excluded in an earlier pass remains excluded.

5. Reconstruct every final probability-panel incidence, including shared endpoints. Check the stored incidence counts and require every stored panel ceiling to enclose the corresponding recomputed maximum.

6. Report the resulting maxima and run the rejection controls.

Individual descendant boxes are not archived in the object. Thus the comparison is between deterministic recomputation and stored counts and ceilings, not between two stored lists of discarded boxes. The mathematical reason every possible witness is covered is given in Appendix D and Proposition 5.3.

A guide to the numerical sources.

The following is a reading map for the retained implementation. Relative paths beginning with `v5/`, `v6/`, and `v7/` below abbreviate `mathematics/fixed-target-v5/2026-10-27/`, `mathematics/fixed-target-v6/2026-10-27/`, and `mathematics/fixed-target-v7/2026-10-27/`, respectively.

`v7/accept_entry.py`: `shape`, `verify`
Record validation, source pins, ordered replay and output comparisons.

`v6/signed-two-pass-check.py`: `one_pass`, `run`
The two coarse quadratic passes and the first central-gap test in Section D.5.

`v7/signed_support.py`: `support`, `cell_radius`
Support-radius gates from the contact and Gaussian heat estimates in Theorem D.1.

`v7/signed_core.py`: `stage`
Signed subdivisions, moment constraints, gap and remainder charges, ordered resource updates, and final panel incidences.

`v5/cell-sources/kernel.py`: `sixth`, `universal_high`, `exphi`
Low-frequency constants, the universal actual-event bound and directed exponential enclosures. The replacement high-frequency window routine is `install` in `v5/cell-sources/high_window_core.py`; it compares complete six-term bounds from Proposition A.3.

The dated correspondence reports in `manuscript/compact-v16/publication/certificate-audit-20261030/` derive the coarse coefficients, signed moment and gap bounds, and complete Fourier enclosures from the formulas in this paper. They check the implementation's signs, denominator cases, all-index gates and closed coverage. Separate rational evaluators reconstruct all 48 coarse updates and every Fourier parameter tuple encountered in the full replay. An import-free calculation checks the initial and cellwise support gates; another checks the sharper output constants and delicate endpoint cases. These calculations supplement the continuous proofs. They are not independent implementations of the entire argument.

AB.4 Exact arithmetic and its limits

Rationals are represented by integer numerators and denominators. Outward rounding deliberately weakens a bound. Square-root endpoints are checked by rational square comparisons. For a nonnegative numerator, an upper numerator bound divided by a positive lower denominator bound gives an upper quotient bound. Signed quotients require the endpoint extrema: if $a \in [a_-, a_+]$ and $0 < b_- \le b \le b_+$, then

$$\frac{a}{b} \in \left[\min_{i,j\in\{-,+\}} \frac{a_i}{b_j}, \max_{i,j\in\{-,+\}} \frac{a_i}{b_j} \right].$$

A denominator interval containing zero requires a separate argument; this rule does not permit division by it. The exponential helper uses, for rational $0 \le y \le \min(x, 64)$,

$$e^{-x} \le e^{-y} \le \left(\sum_{j=0}^{64} \frac{y^j}{j!} \right)^{-1},$$

followed by upward rounding. The positive exponential series proves this enclosure. The Gaussian constants, including the narrow interval for C_{E}, are proved in Appendix B.

Numerical precision determines whether an enclosure is sharp enough to prove a strict inequality; extra digits alone do not prove it. Uniformity in n comes from the analytical majorants in the Fourier appendix, not from sampling indices or assuming monotonicity of C_n. The audited checker semantics and the integer/rational implementation form the computational trust boundary. This is a computer-assisted proof using Python, not a formal proof checked by a proof assistant. Hashes identify bytes, not mathematical truth. Subdivision certificates have established methodological precedents [16]; that precedent does not validate this custom format or transfer a proof assistant's guarantee to Python.

AB.5 Three separate checks

Use a fresh output path for each replay. From the repository root, the retained entry is reproduced by

```
python3 -I -B \
  mathematics/fixed-target-v8/2026-10-28/isolated_entry_replay.py \
  --output-dir runs/compact-v16-entry-review
```

The wrapper copies a fixed 13-file source/data set into an isolated source environment. The wrapper status is `PASS_ISOLATED_REPRODUCTION_FIXED_ENTRY_V7`; its numerical output is `PASS_INITIAL42_FIXED_TARGET_COARSE_V7`. It reconstructs 11264 nominal signed boxes, 100 panels, 30 nonempty panels and 14 surviving moment cells. Nineteen checker rejection controls and five wrapper controls test malformed or altered inputs. They supplement rather than replace the coverage proof. A fresh replay on 30 October 2026, using Python 3.12.14 without optimization, returned zero in 26.94 seconds. Its command, input hashes, return code, stdout and stderr are preserved under `manuscript/compact-v16/publication/certificate-audit-20261030/entry/`. The earlier Python 3.9.6 replay is preserved separately. These are observed runtimes, not performance guarantees.

The acceptor's generic numerical limits are weaker than the maxima in Proposition D.5. Check the latter directly against the fixed accepted object with

```
python3 -I -B \
  manuscript/compact-v16/tools/check_entry_claims.py \
  mathematics/fixed-target-v7/2026-10-27/entry-proof.json
```

This verifies the tighter raw, weighted defect, beta, phase and support bounds, and obtains the exact rho endpoint from the surviving original rows. It supplements the full replay; by itself it only reads a claimed object. The proof object's SHA-256 is

`61b7ad43e20046d7cbf7157e3feb61696331731e7be418f6a5fa535c2eee8bae`.

The frozen wrapper still describes its output as conditional on analytic inputs: that is its modular scope. Here those inputs are supplied by Appendices A and D, with the correspondence and coverage checks just described. The resulting Propositions D.3 and D.5 are therefore proved statements. The Python standard library suffices for all commands in this appendix.

The bounded historical checks and the current document's reference structure are reproduced separately by

```
python3 -I -B \
  manuscript/compact-v16/tools/recheck.py \
  compact-v16-bounded-review
```

Here the output path is relative to `manuscript/compact-v16/`; an absolute fresh path with an existing parent can instead be supplied. This runs nine unchanged v16 mathematical

programs. They retain their original parameters, including the former small-noise minorant. They are regression evidence for the retained inputs, not verification of the revised scalar choices and not a replay of the coarse entry.

The changed scalar claims have their own exact checks:

```
python3 -I -B \
  manuscript/compact-v16/tools/check_revision_constants.py \
  --output-dir runs/compact-v16-scalar-review
```

This runs the geometry, count, and foundation checks. They reconstruct the rational comparisons from their definitions: the endpoint probability exclusion, six ray bounds, the new minorant's full rectangle argument, the common error envelope and its two quartic endpoints, the Fourier quadratic identity, and the narrow C_{E} enclosure. The continuous proofs of sign, concavity, monotonicity, and coverage remain in the text. Selected atomic-law examples are diagnostics, not an all-law proof. The revision record reports these executions separately from the inherited entry replay and from document compilation.

References

[1] Logan Bell. Extremizers for the independent Berry–Esseen inequality. Manuscript dated 2026-10-06; collection version 0.7.0; source snapshot f61bc638774bfa1165065462f98fe31efade30bf, 2026. URL https://github.com/epicycloids/independent-berry-esseen-bounds/blob/f61bc638774bfa1165065462f98fe31efade30bf/paper/extremizers/extremizers.pdf.

[2] Logan Bell. An upper bound on the Berry–Esseen constant for independent summands. Manuscript dated 2026-10-06; collection version 0.7.0; source snapshot f61bc638774bfa1165065462f98fe31efade30bf, 2026. URL https://github.com/epicycloids/independent-berry-esseen-bounds/blob/f61bc638774bfa1165065462f98fe31efade30bf/paper/independent-berry-esseen.pdf.

[3] Andrew C. Berry. The accuracy of the Gaussian approximation to the sum of independent variates. *Transactions of the American Mathematical Society*, 49(1):122–136, 1941. doi: 10.1090/S0002-9947-1941-0003498-3.

[4] Carl-Gustav Esseen. On the Liapunoff limit of error in the theory of probability. *Arkiv för Matematik, Astronomi och Fysik*, 28A(9):1–19, 1942.

[5] Carl-Gustav Esseen. Fourier analysis of distribution functions. A mathematical study of the Laplace–Gaussian law. *Acta Mathematica*, 77:1–125, 1945. doi: 10.1007/BF02392223.

[6] Carl-Gustav Esseen. A moment inequality with an application to the central limit theorem. *Scandinavian Actuarial Journal*, 1956(2):160–170, 1956. doi: 10.1080/03461238.1956.10414946.

[7] Hengzhi He and Guang Cheng. The Berry–Esseen constant conjecture is eventually true, 2026. URL https://arxiv.org/abs/2609.06358v3. Version 3, revised 30 September 2026.

[8] Bartosz Kołodziejek and Kacper Rodziewicz. The sharp iid Berry–Esseen inequality: Analytic reductions and exact certificate proof. Hexagon preprint hexagon:2610.00590v1, 2026. URL https://hexagonmath.org/2610.00590v1. Version 1, posted 29 October 2026; manuscript dated 27 October 2026; source commit 3d053204f30eab39da39bfb7b85e380d83ec0646.

[9] Yongxi Lin. Berry–Esseen bounds in Lean. GitHub repository, 2026. URL https://github.com/CoolRmal/BerryEsseen. Accompanying proof document: The i.i.d. Berry–Esseen constant is at most 0.423: a machine-checked proof, 13 September 2026, paper/upper-0423.pdf. Commit bfb22608ad18b4fa9fda8dd9a1ed2c2b6fbc29e7; accessed 27 October 2026.

[10] Håkan Prawitz. Limits for a distribution, if the characteristic function is given in a finite domain. *Scandinavian Actuarial Journal*, 1972(2):138–154, 1972. doi: 10.1080/03461238.1972.10404645.

[11] Kacper Rodziewicz and Bartosz Kołodziejek. Moment defects and sharp iid Berry–Esseen bounds. Hexagon preprint hexagon:2610.00147v3, 2026. URL https://hexagonmath.org/2610.00147v3. Working draft, version 3, 17 October 2026; accessed 27 October 2026.

[12] Jona Schulz. *The optimal Berry–Esseen constant in the binomial case*. PhD thesis, Universität Trier, 2016. URL https://ubt.opus.hbz-nrw.de/frontdoor/index/index/docId/732.

[13] Irina Shevtsova. On the absolute constants in the Berry–Esseen type inequalities for identically distributed summands, 2011. URL https://arxiv.org/abs/1111.6554v1.

[14] Irina Shevtsova. On the accuracy of the approximation of the complex exponent by the first terms of its Taylor expansion with applications. *Journal of Mathematical Analysis and Applications*, 418(1):185–210, 2014. doi: 10.1016/j.jmaa.2014.03.075. URL https://arxiv.org/abs/1301.2783v3.

[15] Irina G. Shevtsova. On the absolute constants in the Berry–Esseen inequality and its structural and nonuniform improvements. *Informatics and its Applications*, 7(1):124–125, 2013. URL https://www.mathnet.ru/eng/ia252. In Russian.

[16] Alexey Solovyev and Thomas C. Hales. Formal verification of nonlinear inequalities with Taylor interval approximations. In *NASA Formal Methods*, volume 7871 of *Lecture Notes in Computer Science*, pages 383–397, 2013. doi: 10.1007/978-3-642-38088-4_26.

[17] I. S. Tyurin. Refinement of the upper bounds of the constants in Lyapunov's theorem. *Russian Mathematical Surveys*, 65(3):586–588, 2010. doi: 10.1070/RM2010v065n03ABEH004688.

[18] Paul van Beek. An application of Fourier methods to the problem of sharpening the Berry–Esseen inequality. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 23: 187–196, 1972. doi: 10.1007/BF00536558.

[19] Haonan Xiao and Chunlin Li. A 0.4395 upper bound for the i.i.d. Berry–Esseen constant. GitHub repository, 2026. URL https://github.com/haonan-xiao/iid-berry-esseen. Accompanying paper: paper/berry-esseen-04395.pdf. Commit f00d781c8c42908bf4aef2739129442a0919fa38; accessed 27 October 2026.

[20] Vladimir M. Zolotarev. An absolute estimate of the remainder term in the central limit theorem. *Theory of Probability and its Applications*, 11(1):95–105, 1966. doi: 10.1137/1111005.

[21] Vladimir M. Zolotarev. A sharpening of the inequality of Berry–Esseen. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 8:332–342, 1967. doi: 10.1007/BF00531598.